

Mathematical Methods for Physics: Group and Representation Theories

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ABSTRACT: These are the lecture notes for the course “Mathematical Methods for Physics” at the University of Milano-Bicocca. They offer a self-contained introduction to group theory and representation theory for physicists, organised into four main parts:

1. An introduction to groups.
2. The representation theory of finite groups.
3. Matrix Lie groups, Lie algebras, and their representations.
4. The general theory of semisimple Lie algebras.

Applications to quantum mechanics and particle physics are developed throughout. The latest version of the notes can always be downloaded from

<https://nmekareeya.github.io/grouptheory>

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1 Introduction

These lecture notes have been prepared for the course “Mathematical Methods for Physics” at the University of Milano-Bicocca. They are organised into four main parts:

1. An introduction to groups.
2. The representation theory of finite groups.
3. Matrix Lie groups, Lie algebras, and their representations.
4. The general theory of semisimple Lie algebras.

The notes are designed to be suitable for **self-study**. The material is developed in a self-contained manner: complete proofs are given whenever they are of reasonable length, and precise references are provided for the few results whose proofs would take us too far afield. Worked examples — above all the symmetric groups, $SU(2)$ and $\mathfrak{sl}(2, \mathbb{C})$, and $SU(3)$ and $\mathfrak{sl}(3, \mathbb{C})$ — run throughout the text and anchor each new abstract concept.

Three appendices supplement the main text. Appendix [A](#) works out further character tables (the quaternion group, D_5 , A_4 , and the general dihedral groups D_n); Appendix [B](#) develops the criterion for reality and pseudoreality of irreducible representations; and Appendix [C](#) contains the detailed proofs behind the theorem of the highest weight for $\mathfrak{sl}(3, \mathbb{C})$. A dictionary summarising the general structures of semisimple Lie algebras, together with their concrete realisations for $\mathfrak{sl}(3, \mathbb{C})$, is given at the end of Section [10](#).

These notes are intended for pedagogical purposes and should be used to supplement, not replace, the primary references. The content presented here is based heavily on the cited sources, to which all credit for the original material is due.

References. The primary references for each part are as follows: Part [1](#) draws from [\[1–3\]](#); Part [2](#) from [\[4, 5\]](#); and Parts [3](#) and [4](#) from [\[4, 6\]](#). Applications to quantum mechanics are drawn from [\[7\]](#). Readers are also referred to Appendix B and Sections VII.5 and VII.7 of [\[8\]](#) for more applications in quantum field theory, and to [\[9\]](#) as a very useful reference for representations of Lie algebras.

Remarks. Please consider all material in these lecture notes to be examinable, unless I explicitly say otherwise during a lecture. We may not have time to cover every proof and example in class, but the notes are written so that any omitted material can be studied on your own. I will be updating these notes regularly, so I encourage you to check often for the most up-to-date version.

2 Groups and subgroups

Definition 2.1. A pair $(G, *)$ is a **group** if G is a set and $*$ is a binary operation satisfying the following axioms:

- (i) If $a, b \in G$, then $a * b \in G$. (Closure)
- (ii) If $a, b, c \in G$, then $(a * b) * c = a * (b * c)$. (Associativity)
- (iii) There exists an element $e \in G$ such that $e * a = a * e = a$ for all $a \in G$. (Identity)
- (iv) For each $a \in G$, there exists an element $a^{-1} \in G$ such that $a^{-1} * a = a * a^{-1} = e$. (Inverse)

Example 2.2. (i) The associativity axiom means that expressions can be bracketed in any way. Ordinary subtraction is not associative, since $(3 - 4) - 5 = -6 \neq 4 = 3 - (4 - 5)$. Thus, $(\mathbb{Z}, -)$ is not a group. (ii) $(\mathbb{Z}, +)$ is a group. (iii) $(\mathbb{R}^n, +)$ with vector addition is a group.

Proposition 2.3. If $(G, *)$ is a group and $a, b \in G$, then $(a * b)^{-1} = b^{-1} * a^{-1}$.

Proof. We verify that $b^{-1} * a^{-1}$ is the inverse of $a * b$. Using associativity, we have:

$$(b^{-1} * a^{-1}) * (a * b) = b^{-1} * (a^{-1} * a) * b = b^{-1} * e * b = b^{-1} * b = e. \quad (2.1)$$

A similar calculation shows that $(a * b) * (b^{-1} * a^{-1}) = e$. □

Definition 2.4. Let $(G, *)$ be a group. A non-empty subset $H \subseteq G$ is a **subgroup** of G if:

- (i) for all $x, y \in H$, their product $x * y$ is also in H , and
- (ii) for every $x \in H$, its inverse x^{-1} is also in H .

Proposition 2.5. If $(H, *)$ is a subgroup of $(G, *)$, then $(H, *)$ is itself a group.

Proof. The associative law holds for all elements of G , so it automatically holds for all elements of H . Condition (i) ensures that H is closed under the group operation. Since H is non-empty, it contains some element x . By (ii), x^{-1} is in H , and by (i), the identity element $e = x * x^{-1}$ must be in H . Finally, condition (ii) guarantees that every element in H has its inverse in H . □

Definition 2.6. Let G be a group. If the set G is finite, the **order** of G , denoted $|G|$, is the number of elements in G . If G is an infinite set, it is said to have **infinite order**.

2.1 Some examples of groups

A fundamental class of groups is the cyclic groups.

- The set $\mathbb{Z}_n = \{0, 1, 2, \dots, n-1\}$ under addition modulo n forms a group, $(\mathbb{Z}_n, +)$. Closure and associativity are clear. The identity element is 0, and the inverse of any $k \in \mathbb{Z}_n$ is $n-k \pmod{n}$. This is the **cyclic group** of order n .
- Equivalently, the cyclic group of order n , denoted C_n , can be written as the set $\{1, x, x^2, \dots, x^{n-1}\}$ under multiplication, where $x^n = 1$.

This leads to a formal definition.

Definition 2.7. A group G is **cyclic**, or a **cyclic group** (with an operation $*$) if there exists an element $x \in G$ such that every element of G is of the form x^r for some integer r , where $x^r = x * x * \dots * x$ (r times) for $r > 0$, $x^0 = e$, and $x^{-r} = (x^{-1})^r$. We say that G is **generated** by x .

Notation. Henceforth, we will often refer to a group simply as G instead of $(G, *)$, and write ab for the product $a * b$. The identity element may be denoted by $e, I, 1, \mathbf{1}$, or 0 , depending on the operation and the context.

Definition 2.8. Let g be an element of a group G . We say g has **finite order** if $g^m = e$ for some integer $m \geq 1$. The **order** of g is the smallest positive integer k such that $g^k = e$. If no such positive integer exists, g has **infinite order**.

Example 2.9. (i) Under multiplication, C_4 can be represented as $\{1, i, -1, -i\}$. The elements $1, i, -1, -i$ have orders $1, 4, 2, 4$, respectively.

(ii) A cyclic group of order n always contains an element of order n (a generator).

The converse of (ii) is worth recording, as we will use it repeatedly.

Lemma 2.10. Let G be a finite group of order n . If G contains an element g of order n , then G is cyclic and generated by g ; it can therefore be identified with C_n .

Proof. The powers $e, g, g^2, \dots, g^{n-1}$ are distinct: if $g^i = g^j$ with $0 \leq i < j \leq n-1$, then $g^{j-i} = e$ with $1 \leq j-i \leq n-1$, contradicting the fact that the order of g is n . These are therefore n distinct elements of the n -element set G , so $G = \{e, g, g^2, \dots, g^{n-1}\}$. $\square$

Given two groups, we can construct their direct product.

Example 2.11. (Product group.) Let $(G, \star)$ and $(H, *)$ be groups. The product group $(G \times H, \cdot)$ is the set of ordered pairs $\{(g, h) | g \in G, h \in H\}$, with the operation defined component-wise:

$$(g_1, h_1) \cdot (g_2, h_2) = (g_1 \star g_2, h_1 * h_2). \quad (2.2)$$

It is straightforward to verify that this construction yields a group. The order of the product group is $|G \times H| = |G||H|$.

Example 2.12. (i) Consider the cyclic groups $C_2 = (\{1, -1\}, \times)$ and $C_3 = (\{1, \omega, \omega^2\}, \times)$, where $\omega = e^{2\pi i/3}$. The product group $C_2 \times C_3$ contains six distinct elements:

$$(1, 1), (1, \omega), (1, \omega^2), (-1, 1), (-1, \omega), (-1, \omega^2). \quad (2.3)$$

The group $C_2 \times C_3$ is itself cyclic, generated by $x = (-1, \omega)$. The elements above correspond to $x^6, x^4, x^2, x^3, x, x^5$, respectively. Thus, $C_2 \times C_3$ can be identified with C_6 .

(ii) The group $C_2 \times C_4$ has order 8. However, it is not cyclic because it contains no element of order 8. For any element $(g, h) \in C_2 \times C_4$, we have $(g, h)^4 = (g^4, h^4) = (1, 1)$, since $g^2 = 1$ and $h^4 = 1$. Thus, every element has order at most four.

Example 2.13. (Matrix groups.) Let $\mathbb{F}$ be $\mathbb{R}$ or $\mathbb{C}$. The following sets of $n \times n$ matrices with entries in $\mathbb{F}$ form groups under matrix multiplication.

- **General linear group:** $\text{GL}(n, \mathbb{F}) = \{A \in \text{Mat}_n(\mathbb{F}) | \det A \neq 0\}$.
- **Special linear group:** $\text{SL}(n, \mathbb{F}) = \{A \in \text{Mat}_n(\mathbb{F}) | \det A = 1\}$.
- **Orthogonal group:** $\text{O}(n) = \text{O}(n, \mathbb{R}) = \{A \in \text{GL}(n, \mathbb{R}) | A^T A = \mathbf{1}\}$.
- **Special orthogonal group:** $\text{SO}(n) = \text{SO}(n, \mathbb{R}) = \{A \in \text{GL}(n, \mathbb{R}) | A^T A = \mathbf{1}, \det A = 1\}$.
- **Unitary group:** $\text{U}(n) = \{A \in \text{GL}(n, \mathbb{C}) | A^\dagger A = \mathbf{1}\}$, where $\dagger$ denotes the conjugate transpose.
- **Special unitary group:** $\text{SU}(n) = \{A \in \text{GL}(n, \mathbb{C}) | A^\dagger A = \mathbf{1}, \det A = 1\}$.

Matrix groups such as $\text{GL}(n, \mathbb{R})$ for $n \geq 2$ are typically non-commutative.

Definition 2.14. A group G is **abelian** (or **commutative**) if $gh = hg$ for all $g, h \in G$.

Example 2.15. (i) $(\mathbb{Z}, +)$, $(\mathbb{R}^n, +)$, and all cyclic groups are abelian. (ii) $\text{O}(2)$ and $\text{SO}(3)$ are not abelian. (iii) If G and H are abelian, their product $G \times H$ is also abelian.

Points (i) and (iii) are straightforward. For (ii):

- For $O(2)$, consider a reflection about the x -axis and a rotation about the origin:

$$\sigma = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}, \quad R(\theta) = \begin{pmatrix} \cos \theta & -\sin \theta \\ \sin \theta & \cos \theta \end{pmatrix}. \quad (2.4)$$

A direct calculation shows that $\sigma R(\theta) = R(-\theta)\sigma$. For general θ , this is not equal to $R(\theta)\sigma$, so $O(2)$ is not abelian.

- For $SO(3)$, consider rotations about the z - and x -axes:

$$R_z(\theta) = \begin{pmatrix} \cos \theta & -\sin \theta & 0 \\ \sin \theta & \cos \theta & 0 \\ 0 & 0 & 1 \end{pmatrix}, \quad R_x(\phi) = \begin{pmatrix} 1 & 0 & 0 \\ 0 & \cos \phi & -\sin \phi \\ 0 & \sin \phi & \cos \phi \end{pmatrix}. \quad (2.5)$$

In general, $R_z(\theta)R_x(\phi)$ is not equal to $R_x(\phi)R_z(\theta)$.

We conclude this section with a theorem relating the order of a group to the orders of its elements.

Theorem 2.16. *Any group G of even order contains an element of order two.*¹

Proof. Pair each element of G with its inverse. The elements satisfying $g \neq g^{-1}$ are thereby partitioned into two-element sets $\{g, g^{-1}\}$, so there is an even number of them. Hence the number of elements satisfying $g = g^{-1}$, i.e., $g^2 = e$, is even as well. This set contains the identity, so it must contain at least one further element $g \neq e$; any such g has order two. $\square$

2.2 Permutation groups

We now discuss another important class of groups, those formed by permutations of a set.

Definition 2.17. *A **permutation** of a set X is an invertible map (a bijection) $f : X \rightarrow X$. The set of all permutations of X is denoted $P(X)$.*

Theorem 2.18. *The set $P(X)$ of permutations of a non-empty set X forms a group under the operation of function composition.*

Proof. The composition of two bijections is a bijection, so $P(X)$ is closed under composition. Composition of maps is associative — the one axiom worth checking explicitly: for any $x \in X$, $(h(gf))(x) = h(g(f(x))) = ((hg)f)(x)$, so $h(gf) = (hg)f$. The identity map $I(x) = x$ is the identity element and, since each $f \in P(X)$ is invertible, the inverse function f^{-1} is a permutation satisfying $ff^{-1} = f^{-1}f = I$. $\square$

Definition 2.19. *The **symmetric group** S_n is the group of permutations of the set $\{1, 2, \dots, n\}$.*

¹This is the special case $p = 2$ of *Cauchy's theorem*: if a prime p divides $|G|$, then G contains an element of order p . We will not need the general case in this course.

2.2.1 Permutation cycles

A particularly important type of permutation is a cycle.

Definition 2.20. A **cycle** $(n_1 \ n_2 \ \cdots \ n_q)$ is a permutation that maps $n_i \rightarrow n_{i+1}$ for $i < q$ and $n_q \rightarrow n_1$, while fixing all other elements $x \notin \{n_1, n_2, \dots, n_q\}$. The integer q is the **length** of the cycle, and the permutation is called a **q -cycle**.

Example 2.21. (i) The cycle $(1 \ 3 \ 5)$ is the permutation that maps $1 \rightarrow 3$, $3 \rightarrow 5$, and $5 \rightarrow 1$.

(ii) The permutation $(1 \ 2 \ 4)(3)$ fixes the element 3. By convention, 1-cycles representing fixed points are often omitted, so this is written simply as $(1 \ 2 \ 4)$.

(iii) **Permutation multiplication is performed right to left.** Consider $\rho_1 = (1 \ 3 \ 7 \ 4)(2 \ 5)(6 \ 8)$ and $\rho_2 = (1 \ 2 \ 5)(3 \ 6 \ 7 \ 4 \ 8)$ in S_8 . To compute $\rho_2\rho_1$, we trace the path of each element, applying ρ_1 first, then ρ_2 . For example, $1 \xrightarrow{\rho_1} 3 \xrightarrow{\rho_2} 6$. Continuing this process gives:

$$\rho_2\rho_1 = (1 \ 6 \ 3 \ 4 \ 2)(5)(7 \ 8) = (1 \ 6 \ 3 \ 4 \ 2)(7 \ 8). \quad (2.6)$$

Similarly, $\rho_1\rho_2 = (1 \ 5 \ 3 \ 8 \ 7)(2)(4 \ 6) = (1 \ 5 \ 3 \ 8 \ 7)(4 \ 6)$. Since $\rho_1\rho_2 \neq \rho_2\rho_1$, this example illustrates that S_n is non-abelian for $n \geq 3$.

Definition 2.22. Two permutations α and β are **disjoint** if every element moved by one is fixed by the other. That is, for any $k \in \{1, \dots, n\}$, $\alpha(k) \neq k$ implies $\beta(k) = k$.

Theorem 2.23. If α and β are disjoint permutations, they commute: $\alpha\beta = \beta\alpha$.

Proof. Let $k \in \{1, \dots, n\}$. If both α and β fix k , then $(\alpha\beta)(k) = k = (\beta\alpha)(k)$. Assume α moves k , so $\alpha(k) \neq k$. By disjointness, β must fix k , so $\beta(k) = k$. We must also show that β fixes $\alpha(k)$. Since α moves k , it cannot fix $\alpha(k)$ (otherwise $\alpha(\alpha(k)) = \alpha(k)$ implies $\alpha(k) = k$, a contradiction). Thus, by disjointness, β must fix $\alpha(k)$. Then, $(\alpha\beta)(k) = \alpha(\beta(k)) = \alpha(k)$, and $(\beta\alpha)(k) = \beta(\alpha(k)) = \alpha(k)$. The argument is symmetric if β moves k . $\square$

Theorem 2.24. Every permutation in S_n can be written as a product of disjoint cycles.

Proof. We prove it as follows:

- For any permutation ρ of $[n] = \{1, 2, \dots, n\}$ and any $k \in [n]$, follow the trajectory $k, \rho(k), \rho^2(k), \dots$. Since $[n]$ is finite, some element must eventually repeat: $\rho^p(k) = \rho^q(k)$ for some $p > q \geq 0$. Applying ρ^{-q} to both sides gives $\rho^{p-q}(k) = k$. Thus, the first element to reappear is the starting point k itself: the trajectory is a closed loop. Taking the smallest $u_k \geq 1$ such that $\rho^{u_k}(k) = k$, the set of distinct elements

$$\text{Orb}(k) = \{k, \rho(k), \rho^2(k), \dots, \rho^{u_k-1}(k)\} \quad (2.7)$$

is called the **orbit** of k under ρ . By construction, ρ acts on $\text{Orb}(k)$ as the cycle $(k, \rho(k), \dots, \rho^{u_k-1}(k))$.

- A loop looks the same from any of its points. Explicitly, if $m \in \text{Orb}(k)$, say $m = \rho^i(k)$, then the elements $m, \rho(m), \dots, \rho^{u_k-1}(m)$ are the elements $\rho^i(k), \rho^{i+1}(k), \dots, \rho^{i+u_k-1}(k)$ of the same loop, read off from a different starting point; hence $\text{Orb}(m) = \text{Orb}(k)$. Consequently, any two orbits with a common element m are both equal to $\text{Orb}(m)$: *two orbits are either identical or disjoint*.
- The set $[n] = \{1, \dots, n\}$ is therefore partitioned into a union of *disjoint* orbits:

$$[n] = \text{Orb}(k_1) \cup \dots \cup \text{Orb}(k_m), \quad (2.8)$$

for some m , with $\text{Orb}(k_i) \neq \text{Orb}(k_j)$ for $i \neq j$. This is the **orbit decomposition** of the set under ρ . The permutation ρ acts on each $\text{Orb}(k_j)$ as the cycle $\rho_j = (k_j, \rho(k_j), \dots, \rho^{u_{k_j}-1}(k_j))$. These cycles are disjoint by construction, and since every element of $[n]$ lies in exactly one orbit, $\rho = \rho_1 \rho_2 \dots \rho_m$.

- The decomposition is also unique up to the ordering of the cycles: in any expression of ρ as a product of disjoint cycles, the cycle containing a given k must map $k \rightarrow \rho(k) \rightarrow \rho^2(k) \rightarrow \dots$, and is therefore precisely the loop (2.7).

□

Example 2.25. If ρ is a cycle of length q , then its order is q . If σ is the product of disjoint cycles of lengths $n_1, n_2, \dots, n_k$, then the order of σ is the least common multiple of these lengths, $\text{lcm}(n_1, n_2, \dots, n_k)$. For instance, if $\rho = (1\ 3\ 4)(2\ 9\ 5\ 6)(7\ 8)$, its order is $\text{lcm}(3, 4, 2) = 12$.

2.2.2 The sign of a permutation

A 2-cycle, which swaps two elements, is called a **transposition**.

Theorem 2.26. Every permutation is a product of transpositions.

Proof. Every permutation is a product of cycles. Any cycle can be written as a product of transpositions:

$$(a_1\ a_2\ \dots\ a_p) = (a_1\ a_p) \dots (a_1\ a_3)(a_1\ a_2). \quad (2.9)$$

The result follows. □

Theorem 2.27. A permutation $\rho \in S_n$ that partitions $\{1, \dots, n\}$ into m disjoint orbits can be expressed as a product of $n - m$ transpositions.

Proof. Let $\rho = \rho_1 \cdots \rho_m$ be the decomposition into disjoint cycles, where the cycle ρ_j has length n_j . Thus $\sum n_j = n$. From (2.9), each cycle ρ_j of length $n_j \geq 2$ can be written as $n_j - 1$ transpositions. A 1-cycle (a fixed point) requires $1 - 1 = 0$ transpositions. The total number of transpositions is $\sum_j (n_j - 1) = (\sum_j n_j) - m = n - m$. $\square$

While a permutation can be written as a product of transpositions in many ways, the parity of the number of transpositions is always the same.

Proposition 2.28. *If the identity permutation I is expressed as a product of m transpositions, then m must be even.*

Proof. Consider the polynomial in n variables

$$\Delta(x_1, \dots, x_n) = \prod_{1 \leq i < j \leq n} (x_i - x_j) .$$

For a permutation $\rho \in S_n$, define $\rho \cdot \Delta := \Delta(x_{\rho(1)}, \dots, x_{\rho(n)}) = \prod_{i < j} (x_{\rho(i)} - x_{\rho(j)})$. Since ρ is a bijection, every unordered pair $\{k, l\}$ appears exactly once in this product, possibly with the order of subtraction reversed; hence $\rho \cdot \Delta = \pm \Delta$.

We claim that $\tau \cdot \Delta = -\Delta$ for any transposition $\tau = (a \ b)$, with $a < b$. Indeed, the factor $(x_a - x_b)$ changes sign; factors containing neither a nor b are untouched; for $c < a$ or $c > b$, the two factors containing the pairs $\{c, a\}$ and $\{c, b\}$ are exchanged with no sign change; and for $a < c < b$, the factors $(x_a - x_c)$ and $(x_c - x_b)$ are exchanged with two compensating sign changes. The net effect is exactly one sign change.

Now suppose $I = \tau_1 \cdots \tau_m$. Substituting the variables one transposition at a time and using the claim at each step, we find

$$\Delta = I \cdot \Delta = (-1)^m \Delta ,$$

which forces m to be even. $\square$

Theorem 2.29. *If a permutation can be expressed as a product of p transpositions and also as a product of q transpositions, then p and q have the same parity (both are even or both are odd).*

Proof. Suppose $\tau_1 \cdots \tau_p = \sigma_1 \cdots \sigma_q$. Then $I = (\sigma_1 \cdots \sigma_q)^{-1} (\tau_1 \cdots \tau_p) = \sigma_q \cdots \sigma_1 \tau_1 \cdots \tau_p$. This expresses I as a product of $p + q$ transpositions. By Proposition 2.28, $p + q$ must be even, which implies p and q have the same parity. $\square$

Definition 2.30. The **sign** of a permutation ρ , denoted $\text{sgn}(\rho)$, is $(-1)^q$, where ρ is a product of q transpositions. A permutation is an even permutation if $\text{sgn}(\rho) = 1$ and an odd permutation if $\text{sgn}(\rho) = -1$.

Corollary 2.31. *A cycle of length p is a product of $p - 1$ transpositions, so its sign is $(-1)^{p-1}$; thus a cycle of even length is odd, and a cycle of odd length is even.*

If the permutations α and β can be expressed as products of p and q transpositions, respectively, then the composition $\alpha\beta$ can be expressed as a product of $p + q$ transpositions; thus the next two results are clear.

Theorem 2.32. *For any permutations α, β , we have $\text{sgn}(\alpha\beta) = \text{sgn}(\alpha)\text{sgn}(\beta)$. In particular, $\text{sgn}(\alpha) = \text{sgn}(\alpha^{-1})$.*

Theorem 2.33. *The product of two even permutations is an even permutation. The inverse of an even permutation is an even permutation. More generally, the set of even permutations in S_n is a subgroup of S_n .*

Definition 2.34. *The alternating group A_n is the group of all even permutations in S_n .*

Theorem 2.35. *The symmetric group S_n has order $n!$, and the alternating group A_n (with $n \geq 2$) has order $n!/2$.*

Proof. The number of permutations of n elements is $n!$. Let $\tau = (1\ 2)$ be a fixed transposition. The map $f : S_n \rightarrow S_n$ defined by $f(\rho) = \tau\rho$ is invertible, since $f^{-1} = f$ due to the fact that $f(f(\rho)) = f(\tau\rho) = \tau\tau\rho = \rho$. It maps the set of even permutations to the set of odd permutations, and vice versa. Therefore, the number of even and odd permutations must be equal, each being $n!/2$. $\square$

For example, the group A_4 consists of the 12 even permutations in S_4 :

$$A_4 = \{I, (12)(34), (13)(24), (14)(23), (123), (132), (124), (142), (134), (143), (234), (243)\}. \quad (2.10)$$

Example 2.36. (i) S_n is generated by the transpositions $(1\ 2), (1\ 3), \dots, (1\ n)$.

(ii) For $n \geq 3$, A_n is generated by the 3-cycles $(1\ 2\ 3), (1\ 2\ 4), \dots, (1\ 2\ n)$.

Proof. For (i), we recall that every permutation is a product of transpositions. It only remains to express the general transposition in terms of those of the form $(1\ i)$. If $i, j \neq 1$, this is achieved by

$$(i\ j) = (1\ i)(1\ j)(1\ i). \quad (2.11)$$

Together with the relation $(i\ 1) = (1\ i)$, we establish (i).

Let us now prove (ii). The formulae

$$(k\ l)(i\ j) = (k\ i\ l)(i\ j\ k), \quad (i\ k)(i\ j) = (i\ j\ k) \quad (2.12)$$

show that A_n is generated by all 3-cycles. Now we use the formulae

$$\begin{aligned}(i \ j \ k) &= (1 \ 2 \ i)(2 \ j \ k)(1 \ 2 \ i)^{-1}, & (2 \ j \ k) &= (1 \ 2 \ j)(1 \ 2 \ k)(1 \ 2 \ j)^{-1}, \\ (1 \ j \ k) &= (1 \ 2 \ k)^{-1}(1 \ 2 \ j)(1 \ 2 \ k)\end{aligned}\tag{2.13}$$

where i, j, k are distinct elements of $\{3, \dots, n\}$. Together with $(1 \ k \ 2) = (1 \ 2 \ k)^{-1}$, these formulae express every 3-cycle in terms of the $(1 \ 2 \ i)$. $\square$

2.3 Cosets

A subgroup H of a group G naturally acts on G by left multiplication, where an element $h \in H$ maps an element $g \in G$ to hg . This concept leads to the definition of a coset.

Definition 2.37. *If H is a subgroup of a group G and $g \in G$, the **left coset** of H containing g is the set*

$$gH = \{gh \mid h \in H\}.\tag{2.14}$$

*Similarly, the **right coset** of H containing g is the set $Hg = \{hg \mid h \in H\}$.*

If the group is abelian, then $gH = Hg$ for all $g \in G$. However, this is not true in general.

Example 2.38. *Let $G = S_3 = \{I, (1 \ 2), (2 \ 3), (3 \ 1), (1 \ 2 \ 3), (1 \ 3 \ 2)\}$. Let $H = \{I, (1 \ 2)\}$ and $g = (1 \ 3)$. Then the left coset is $gH = \{(1 \ 3)I, (1 \ 3)(1 \ 2)\} = \{(1 \ 3), (1 \ 2 \ 3)\}$, whereas the right coset is $Hg = \{I(1 \ 3), (1 \ 2)(1 \ 3)\} = \{(1 \ 3), (1 \ 3 \ 2)\}$. Clearly, $gH \neq Hg$.*

Example 2.39. *Let $G = (\mathbb{C} - \{0\}, \times)$, and let H be the subgroup $\{z : |z| = 1\}$ (the unit circle). For any $w \in G$, the coset wH is the circle with centre 0 and radius $|w|$. The cosets are concentric circles that are either identical or disjoint, and their union is all of G .*

Example 2.40. *Consider the group $G = (\mathbb{Z}, +)$ and the subgroup $H = 3\mathbb{Z}$ of all multiples of 3. The cosets are:*

$$\begin{aligned}0 + H &= \{\dots, -6, -3, 0, 3, 6, \dots\} \\ 1 + H &= \{\dots, -5, -2, 1, 4, 7, \dots\} \\ 2 + H &= \{\dots, -4, -1, 2, 5, 8, \dots\}.\end{aligned}$$

Any integer m can be written as $m = 3k + a$ where $a \in \{0, 1, 2\}$, which implies $m + H = a + H$. Again, the cosets partition the group into disjoint subsets.

The partitioning seen in these examples is an instance of a general and very useful concept: an equivalence relation. Recall that an **equivalence relation** on a set X is a relation $\sim$ that is:

1. *Reflexive*: For all $x \in X$, $x \sim x$.
2. *Symmetric*: If $x \sim y$, then $y \sim x$.
3. *Transitive*: If $x \sim y$ and $y \sim z$, then $x \sim z$.

The **equivalence class** of an element $x \in X$ is the set $E(x) = \{y \in X : x \sim y\}$.

Theorem 2.41. *Let $\sim$ be an equivalence relation on a non-empty set X . Then any two equivalence classes are either equal or disjoint, and X is the union of all its distinct equivalence classes.*

Proof. Suppose $x, y \in X$ and the equivalence classes $E(x) \cap E(y) \neq \emptyset$. Let $z \in E(x) \cap E(y)$. By definition, this means $x \sim z$ and $y \sim z$. Let $y' \in E(y)$. Then $y' \sim y$. By transitivity, $y' \sim y$ and $y \sim z$ implies $y' \sim z$. By symmetry, $x \sim z$ implies $z \sim x$. Again by transitivity, $y' \sim z$ and $z \sim x$ implies $y' \sim x$. Thus, $y' \in E(x)$, which shows $E(y) \subseteq E(x)$. A symmetric argument shows $E(x) \subseteq E(y)$. Therefore, $E(x) = E(y)$. This proves that if two equivalence classes are not disjoint, they must be identical. Finally, for any $x \in X$, reflexivity implies $x \sim x$, so $x \in E(x)$. Thus, every element belongs to an equivalence class, and $X = \bigcup_{x \in X} E(x)$. $\square$

We have already met one instance of this principle: the orbits of a permutation ρ partition the set $[n]$, the relevant relation being $x \sim y$ if and only if $y = \rho^k(x)$ for some integer k . The cosets of a subgroup provide a second instance, as we now show, and conjugacy classes (Section 2.7) will provide a third.

We can now state the fundamental result about cosets.

Theorem 2.42. *Let H be a subgroup of a group G . Then,*

- (i) *G is the union of its left cosets: $G = \bigcup_{g \in G} gH$.*
- (ii) *Any two left cosets are either equal or disjoint. Furthermore, $fH = gH$ if and only if $f^{-1}g \in H$.*
- (iii) *All left cosets have the same size: $|gH| = |H|$ for any $g \in G$.*

Similar statements hold for right cosets.

Proof. (i) and (ii): Define a relation on G by $f \sim g$ if and only if $f^{-1}g \in H$. Since H is a subgroup, this is an equivalence relation: it is reflexive because $f^{-1}f = e \in H$; symmetric because $g^{-1}f = (f^{-1}g)^{-1} \in H$; and transitive because $f^{-1}h = (f^{-1}g)(g^{-1}h) \in H$. The equivalence class of f is $E(f) = \{g \in G : f^{-1}g \in H\} = fH$. Statements (i) and (ii) now follow from Theorem 2.41, together with the fact that $E(f) = E(g)$ if and only if $f \sim g$, i.e., $f^{-1}g \in H$.

(iii) For any $g \in G$, the map $\phi : H \rightarrow gH$ defined by $\phi(h) = gh$ is a bijection, with inverse $\phi^{-1}(x) = g^{-1}x$. Therefore, H and gH have the same number of elements. $\square$

Theorem 2.42 shows that the left (or right) cosets of a subgroup H partition the group G into disjoint subsets of equal size. This is called the **coset decomposition** of G with respect to H .

2.4 Lagrange's theorem

The following theorem establishes a fundamental relationship between the order of a finite group and its subgroups.

Theorem 2.43. (Lagrange's theorem.) *Let H be a subgroup of a finite group G . Then the order of H divides the order of G . The ratio $|G|/|H|$ is the number of distinct left (or right) cosets of H in G .*

Proof. By Theorem 2.42, G is the union of a finite number of pairwise disjoint left cosets, say $G = g_1H \cup \cdots \cup g_rH$. Consequently,

$$|G| = |g_1H| + \cdots + |g_rH|.$$

Since all cosets have the same size, $|g_jH| = |H|$ for all j , it follows that $|G| = r|H|$. $\square$

The number of distinct left cosets of H in G is called the **index** of H in G , denoted $[G : H]$; thus $[G : H] = |G|/|H|$.

Corollary 2.44. *If G is a group of prime order p , its only subgroups are $\{e\}$ and G itself.*

Proof. If H is a subgroup of G , its order $|H|$ must divide $|G| = p$. Since p is prime, the only positive divisors are 1 and p . Thus, $|H| = 1$ (so $H = \{e\}$) or $|H| = p$ (so $H = G$). $\square$

Another important consequence of Lagrange's theorem is the following.

Theorem 2.45. *Let g be an element of a finite group G . Then the order of g divides the order of G .*

Proof. Let g be an element of a finite group G , and let its order be d . The set $\{e, g, \dots, g^{d-1}\}$ forms a cyclic subgroup of G of order d . By Lagrange's theorem, the order of this subgroup, d , must divide the order of G . $\square$

We end this section by demonstrating that the *converse of Lagrange's theorem is false*. It is not guaranteed that if an integer q divides $|G|$, then G must have a subgroup of order q .

First, we record a simple consequence of the coset decomposition.

Lemma 2.46. *Let H be a subgroup of index 2 in a group G . Then $g^2 \in H$ for every $g \in G$.*

Proof. If $g \in H$, this is clear. If $g \notin H$, then $G = H \cup gH$ is a disjoint union, and g^2 lies in one of the two cosets. If $g^2 \in gH$, then $g^2 = gh$ for some $h \in H$, so $g = h \in H$, a contradiction. Hence $g^2 \in H$. $\square$

Example 2.47. The alternating group A_4 has order $|A_4| = 12$, but it has no subgroup of order six.

Proof. Assume that H is a subgroup of A_4 with $|H| = 6$, i.e., of index 2. By Lemma 2.46, $g^2 \in H$ for every $g \in A_4$. But every 3-cycle in A_4 is a square of another 3-cycle: $(a\ b\ c) = (a\ c\ b)^2$. Hence all eight 3-cycles of A_4 lie in H , contradicting $|H| = 6$. $\square$

2.5 Isomorphisms

In many examples, we see that the same underlying group structure can appear in different forms, such as $(\mathbb{Z}_n, +)$ and C_n , or $C_2 \times C_3$ and C_6 (Example 2.12). To formalise the notion of when two groups are structurally identical, we introduce the concept of an isomorphism.

Definition 2.48. Let G and G' be groups. A map $\theta : G \rightarrow G'$ is a **homomorphism** if it preserves the group operation, i.e., for all $g, h \in G$, $\theta(gh) = \theta(g)\theta(h)$. If a homomorphism θ is also a bijection (one-to-one and onto), then θ is an **isomorphism**. In this case, we say that G and G' are **isomorphic** groups and write $G \cong G'$.

Isomorphic groups share all their group-theoretic properties. Some examples are given below.

Theorem 2.49. Suppose that $\theta : G \rightarrow G'$ is an isomorphism. Then,

1. If G is abelian, then G' is abelian.
2. $\theta(e) = e'$, where e and e' are the identity elements of G and G' , respectively.
3. For all $g \in G$, $\theta(g^{-1}) = [\theta(g)]^{-1}$.
4. The order of any element $g \in G$ is equal to the order of its image $\theta(g) \in G'$.

Proof. (1) Suppose G is abelian. For any $g', h' \in G'$, since θ is surjective, there exist $g, h \in G$ such that $\theta(g) = g'$ and $\theta(h) = h'$. Then $g'h' = \theta(g)\theta(h) = \theta(gh) = \theta(hg) = \theta(h)\theta(g) = h'g'$. Thus, G' is abelian.

(2) We have $\theta(e)\theta(e) = \theta(ee) = \theta(e)$; multiplying both sides by $[\theta(e)]^{-1}$ gives $\theta(e) = e'$.

(3) For any $g \in G$, we have $\theta(g)\theta(g^{-1}) = \theta(gg^{-1}) = \theta(e) = e'$. Similarly, $\theta(g^{-1})\theta(g) = e'$. Therefore, $\theta(g^{-1})$ is the inverse of $\theta(g)$.

(4) For any integer m , the homomorphism property implies $\theta(g^m) = (\theta(g))^m$. Since θ is a bijection, $g^m = e$ if and only if $\theta(g^m) = e'$, which means $(\theta(g))^m = e'$. Thus, g and $\theta(g)$ have the same order. $\square$

Note that the proofs of (2) and (3) use only the property $\theta(gh) = \theta(g)\theta(h)$, and not bijectivity; these two statements therefore hold for *any* homomorphism. We will use this in Section 2.10.

To prove that two groups are isomorphic, one must construct an isomorphism between them. To prove they are not isomorphic, it suffices to find a single group-theoretic property that one possesses but the other does not.

Example 2.50. (i) $C_2 \times C_4$ and C_8 are both groups of order 8, but they are not isomorphic because C_8 has an element of order 8, while every element in $C_2 \times C_4$ has an order of at most 4. (ii) $C_2 \times C_2$ is not isomorphic to C_4 because every non-identity element in $C_2 \times C_2$ has order 2, whereas C_4 contains an element of order 4. (iii) It follows from the next theorem that $C_6 \cong C_2 \times C_3$.

Theorem 2.51. Let m and n be coprime integers. Then $C_{mn} \cong C_m \times C_n$.

Proof. This is also known as the *Chinese remainder theorem*. Consider the element $x = (1, 1)$ of $\mathbb{Z}_m \times \mathbb{Z}_n$, written additively. The order of x is the smallest integer $k \geq 1$ such that $kx = (k \bmod m, k \bmod n) = (0, 0)$, i.e., the smallest k divisible by both m and n , which is $\text{lcm}(m, n)$. Since $\gcd(m, n) = 1$, we have $\text{lcm}(m, n) = mn = |\mathbb{Z}_m \times \mathbb{Z}_n|$. Hence, by Lemma 2.10, $\mathbb{Z}_m \times \mathbb{Z}_n$ is cyclic, generated by $(1, 1)$, and is therefore isomorphic to C_{mn} . $\square$

Theorem 2.52. Any group of prime order p is isomorphic to the cyclic group C_p . Furthermore, it is generated by any of its non-identity elements.

Proof. Let G be a group with $|G| = p$, where p is prime. Let $g \in G$ with $g \neq e$. By Lagrange's theorem, the order of g , say k , must divide p . Since $g \neq e$, its order $k \neq 1$. Because p is prime, the only other possibility is $k = p$. By Lemma 2.10, G is cyclic and generated by g . $\square$

Isomorphism defines an equivalence relation on the class of all groups.

Theorem 2.53. Let K be the class of all groups. The relation $\sim$ on K defined by $G_1 \sim G_2$ if and only if G_1 is isomorphic to G_2 is an equivalence relation.

This theorem formalises the idea that isomorphic groups are algebraically identical. The equivalence class of a group G consists of all groups that share its structure.

Proof of Theorem 2.53. *Reflexivity:* the identity map $\text{id}_G : G \rightarrow G$ is an isomorphism. *Symmetry:* if $\theta : G_1 \rightarrow G_2$ is an isomorphism, its inverse θ^{-1} is a bijection, and it is a homomorphism: for any $x, y \in G_2$, we have $\theta(\theta^{-1}(x)\theta^{-1}(y)) = xy$, and applying θ^{-1} to both sides gives $\theta^{-1}(x)\theta^{-1}(y) = \theta^{-1}(xy)$. *Transitivity:* the composition $\theta_2 \circ \theta_1$ of two bijective homomorphisms is bijective and is a homomorphism, since $(\theta_2 \circ \theta_1)(gh) = \theta_2(\theta_1(g)\theta_1(h)) = (\theta_2 \circ \theta_1)(g)(\theta_2 \circ \theta_1)(h)$. $\square$

Isomorphism allows us to classify finite groups up to a certain order.

Theorem 2.54. *Let G be a finite group in which every non-identity element has order two. Then G is abelian and is isomorphic to a direct product of cyclic groups of order 2. In particular, $|G| = 2^m$ for some integer $m \geq 0$.*

Proof. First, we show G is abelian. For any $x, y \in G$, we have $(xy)^2 = e$, so $xyxy = e$. Multiplying on the right by yx gives $xy = yx$.

Now observe that an abelian group in which every element satisfies $x^2 = e$ is precisely a vector space over the field $\mathbb{Z}_2 = \{0, 1\}$: the group operation plays the role of vector addition, and the scalars act by $0 \cdot x := e$ and $1 \cdot x := x$. The vector space axioms follow from the group axioms together with commutativity; note in particular that distributivity requires $(1 + 1) \cdot x = 0 \cdot x$, i.e., $x^2 = e$, which is exactly our hypothesis. Since G is finite, this vector space has a finite basis $a_1, \dots, a_m$, so every element of G can be written uniquely as $a_1^{p_1} \cdots a_m^{p_m}$ with $p_i \in \{0, 1\}$. The map $a_1^{p_1} \cdots a_m^{p_m} \mapsto (p_1, \dots, p_m)$ is then an isomorphism $G \cong (C_2)^m$, and, in particular, $|G| = 2^m$. $\square$

Theorem 2.55. *Every group of order four is isomorphic to either C_4 or $C_2 \times C_2$. In particular, every group of order four is abelian.*

Proof. Let G be a group of order 4. By Lagrange's theorem, any non-identity element must have order 2 or 4. If there is an element of order 4, then $G \cong C_4$ by Lemma 2.10. If there is no element of order 4, then every non-identity element must have order 2. By Theorem 2.54, G must be isomorphic to $C_2 \times C_2$. $\square$

2.6 The dihedral group

The **dihedral group** D_n is the group of symmetries of a regular n -gon: the distance-preserving maps (isometries) of $\mathbb{R}^2$ that map the set of vertices of the polygon to itself.

- Any symmetry of the n -gon fixes its centre, since this is the only point with equal distance from each of the vertices. Therefore, D_n is a subgroup of the orthogonal group $O(2)$, and its elements are rotations or reflections; see (2.4).
- The rotations in D_n form a cyclic subgroup $R = \{r^k\}_{k=0}^{n-1} \cong C_n$, generated by the rotation r by angle $2\pi/n$ about the centre. Let σ be the reflection across a line through the centre and a vertex. Then every reflection $\rho \in D_n$ lies in the coset σR : indeed, $\sigma\rho$ is a rotation in D_n , so $\sigma\rho = r^k$ and $\rho = \sigma r^k$ for some k . Hence $D_n = R \cup \sigma R$, and the order of D_n is $2n$.
- Thus, D_n is generated by r and σ , which satisfy the relations

$$r^n = I, \quad \sigma^2 = I, \quad \sigma r \sigma^{-1} = r^{-1}; \quad (2.15)$$

since $\sigma^2 = I$, the last relation is often written as $\sigma r = r^{-1}\sigma$. In summary, $D_n = \langle r, \sigma \mid r^n = I, \sigma^2 = I, \sigma r = r^{-1}\sigma \rangle$, with the $2n$ distinct elements $\{I, r, \dots, r^{n-1}, \sigma, \sigma r, \dots, \sigma r^{n-1}\}$.

Notation. In the literature, the same group is denoted by both D_n and D_{2n} . In this text, D_n denotes the symmetry group of the regular n -gon, which has order $2n$.

Example 2.56. (i) For $n \geq 3$, D_n is non-abelian, since $\sigma r = r^{-1}\sigma \neq r\sigma$. Thus, D_3 is not isomorphic to the abelian group C_6 .

(ii) D_2 is a group of order 4 where every non-identity element has order 2. Thus, $D_2 \cong C_2 \times C_2$.

(iii) $D_3 \cong S_3$. The six symmetries of an equilateral triangle correspond precisely to the six permutations of its three vertices. The rotation r corresponds to the 3-cycle $(1\ 2\ 3)$ and the reflection σ to the transposition $(1\ 2)$.

Theorem 2.57. Let G be a group of order $2p$, where $p \geq 3$ is a prime number. Then G is isomorphic to either the cyclic group C_{2p} or the dihedral group D_p .

Proof. By Lagrange's theorem, element orders in G can only be 1, 2, p , or $2p$. Since $|G|$ is even, there must exist an element a of order 2. Since $|G| = 2p$ and p is an odd prime, $|G|$ is not a power of 2. By Theorem 2.54, not every non-identity element can have order 2. Thus, there must exist an element of order p or an element of order $2p$. In either case, G contains an element b of order p .

Let $H = \langle b \rangle = \{e, b, \dots, b^{p-1}\}$ be the cyclic subgroup of order p . Since a has order 2, it cannot be in H (as every non-identity element of H has order p). Thus, the coset aH is disjoint from H . Since $|G| = 2p$ and $|H| = p$, these two cosets partition G :

$$G = H \cup aH = \{e, b, \dots, b^{p-1}\} \cup \{a, ab, \dots, ab^{p-1}\}.$$

Since $G = H \cup aH$ and also $G = H \cup Ha$ (both disjoint unions), we have $aH = G \setminus H = Ha$, and hence $aHa^{-1} = H$. This implies aba^{-1} is some power of b , say $aba^{-1} = b^k$ for some $k \in \{1, \dots, p-1\}$. Applying the conjugation by a twice, we get $b = a^2ba^{-2} = a(aba^{-1})a^{-1} = a(b^k)a^{-1} = (aba^{-1})^k = (b^k)^k = b^{k^2}$. This means $b^{k^2-1} = e$. Since the order of b is p , we must have $p \mid (k^2 - 1)$, so $p \mid (k-1)(k+1)$. As p is prime, this implies $p \mid (k-1)$ or $p \mid (k+1)$. Since $1 \leq k \leq p-1$, the only possibilities are $k = 1$ or $k = p-1$.

Case 1: $k = 1$. Then $aba^{-1} = b$, so $ab = ba$. Since a and b commute, $(ab)^m = a^m b^m$, which equals e if and only if $a^m = e = b^m$ (as $a^m \in \langle a \rangle$ and $b^m \in \langle b \rangle$ can only coincide at e , the two subgroups having coprime orders). Hence the order of ab is $\text{lcm}(2, p) = 2p$. Thus, $G \cong C_{2p}$ by Lemma 2.10.

Case 2: $k = p-1$. Then $aba^{-1} = b^{p-1} = b^{-1}$. This gives the relation $ab = b^{-1}a$. The group G is generated by elements a and b with relations $a^2 = e, b^p = e, ab = b^{-1}a$. These are the defining relations for the dihedral group D_p . Thus, $G \cong D_p$. $\square$

Example 2.58. *The Platonic solids provide further examples of finite groups.*

(i) *The full symmetry group of the regular tetrahedron is isomorphic to S_4 , permuting the four vertices.*

(ii) *The group of rotational symmetries of the cube, denoted G_+ , has 24 elements. These consist of the following:*

1. *The identity (1).*
2. *Rotations about axes through opposite face centres (9): (three choices for $L_{1,2,3}$) $\times$ (three non-identity elements in C_4 for each L_i).*
3. *Rotations about axes through opposite vertices (8): (four choices for $d_{1,2,3,4}$) $\times$ (two non-identity elements in C_3 for each d_i); see Figure 2 for an illustration.²*
4. *Rotations about a line through the midpoints of opposite edges (6): (six choices for the axis between d_i and d_j with $i < j$) $\times$ (one non-identity element in C_2 for each such axis).*

In fact, $G_+ \cong S_4$. Each rotation permutes the four main diagonals of the cube. This action defines a homomorphism from G_+ to S_4 , which we now show to be an isomorphism. For example, a 180-degree rotation about an axis through opposite edge midpoints corresponds to a transposition of two diagonals.³ By the symmetry of the cube, the six edge axes give rise in the same way to the six transpositions $(d_i d_j)$. Since transpositions generate S_4 , the map is surjective. As both groups have order 24, it is an isomorphism.

(iii) *The full symmetry group of the cube includes reflections. The inversion through the centre, σ , commutes with all rotations. The full symmetry group is isomorphic to $G_+ \times \langle \sigma \rangle \cong S_4 \times C_2$.*

²Let us demonstrate that the rotations about a main diagonal are by $\pm 120^\circ$. Consider the diagonal d_1 through $(1, 1, 1)$ and $(-1, -1, -1)$. The cyclic permutation of the coordinates, $R(x, y, z) = (z, x, y)$, is an element of $\text{SO}(3)$ that maps the cube to itself and fixes d_1 , so it is a rotation about d_1 . Since $R^3 = \mathbf{1}$ and $R \neq \mathbf{1}$, the rotation angle is 120° or 240° . (Indeed, using the fact that a rotation by an angle θ in three dimensions has trace $1 + 2 \cos \theta$, and noting $\text{tr } R = 0$, we find $\theta = 120^\circ$.) Geometrically, the three vertices adjacent to $(1, 1, 1)$, namely $(1, 1, -1)$, $(-1, 1, 1)$ and $(1, -1, 1)$, form an equilateral triangle lying in a plane perpendicular to d_1 , and R permutes them cyclically: $(1, 1, -1) \mapsto (-1, 1, 1) \mapsto (1, -1, 1) \mapsto (1, 1, -1)$; see Figure 2. The two non-identity rotations about d_1 are R and $R^2 = R^{-1}$, which together with the identity form a group C_3 .

³Explicitly, place the vertices at $(\pm 1, \pm 1, \pm 1)$ and label the main diagonals as in Figure 1: d_1 joins $(-1, -1, -1)$ to $(1, 1, 1)$, d_2 joins $(1, -1, -1)$ to $(-1, 1, 1)$, d_3 joins $(-1, 1, -1)$ to $(1, -1, 1)$, and d_4 joins $(1, 1, -1)$ to $(-1, -1, 1)$. Consider the 180° rotation R about the axis through the midpoints $(1, 1, 0)$ and $(-1, -1, 0)$ of two opposite edges, drawn in red in Figure 1. Using $R(v) = 2(n \cdot v)n - v$ with the unit axis vector $n = \frac{1}{\sqrt{2}}(1, 1, 0)$, we find $R(x, y, z) = (y, x, -z)$. Then R sends $(1, 1, 1) \mapsto (1, 1, -1)$ and $(-1, -1, -1) \mapsto (-1, -1, 1)$, so $d_1 \mapsto d_4$ and, likewise, $d_4 \mapsto d_1$. On the other hand, $(1, -1, -1) \mapsto (-1, 1, 1)$ and $(-1, 1, -1) \mapsto (1, -1, 1)$, so d_2 and d_3 are each mapped to themselves, with their two endpoints exchanged. Hence, R acts on the diagonals as the transposition $(d_1 d_4)$.

(iv) The cube and the regular octahedron are dual solids: the centres of the faces of one form the vertices of the other. Any symmetry of the cube induces a corresponding symmetry of the inscribed octahedron, and vice versa. Therefore, their symmetry groups are isomorphic.

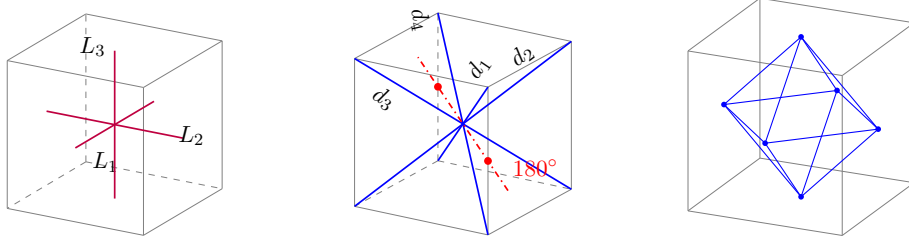


Figure 1: The cube. Left: the axes L_1, L_2, L_3 through opposite face centres. Middle: the main diagonals $d_1, \dots, d_4$, and (in red) the 180° rotation axis through the midpoints of two opposite edges. Right: the octahedron whose vertices are the face centres of the cube.

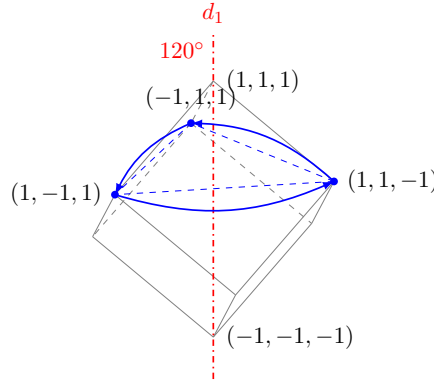


Figure 2: The rotation by 120° about the main diagonal d_1 cyclically permutes the three vertices adjacent to $(1, 1, 1)$.

2.7 Conjugacy classes

We now introduce a way to relate elements and subgroups to each other within a group.

Definition 2.59. (i) Two elements x, y in a group G are said to be **conjugate** if there exists an element $g \in G$ such that $x = gyg^{-1}$.

(ii) Two subgroups H and K of a group G are said to be **conjugate** if there exists an element $g \in G$ such that $H = gKg^{-1}$.

Conjugacy of elements defines an equivalence relation.

Theorem 2.60. For any elements x, y, z in a group G , the conjugacy relation $\sim$ is:

- (a) *Reflexive:* $x \sim x$.
- (b) *Symmetric:* If $x \sim y$, then $y \sim x$.
- (c) *Transitive:* If $x \sim y$ and $y \sim z$, then $x \sim z$.

Therefore, by Theorem 2.41, conjugacy partitions the group G into disjoint equivalence classes, known as **conjugacy classes**.

Proof. (a) $x = exe^{-1}$, so $x \sim x$. (b) If $x = aya^{-1}$, then $y = a^{-1}x(a^{-1})^{-1}$, so $y \sim x$. (c) If $x = aya^{-1}$ and $y = bzb^{-1}$, then $x = (ab)z(ab)^{-1}$, so $x \sim z$. $\square$

The structure of conjugacy classes reveals important properties of a group.

Theorem 2.61. *In an abelian group, every element forms its own conjugacy class.*

Proof. If G is abelian, then $aga^{-1} = gaa^{-1} = g$ for all $a \in G$, so the conjugacy class of g is $\{g\}$. $\square$

Theorem 2.62. *Conjugate elements have the same order.*

Proof. If $a = hbh^{-1}$, then $a^n = (hbh^{-1})^n = hb^n h^{-1}$ for any integer n ; hence $a^n = e$ if and only if $b^n = e$, and a and b have the same order. $\square$

The structure of conjugacy classes in the symmetric group is particularly elegant.

Theorem 2.63. *Two elements of S_n are conjugate if and only if they decompose into the same number of disjoint cycles of each length.*

For a permutation, we can produce a unique string $1^{a_1}2^{a_2}\dots n^{a_n}$, where a_i is the number of disjoint cycles of length i . We call such a string the **cycle type** or **cycle structure** of a permutation. Hence, two permutations are in the same conjugacy class if and only if they have the same cycle type.

Proof. We will show that for any permutation $\sigma \in S_n$, conjugation acts on cycles as follows:

$$\sigma(a_1a_2\dots a_m)\sigma^{-1} = (\sigma(a_1)\sigma(a_2)\dots\sigma(a_m)). \quad (2.16)$$

More generally, if a permutation is a product of disjoint cycles, conjugation distributes over the product:

$$\sigma \left(\prod_{j=1}^s (a_{1j}a_{2j}\dots a_{m(j)j}) \right) \sigma^{-1} = \prod_{j=1}^s (\sigma(a_{1j})\sigma(a_{2j})\dots\sigma(a_{m(j)j})). \quad (2.17)$$

To prove (2.16), let $\tau = (a_1a_2\dots a_m)$. We examine the action of $\sigma\tau\sigma^{-1}$ on an element of the form $\sigma(a_i)$.

$$(\sigma\tau\sigma^{-1})(\sigma(a_i)) = \sigma(\tau(a_i)) = \sigma(a_{i+1}) \quad (\text{indices mod } m).$$

If an element y is not of the form $\sigma(a_i)$, then $\sigma^{-1}(y)$ is not in $\{a_1, \dots, a_m\}$, so it is fixed by τ . Thus $(\sigma\tau\sigma^{-1})(y) = \sigma(\tau(\sigma^{-1}(y))) = \sigma(\sigma^{-1}(y)) = y$. This shows that $\sigma\tau\sigma^{-1}$ acts as the cycle $(\sigma(a_1)\sigma(a_2)\dots\sigma(a_m))$. The second identity, (2.17), follows because conjugation is a homomorphism, namely $\sigma(\tau_1 \dots \tau_s)\sigma^{-1} = (\sigma\tau_1\sigma^{-1}) \dots (\sigma\tau_s\sigma^{-1})$. $\square$

Proposition 2.64. *Let $\sigma \in S_n$ with cycle type $1^{a_1}2^{a_2}\dots n^{a_n}$. The size of the conjugacy class containing σ is*

$$\frac{n!}{\prod_{k=1}^n a_k! k^{a_k}}.$$

Proof. The total number of permutations in S_n is $n!$. For each of the a_k cycles of length k , we must divide by $a_k!$, since the order of disjoint cycles of the same length does not matter. We must also divide by k^{a_k} because the cyclic order within each of the k -cycles does not matter; for example, (123) is the same as (231) or (312). $\square$

Example 2.65. *Consider the dihedral group $D_n = \langle r, \sigma \mid r^n = I, \sigma^2 = I, \sigma r = r^{-1}\sigma \rangle$.*

- *When n is odd, D_n has $(n+3)/2$ conjugacy classes:*

$$\{I\}, \quad \{\sigma r^s \mid 0 \leq s \leq n-1\}, \quad \{r^k, r^{-k}\} \text{ for } 1 \leq k \leq (n-1)/2.$$

- *When $n = 2m$ is even, D_n has $m+3$ conjugacy classes:*

$$\{I\}, \quad \{\sigma r^s \mid s \text{ even}\}, \quad \{\sigma r^s \mid s \text{ odd}\}, \quad \{r^m\}, \quad \{r^k, r^{-k}\} \text{ for } 1 \leq k \leq m-1.$$

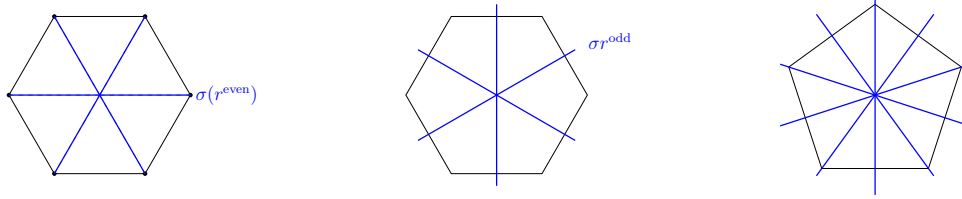
Proof. Let $R = \langle r \rangle$ be the cyclic subgroup of rotations.

The conjugacy class of an element $r^k \in R$ is $C(r^k) = \{gr^k g^{-1} \mid g \in D_n\}$. If $g \in R$, then $gr^k g^{-1} = r^k$ since R is abelian. If $g \notin R$, then $g = \sigma r^p$ for some p . The conjugate is $(\sigma r^p)r^k(\sigma r^p)^{-1} = \sigma r^p r^k r^{-p} \sigma^{-1} = \sigma r^k \sigma = r^{-k}$. Thus, the conjugacy class of r^k is $\{r^k, r^{-k}\}$. This class has two elements, unless $r^k = r^{-k}$ (i.e., $r^{2k} = I$), which happens only if n is even and $k = n/2$.

Now consider the conjugacy class of a reflection σr^k . If we conjugate by a rotation r^p , we get $r^p(\sigma r^k)r^{-p} = \sigma r^{-p} r^k r^{-p} = \sigma r^{k-2p}$. If we conjugate by a reflection σr^p , we get $(\sigma r^p)(\sigma r^k)(\sigma r^p)^{-1} = r^{-p} r^k r^{-p} \sigma = r^{k-2p} \sigma = \sigma r^{-(k-2p)}$. The set of all conjugates of σr^k is $\{\sigma r^{k-2p} \mid p \in \mathbb{Z}\}$.

Recall that the power of r has to be taken modulo n . **If n is odd**, $\gcd(2, n) = 1$, so as p varies, $k - 2p$ covers all exponents modulo n . Thus all reflections are in one class. **If n is even**, $k - 2p$ has the same parity as k . This splits the reflections into two classes: those with even powers of r and those with odd powers. $\square$

This algebraic result has a clear geometric interpretation. For n even (such as D_6), the reflections that fix two opposite vertices form one conjugacy class, while the reflections that fix two opposite edges form another. For n odd (such as D_5), all reflections are conjugate since each reflection fixes a vertex and the opposite edge.



2.8 Normal subgroups and quotient groups

We have seen that for a subgroup H of G , the left coset gH and the right coset Hg need not be the same. Cases where they do coincide are of special importance.

Definition 2.66. A subgroup H of a group G is **normal**, denoted $H \triangleleft G$, if $gH = Hg$ for all $g \in G$.

The following conditions are equivalent to H being a normal subgroup of G :

1. H is invariant under conjugation: $gHg^{-1} = H$ for all $g \in G$.
2. For every $h \in H$ and every $g \in G$, the conjugate element ghg^{-1} is in H .

Example 2.67. Any subgroup of an abelian group is normal. This is because for any $g \in G$ and $h \in H$, $ghg^{-1} = gg^{-1}h = h \in H$.

Theorem 2.68. Any subgroup of index 2 is normal.

Proof. Let H be a subgroup of G with $[G : H] = 2$. There are exactly two left cosets, H and gH for any $g \notin H$, and two right cosets, H and Hg . Since the cosets in each case partition G , we must have $gH = G \setminus H$ and $Hg = G \setminus H$.⁴ Thus $gH = Hg$ for any $g \notin H$. For any $h \in H$, we already know $hH = H = Hh$. Therefore, $gH = Hg$ for all $g \in G$. (Compare Lemma 2.46, which was obtained from the same two-coset decomposition.) $\square$

Example 2.69. In the dihedral group D_n , the cyclic subgroup of rotations $C_n = \langle r \rangle$ has order n in a group of order $2n$. It has index 2 and is therefore a normal subgroup.

The set of cosets of a normal subgroup can itself be given a group structure.

Definition 2.70. If H is a subgroup of a group G , we define G/H to be the set of left cosets of H in G , i.e.

$$G/H = \{gH : g \in G\}.$$

This set is usually called a **coset space**.

Corollary 2.71. If H is a subgroup of a group G , the size of the set G/H is the index $[G : H]$ of H in G . By Lagrange's Theorem,

$$|G/H| = [G : H] = |G|/|H|.$$

⁴The notation $A \setminus B$ denotes the set consisting of elements that are in A but not in B .

Note that for an arbitrary subgroup H of G , the coset space G/H may **not** be a group. However, if H is a **normal** subgroup of G , then it is possible to make G/H a group.

Definition 2.72. If A, B are subsets of a group G , then we define a product AB of two subsets of the group G as

$$AB = \{ab : a \in A, b \in B\}.$$

(This should not be confused with the product group $A \times B$.)

Lemma 2.73. Let H be a normal subgroup of a group G and let $a, b \in G$. Then, the product (defined as in Definition 2.72) of two cosets aH and bH is again a coset. In fact,

$$(aH)(bH) = abH.$$

Proof. Since H is normal, $bH = Hb$. Since H is a subgroup, $HH = H$. Then, it follows that

$$(aH)(bH) = a(Hb)H = a(bH)H = ab(HH) = abH,$$

as claimed. □

Theorem 2.74. Let H be a normal subgroup of a group G . Then, G/H is a group under the operation given in Lemma 2.73. This is called the **quotient group**.

Proof. Let $a, b, c \in G$. We have $(aH bH) cH = abcH = aH (bH cH)$. Also, $H = eH$ is the identity, and $aHa^{-1}H = H$. □

QUOTIENT GROUPS ARE DEFINED ONLY FOR NORMAL SUBGROUPS.

Example 2.75. $C_6/C_3 \cong C_2$. Let $C_6 = \langle a \mid a^6 = 1 \rangle$ and let $H = \langle a^2 \rangle = \{1, a^2, a^4\}$ be a (normal) subgroup isomorphic to C_3 . The distinct cosets are H and aH . The quotient $C_6/H = \{H, aH\}$ is a group of order 2, which must be isomorphic to C_2 .

Example 2.76. For any integer $n > 1$, $\mathbb{Z}/n\mathbb{Z} \cong \mathbb{Z}_n$. The cosets of the subgroup $n\mathbb{Z}$ in $\mathbb{Z}$ are of the form $a + n\mathbb{Z}$, where $a \in \{0, 1, \dots, n-1\}$. The addition of cosets $(a + n\mathbb{Z}) + (b + n\mathbb{Z}) = (a + b) + n\mathbb{Z}$ corresponds exactly to addition modulo n .

Example 2.77. Let $(\mathbb{Q}, +)$ be the group of rational numbers under addition. The integers $\mathbb{Z}$ form a normal subgroup. The quotient group $\mathbb{Q}/\mathbb{Z}$ consists of cosets $q + \mathbb{Z}$. We can add or subtract a multiple of 1 from any rational number q until it falls into the interval $[0, 1)$. Hence, $\mathbb{Q}/\mathbb{Z}$ can be identified with $\mathbb{Q} \cap [0, 1)$. It is clear that every element of $\mathbb{Q}/\mathbb{Z}$ can be represented by a fraction in lowest terms in the interval $[0, 1)$.

While $\mathbb{Q}/\mathbb{Z}$ is an infinite group, every element has finite order. An element represented by p/q (in lowest terms) has order q , since $q(p/q) = p \in \mathbb{Z}$, which corresponds to the identity element in the quotient group.

2.9 The classification of groups of small order

We begin by collecting several key theorems that will be used in the classification.

- Theorem 2.52: If G is a group of prime order p , then $G \cong C_p$.
- Theorem 2.54: If every non-identity element in a finite group G has order 2, then G is isomorphic to a direct product of cyclic groups of order 2.
- Theorem 2.16: Any group of even order contains an element of order two.
- Theorem 2.55: Any group of order 4 is isomorphic to either C_4 or $C_2 \times C_2$.
- Theorem 2.57: If G is a group of order $2p$ where $p \geq 3$ is prime, then G is isomorphic to either C_{2p} or D_p .
- Theorem 2.51: If integers r and s are coprime, then $C_{rs} \cong C_r \times C_s$.

Another useful theorem, stated here without proof, concerns groups of order p^2 .

Theorem 2.78. *Every group of order p^2 , where p is a prime, is isomorphic to either the cyclic group C_{p^2} or the direct product $C_p \times C_p$.*

Proof. See Chapter 6, (1.14) of [2]. □

We will now classify the groups of order 8 up to isomorphism. First, we introduce an important example.

Example 2.79. *Consider the following 2×2 matrices:*

$$I = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}, \quad \mathbf{i} = \begin{pmatrix} i & 0 \\ 0 & -i \end{pmatrix}, \quad \mathbf{j} = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}, \quad \mathbf{k} = \begin{pmatrix} 0 & i \\ i & 0 \end{pmatrix}.$$

The set

$$\mathbb{H} = \{I, -I, \mathbf{i}, -\mathbf{i}, \mathbf{j}, -\mathbf{j}, \mathbf{k}, -\mathbf{k}\}$$

forms a subgroup of $\text{GL}(2, \mathbb{C})$ of order 8, known as the **quaternion group**. The elements satisfy the relations:

$$\mathbf{i}^2 = \mathbf{j}^2 = \mathbf{k}^2 = -I, \quad \mathbf{ij} = -\mathbf{ji} = \mathbf{k}, \quad \mathbf{jk} = -\mathbf{kj} = \mathbf{i}, \quad \mathbf{ki} = -\mathbf{ik} = \mathbf{j}.$$

Abstractly, $\mathbb{H}$ can be described as a group generated by two distinct elements x and y of order 4, subject to the relations $x^4 = 1$, $y^2 = x^2$ and $xyx^{-1} = x^{-1}$. (One can take x and y to be any pair of distinct elements from $\{\pm\mathbf{i}, \pm\mathbf{j}, \pm\mathbf{k}\}$.)

Note that all subgroups of $\mathbb{H}$ are normal. We can see this as follows: The subgroups $\{I\}$, $\{I, -I\}$, and $\mathbb{H}$ are normal. The other proper subgroups, such as $\langle \mathbf{i} \rangle = \{\pm I, \pm \mathbf{i}\}$, have index 2 and are therefore also normal.

Theorem 2.80. *There are exactly five isomorphism classes of groups of order 8, represented by C_8 , $C_2 \times C_4$, $C_2 \times C_2 \times C_2$, D_4 , and $\mathbb{H}$.*

Part 1: The abelian groups. Let G be an abelian group of order 8. By Lagrange's theorem, the order of any element in G must be 1, 2, 4, or 8.

- If G contains an element of order 8, then $G \cong C_8$ by Lemma 2.10.
- If all non-identity elements of G have order 2, then by Theorem 2.54, $G \cong C_2 \times C_2 \times C_2$.
- Suppose G is not cyclic but contains an element x of order 4. Let $H = \langle x \rangle = \{1, x, x^2, x^3\}$, a subgroup of index 2, and let $y \in G \setminus H$. By Lemma 2.46, $y^2 \in H$. If $y^2 = x$ or $y^2 = x^3$, then y would have order 8 and G would be cyclic (Lemma 2.10); hence $y^2 = 1$ or $y^2 = x^2$. In the latter case, we replace y by $yx \in G \setminus H$, which satisfies $(yx)^2 = y^2x^2 = x^4 = 1$ since G is abelian. In either case, we obtain an element $z \in G \setminus H$ of order 2, and $G = H \cup zH = \{z^i x^j \mid i \in \{0, 1\}, j \in \{0, 1, 2, 3\}\}$. The map $\phi : G \rightarrow C_2 \times C_4$ defined by $\phi(z^i x^j) = (i, j)$ is a bijective homomorphism (using again that G is abelian). Thus, $G \cong C_2 \times C_4$.

We conclude that there are three non-isomorphic abelian groups of order 8: C_8 , $C_2 \times C_4$, and $C_2 \times C_2 \times C_2$. $\square$

Part 2: The non-abelian groups. Let G be a non-abelian group of order 8. G cannot have an element of order 8, as it would then be cyclic and thus abelian. Therefore, all non-identity elements have order 2 or 4. Not all can have order 2, or G would be abelian. Thus, G must contain an element x of order 4. Let $H = \langle x \rangle = \{1, x, x^2, x^3\}$. Since $[G : H] = 2$, H is a normal subgroup of G .

Let y be any element in $G \setminus H$. Since H is normal, $xyx^{-1} \in H$. As conjugation preserves order, xyx^{-1} must be an element of order 4 in H . The only elements of order 4 in H are x and $x^3 = x^{-1}$. If $xyx^{-1} = x$, then $yx = xy$, meaning y commutes with x . Since y also commutes with all powers of x , and $G = H \cup yH$, it follows that G is abelian, which is a contradiction. Therefore, we must have $xyx^{-1} = x^{-1}$. Now we consider the order of y .

- Suppose y has order 2. Then G is generated by x and y with the relations $x^4 = 1, y^2 = 1$, and $xyx^{-1} = x^{-1}$. These are the defining relations for the dihedral group D_4 . Thus, $G \cong D_4$.
- Suppose y has order 4. Since H has index 2 in G , we have $y^2 \in H$ by Lemma 2.46; as y^2 has order 2 and the only element of order 2 in H is x^2 , it follows that $y^2 = x^2$. Then G is generated by x and y with the relations $x^4 = 1, y^2 = x^2$, and $xyx^{-1} = x^{-1}$. These are the defining relations for the quaternion group $\mathbb{H}$. Thus, $G \cong \mathbb{H}$.

Therefore, any non-abelian group of order 8 is isomorphic to either D_4 or $\mathbb{H}$. $\square$

Using the theorems discussed, we can list the distinct groups for orders up to 11.

Example 2.81. *Up to isomorphism, the groups of order 11 or less are:*

1. Order 1: $\{e\}$ (isomorphic to C_1)
2. Order 2: C_2
3. Order 3: C_3
4. Order 4: $C_4, C_2 \times C_2$
5. Order 5: C_5
6. Order 6: C_6 (isomorphic to $C_2 \times C_3$), D_3 (isomorphic to S_3)
7. Order 7: C_7
8. Order 8: $C_8, C_2 \times C_4, C_2 \times C_2 \times C_2, D_4, \mathbb{H}$
9. Order 9: $C_9, C_3 \times C_3$
10. Order 10: C_{10}, D_5
11. Order 11: C_{11}

All groups listed for a given order are non-isomorphic. For a more extensive list, see [Wikipedia's list of small groups](#).

2.10 Homomorphisms

We now explore maps between groups that preserve the group structure without necessarily being bijective. Recall from Section 2.5 that a map $f : G \rightarrow H$ between groups $(G, *)$ and $(H, \cdot)$ is a **homomorphism** if for all $x, y \in G$,

$$f(x * y) = f(x) \cdot f(y) ,$$

and that a bijective homomorphism is an isomorphism.

Lemma 2.82. *Let $f : G \rightarrow H$ be a homomorphism.*

- (i) *f maps the identity of G to the identity of H : $f(e_G) = e_H$.*
- (ii) *f preserves inverses: for any $x \in G$, $f(x^{-1}) = [f(x)]^{-1}$.*

Proof. Both statements were established in the proof of Theorem 2.49 (points 2 and 3), where bijectivity was not used. $\square$

Henceforth, we will often write e for both identity elements, unless clarity demands otherwise.

Definition 2.83. If $\theta : G \rightarrow H$ is a homomorphism, we define its **image**, $\text{Im}(\theta)$, and its **kernel**, $\ker(\theta)$, as follows:

$$\text{Im}(\theta) = \theta(G) = \{\theta(g) : g \in G\},$$

$$\ker(\theta) = \{g \in G : \theta(g) = e\}.$$

Example 2.84. The determinant map, $\det : \text{GL}(n, \mathbb{R}) \rightarrow (\mathbb{R} \setminus \{0\}, \times)$, is a homomorphism. Its kernel is the special linear group, $\text{SL}(n, \mathbb{R})$.

Lemma 2.85. A homomorphism $\theta : G \rightarrow H$ is injective if and only if its kernel is trivial, i.e., $\ker(\theta) = \{e_G\}$.

Proof. Suppose θ is injective. If $g \in \ker(\theta)$, then $\theta(g) = e_H = \theta(e_G)$. By injectivity, $g = e_G$. Thus, $\ker(\theta) = \{e_G\}$. Conversely, suppose $\ker(\theta) = \{e_G\}$. If $\theta(g_1) = \theta(g_2)$ for some $g_1, g_2 \in G$, then $\theta(g_1 g_2^{-1}) = \theta(g_1) \theta(g_2^{-1}) = \theta(g_1) [\theta(g_2)]^{-1} = e_H$. This implies $g_1 g_2^{-1} \in \ker(\theta)$, so $g_1 g_2^{-1} = e_G$, which gives $g_1 = g_2$. Thus, θ is injective. $\square$

Lemma 2.86. Let $\theta : G \rightarrow H$ be a homomorphism.

- (i) The image $\text{Im}(\theta)$ is a subgroup of H .
- (ii) The kernel $\ker(\theta)$ is a subgroup of G .
- (iii) The kernel $\ker(\theta)$ is a normal subgroup of G .

Proof. (i) Since $e_H = \theta(e_G)$, the image contains the identity. If $h = \theta(g) \in \text{Im}(\theta)$, then $h^{-1} = \theta(g^{-1}) \in \text{Im}(\theta)$. If $h_1 = \theta(g_1)$ and $h_2 = \theta(g_2)$ are in the image, then $h_1 h_2 = \theta(g_1 g_2) \in \text{Im}(\theta)$. Thus, $\text{Im}(\theta)$ is a subgroup of H .

(ii) Since $\theta(e_G) = e_H$, $e_G \in \ker(\theta)$. If $g \in \ker(\theta)$, then $\theta(g^{-1}) = [\theta(g)]^{-1} = e_H^{-1} = e_H$, so $g^{-1} \in \ker(\theta)$. If $g_1, g_2 \in \ker(\theta)$, then $\theta(g_1 g_2) = \theta(g_1) \theta(g_2) = e_H e_H = e_H$, so $g_1 g_2 \in \ker(\theta)$. Thus, $\ker(\theta)$ is a subgroup of G .

(iii) We must show that for any $k \in \ker(\theta)$ and any $g \in G$, the conjugate $g k g^{-1}$ is also in the kernel. We compute its image: $\theta(g k g^{-1}) = \theta(g) \theta(k) \theta(g^{-1}) = \theta(g) e_H [\theta(g)]^{-1} = e_H$. Thus, $g k g^{-1} \in \ker(\theta)$, which proves that the kernel is a normal subgroup. $\square$

2.11 The Isomorphism Theorem

The concepts of homomorphism, kernel, and quotient group are elegantly connected by the following fundamental theorem.

Theorem 2.87. (The First Isomorphism Theorem.) *If $\theta : G \rightarrow H$ is a homomorphism, then the quotient group $G/\ker(\theta)$ is isomorphic to the image $\theta(G)$.*

$$G/\ker(\theta) \cong \text{Im}(\theta).$$

Proof. Let $K = \ker(\theta)$. We define a map $\phi : G/K \rightarrow \theta(G)$ by $\phi(gK) = \theta(g)$. The chain of equivalences

$$g_1K = g_2K \iff g_2^{-1}g_1 \in K \iff \theta(g_2)^{-1}\theta(g_1) = e_H \iff \theta(g_1) = \theta(g_2)$$

shows, read from left to right, that ϕ is well defined and, read from right to left, that ϕ is injective. It is surjective by construction, since any element of $\theta(G)$ has the form $\theta(g) = \phi(gK)$. Finally, ϕ is a homomorphism: $\phi((g_1K)(g_2K)) = \phi(g_1g_2K) = \theta(g_1g_2) = \theta(g_1)\theta(g_2) = \phi(g_1K)\phi(g_2K)$. Since ϕ is a bijective homomorphism, it is an isomorphism. $\square$

Example 2.88. *The map $\det : \text{GL}(n, \mathbb{R}) \rightarrow (\mathbb{R} \setminus \{0\}, \times)$ is a surjective homomorphism with kernel $\text{SL}(n, \mathbb{R})$. By the Isomorphism Theorem, $\text{GL}(n, \mathbb{R})/\text{SL}(n, \mathbb{R}) \cong \mathbb{R} \setminus \{0\}$.*

Example 2.89. *Consider the homomorphism $\theta : (\mathbb{Z}, +) \rightarrow (C_n, \cdot)$ given by $\theta(r) = a^r$, where a is a generator of C_n . The image is all of C_n , and the kernel is the set of integers r such that $a^r = e$, which is precisely $n\mathbb{Z}$. Thus, $\mathbb{Z}/n\mathbb{Z} \cong C_n$.*

Example 2.90. *The map $f : (\mathbb{R}, +) \rightarrow (\text{U}(1), \times)$ given by $f(x) = e^{2\pi i x}$ is a surjective homomorphism. Its kernel is $\mathbb{Z}$. Therefore, $\mathbb{R}/\mathbb{Z} \cong \text{U}(1)$.*

Example 2.91. *Consider the map $\theta : D_3 \rightarrow C_2$ defined by $\theta(r^s\sigma^t) = (-1)^t$. This is a homomorphism where rotations are mapped to the identity and reflections are mapped to the non-identity element. The kernel is the subgroup of rotations, $\langle r \rangle \cong C_3$, and the image is C_2 . Thus, $D_3/C_3 \cong C_2$. Note that this does not imply $D_3 \cong C_3 \times C_2$, as D_3 is non-abelian.*

Theorem 2.92. *If H and K are normal subgroups of a group G such that $H \cap K = \{e\}$ and $HK = G$, then $G \cong H \times K$.*

Proof. First, we show that elements of H and K commute. For any $h \in H$ and $k \in K$, consider the element $hkh^{-1}k^{-1}$. Since K is normal, $hkh^{-1} \in K$, so $hkh^{-1}k^{-1} \in K$. Since H is normal, $khk^{-1} \in H$, so its inverse $kh^{-1}k^{-1} \in H$, which means $h(kh^{-1}k^{-1}) \in H$. The element $hkh^{-1}k^{-1}$ is therefore in $H \cap K$. Since the intersection is trivial, $hkh^{-1}k^{-1} = e$, which implies $hk = kh$.

Now, define the map $\phi : H \times K \rightarrow G$ by $\phi(h, k) = hk$. This map is a homomorphism because elements of H and K commute: $\phi((h_1, k_1)(h_2, k_2)) = \phi(h_1h_2, k_1k_2) = h_1h_2k_1k_2 = h_1k_1h_2k_2 = \phi(h_1, k_1)\phi(h_2, k_2)$. The map is surjective since $HK = G$. To see it is injective, suppose $\phi(h, k) = e$. Then $hk = e$, so $h = k^{-1}$. Since $h \in H$ and $k^{-1} \in K$, this element is in $H \cap K = \{e\}$. Thus $h = e$ and $k = e$, so the kernel is trivial. Hence, ϕ is an isomorphism. $\square$

Example 2.93. (i) For $n \geq 2$, $O(n)/SO(n) \cong C_2$. The determinant is a homomorphism from $O(n)$ to $\{1, -1\}$ with kernel $SO(n)$. (ii) For odd $n \geq 3$, $O(n) \cong SO(n) \times C_2$.

Proof. (ii) Let $C_2 = \{1, -1\}$. This is a normal subgroup of $O(n)$, since $R1R^{-1} = 1 \in C_2$ and $R(-1)R^{-1} = -1 \in C_2$ for any $R \in O(n)$. For odd n , $\det(-1) = (-1)^n = -1$, so $SO(n) \cap C_2 = \{1\}$. We claim that, for odd n , any matrix $A \in O(n)$ can be written as B or $(-1)B$ for some $B \in SO(n)$. If $\det(A) = 1$, then $A \in SO(n)$. But, if $\det(A) = -1$, then $A = (-1)[-1A]$, with $-1A \in SO(n)$, since $\det(-1A) = \det(-1)\det(A) = 1$. Thus $O(n) = SO(n)C_2$. By Theorem 2.92, the result follows. This does not hold for even n , because in that case $-1 \in SO(n)$. $\square$

2.12 Group actions and the orbit-stabiliser theorem

We now formalise the idea of a group acting on a set.

Definition 2.94. An **action** of a group G on a non-empty set X is a map from $G \times X$ to X , denoted by $(g, x) \mapsto gx$, that satisfies two axioms:

(i) $(gh)x = g(hx)$ for all $g, h \in G$ and all $x \in X$. (Associativity)

(ii) $ex = x$ for all $x \in X$, where e is the identity element of G . (Identity)

Example 2.95. (i) The symmetric group $G = S_n$ acts on the set $X = \{1, \dots, n\}$ by permutation: for $\sigma \in S_n$ and $i \in X$, the action is $\sigma i = \sigma(i)$.

(ii) The additive group of integers $G = (\mathbb{Z}, +)$ acts on the set of real numbers $X = \mathbb{R}$ by translation: for $n \in \mathbb{Z}$ and $x \in \mathbb{R}$, the action is $nx = x + n$.

Here are two important consequences of this definition:

- An action of G on X is equivalent to a homomorphism $\pi : G \rightarrow P(X)$, where $P(X)$ is the group of all permutations (bijections) of X . For each $g \in G$, the map $\pi(g) : X \rightarrow X$ defined by $\pi(g)(x) = gx$ is a permutation of X . Its inverse is $\pi(g^{-1})$, since $\pi(g^{-1})(\pi(g)(x)) = (g^{-1}g)x = ex = x$. The group axioms for an action correspond directly to the homomorphism property for π . Such a homomorphism is called a **permutation representation** of G .
- A group action induces an equivalence relation on the set X : define $x \sim y$ if and only if $gx = y$ for some $g \in G$. This relation is reflexive ($ex = x$), symmetric ($gx = y$ implies $x = g^{-1}y$) and transitive ($gx = y$ and $hy = z$ imply $(hg)x = z$).

Definition 2.96. Suppose a group G acts on a set X . For any $x \in X$, the **orbit** of x under G , denoted $\text{Orb}(x)$, is the set of all elements in X that can be reached from x by the action of G :

$$\text{Orb}(x) = \{gx : g \in G\}.$$

The group action is called **transitive** if it has only one orbit, i.e. there is an $x \in X$ such that $X = \text{Orb}(x)$.

The orbits are precisely the equivalence classes of the relation defined by the group action. This immediately implies that the orbits partition the set X .

Theorem 2.97. Suppose a group G acts on a set X .

- (i) The set X is the union of all orbits: $X = \bigcup_{x \in X} \text{Orb}(x)$.
- (ii) For any two elements $x, y \in X$, their orbits are either identical or disjoint: $\text{Orb}(x) = \text{Orb}(y)$ or $\text{Orb}(x) \cap \text{Orb}(y) = \emptyset$.
- (iii) If X is finite and the distinct orbits are $\text{Orb}(x_1), \dots, \text{Orb}(x_m)$, then

$$|X| = |\text{Orb}(x_1)| + \dots + |\text{Orb}(x_m)|.$$

Definition 2.98. Suppose a group G acts on a set X . For any $x \in X$, the **stabiliser** of x in G , denoted $\text{Stab}_G(x)$, is the set of all elements in G that fix x :

$$\text{Stab}_G(x) = \{g \in G : gx = x\}.$$

Example 2.99. (i) For any $x \in X$, $\text{Stab}_G(x)$ is a subgroup of G . To see this: if $g, h \in \text{Stab}_G(x)$, then $(gh)x = g(hx) = gx = x$, so $gh \in \text{Stab}_G(x)$. Also, if $gx = x$, then $x = g^{-1}x$, so $g^{-1} \in \text{Stab}_G(x)$.

(ii) The stabiliser is not necessarily a normal subgroup. For example, consider the action of S_3 on $\{1, 2, 3\}$. The stabiliser of $1 \in \{1, 2, 3\}$ is the order two cyclic subgroup generated by (23) , which is obviously not normal in S_3 .

(iii) Let H be a subgroup of a group G . The group G acts transitively on the coset space G/H , since $G/H = \text{Orb}(eH) = \text{Orb}(H)$. The stabiliser of the coset $eH = H$ is the subgroup H of G , since $hH = H$ for any $h \in H$.

The sizes of the orbit and stabiliser are related by a fundamental theorem.

Theorem 2.100. (The Orbit-Stabiliser Theorem.) Suppose a group G acts on a set X , and let $x \in X$.

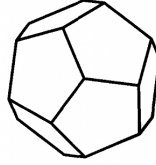
- (i) There is a bijection between the orbit of x and the set of left cosets of the stabiliser of x .
- (ii) If G is finite, then $|\text{Orb}(x)| = |G|/|\text{Stab}_G(x)|$.
- (iii) If G is finite, the size of any orbit divides the order of the group.

Proof. (i) Let $H = \text{Stab}_G(x)$ and define $\phi : G/H \rightarrow \text{Orb}(x)$ by $\phi(gH) = gx$. The chain of equivalences

$$g_1H = g_2H \iff g_2^{-1}g_1 \in H \iff (g_2^{-1}g_1)x = x \iff g_1x = g_2x$$

shows, read from left to right, that ϕ is well defined and, read from right to left, that ϕ is injective. It is surjective since any $y \in \text{Orb}(x)$ has the form $y = gx = \phi(gH)$. Since ϕ is a bijection, the two sets have the same size. (ii) From (i), we have $|\text{Orb}(x)| = |G/H| = |G|/|\text{Stab}_G(x)|$. (iii) This is a direct consequence of (ii). $\square$

Example 2.101. Consider the group G_+ of rotational symmetries of a regular dodecahedron D , which is a polyhedron with 12 regular pentagon faces and 20 vertices. We will compute $|G_+|$ using the orbit-stabiliser theorem.



Consider the action of G_+ on the set S of the faces. The stabiliser of a face s is the group of rotations by multiples of $2\pi/5$ about a perpendicular through the centre of s ; thus, $|\text{Stab}_{G_+}(s)| = 5$. There are 12 faces, and G_+ acts transitively on them; thus, $|\text{Orb}(s)| = 12$. By the orbit-stabiliser theorem, $|G_+| = |\text{Orb}(s)| \cdot |\text{Stab}_{G_+}(s)| = 60$.

Alternatively, G_+ acts transitively on the set containing each vertex v of D ; thus $|\text{Orb}(v)| = 20$, since there are 20 vertices. There are three rotations, including 1, that fix v ; thus, $|\text{Stab}_{G_+}(v)| = 3$. By the orbit-stabiliser theorem, $|G_+| = |\text{Orb}(v)| \cdot |\text{Stab}_{G_+}(v)| = 60$.

Note that the inversion σ through the centre commutes with all rotations. The full symmetry group of D is therefore $G_+ \times \langle \sigma \rangle$, which has order 120.

The regular dodecahedron and the regular icosahedron, which is a regular polyhedron with 20 equilateral triangle faces and 12 vertices, are dual solids. Therefore, their symmetry groups are isomorphic.

2.13 Cayley's theorem

A particularly important action is that of a group on itself. Consider a group G acting on the set $X = G$ by left multiplication: for $g, x \in G$, the action is defined as $g \cdot x = gx$.

- For any $g \in G$, the map $x \mapsto gx$ is a **bijection** on G . It is injective because $gx = gy$ implies $x = y$. It is surjective because for any $z \in G$, we have $z = g(g^{-1}z)$. Thus, each $g \in G$ defines a permutation of G , and we obtain a map $\pi : G \rightarrow P(G)$ sending g to the permutation $x \mapsto gx$. Moreover, π is a **homomorphism**: $\pi(gh)$ sends x to $(gh)x = g(hx)$, so $\pi(gh) = \pi(g)\pi(h)$.

- This action is **transitive**. For any $x, y \in G$, the element $g = yx^{-1}$ satisfies $g \cdot x = (yx^{-1})x = y$. Thus, there is only one orbit, which is the entire group G .
- The action is also **faithful**, meaning the only element that fixes everything is the identity. If $gx = x$ for all $x \in G$, then $gx(x^{-1}) = x(x^{-1})$ and so $g = e$. This means the kernel of the homomorphism $\pi : G \rightarrow P(G)$ contains only the identity; it is thus injective.

This leads to a remarkable theorem showing that every finite group can be viewed as a group of permutations.

Theorem 2.102. (Cayley's Theorem.) *Every finite group G is isomorphic to a subgroup of a permutation group. If $|G| = n$, then G is isomorphic to a subgroup of S_n .*

Proof. Consider the action of G on itself by left multiplication. This defines a homomorphism $\pi : G \rightarrow P(G)$, where $P(G)$ is the group of permutations of the set G . As shown above, this action is faithful, implying $\ker(\pi) = \{e\}$, and so π is injective. An injective homomorphism is an isomorphism onto its image $\pi(G)$. Therefore, $G \cong \pi(G)$, which is a subgroup of $P(G)$. If $|G| = n$, then $P(G) \cong S_n$. $\square$

While Cayley's Theorem is of great theoretical importance, it is not always practical for computation, as the order of S_n is $n!$, which is typically much larger than the order of G .

3 Representations

Representation theory provides a powerful way to study abstract groups by realising their actions on a vector space. This perspective allows us to apply the tools of linear algebra to understand group structures. In the following, we let $\mathbb{F}$ be $\mathbb{R}$ or $\mathbb{C}$.

Definition 3.1. *A **representation** of a group G is a finite-dimensional vector space V over $\mathbb{F}$ equipped with a linear action of G . That is, for every $g \in G$ and $v \in V$, a vector $gv \in V$ is defined, satisfying:*

- (i) $(g_1g_2)v = g_1(g_2v)$ and $ev = v$ for all $g_1, g_2 \in G$ and $v \in V$.
- (ii) $g(v_1 + v_2) = gv_1 + gv_2$ and $g(\lambda v) = \lambda(gv)$ for all $g \in G$, $v, v_1, v_2 \in V$, and $\lambda \in \mathbb{F}$.

The **dimension** of the representation is the dimension of the vector space V over $\mathbb{F}$. The representation is said to be **faithful** if $gv = v$ for all $v \in V$ implies that $g = e$.

Example 3.2. (i) The **trivial representation** of a group G is a one-dimensional vector space $U \cong \mathbb{F}$ with the action $gu = u$ for all $g \in G, u \in U$.

(ii) The **sign representation** of S_n is a space $V \cong \mathbb{F}$ with $\sigma v = \text{sgn}(\sigma)v$.

(iii) For S_n with $n \geq 2$, the trivial representation is not faithful; for $n \geq 3$, the sign representation is also not faithful.

Mathematicians and physicists sometimes adopt the following definition of a representation.

Definition 3.3. (i) For a finite-dimensional vector space V over $\mathbb{F}$, let $\text{GL}(V)$ be the group of all invertible linear transformations from V to itself (automorphisms on V). A **representation** of a group G on V is a homomorphism $\rho : G \rightarrow \text{GL}(V)$.

(ii) The **dimension** of the representation is $\dim_{\mathbb{F}} V$.

(iii) The representation ρ is **faithful** if ρ is injective.

Some remarks are in order.

Remark 3.4. (i) V is a real (or complex) representation if it is a vector space over $\mathbb{R}$ (or $\mathbb{C}$).

(ii) In Definition 3.1, a group element g is treated as an invertible linear transformation from V to itself. The abstract map $\rho(g)$ from Definition 3.3 is identified with the action of g on V . When there is no ambiguity, one can drop ρ and talk about the action of g itself.

(iii) Choosing a basis $\mathcal{B}$ for V allows us to obtain a matrix representation $[g]_{\mathcal{B}}$ for each $g \in G$. A different basis choice yields a conjugate matrix representation: $[g]_{\mathcal{B}'} = P^{-1}[g]_{\mathcal{B}}P$ for some matrix P . Our main goal is to study representations without referring to a particular basis.

Here is a formal definition of the permutation representation.

Definition 3.5. Let G be a group acting on a finite set X . The **permutation representation** of G acting on X is the vector space with basis $\{e_x : x \in X\}$ and a linear group action defined by its effect on the basis vectors: $g \cdot e_x = e_{gx}$. This extends linearly to all vectors:

$$g \left(\sum_{x \in X} a_x e_x \right) = \sum_{x \in X} a_x e_{gx}.$$

Example 3.6. For $G = S_3$, the matrix representation of each element with respect

to the standard basis $\{e_1, e_2, e_3\}$ is

$$\begin{aligned} [e] &= \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}, & [(123)] &= \begin{pmatrix} 0 & 0 & 1 \\ 1 & 0 & 0 \\ 0 & 1 & 0 \end{pmatrix}, & [(132)] &= \begin{pmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 1 & 0 & 0 \end{pmatrix} \\ [(12)] &= \begin{pmatrix} 0 & 1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 1 \end{pmatrix}, & [(23)] &= \begin{pmatrix} 1 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & 1 & 0 \end{pmatrix}, & [(13)] &= \begin{pmatrix} 0 & 0 & 1 \\ 0 & 1 & 0 \\ 1 & 0 & 0 \end{pmatrix} \end{aligned}$$

A special case is when the group acts on itself.

Definition 3.7. The **regular representation** of a finite group G is the permutation representation associated with the action of G on itself by left multiplication. It is the vector space with basis $\{e_g : g \in G\}$ and the action

$$h \left(\sum_{g \in G} a_g e_g \right) = \sum_{g \in G} a_g e_{hg}.$$

Corollary 3.8. (i) The dimension of the permutation representation acting on a finite set X is $|X|$. (ii) The dimension of the regular representation of a finite group G is $|G|$.

3.1 Direct sums and decomposability

Definition 3.9. (i) A subspace W of a representation V is a **subrepresentation** if it is invariant under the action of G ; that is, for all $w \in W$ and $g \in G$, the vector gw is also in W . We also call W a **G -invariant subspace**.

(ii) A representation V is **irreducible** if its only G -invariant subspaces are $\{0\}$ and V itself. Otherwise, V is **reducible**.

Example 3.10. One-dimensional representations are always irreducible.

Example 3.11. The cyclic group $C_3 = \{e, a, a^2\}$ has three distinct one-dimensional irreducible complex representations, denoted U_0, U_1, U_2 . In any one-dimensional representation, the element a must act as multiplication by a scalar λ such that $\lambda^3 = 1$. The three possible values for λ are $1, \omega$, and ω^2 , where $\omega = e^{2\pi i/3}$. The action of the group elements on each representation is summarised in the following table:

C_3	e	a	a^2
U_0	1	1	1
U_1	1	ω	ω^2
U_2	1	ω^2	ω

(3.1)

As we will see, every irreducible complex representation of a finite abelian group is one-dimensional.

Definition 3.12. A representation V is **decomposable** if it can be written as a direct sum of proper, non-zero subrepresentations, $V = U \oplus W$.

If a representation V is decomposable as $V = U \oplus W$, we can choose a basis for V that is a union of a basis for U and a basis for W . In such a basis $\mathcal{B}$, the matrix for any group element g takes a **block diagonal form**:

$$[g]_{\mathcal{B}} = \begin{pmatrix} [g|_U]_{\mathcal{B}_1} & 0 \\ 0 & [g|_W]_{\mathcal{B}_2} \end{pmatrix},$$

where $g|_U$ and $g|_W$ are the restrictions of the action of g to the subspaces U and W , and $\mathcal{B}_1, \mathcal{B}_2$ are the bases for U and W .

Example 3.13. Consider a two dimensional complex representation V of $C_3 = \{e, a, a^2\}$ such that $a^3 = e$, with the following matrix representations with respect to some basis of V .

$$e = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}, \quad a = \frac{1}{2} \begin{pmatrix} -1 & \sqrt{3} \\ -\sqrt{3} & -1 \end{pmatrix}, \quad a^2 = \frac{1}{2} \begin{pmatrix} -1 & -\sqrt{3} \\ \sqrt{3} & -1 \end{pmatrix}.$$

The eigenvalues of a are ω and ω^2 , where $\omega = \exp(2\pi i/3)$. Since the eigenvalues are distinct, there exists a basis $\mathcal{B}$ of V such that a becomes diagonal: $a = \text{diag}(\omega, \omega^2)$. The matrices for the group elements are then

$$[e] = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}, \quad [a] = \begin{pmatrix} \omega & 0 \\ 0 & \omega^2 \end{pmatrix}, \quad [a^2] = \begin{pmatrix} \omega^2 & 0 \\ 0 & \omega \end{pmatrix}.$$

This shows that the representation decomposes into a direct sum of the one-dimensional irreducible representations found in Example 3.11: $V \cong U_1 \oplus U_2$.

3.2 Complete reducibility and Schur's Lemma

A cornerstone of the representation theory of finite groups is the following result.

Theorem 3.14. (Maschke's Theorem.) Let V be a representation of a finite group G over $\mathbb{R}$ or $\mathbb{C}$. If W is a subrepresentation of V , then there exists a complementary G -invariant subspace W' such that $V = W \oplus W'$.

An immediate and powerful consequence of Maschke's Theorem is the property of complete reducibility.

Corollary 3.15. (Complete Reducibility.) Every finite-dimensional representation of a finite group over $\mathbb{C}$ or $\mathbb{R}$ can be written as a direct sum of irreducible representations.

This property also holds for compact groups, such as $U(1)$ and $SU(n)$.

The uniqueness of this decomposition is a consequence of another fundamental result.

Theorem 3.16. (Schur's Lemma.) *Let V and W be irreducible representations of a group G , and let $\phi : V \rightarrow W$ be a homomorphism such that $\phi g(v) = g\phi(v)$ for all $g \in G, v \in V$. (The map ϕ is also known as an **intertwining map**.)*

(i) *Then either ϕ is an isomorphism, or ϕ is the zero map.*

(ii) *If V and W are complex representations and $V = W$, then ϕ must be a scalar multiple of the identity map: $\phi = \lambda \mathbf{1}$ for some $\lambda \in \mathbb{C}$.*

Proof. (i) The kernel, $\ker(\phi)$, is a subrepresentation of V . Since V is irreducible, $\ker(\phi)$ must be either $\{0\}$ (so ϕ is injective) or V (so $\phi = 0$). Similarly, the image, $\text{Im}(\phi)$, is a subrepresentation of W . Since W is irreducible, $\text{Im}(\phi)$ must be either $\{0\}$ (so $\phi = 0$) or W (so ϕ is surjective). Combining these, ϕ must either be the zero map or a bijection (and thus an isomorphism).

(ii) If V is a complex vector space, the linear map $\phi : V \rightarrow V$ must have at least one eigenvalue, $\lambda \in \mathbb{C}$. Consider the map $\phi' = \phi - \lambda \mathbf{1}$. This is also an intertwining map. Since there is a non-zero eigenvector v such that $\phi(v) = \lambda v$, we have $\phi'(v) = 0$, so $\ker(\phi')$ is non-zero. By part (i), since V is irreducible, we must have $\phi' = 0$. Therefore, $\phi = \lambda \mathbf{1}$. $\square$

The decomposition of a representation into irreducibles is unique in the following sense.

Theorem 3.17. *For any representation V of a finite group G , there is a decomposition*

$$V \cong V_1^{\oplus a_1} \oplus \cdots \oplus V_k^{\oplus a_k},$$

*where the V_i are distinct irreducible representations. The irreducible representations V_i that occur, and their multiplicities a_i , are **uniquely determined** by V .*

The existence of such a decomposition is Corollary 3.15. We postpone the proof of uniqueness to Section 3.6: Corollary 3.38 there expresses each multiplicity a_i as an inner product of characters, a quantity which depends only on V and V_i and not on the chosen decomposition.

Convention 3.18. *The decomposition $V = V_1^{\oplus a_1} \oplus \cdots \oplus V_k^{\oplus a_k}$ is sometimes written as $V = a_1 V_1 \oplus \cdots \oplus a_k V_k$ to emphasise the multiplicities.*

3.3 Applications to abelian groups

An immediate consequence of Schur's Lemma is a fundamental fact about the representations of abelian groups.

Corollary 3.19. *The irreducible complex representations of a finite abelian group are all one-dimensional.*

Proof. Let V be an irreducible complex representation of an abelian group G . For any fixed element $g \in G$, the map $\phi_g : V \rightarrow V$ defined by $\phi_g(v) = gv$ is a linear map. Since G is abelian, this map commutes with the action of any other element $h \in G$:

$$\phi_g(hv) = g(hv) = (gh)v = (hg)v = h(gv) = h(\phi_g(v)).$$

Thus, ϕ_g is an intertwining map from V to itself, and by Schur's Lemma (part ii), $\phi_g = \lambda_g \mathbf{1}$ for some $\lambda_g \in \mathbb{C}$. Every element of G therefore acts on V by a scalar, so *every* subspace of V is G -invariant. Irreducibility then forces V to have no proper non-zero subspaces, i.e., $\dim V = 1$. $\square$

Proposition 3.20. *A finite abelian group $G = C_{n_1} \times \dots \times C_{n_r}$ has exactly $|G| = \prod_{j=1}^r n_j$ distinct irreducible complex representations.*

Proof. By the preceding corollary, any irreducible complex representation V is one-dimensional. The generator x_j of the factor C_{n_j} acts on V as multiplication by a scalar λ_j satisfying $\lambda_j^{n_j} = 1$, i.e., an n_j -th root of unity, and a general element $(x_1^{k_1}, \dots, x_r^{k_r})$ then acts as multiplication by $\lambda_1^{k_1} \dots \lambda_r^{k_r}$. The representation is thus uniquely determined by the r -tuple $(\lambda_1, \dots, \lambda_r)$, and every such tuple defines one. Since there are n_j choices for each λ_j , there are $n_1 \dots n_r = |G|$ distinct irreducible representations. $\square$

Example 3.21. (i) *The cyclic group C_n has n distinct one-dimensional irreducible complex representations. This was demonstrated for C_3 in Example 3.11.*

(ii) *Consider the Klein four-group $V_4 \cong C_2 \times C_2 = \{e, x_1, x_2, x_1x_2\}$. It has 4 irreducible representations, determined by whether x_1 and x_2 act as $+1$ or -1 .*

$C_2 \times C_2$	e	x_1	x_2	x_1x_2
U_0	1	1	1	1
U_1	1	1	-1	-1
U_2	1	-1	1	-1
U_3	1	-1	-1	1

(3.2)

3.4 Inner product and unitary representations

Definition 3.22. *A Hermitian inner product on a complex vector space V is a map $\langle \cdot, \cdot \rangle : V \times V \rightarrow \mathbb{C}$ such that for all $u, v, w \in V$ and all $\lambda, \mu \in \mathbb{C}$:*

(i) *Hermitian symmetry:* $\langle v, w \rangle = \overline{\langle w, v \rangle}$.

(ii) *Linearity in the second argument:* $\langle w, \lambda u + \mu v \rangle = \lambda \langle w, u \rangle + \mu \langle w, v \rangle$. (Together with (i), this implies conjugate-linearity in the first argument. This is the convention used throughout these notes.)

(iii) *Positive-definiteness:* $\langle v, v \rangle > 0$ if $v \neq 0$.

Definition 3.23. A Hermitian inner product $\langle \cdot, \cdot \rangle$ on a representation V of a group G is **G -invariant** if $\langle gu, gv \rangle = \langle u, v \rangle$ for all $u, v \in V$ and $g \in G$.

Lemma 3.24. Let V be a representation of G with a G -invariant Hermitian inner product $\langle \cdot, \cdot \rangle$, and let W be a subrepresentation of V . Then the orthogonal complement

$$W^\perp = \{u \in V : \langle u, w \rangle = 0 \text{ for all } w \in W\}$$

is also a subrepresentation of V , and $V = W \oplus W^\perp$.

Proof. From linear algebra, $V = W \oplus W^\perp$ as vector spaces, so we only need to show that $W^\perp$ is G -invariant. Let $u \in W^\perp$, $g \in G$ and $w \in W$. Since W is a subrepresentation, $g^{-1}w \in W$, and by the G -invariance of the inner product,

$$\langle gu, w \rangle = \langle gu, g(g^{-1}w) \rangle = \langle u, g^{-1}w \rangle = 0.$$

Hence $gu \in W^\perp$, and $W^\perp$ is a subrepresentation. □

Theorem 3.25. (Weyl's Unitary Trick.) Any complex representation V of a finite group G admits a G -invariant Hermitian inner product.

Proof. Begin with any arbitrary Hermitian inner product on V , denoted $\{\cdot, \cdot\}$. We can construct a new, G -invariant inner product $\langle \cdot, \cdot \rangle$ by averaging over the group:

$$\langle v, w \rangle = \frac{1}{|G|} \sum_{g \in G} \{gv, gw\}.$$

This is still a valid Hermitian inner product. It is G -invariant: for any $h \in G$,

$$\langle hv, hw \rangle = \frac{1}{|G|} \sum_{g \in G} \{(gh)v, (gh)w\} = \langle v, w \rangle,$$

since $g' = gh$ runs over all of G as g does. □

This result provides a direct proof of Maschke's Theorem.

Proof of Maschke's Theorem 3.14. Let W be a subrepresentation of a representation V of a finite group G . By Theorem 3.25, we can equip V with a G -invariant Hermitian inner product. By Lemma 3.24, the orthogonal complement $W^\perp$ is also a G -invariant subspace. Setting $W' = W^\perp$, we obtain the direct sum decomposition $V = W \oplus W'$, where both W and W' are subrepresentations. □

A representation equipped with a G -invariant inner product is called a unitary representation.

Corollary 3.26. *Any complex representation of a finite group G is equivalent to a unitary representation. That is, any finite subgroup of $\mathrm{GL}(n, \mathbb{C})$ is conjugate to a subgroup of the unitary group $\mathrm{U}(n)$.*

Proof. Let $\{\cdot, \cdot\}$ be the standard inner product on $\mathbb{C}^n$; a matrix A is unitary if $\{Ax, Ay\} = \{x, y\}$ for all x, y . By Weyl's unitary trick, there is a G -invariant inner product $\langle \cdot, \cdot \rangle$ on $\mathbb{C}^n$. Choose, by the Gram–Schmidt process, a basis orthonormal with respect to $\langle \cdot, \cdot \rangle$, and let P be the change-of-basis matrix from the standard basis, so that $\langle Pu, Pv \rangle = \{u, v\}$. Then, for $g' = P^{-1}gP$,

$$\{g'u, g'v\} = \langle gPu, gPv \rangle = \langle Pu, Pv \rangle = \{u, v\}, \quad (3.3)$$

using the G -invariance of $\langle \cdot, \cdot \rangle$ in the second equality. Thus, g' is unitary for all $g \in G$. $\square$

3.5 Characters

Definition 3.27. *Let V be a representation of a group G . The **character** χ_V of the representation is the complex-valued function on the group defined by the trace of the linear operators:*

$$\chi_V(g) = \mathrm{tr}(g|_V).$$

Characters have several fundamental properties.

Proposition 3.28. *Let V and W be representations of a finite group G .*

- (i) $\chi_V(e) = \dim(V)$.
- (ii) *Characters are constant on conjugacy classes: $\chi_V(hgh^{-1}) = \chi_V(g)$ for all $g, h \in G$. A function with this property is called a **class function**.*
- (iii) *For any $g \in G$, $\chi_V(g^{-1}) = \overline{\chi_V(g)}$.*
- (iv) *The character of a direct sum is the sum of the characters: $\chi_{V \oplus W} = \chi_V + \chi_W$.*

Proof. (i) The trace of the identity operator on V is its dimension. (ii) Using the cyclic property of the trace, $\chi_V(hgh^{-1}) = \mathrm{tr}(hgh^{-1}) = \mathrm{tr}(gh^{-1}h) = \mathrm{tr}(g) = \chi_V(g)$. (iii) Since G is finite, any element g has finite order, so its action on V is diagonalisable with eigenvalues $\{\lambda_i\}$ that are roots of unity. The action of g^{-1} has eigenvalues $\{\lambda_i^{-1}\} = \{\overline{\lambda_i}\}$. Therefore, $\chi_V(g^{-1}) = \sum \lambda_i^{-1} = \sum \overline{\lambda_i} = \overline{\sum \lambda_i} = \overline{\chi_V(g)}$. (iv) In a basis adapted to the decomposition $V \oplus W$, the matrix for any g is block-diagonal. The trace of a block-diagonal matrix is the sum of the traces of the blocks. $\square$

Proposition 3.29. (The fixed-point formula.) *If V is the permutation representation associated with the action of a group G on a finite set X , then $\chi_V(g)$ is the number of elements in X fixed by g .*

Proof. Let $\{e_x\}_{x \in X}$ be the basis for V . The action of g sends e_x to e_{gx} . The matrix of this transformation in this basis has a 1 in the (gx, x) position for each $x \in X$, and 0s elsewhere. The trace is the sum of the diagonal entries. A diagonal entry $(g)_{xx}$ is 1 if $gx = x$ and 0 otherwise. Therefore, the trace is the count of elements x such that $gx = x$. $\square$

Example 3.30. Consider the **permutation representation** of S_3 on $X = \{1, 2, 3\}$. The space is $\mathbb{C}^3$. The conjugacy classes are represented by $e, (12), (123)$.

- e fixes all 3 elements, so $\chi_{\text{perm}}(e) = 3$.
- A transposition like (12) fixes one element (3) , so $\chi_{\text{perm}}(12) = 1$.
- A 3-cycle like (123) fixes no elements, so $\chi_{\text{perm}}(123) = 0$.

Definition 3.31. A **character table** of a finite group G is a table whose rows are indexed by the irreducible representations V_i , whose columns are indexed by the conjugacy classes C_j , and whose (i, j) -entry is the value of the character χ_{V_i} on any element of the class C_j . The size of each conjugacy class is typically written above the column header.

Example 3.32. (The character table of C_n .) The group $C_n = \langle a \mid a^n = e \rangle$ is abelian, so it has n conjugacy classes, $\{a^k\}$, and n one-dimensional irreducible representations, U_j . In the representation U_j , the generator a acts as multiplication by ω^j , where $\omega = e^{2\pi i/n}$. The element a^k acts as multiplication by ω^{jk} . The character is simply this value.

C_n	e	a	$\dots$	a^{n-1}
U_0	1	1	$\dots$	1
U_1	1	ω	$\dots$	ω^{n-1}
$\vdots$	$\vdots$	$\vdots$	$\ddots$	$\vdots$
U_{n-1}	1	ω^{n-1}	$\dots$	$\omega^{(n-1)^2}$

(3.4)

3.6 More Properties of Characters

Definition 3.33. The set of **class functions** on a group G is the complex vector space of functions $f : G \rightarrow \mathbb{C}$ that are constant on conjugacy classes:

$$\mathcal{C}(G) = \{f : G \rightarrow \mathbb{C} \mid f(hgh^{-1}) = f(g) \text{ for all } g, h \in G\}.$$

Characters are class functions, so $\chi_V \in \mathcal{C}(G)$ for any representation V .

Lemma 3.34. The dimension of the space of class functions $\mathcal{C}(G)$ is equal to the number of conjugacy classes of G .

Proof. Let $C_1, \dots, C_k$ be the conjugacy classes of G . For each $j \in \{1, \dots, k\}$, define the indicator function δ_j by $\delta_j(g) = 1$ if $g \in C_j$ and 0 otherwise. These k functions are linearly independent and form a basis for $\mathcal{C}(G)$. $\square$

Definition 3.35. *The standard Hermitian inner product on the space of class functions $\mathcal{C}(G)$ is defined as*

$$(f_1, f_2) = \frac{1}{|G|} \sum_{g \in G} \overline{f_1(g)} f_2(g).$$

The most important result in character theory relates irreducible characters to this inner product.

Theorem 3.36. (Orthogonality Relations for Characters.) *(i) The characters of the irreducible representations of a finite group G are orthonormal with respect to this inner product. If V and W are irreducible representations, then*

$$(\chi_V, \chi_W) = \frac{1}{|G|} \sum_{g \in G} \overline{\chi_V(g)} \chi_W(g) = \begin{cases} 1 & \text{if } V \cong W \\ 0 & \text{if } V \not\cong W. \end{cases} \quad (3.5)$$

(ii) The set of irreducible characters forms an orthonormal basis for the space of class functions $\mathcal{C}(G)$.

Proof. See [4, Section 2.2 and Proposition 2.30]. $\square$

Corollary 3.37. *The number of non-isomorphic irreducible representations of a finite group G is equal to the number of its conjugacy classes.*

This orthogonality is extremely powerful for decomposing representations.

Corollary 3.38. (Any complex representation V of a finite group G is uniquely determined by its character.) *If the decomposition of V into distinct irreducible representations V_i is*

$$V \cong a_1 V_1 \oplus a_2 V_2 \oplus \dots \oplus a_k V_k,$$

then the character of V is $\chi_V = \sum_{i=1}^k a_i \chi_{V_i}$. The multiplicity a_i of each irreducible representation is given by the inner product:

$$a_i = (\chi_{V_i}, \chi_V) = \frac{1}{|G|} \sum_{g \in G} \overline{\chi_{V_i}(g)} \chi_V(g). \quad (3.6)$$

This leads to a simple test for irreducibility.

Corollary 3.39. (Irreducibility Criterion) *A representation V is irreducible if and only if its character has norm 1:*

$$(\chi_V, \chi_V) = \frac{1}{|G|} \sum_{g \in G} |\chi_V(g)|^2 = 1. \quad (3.7)$$

Proof. If V is irreducible, the result follows from the orthogonality relations. Conversely, if $(\chi_V, \chi_V) = 1$, let the decomposition of V be $\bigoplus a_i V_i$. Then $\chi_V = \sum a_i \chi_{V_i}$. The inner product becomes $(\chi_V, \chi_V) = \sum a_i^2$. If this sum of squares of non-negative integers is 1, exactly one a_i must be 1 and all others must be 0. Thus, V is irreducible. $\square$

Lemma 3.40. *The character of the regular representation R of a group G is*

$$\chi_R(g) = \begin{cases} |G| & \text{if } g = e \\ 0 & \text{if } g \neq e. \end{cases}$$

Proof. The regular representation is the permutation representation of G acting on itself. The character $\chi_R(g)$ is the number of elements fixed by g . If $g \neq e$, then $gx \neq x$ for all $x \in G$, so the number of fixed points is 0. If $g = e$, it fixes all $|G|$ elements. $\square$

Lemma 3.41. *The multiplicity of any irreducible representation V_i in the regular representation R is equal to its dimension, $\dim(V_i)$.*

Proof. Let the multiplicity be a_i . Using the formula from Corollary 3.38:

$$a_i = (\chi_{V_i}, \chi_R) = \frac{1}{|G|} \sum_{g \in G} \overline{\chi_{V_i}(g)} \chi_R(g).$$

Since $\chi_R(g)$ is non-zero only at $g = e$, the sum reduces to one term:

$$a_i = \frac{1}{|G|} \overline{\chi_{V_i}(e)} \chi_R(e) = \frac{1}{|G|} (\dim V_i) |G| = \dim V_i.$$

$\square$

This gives a remarkable formula relating the dimensions of the irreducible representations.

Corollary 3.42. *For any finite group G , the sum of the squares of the dimensions of its irreducible representations is equal to the order of the group:*

$$|G| = \sum_i (\dim V_i)^2. \quad (3.8)$$

Proof. From Lemma 3.41, the regular representation decomposes as $R \cong \bigoplus_i (\dim V_i) V_i$. Taking the dimension of both sides gives $\dim(R) = \sum_i (\dim V_i) \dim(V_i) = \sum_i (\dim V_i)^2$. Since $\dim(R) = |G|$, the result follows. $\square$

Corollary 3.43. *For any non-identity element $g \in G$, the characters of the irreducible representations satisfy:*

$$\sum_i (\dim V_i) \chi_{V_i}(g) = 0. \quad (3.9)$$

Proof. This follows from evaluating the character of the regular representation at $g \neq e$. We have $\chi_R(g) = 0$. Also, from its decomposition, $\chi_R(g) = \sum_i (\dim V_i) \chi_{V_i}(g)$. Equating the two gives the result. $\square$

3.7 The character table of S_3 (and D_3)

The group S_3 has three conjugacy classes, represented by $e, (12), (123)$. It therefore has three irreducible representations.

- **The trivial representation U or $\square\square\square$:** This is one-dimensional, with character $\chi_U = (1, 1, 1)$ on the conjugacy classes $[e], [(12)], [(123)]$.
- **The sign representation U' or $\begin{smallmatrix} \square \\ \square \\ \square \end{smallmatrix}$:** This is one-dimensional, with character $\chi_{U'} = (1, -1, 1)$ on the conjugacy classes $[e], [(12)], [(123)]$.
- **The standard representation V or $\begin{smallmatrix} \square & \square \\ \square & \square \end{smallmatrix}$:** From $|S_3| = \sum (\dim V_i)^2$, we have $6 = 1^2 + 1^2 + (\dim V)^2$, so $\dim V = 2$. Let its character be $\chi_V = (a, b, c)$ on the classes $[e], [(12)], [(123)]$. We know $a = \dim V = 2$. Using the orthogonality relation from (3.9) on the class of (12) : $1 \cdot \chi_U(12) + 1 \cdot \chi_{U'}(12) + 2 \cdot \chi_V(12) = 1(1) + 1(-1) + 2b = 0$, which gives $b = 0$. For the class of (123) : $1(1) + 1(1) + 2c = 0$, which gives $c = -1$. Thus, $\chi_V = (2, 0, -1)$.

The character table for S_3 is:

S_3	1 e	3 (12)	2 (123)
trivial U $\square\square\square$	1	1	1
sign U' $\begin{smallmatrix} \square \\ \square \\ \square \end{smallmatrix}$	1	-1	1
standard V $\begin{smallmatrix} \square & \square \\ \square & \square \end{smallmatrix}$	2	0	-1

(3.10)

Remark 3.44. The character of the standard representation V can also be found from its geometric interpretation as the symmetries of an equilateral triangle (D_3). A rotation by $2\pi/3$ (a 3-cycle) and a reflection (a transposition) can be represented by matrices

$$(123) \rightarrow \begin{pmatrix} \cos(2\pi/3) & -\sin(2\pi/3) \\ \sin(2\pi/3) & \cos(2\pi/3) \end{pmatrix}, \quad (12) \rightarrow \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}.$$

The traces are -1 and 0 , respectively, matching the character table.

Remark 3.45. The 3-dimensional permutation representation $\mathbb{C}^3$ of S_3 has character $\chi_{\text{perm}} = (3, 1, 0)$, as found in Example 3.30. Using the character table, we see that $\chi_{\text{perm}} = \chi_U + \chi_V$. This shows that the permutation representation decomposes as $\mathbb{C}^3 \cong U \oplus V$.

3.8 The character table of D_4

The dihedral group D_4 of order 8 is the symmetry group of a square. It is generated by r and s , subject to the relations $r^4 = e, s^2 = e, srs = r^{-1}$.

Conjugacy Classes

There are 5 conjugacy classes:

$$\{e\}, \quad \{r^2\}, \quad \{r, r^3\}, \quad \{s, r^2s\}, \quad \{rs, r^3s\}. \quad (3.11)$$

The sizes of these classes are 1, 1, 2, 2, 2, respectively.

Irreducible Representations and Characters

Since there are 5 conjugacy classes, there must be 5 irreducible representations. The sum of the squares of their dimensions must be $|D_4| = 8$. The only integer solution to $d_1^2 + d_2^2 + d_3^2 + d_4^2 + d_5^2 = 8$ is $1^2 + 1^2 + 1^2 + 1^2 + 2^2 = 8$. Thus, D_4 has four 1-dimensional and one 2-dimensional irreducible representations.

- **The 2D representation V :** This is the standard geometric representation of D_4 on the plane. In a suitable basis, the generators can be represented by matrices:

$$r \mapsto \begin{pmatrix} i & 0 \\ 0 & -i \end{pmatrix}, \quad s \mapsto \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}.$$

From these matrices, we can find the matrices for all group elements and compute their traces to find the character χ_V .

- **The 1D representations U_i :** These are homomorphisms from D_4 to $\mathbb{C}^*$. In any 1D representation, the relations imply that the image of r must square to 1, and the image of s must square to 1. This gives four possibilities for the pair $(\chi(r), \chi(s))$, which define the four 1D representations.

The complete character table is:

D_4	1 e	1 r^2	2 r	2 s	2 rs
U_1	1	1	1	1	1
U_2	1	1	1	-1	-1
U_3	1	1	-1	1	-1
U_4	1	1	-1	-1	1
V	2	-2	0	0	0

(3.12)

Further examples of character tables — for the quaternion group, D_5 , A_4 and the general dihedral groups D_n — are collected in [Appendix A](#).

4 New representations from old

Our goal in this section is to construct new representations from existing ones.

4.1 Tensor product

We begin with the tensor product. Let U and V be finite-dimensional vector spaces over $\mathbb{F}$ (either $\mathbb{R}$ or $\mathbb{C}$).

- The tensor product space $U \otimes V$ is formally defined as the vector space spanned by symbols of the form $u \otimes v$, where $u \in U$ and $v \in V$.
- This product $\otimes$ is required to be bilinear, meaning it satisfies the following relations for all $u, u_1, u_2 \in U$, $v, v_1, v_2 \in V$, and $a \in \mathbb{F}$:

$$\begin{aligned} (u_1 + au_2) \otimes v &= u_1 \otimes v + a(u_2 \otimes v) \\ u \otimes (v_1 + av_2) &= u \otimes v_1 + a(u \otimes v_2) \end{aligned} \tag{4.1}$$

- If $\{e_1, \dots, e_n\}$ is a basis for U and $\{f_1, \dots, f_m\}$ is a basis for V , then from bilinearity, the set

$$\{e_j \otimes f_k \mid 1 \leq j \leq n, 1 \leq k \leq m\} \tag{4.2}$$

forms a basis for $U \otimes V$. In particular, the dimension of the tensor product space is

$$\dim(U \otimes V) = (\dim U)(\dim V). \tag{4.3}$$

These properties define the tensor product.

Proposition 4.1. *If V and W are representations of a group G , then their tensor product $V \otimes W$ is also a representation of G , with the group action defined by*

$$g(v \otimes w) = (gv) \otimes (gw) \tag{4.4}$$

for all $v \in V$, $w \in W$, and $g \in G$. The character of this representation is the product of the individual characters:

$$\chi_{V \otimes W}(g) = \chi_V(g)\chi_W(g). \tag{4.5}$$

Proof. Let $g \in G$. Given (4.4), it is clear that $g : \sum_{i,j} \lambda_{ij} v_i \otimes w_j \mapsto \sum_{i,j} \lambda_{ij} (gv_i) \otimes (gw_j)$. This map is a homomorphism $G \rightarrow \text{GL}(V \otimes W)$, and so $V \otimes W$ is a representation of G .

To prove the character formula, let $g \in G$. Since the action of g on V and W is by a matrix of finite order, we can choose a basis $\{v_i\}$ for V and a basis $\{w_j\}$ for W consisting of eigenvectors for the actions of g on these spaces.⁵ Writing $gv_i = \lambda_i v_i$ and $gw_j = \mu_j w_j$, the basis vectors of $V \otimes W$ satisfy

$$g(v_i \otimes w_j) = (gv_i) \otimes (gw_j) = (\lambda_i \mu_j)(v_i \otimes w_j), \tag{4.6}$$

⁵Let $g \in \text{GL}(V)$, where V is a finite-dimensional vector space over $\mathbb{C}$ and $g^m = 1$ for some positive integer m . Then g is diagonalisable, and there exists a basis of eigenvectors of g .

so they are eigenvectors of g with eigenvalues $\lambda_i \mu_j$. The character is the sum of the eigenvalues:

$$\chi_{V \otimes W}(g) = \sum_{i,j} \lambda_i \mu_j = \left(\sum_i \lambda_i \right) \left(\sum_j \mu_j \right) = \chi_V(g) \chi_W(g)$$

as required. $\square$

We can generalise this to define the n -th tensor power of a representation V :

$$V^{\otimes n} = V \otimes V \otimes \cdots \otimes V \quad (n \text{ times}). \quad (4.7)$$

If $\dim_{\mathbb{F}} V = d$ with basis $\{v_i\}$, then $V^{\otimes n}$ has a basis $\{v_{i_1} \otimes \cdots \otimes v_{i_n} \mid i_k \in \{1, \dots, d\}\}$. The symmetric group S_n acts naturally on this space by permuting the factors in a tensor product. For any $\sigma \in S_n$, we can define an action $\sigma : V^{\otimes n} \rightarrow V^{\otimes n}$ by

$$v_1 \otimes \cdots \otimes v_n \mapsto v_{\sigma^{-1}(1)} \otimes \cdots \otimes v_{\sigma^{-1}(n)}, \quad (4.8)$$

so that the vector in the i -th slot is moved to the $\sigma(i)$ -th slot. (The inverse ensures that $\tau(\sigma x) = (\tau\sigma)x$.) This gives rise to a representation of S_n on $V^{\otimes n}$. Furthermore, if V is a representation of a group G , then $V^{\otimes n}$ is also a representation of G via the diagonal action:

$$g(v_1 \otimes \cdots \otimes v_n) = (gv_1) \otimes \cdots \otimes (gv_n). \quad (4.9)$$

The actions of G and S_n on $V^{\otimes n}$ commute with each other. This allows us to define important subrepresentations.

Definition 4.2. *Let V be a representation of a group G .*

1. *The **n -th symmetric power of V** , denoted $\text{Sym}^n V$, is the subspace of $V^{\otimes n}$ consisting of tensors that are invariant under the action of S_n :*

$$\text{Sym}^n V = \{x \in V^{\otimes n} \mid \sigma(x) = x \text{ for all } \sigma \in S_n\}. \quad (4.10)$$

2. *The **n -th exterior power of V** , denoted $\wedge^n V$, is the subspace of $V^{\otimes n}$ consisting of tensors that transform according to the sign of the permutation:*

$$\wedge^n V = \{x \in V^{\otimes n} \mid \sigma(x) = (\text{sgn } \sigma)x \text{ for all } \sigma \in S_n\}. \quad (4.11)$$

If $\{v_1, \dots, v_d\}$ is a basis for V , then bases for these subspaces can be constructed. Their dimensions are given by:⁶

$$\dim_{\mathbb{F}} \text{Sym}^n V = \binom{n+d-1}{n}, \quad \dim_{\mathbb{F}} \wedge^n V = \binom{d}{n}. \quad (4.12)$$

⁶Note that $\dim_{\mathbb{F}} \text{Sym}^n V$ is equal to the number of non-decreasing sequences of n integers in $\{1, 2, \dots, d\}$. Let $x_1 = i_1 - 1$, $x_j = i_j - i_{j-1}$ for $j = 2, \dots, n$, and $x_{n+1} = d - i_n$. Therefore, such non-decreasing sequences are in one-to-one correspondence with the non-negative integer solutions of $x_1 + x_2 + \dots + x_{n+1} = d - 1$. The number of such solutions can be solved using the “stars and bars” method which amounts to counting the permutations of $d - 1$ identical stars and n identical bars. This gives $\frac{(n+d-1)!}{n!(d-1)!}$, which is the first equality of (4.12).

For the special case $n = 2$, we have the decomposition

$$V \otimes V = \text{Sym}^2 V \oplus \wedge^2 V. \quad (4.13)$$

Proof of (4.13). Any tensor $x \in V \otimes V$ can be uniquely written as a sum of a symmetric and an anti-symmetric part:

$$x = \frac{1}{2}(x + \sigma(x)) + \frac{1}{2}(x - \sigma(x)),$$

where σ is the transposition (12). The first term is in $\text{Sym}^2 V$ and the second is in $\wedge^2 V$. $\square$

The characters of these representations are given by the following useful formulae.

$$\chi_{\text{Sym}^2 V}(g) = \frac{1}{2} (\chi_V(g)^2 + \chi_V(g^2)), \quad \chi_{\wedge^2 V}(g) = \frac{1}{2} (\chi_V(g)^2 - \chi_V(g^2)). \quad (4.14)$$

Proof of (4.14). Let $\{v_i\}$ be a basis of eigenvectors for the action of g on V , with $gv_i = \lambda_i v_i$. The vectors $v_i \otimes v_j - v_j \otimes v_i$ (with $i < j$) form an eigenbasis of $\wedge^2 V$ with eigenvalues $\lambda_i \lambda_j$, so $\chi_{\wedge^2 V}(g) = \sum_{i < j} \lambda_i \lambda_j$. Then

$$\chi_V(g)^2 = \left(\sum_i \lambda_i \right)^2 = \sum_i \lambda_i^2 + 2 \sum_{i < j} \lambda_i \lambda_j = \chi_V(g^2) + 2\chi_{\wedge^2 V}(g),$$

where we used $\sum_i \lambda_i^2 = \text{tr}(g^2|_V) = \chi_V(g^2)$. Rearranging gives the formula for $\chi_{\wedge^2 V}(g)$; the formula for $\chi_{\text{Sym}^2 V}(g)$ then follows from $\chi_{\text{Sym}^2 V} + \chi_{\wedge^2 V} = \chi_{V \otimes V} = \chi_V^2$. $\square$

4.2 The character table of S_4

The group S_4 has 5 conjugacy classes, represented by $e, (12), (123), (1234), (12)(34)$. Therefore, it has 5 irreducible representations.

- **The trivial representation** U , $\square\square\square\square$: A 1D representation with character $\chi_U = (1, 1, 1, 1, 1)$.

- **The sign representation** U' , $\begin{smallmatrix} \square \\ \square \\ \square \\ \square \end{smallmatrix}$: A 1D representation with character $\chi_{U'} = (1, -1, 1, -1, 1)$.

- **The standard representation** V , $\begin{smallmatrix} \square & \square \\ \square & \square \end{smallmatrix}$: This can be found by decomposing the 4D permutation representation $\mathbb{C}^4$. The character of the permutation representation is found using the fixed-point formula: $\chi_{\text{perm}} = (4, 2, 1, 0, 0)$. Since the permutation representation contains one copy of the trivial representation ($\mathbb{C}^4 = U \oplus V$), the character of V is

$$\chi_V = \chi_{\text{perm}} - \chi_U = (3, 1, 0, -1, -1). \quad (4.15)$$

We have $\dim V = \chi_V(e) = 3$, and it is irreducible since $(\chi_V, \chi_V) = 1$.

The first three rows of the character table of S_4 looks like

S_4	1 e	6 (12)	8 (123)	6 (1234)	3 (12)(34)
trivial U	1	1	1	1	1
sign U'	1	-1	1	-1	1
standard V	3	1	0	-1	-1

(4.16)

The sum of squares of dimensions so far is $1^2 + 1^2 + 3^2 = 11$. Since $|S_4| = 24$, the sum of the squares of the remaining two irreducible representations must be $24 - 11 = 13$. The only way to write 13 as a sum of two squares is $2^2 + 3^2$. So, the remaining representations have dimensions 2 and 3.

- Let the 3D representation be V' . We can construct it by taking the tensor product of the standard representation V with the sign representation U' . Its character is

$$\chi_{V'} = \chi_V \cdot \chi_{U'} = (3 \cdot 1, 1 \cdot -1, 0 \cdot 1, -1 \cdot -1, -1 \cdot 1) = (3, -1, 0, 1, -1). \quad (4.17)$$

We can check its irreducibility: $(\chi_{V'}, \chi_{V'}) = \frac{1}{24}[1(3^2) + 6(-1)^2 + 8(0)^2 + 6(1)^2 + 3(-1)^2] = \frac{1}{24}(9 + 6 + 0 + 6 + 3) = 1$.

- Let the 2D representation be W . Let its character be $\chi_W = (2, x_2, x_3, x_4, x_5)$. Using the orthogonality relation $\sum_i (\dim V_i) \chi_{V_i}(g) = 0$ for $g \neq e$, we can solve for the unknown character values. For the class of (12),

$$1(1) + 1(-1) + 3(1) + 3(-1) + 2x_2 = 0 \implies x_2 = 0,$$

and, similarly, $x_3 = -1$, $x_4 = 0$ and $x_5 = 2$ for the classes of (123), (1234) and (12)(34), respectively. So, $\chi_W = (2, 0, -1, 0, 2)$. Alternatively, W can be constructed directly: the Klein four-group $V_4 = \{e, (12)(34), (13)(24), (14)(23)\}$ is a normal subgroup of S_4 with $S_4/V_4 \cong S_3$, and composing the quotient map $S_4 \rightarrow S_3$ with the 2D standard representation of S_3 yields W .⁷ Since the standard representation of S_3 has character $(2, 0, -1)$ on the classes of the identity, transpositions and 3-cycles, respectively, we recover the character above. For example, (12)(34) lies in V_4 , so it maps to the identity of S_3 ; hence $\chi_W = \chi_{S_3}(e) = 2$. As for (1234), its square is (13)(24) $\in V_4$, so its image in the quotient is a non-identity element of order 2 — a transposition in S_3 ; hence $\chi_W = \chi_{S_3}(\text{transposition}) = 0$.

⁷One can see the isomorphism $S_4/V_4 \cong S_3$ concretely using the following explicit realisation: S_4 permutes the three pairings of $\{1, 2, 3, 4\}$ into two pairs, namely $P_1 = \{12|34\}$, $P_2 = \{13|24\}$ and $P_3 = \{14|23\}$, and this action is the quotient map (its kernel is exactly V_4). Then (12)(34) fixes all three pairings (the identity in S_3), while (1234) swaps $P_1 \leftrightarrow P_3$ and fixes P_2 (a transposition). The other classes check out the same way: (12) acts as a transposition ($\chi_W = 0$) and (123) as a 3-cycle ($\chi_W = -1$).

The complete character table for S_4 is:

S_4	1 e	6 (12)	8 (123)	6 (1234)	3 (12)(34)
trivial U 	1	1	1	1	1
sign U' 	1	-1	1	-1	1
standard V 	3	1	0	-1	-1
V' 	3	-1	0	1	-1
W 	2	0	-1	0	2

(4.18)

As an application, let us decompose the tensor square of the standard representation V . We first compute the characters for $\text{Sym}^2 V$ and $\wedge^2 V$ using the formulae from (4.14).

S_4	1 e	6 (12)	8 (123)	6 (1234)	3 (12)(34)
$\chi_V(g)^2$	9	1	0	1	1
$\chi_V(g^2)$	3	3	0	-1	3
$\chi_{\text{Sym}^2 V}$	6	2	0	0	2
$\chi_{\wedge^2 V}$	3	-1	0	1	-1

(4.19)

By inspecting the character table, we see that $\chi_{\wedge^2 V}$ is precisely the character of the irreducible representation V' . For $\text{Sym}^2 V$, we can use the inner product to find its decomposition. For example, $(\chi_{\text{Sym}^2 V}, \chi_U) = \frac{1}{24}[1(6 \cdot 1) + 6(2 \cdot 1) + \dots] = 1$. Continuing this process, we find:

$$\text{Sym}^2 V \cong U \oplus V \oplus W, \quad \wedge^2 V \cong V'. \quad (4.20)$$

4.3 Dual representations

Given a vector space V over $\mathbb{F}$, its **dual vector space** V^* is the space of all linear maps from V to $\mathbb{F}$, i.e., $V^* = \text{Hom}(V, \mathbb{F})$.

Definition 4.3. *Let V be a representation of a group G . The **dual representation** V^* is defined by the action of $g \in G$ on a linear functional $\theta \in V^*$. For any $v \in V$, the action is given by:*

$$(g\theta)(v) = \theta(g^{-1}v). \quad (4.21)$$

In terms of matrices, if the action of g on V is given by a matrix $[g]$, the action on V^* is given by a matrix $[g^{-1}]^T$.

Proof. Let $v \in V$ and $\theta \in V^*$. We view the elements of V and V^* as **column vectors**. Since $\theta v \in \mathbb{F}$, the matrix representation $[\theta]$ of θ is a column vector, so $\theta(v) = [\theta]^T v \in \mathbb{F}$. The relation (4.21) can be written as $[g\theta]^T(v) = [\theta]^T[g^{-1}](v) = ([g^{-1}]^T[\theta])^T(v)$. Clearly, the action of g on V^* is given by the matrix $[g^{-1}]^T$. $\square$

Lemma 4.4. *The dual representation is a representation, and its character is the complex conjugate of the original character: $\chi_{V^*} = \overline{\chi_V}$.*

Proof. For $g, h \in G$, we have $((gh)\theta)(v) = \theta(h^{-1}g^{-1}v) = (h\theta)(g^{-1}v) = (g(h\theta))(v)$, so $(gh)\theta = g(h\theta)$ and V^* is a representation. For the character, let $\{v_i\}$ be a basis of eigenvectors, $gv_i = \lambda_i v_i$, and let $\{\theta_j\}$ be the dual basis, $\theta_j(v_i) = \delta_{ij}$. Then $(g\theta_j)(v_i) = \theta_j(g^{-1}v_i) = \lambda_i^{-1}\delta_{ij}$, i.e., $g\theta_j = \lambda_j^{-1}\theta_j$. Since g has finite order, each λ_j is a root of unity, so $\lambda_j^{-1} = \overline{\lambda_j}$. Thus,

$$\chi_{V^*}(g) = \sum_j \lambda_j^{-1} = \sum_j \overline{\lambda_j} = \overline{\chi_V(g)}.$$

$\square$

Definition 4.5. *A representation V is **self-dual** if it is isomorphic to its dual, $V \cong V^*$.*

A complex representation is self-dual if and only if its character is real-valued, since $V \cong V^*$ is equivalent to $\chi_V = \chi_{V^*}$, which means $\chi_V = \overline{\chi_V}$.

Example 4.6. 1. Let $G = C_3 = \langle x \rangle$ and consider the 1D representation where x acts as multiplication by $\omega = e^{2\pi i/3}$. In the dual representation, x acts as multiplication by $\overline{\omega} = \omega^2$. Since $\omega \neq \omega^2$, this representation is not self-dual.

2. For the symmetric group S_n , any element g is conjugate to its inverse g^{-1} . Since characters are class functions, $\chi_V(g) = \chi_V(g^{-1}) = \overline{\chi_V(g)}$. This means all characters of S_n are real-valued, and therefore all of its irreducible representations are self-dual.

4.4 Young diagrams

For S_n , each conjugacy class is uniquely specified by an integer partition $n = n_1 + n_2 + \dots + n_k$, with $n_1 \geq n_2 \geq \dots \geq n_k > 0$, since this corresponds to a partition of the set $\{1, 2, \dots, n\}$ into a product of disjoint cycles of length $n_1, n_2, \dots, n_k$. This can be represented by a diagram with n_j boxes in the j -th row (with $j = 1, \dots, k$). This is called a **Young diagram**.

Since the number of irreducible representations equals the number of conjugacy classes, the Young diagrams with n boxes are equal in number to the irreducible representations of S_n . In fact, there is a natural bijection between the two, which we describe below.

For S_3 , the Young diagrams with 3 boxes are $\begin{array}{|c|c|c|}\hline\Box & \Box & \Box \\ \hline\end{array}$, $\begin{array}{|c|c|}\hline\Box & \Box \\ \hline\Box \\ \hline\end{array}$ and $\begin{array}{|c|}\hline\Box \\ \hline\Box \\ \hline\Box \\ \hline\end{array}$. For S_4 , the Young diagrams with 4 boxes are $\begin{array}{|c|c|c|c|}\hline\Box & \Box & \Box & \Box \\ \hline\end{array}$, $\begin{array}{|c|c|c|}\hline\Box & \Box & \Box \\ \hline\Box \\ \hline\end{array}$, $\begin{array}{|c|c|}\hline\Box & \Box \\ \hline\Box & \Box \\ \hline\end{array}$, $\begin{array}{|c|c|}\hline\Box & \Box \\ \hline\Box & \Box \\ \hline\Box \\ \hline\end{array}$, and $\begin{array}{|c|}\hline\Box \\ \hline\Box \\ \hline\Box \\ \hline\Box \\ \hline\end{array}$; these are 5 irreducible representations of S_4 .

Here is how to associate a Young diagram to each irreducible representation.

- For the Young diagram with n boxes, fill each diagram with the integers $1, \dots, n$ such that the numbers in each row are increasing from left to right and the numbers in each column are increasing from top to bottom. These are called the standard **tableaux**.

For example, for S_3 , we have

$$\begin{array}{|c|c|c|}, \quad \begin{array}{|c|}\hline 1 \\ \hline 2 \\ \hline 3 \\ \hline\end{array}, \quad \begin{array}{|c|c|}\hline 1 & 2 \\ \hline 3 & \\ \hline\end{array} \quad \text{or} \quad \begin{array}{|c|c|}\hline 1 & 3 \\ \hline 2 & \\ \hline\end{array} \quad (4.22)$$

The **dimension** of an irreducible representation is equal to the number of corresponding standard tableaux. (An efficient way to compute this number, the *hook length formula*, is given at the end of this section.)

- The numbers in a given **row** are first subject to **symmetrisation**. Those appearing in a given **column** are subject to **antisymmetrisation**.
- Young tableau $\begin{array}{|c|c|c|}\hline 1 & 2 & 3 \\ \hline\end{array}$ gives rise to a basis element of U :

$$c_{\text{trivial}} = e_1 + e_{(12)} + e_{(13)} + e_{(23)} + e_{(123)} + e_{(132)} . \quad (4.23)$$

Young tableau $\begin{array}{|c|}\hline 1 \\ \hline 2 \\ \hline 3 \\ \hline\end{array}$ gives rise to a basis element of U' :

$$c_{\text{sign}} = e_1 - e_{(12)} - e_{(13)} - e_{(23)} + e_{(13)(12)=(123)} + e_{(12)(13)=(132)} . \quad (4.24)$$

Young tableaux $\begin{array}{|c|c|}\hline 1 & 2 \\ \hline 3 & \\ \hline\end{array}$ and $\begin{array}{|c|c|}\hline 1 & 3 \\ \hline 2 & \\ \hline\end{array}$ give rise to two basis elements of V :

$$\begin{aligned} c_{\text{std}}^{(1)} &= e_1 + e_{(12)} - e_{(13)} - e_{(12)(13)=(132)} , \\ c_{\text{std}}^{(2)} &= e_1 + e_{(13)} - e_{(12)} - e_{(13)(12)=(123)} . \end{aligned} \quad (4.25)$$

- For S_3 , the character of each irreducible representation can be read off from the coefficients of the sum of the corresponding basis elements: for example, $c_{\text{std}}^{(1)} + c_{\text{std}}^{(2)} = 2e_1 + 0e_{(12)} + 0e_{(13)} + 0e_{(23)} - e_{(123)} - e_{(132)}$ reproduces $\chi_V = (2, 0, -1)$. (This is a feature of S_3 rather than a general rule. In general, the element of the group algebra whose coefficients are the character values is the central

element $\frac{\dim V}{|G|} \sum_{g \in G} \chi_V(g^{-1}) e_g$, which acts as the projection onto the V -isotypic component of any representation.)

Since an irreducible representation is uniquely specified by its character, we can use the Young diagram to denote each irreducible representation of S_n in a well-defined and unique way.

Application in quantum mechanics [7, Section 63]

Consider the spin wavefunctions $\chi(\sigma_1, \sigma_2, \sigma_3)$ for a system with **three electrons**, where $\sigma_i = \pm \frac{1}{2}$. The system has the symmetry S_3 permuting the identical electrons. We consider the 3 irreducible representations:

1. For the trivial representation $\square\square\square$, from c_{trivial} in (4.23), we have the totally symmetric spin wavefunction (up to a normalisation constant):

$$\chi(\sigma_1, \sigma_2, \sigma_3) + \chi(\sigma_2, \sigma_1, \sigma_3) + \chi(\sigma_3, \sigma_2, \sigma_1) + \chi(\sigma_1, \sigma_3, \sigma_2) + \chi(\sigma_3, \sigma_1, \sigma_2) + \chi(\sigma_2, \sigma_3, \sigma_1),$$

corresponding to a total spin $S = \frac{3}{2}$, with the total spin components $m_S = -\frac{3}{2}, -\frac{1}{2}, \frac{1}{2}, \frac{3}{2}$; this is a **spin quartet state**.

The representation is 1D, so there is no permutational degeneracy in this case.

2. For the sign representation $\begin{smallmatrix} \square \\ \square \end{smallmatrix}$, the basis element c_{sign} in (4.24) gives the totally antisymmetric spin wavefunction:

$$\chi(\sigma_1, \sigma_2, \sigma_3) - \chi(\sigma_2, \sigma_1, \sigma_3) - \chi(\sigma_3, \sigma_2, \sigma_1) - \chi(\sigma_1, \sigma_3, \sigma_2) + \chi(\sigma_3, \sigma_1, \sigma_2) + \chi(\sigma_2, \sigma_3, \sigma_1),$$

but this **vanishes identically**, since $\sigma_i = \pm 1/2$ and we antisymmetrise over 3 variables, two of them must take the same value, so that we have zero identically.⁸

3. For the 2D standard representation $\begin{smallmatrix} \square & \square \end{smallmatrix}$, there is a 2-fold permutational degeneracy. From c_{std} in (4.25), the spin wavefunctions (up to a normalisation constant) are

$$\begin{aligned} & \chi(\sigma_1, \sigma_2, \sigma_3) + \chi(\sigma_2, \sigma_1, \sigma_3) - \chi(\sigma_3, \sigma_2, \sigma_1) - \chi(\sigma_2, \sigma_3, \sigma_1), \\ & \chi(\sigma_1, \sigma_2, \sigma_3) + \chi(\sigma_3, \sigma_2, \sigma_1) - \chi(\sigma_2, \sigma_1, \sigma_3) - \chi(\sigma_3, \sigma_1, \sigma_2). \end{aligned} \quad (4.26)$$

Since $\begin{smallmatrix} a & c \\ b \end{smallmatrix}$ antisymmetrises the pair (a, b) and leaves c alone, effectively the total spin S is equal to the spin of c , namely $S = 1/2$,⁹ with the total spin components $m_S = -\frac{1}{2}, \frac{1}{2}$; this is a **spin doublet state**.

⁸For this reason, for a system of electrons, the Young diagrams for the spin wavefunctions can contain columns of *at most two* boxes.

⁹For a system of N electrons, the Young diagrams for spin wavefunctions with N_1 boxes in the first row and N_2 boxes in the second row (with $N = N_1 + N_2$) correspond to a total spin of $S = \frac{1}{2}(N_1 - N_2)$. This is because a spinor of rank N reduces to a symmetrical spinor of rank $N_1 - N_2$, corresponding to the total spin $S = \frac{1}{2}(N_1 - N_2)$.

By the Pauli exclusion principle, for a system of half-integral spin particles (fermions), the total wavefunction, which is the product of the coordinate wavefunction ϕ and the spin wavefunction χ , must be **antisymmetrical** with respect to all particles. Hence, ϕ and χ must transform in the *transpose* representations, i.e. interchanging rows and columns,¹⁰ as shown below.

$$\begin{array}{ccc}
 \phi & \begin{array}{|c|} \hline \square \\ \hline \square \\ \hline \square \\ \hline \end{array} & \begin{array}{|c|c|} \hline \square & \square \\ \hline \square & \square \\ \hline \end{array} \\
 \chi & \begin{array}{|c|c|c|} \hline \square & \square & \square \\ \hline \end{array} & \begin{array}{|c|c|} \hline \square & \square \\ \hline \square & \square \\ \hline \end{array} \\
 & S = 3/2 & S = 1/2
 \end{array} \tag{4.27}$$

For a system of **four electrons**, the coordinate wavefunction ϕ and the spin wavefunction χ transform in the representations as shown below.¹¹

$$\begin{array}{ccccc}
 \phi & \begin{array}{|c|} \hline \square \\ \hline \square \\ \hline \square \\ \hline \square \\ \hline \end{array} & \begin{array}{|c|c|} \hline \square & \square \\ \hline \square & \square \\ \hline \square & \square \\ \hline \end{array} & \begin{array}{|c|c|} \hline \square & \square \\ \hline \square & \square \\ \hline \square & \square \\ \hline \end{array} & \\
 \chi & \begin{array}{|c|c|c|c|} \hline \square & \square & \square & \square \\ \hline \end{array} & \begin{array}{|c|c|c|} \hline \square & \square & \square \\ \hline \square & \square & \square \\ \hline \end{array} & \begin{array}{|c|c|} \hline \square & \square \\ \hline \square & \square \\ \hline \square & \square \\ \hline \end{array} & \\
 & S = 2 & S = 1 & S = 0 &
 \end{array} \tag{4.28}$$

The hook length formula for S_n

There is a convenient formula for computing the dimension of irreducible representations of S_n called the **hook length formula** (see [4, Section 4]).

Definition 4.7. *The **hook length** of a box in a Young diagram is the number of squares directly below or directly to the right of the box, including the box once.*

In the following diagram, each box is labelled by its hook length:

$$\begin{array}{|c|c|c|c|} \hline 6 & 4 & 3 & 1 \\ \hline 4 & 2 & 1 & \\ \hline 1 & & & \\ \hline \end{array} \tag{4.29}$$

(We emphasise that the number in each box is the hook length of that particular box. This is not to be confused with a Young tableau.)

Proposition 4.8. (The hook length formula for S_n .) *The dimension of the irreducible representation V_λ of S_n corresponding to the Young diagram λ is given by*

$$\dim V_\lambda = \frac{n!}{\prod (\text{Hook lengths})} . \tag{4.30}$$

¹⁰For a system of integral spin particles (i.e. bosons), the total wavefunction must be symmetrical with respect to all particles. Therefore, the symmetry of the coordinate and the spin wavefunctions must be given by the same Young diagram.

¹¹Please refer to Footnotes 8 and 9.

Example 4.9. For the Young diagram $\begin{array}{|c|c|c|} \hline \square & \square & \square \\ \hline \square & & \\ \hline \square & & \\ \hline \end{array}$ whose hook lengths are given in (4.29), $n = 4 + 3 + 1 = 8$, and the dimension of the corresponding representation of S_8 is therefore

$$\frac{8!}{6 \times 4 \times 4 \times 2 \times 3} = 70 .$$

Example 4.10. As a check, applying the hook length formula to the five Young diagrams with four boxes reproduces the dimensions 1, 3, 2, 3 and 1 of the irreducible representations of S_4 found in Section 4.2. For instance, the hook lengths of $\begin{array}{|c|c|} \hline \square & \square \\ \hline \square & \square \\ \hline \end{array}$ are

$\begin{array}{|c|c|} \hline 3 & 2 \\ \hline 2 & 1 \\ \hline \end{array}$, giving $\dim V_{\begin{array}{|c|c|} \hline \square & \square \\ \hline \square & \square \\ \hline \end{array}} = 4!/(3 \times 2 \times 2 \times 1) = 2$.

5 Applications to quantum mechanics

Group theory enters quantum mechanics because the Schrödinger equation of a physical system, such as an atom or molecule, is invariant under the symmetry transformations of that system.

- If G is the symmetry group of the system, its Hamiltonian $\hat{H}$ must commute with every group element $g \in G$:

$$g\hat{H} = \hat{H}g. \quad (5.1)$$

- Applying a symmetry transformation to an energy eigenstate yields another eigenstate with the same energy.

Justification: If $\hat{H}|\psi\rangle = E|\psi\rangle$, then for any $g \in G$,

$$\hat{H}(g|\psi\rangle) = g(\hat{H}|\psi\rangle) = E(g|\psi\rangle) , \quad (5.2)$$

so $g|\psi\rangle$ is an eigenstate of $\hat{H}$ with the same energy E .

- The set of stationary states corresponding to a given energy level forms a vector space that carries a representation of the symmetry group.

Justification: Let $V(E) = \{|\psi\rangle : \hat{H}|\psi\rangle = E|\psi\rangle\}$ be the space of stationary states with energy E . By (5.2), $g|\psi\rangle \in V(E)$ whenever $|\psi\rangle \in V(E)$, so $V(E)$ is G -invariant and carries a representation of G .

- If V is an irreducible representation of G , then $\hat{H}|_V$ is an intertwining map from V to itself, since $\hat{H}$ commutes with all $g \in G$. By Schur's Lemma, $\hat{H}|_V = \lambda \mathbf{1}$ for some scalar λ .

All state vectors within a given irreducible representation of the symmetry group are degenerate, sharing the same energy eigenvalue.

- In most cases, the representation formed by the stationary states of a given energy level is irreducible. When states belonging to non-isomorphic irreducible representations share the same energy, one speaks of an “**accidental**” **degeneracy** — often a sign of a larger, hidden symmetry group of the Hamiltonian.
- **Example: The Hydrogen Atom.** The electron in a Coulomb potential exhibits accidental degeneracy: all states with the same principal quantum number n but different angular momentum quantum numbers l share the energy $E_n = -\frac{1}{2n^2}$ (in atomic units), a degeneracy of $\sum_{l=0}^{n-1} (2l+1) = n^2$. It is due to an additional conserved quantity, the Laplace–Runge–Lenz (LRL) vector (see the end of Section 36 of [7]):

$$\hat{\mathbf{A}} = \frac{\mathbf{r}}{r} - \frac{1}{2}(\hat{\mathbf{p}} \times \hat{\mathbf{l}} - \hat{\mathbf{l}} \times \hat{\mathbf{p}}),$$

which commutes with the Hamiltonian $\hat{H} = \frac{1}{2}\hat{p}^2 - 1/r$.

The angular momentum operators $\hat{l}_i$ and the rescaled LRL vector operators $\hat{u}_i = \hat{A}_i/\sqrt{-2E}$ satisfy the commutation relations:

$$[\hat{l}_i, \hat{l}_j] = i\epsilon_{ijk}\hat{l}_k, \quad [\hat{l}_i, \hat{u}_j] = i\epsilon_{ijk}\hat{u}_k, \quad [\hat{u}_i, \hat{u}_j] = i\epsilon_{ijk}\hat{l}_k. \quad (5.3)$$

The first relation in (5.3) generates the familiar rotations of $\text{SO}(3)$; the full set generates $\text{SO}(3) \times \text{SO}(3)$, so the LRL vector supplies an additional “accidental” $\text{SO}(3)$ symmetry. We will see that this algebra is isomorphic to that of $\text{SO}(4)$, the rotation group of four-dimensional Euclidean space.

Assuming no accidental degeneracy, the connection between energy levels and representations is direct.

- *To each energy level of the system, there corresponds a unique irreducible representation of its symmetry group. The dimension of this representation determines the degree of degeneracy of that level.*

The irreducible representation to which a state belongs dictates how it transforms under the symmetry operations. Since only non-abelian groups have irreducible representations of dimension greater than one, degeneracy is a direct consequence of non-commuting symmetry operators.

- In the absence of an external magnetic field, physical systems exhibit time-reversal symmetry, which in quantum mechanics implies that if ψ is an energy eigenfunction, its complex conjugate ψ^* is an eigenfunction with the same energy.

If a representation V and its dual representation V^ are non-isomorphic, they must combine to form a single “physically irreducible” representation of dimension $2\dim(V)$, corresponding to a single degenerate energy level.*

Example 5.1. The group C_3 has only one-dimensional complex representations. However, two of these (U_1 and U_2 in Example 3.11) are complex conjugates of each other. Physically, they correspond to a single, doubly degenerate energy level.

In the presence of a magnetic field, time-reversal symmetry is broken, and complex conjugate representations can correspond to distinct energy levels.

- **Perturbation Theory:** Consider a system with Hamiltonian $\hat{H}_0$ and symmetry group G_0 , which is subjected to a perturbation $\hat{V}$ with its own symmetry group G_p .¹² The new Hamiltonian is $\hat{H} = \hat{H}_0 + \hat{V}$, and its symmetry group G is the intersection of G_0 and G_p . How does the perturbation affect the degenerate energy levels of the original system?

(i) If the original symmetry is a subgroup of the perturbation's symmetry ($G_0 \subseteq G_p$), then the overall symmetry is unchanged ($G = G_0$). In this case, no splitting of degenerate levels occurs.

(ii) If the perturbation's symmetry is a subgroup of the original symmetry ($G_p \subseteq G_0$), then the overall symmetry is reduced ($G = G_p$). An irreducible representation of G_0 may become reducible as a representation of the smaller group G_p ; this reduction corresponds to a splitting of the original degenerate energy level.

- **Example: Methane Molecule (CH_4).** The methane molecule — a carbon atom at the centre and four hydrogen atoms at the vertices of a tetrahedron — has symmetry group S_4 . Suppose an electronic state corresponds to the 3-dimensional standard representation V_{S_4} , which is triply degenerate. Its character is:

S_4	e	(12)	(123)	(1234)	(12)(34)
V_{S_4}	3	1	0	-1	-1

(5.4)

Now apply a perturbation with S_3 symmetry. Restricting V_{S_4} to the subgroup S_3 gives a representation X , whose character is read off from the values of $\chi_{V_{S_4}}$ on the elements of S_3 :

S_3	e	(12)	(123)
X	3	1	0

(5.5)

¹²For example, in the case of the energy levels of the d and f shells of ions in a crystal lattice which interact slightly with the surrounding atoms, the perturbation is the field acting on an ion due to the other atoms.

Using the character table of S_3 from Section 3.7 (whose standard representation we denote by V_{S_3}), inspection or the inner product formula gives $\chi_X = \chi_U + \chi_{V_{S_3}}$, i.e., $X \cong U \oplus V_{S_3}$: the triply degenerate level splits into a non-degenerate level (from U) and a doubly degenerate level (from V_{S_3}).

Next, suppose the perturbation has symmetry $C_2 \times C_2 \cong V_4$, identified with the subgroup $\{e, (12), (34), (12)(34)\}$ of S_4 . The restricted representation Y has character:

	1	1	1	1
$C_2 \times C_2$	e	(12)	(34)	(12)(34)
Y	3	1	1	-1

(5.6)

Using the character table (3.2), with (x_1, x_2) identified with $((12), (34))$, we find $\chi_Y = \chi_{U_0} + \chi_{U_1} + \chi_{U_2}$, i.e., $Y \cong U_0 \oplus U_1 \oplus U_2$: the triply degenerate level is completely split into three non-degenerate levels.

6 Lie Groups and Lie Algebras

6.1 Matrix Lie Groups

Definition 6.1. A sequence of complex matrices $\{A_m\}$ in $\text{Mat}_n(\mathbb{C})$ **converges** to a matrix A if each entry $(A_m)_{ij}$ converges to A_{ij} as $m \rightarrow \infty$ for all $1 \leq i, j \leq n$.

Definition 6.2. A **matrix Lie group** is a subgroup G of $\text{GL}(n, \mathbb{C})$ with the property that if $\{A_m\}$ is any sequence of matrices in G that converges to a matrix A , then either $A \in G$ or A is not invertible. Equivalently, a matrix Lie group is a **closed subgroup** of $\text{GL}(n, \mathbb{C})$.

Example 6.3. (i) Many common and important subgroups of $\text{GL}(n, \mathbb{C})$ are matrix Lie groups, including $\text{SL}(n, \mathbb{C})$, $\text{GL}(n, \mathbb{R})$, $\text{SL}(n, \mathbb{R})$, $\text{O}(n)$, $\text{SO}(n)$, $\text{U}(n)$, and $\text{SU}(n)$.

(ii) An example of a subgroup that is not a matrix Lie group is the set of $n \times n$ invertible matrices with rational entries, $\text{GL}(n, \mathbb{Q})$. One can construct a sequence of matrices in $\text{GL}(n, \mathbb{Q})$ that converges to an invertible matrix with irrational entries, e.g., the sequence $(1 + 1/m)^m$ of rational numbers converges to e , which is irrational.

Example 6.4. Here are further examples of matrix Lie groups.

1. The **unitary group** $\text{U}(n)$ is the set of $n \times n$ complex matrices A that preserve the standard inner product on $\mathbb{C}^n$, $\langle x, y \rangle = \sum_i x_i^* y_i$. The condition is $\langle Ax, Ay \rangle = \langle x, y \rangle$ for all $x, y \in \mathbb{C}^n$, which is equivalent to $A^\dagger A = \mathbf{1}$.

2. The **complex orthogonal group** $O(n, \mathbb{C})$ is the set of $n \times n$ complex matrices A that preserve the symmetric bilinear form $(x, y) = \sum_i x_i y_i$ on $\mathbb{C}^n$. The condition is $(Ax, Ay) = (x, y)$ for all $x, y \in \mathbb{C}^n$, which is equivalent to $A^T A = \mathbf{1}$.
3. The **indefinite orthogonal group** $O(n, k)$ is the set of $(n+k) \times (n+k)$ real matrices A that preserve the symmetric bilinear form $[x, y]_{n,k} = \sum_{i=1}^n x_i y_i - \sum_{j=n+1}^{n+k} x_j y_j$ on $\mathbb{R}^{n+k}$. In particular, $O(3, 1)$ is called the **Lorentz group**. The subgroup of $O(n, k)$ with determinant 1 is $SO(n, k)$.
4. Let ω be the skew-symmetric bilinear form on $\mathbb{C}^{2n}$ defined by $\omega(z_1, z_2) = z_1^T \Omega z_2$, where

$$\Omega = \begin{pmatrix} 0 & \mathbf{1}_n \\ -\mathbf{1}_n & 0 \end{pmatrix}.$$

The set of $2n \times 2n$ complex matrices A preserving ω is the **complex symplectic group** $Sp(n, \mathbb{C})$.¹³ The condition $\omega(Az_1, Az_2) = \omega(z_1, z_2)$ is equivalent to $A^T \Omega A = \Omega$.

The **compact symplectic group** $Sp(n)$ is defined as¹⁴

$$Sp(n) = Sp(n, \mathbb{C}) \cap U(2n).$$

Thus, $Sp(n)$ consists of matrices that preserve both the standard inner product and the symplectic form ω . It can be shown that $Sp(1, \mathbb{C}) \cong SL(2, \mathbb{C})$ and $Sp(1) \cong SU(2)$.

Definition 6.5. Let G and H be matrix Lie groups. A map $\Phi : G \rightarrow H$ is a **Lie group homomorphism** if it is (1) a group homomorphism and (2) continuous. If, in addition, Φ is a bijection and its inverse Φ^{-1} is continuous, Φ is a **Lie group isomorphism**. If such a map exists, G and H are **isomorphic**, written $G \cong H$.

Proposition 6.6. The group $SU(2)$ is homeomorphic to the three-sphere $S^3 \subset \mathbb{R}^4$.

Proof. For $U = \begin{pmatrix} a & b \\ c & d \end{pmatrix} \in SU(2)$, the condition $U^\dagger = U^{-1}$, together with $\det(U) = ad - bc = 1$, gives $\begin{pmatrix} a^* & c^* \\ b^* & d^* \end{pmatrix} = \begin{pmatrix} d & -b \\ -c & a \end{pmatrix}$, i.e., $d = a^*$ and $c = -b^*$. Thus,

$$U = \begin{pmatrix} a & b \\ -b^* & a^* \end{pmatrix}, \quad |a|^2 + |b|^2 = \det U = 1.$$

Setting $a = x_0 + ix_1$ and $b = x_2 + ix_3$ with $x_i \in \mathbb{R}$, the constraint reads $x_0^2 + x_1^2 + x_2^2 + x_3^2 = 1$: the equation of the unit 3-sphere S^3 in $\mathbb{R}^4$. $\square$

¹³This group is sometimes denoted $Sp(2n, \mathbb{C})$.

¹⁴This group is sometimes denoted $USp(2n)$.

6.2 Topological Properties

- **Compactness:** A matrix Lie group $G \subset \mathrm{GL}(n, \mathbb{C})$ is **compact** if it is a compact set in the standard topology of $\mathrm{Mat}_n(\mathbb{C}) \cong \mathbb{R}^{2n^2}$. By the Heine-Borel theorem, this is equivalent to G being closed and bounded.
 - **Closed:** If $\{A_m\} \subset G$ and $A_m \rightarrow A$, then $A \in G$.
 - **Bounded:** There is a constant C such that $|A_{ij}| \leq C$ for all $A \in G$ and all i, j .

The groups $\mathrm{O}(n)$, $\mathrm{SO}(n)$, $\mathrm{U}(n)$, $\mathrm{SU}(n)$, and $\mathrm{Sp}(n)$ are compact. In contrast, $\mathrm{SL}(n, \mathbb{R})$ for $n \geq 2$ is not compact, as it contains unbounded elements like $\mathrm{diag}(m, 1/m, 1, \dots, 1)$.

- **Connectedness:** A matrix Lie group G is **(path) connected** if for any two matrices $A, B \in G$, there exists a continuous path $A(t)$ in G for $t \in [0, 1]$ such that $A(0) = A$ and $A(1) = B$.¹⁵
 - The groups $\mathrm{GL}(n, \mathbb{C})$, $\mathrm{SL}(n, \mathbb{C})$, $\mathrm{U}(n)$, $\mathrm{SU}(n)$, and $\mathrm{SO}(n)$ are connected. For proofs, see [6, Section 1.3.2].
 - The group $\mathrm{O}(n)$ is not connected. It consists of two disjoint components determined by the determinant: $\mathrm{O}(n) = \mathrm{SO}(n) \cup \mathrm{O}_-(n)$, where $\mathrm{O}_-(n) = \{M \in \mathrm{O}(n) \mid \det(M) = -1\}$.

For any matrix Lie group G , the **identity component** G_0 is the set of all elements that are path-connected to the identity. For $\mathrm{O}(n)$, the identity component is $\mathrm{SO}(n)$.

Proposition 6.7. *If G is a matrix Lie group, its identity component G_0 is a normal subgroup of G .*

Proof. First, G_0 is a subgroup. If $A, B \in G_0$, there are paths $\alpha(t)$ from $\mathbf{1}$ to A and $\beta(t)$ from $\mathbf{1}$ to B . Then $\alpha(t)\beta(t)$ is a path from $\mathbf{1}$ to AB , and $\alpha(t)^{-1}$ is a path from $\mathbf{1}$ to A^{-1} . So $AB, A^{-1} \in G_0$. Second, G_0 is normal. For any $C \in G$ and $A \in G_0$, the path $C\alpha(t)C^{-1}$ connects $C\mathbf{1}C^{-1} = \mathbf{1}$ to CAC^{-1} , so $CAC^{-1} \in G_0$. $\square$

Theorem 6.8. *There exists a surjective Lie group homomorphism $\Phi : \mathrm{SU}(2) \rightarrow \mathrm{SO}(3)$ that is two-to-one, with $\ker(\Phi) = \{\mathbf{1}, -\mathbf{1}\}$.*

¹⁵In topology, a path-connected space is always connected, but the converse is not guaranteed. However, the two properties are equivalent for matrix Lie groups; see [6, Section 3.8]. Therefore, in the context of matrix Lie groups, we will use these two terms interchangeably.

Proof. Consider the real vector space V of 2×2 traceless self-adjoint matrices:

$$V = \{X \in \text{Mat}_2(\mathbb{C}) \mid X^\dagger = X, \text{tr}(X) = 0\}.$$

Any such matrix can be written as

$$X = \begin{pmatrix} x_1 & x_2 - ix_3 \\ x_2 + ix_3 & -x_1 \end{pmatrix}, \quad x_1, x_2, x_3 \in \mathbb{R}.$$

This establishes an isomorphism $V \cong \mathbb{R}^3$. We define an inner product on V by $\langle X, X' \rangle = \frac{1}{2} \text{tr}(XX')$, which corresponds to the standard dot product on $\mathbb{R}^3$: $\langle X, X' \rangle = x_1x'_1 + x_2x'_2 + x_3x'_3$.

For each $U \in \text{SU}(2)$, define a linear map $\Phi_U : V \rightarrow V$ by $\Phi_U(X) = UXU^\dagger$. This map is well-defined, since $(\Phi_U(X))^\dagger = (UXU^\dagger)^\dagger = UX^\dagger U^\dagger = UXU^\dagger$, and $\text{tr}(\Phi_U(X)) = \text{tr}(UXU^\dagger) = \text{tr}(X) = 0$. The map $\Phi : U \mapsto \Phi_U$ is a group homomorphism since $\Phi_{U_1U_2}(X) = (U_1U_2)X(U_1U_2)^\dagger = U_1(U_2XU_2^\dagger)U_1^\dagger = \Phi_{U_1}(\Phi_{U_2}(X))$.

Each map Φ_U preserves the inner product on V :

$$\langle \Phi_U(X), \Phi_U(X') \rangle = \frac{1}{2} \text{tr}(UXU^\dagger UX'U^\dagger) = \frac{1}{2} \text{tr}(UXX'U^\dagger) = \frac{1}{2} \text{tr}(XX') = \langle X, X' \rangle.$$

Thus, Φ_U is an orthogonal transformation of $V \cong \mathbb{R}^3$, so Φ is a homomorphism from $\text{SU}(2)$ to $\text{O}(3)$. Since $\text{SU}(2)$ is path-connected, its continuous image $\Phi(\text{SU}(2))$ must be a path-connected subgroup of $\text{O}(3)$. As $\Phi(\mathbf{1}) = \mathbf{1}$, the image must lie in the identity component of $\text{O}(3)$, which is $\text{SO}(3)$.

The homomorphism is continuous because matrix multiplication and conjugation are continuous. For any $X \in V$, $\Phi_U(X) = UXU^\dagger = (-U)X(-U)^\dagger = \Phi_{-U}(X)$, so the map is at least two-to-one. The kernel consists of all $U \in \text{SU}(2)$ such that $UXU^\dagger = X$ for all $X \in V$. This implies U must commute with all traceless self-adjoint matrices, which in turn implies U is a multiple of the identity, $U = \lambda \mathbf{1}$. Since $U \in \text{SU}(2)$, we must have $|\lambda|^2 = 1$ and $\det(U) = \lambda^2 = 1$, so $\lambda = \pm 1$. Thus, $\ker(\Phi) = \{\mathbf{1}, -\mathbf{1}\}$.

To show Φ is surjective, let $U = \text{diag}(e^{i\theta/2}, e^{-i\theta/2}) \in \text{SU}(2)$. Then

$$\begin{aligned} \Phi_U(X) &= \begin{pmatrix} e^{i\theta/2} & 0 \\ 0 & e^{-i\theta/2} \end{pmatrix} \begin{pmatrix} x_1 & x_2 - ix_3 \\ x_2 + ix_3 & -x_1 \end{pmatrix} \begin{pmatrix} e^{-i\theta/2} & 0 \\ 0 & e^{i\theta/2} \end{pmatrix} \\ &= \begin{pmatrix} x_1 & e^{i\theta}(x_2 - ix_3) \\ e^{-i\theta}(x_2 + ix_3) & -x_1 \end{pmatrix}. \end{aligned}$$

This corresponds to the vector transformation $x'_1 = x_1$, and $(x'_2 - ix'_3) = e^{i\theta}(x_2 - ix_3)$, which is a rotation by θ in the (x_2, x_3) -plane, i.e., about the x_1 -axis. Similarly, the matrix

$$U' = \begin{pmatrix} \cos(\theta/2) & i \sin(\theta/2) \\ i \sin(\theta/2) & \cos(\theta/2) \end{pmatrix} \in \text{SU}(2)$$

commutes with $\begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$, so $\Phi_{U'}$ fixes the x_2 -axis, and a computation analogous to the one above shows that $\Phi_{U'}$ is a rotation through the angle θ about the x_2 -axis.

Thus, the image of Φ , which is a subgroup of $\text{SO}(3)$, contains all rotations about the x_1 -axis and all rotations about the x_2 -axis. Since every rotation can be written as a product of rotations about two fixed perpendicular axes (Euler angles: $R = R_{x_1}(\phi) R_{x_2}(\theta) R_{x_1}(\psi)$), Φ is surjective. $\square$

Remark 6.9. (i) Since $\text{SU}(2) \cong S^3$, we can view $\text{SO}(3)$ as S^3 with antipodal points identified. This space is the **real projective space** $\mathbb{RP}^3$.

(ii) By the first isomorphism theorem, $\text{SO}(3) \cong \text{SU}(2)/\ker(\Phi) = \text{SU}(2)/\{\mathbf{1}, -\mathbf{1}\} \cong \text{SU}(2)/\mathbb{Z}_2$. Thus $\text{SU}(2)$ and $\text{SO}(3)$ are not isomorphic. We say that $\text{SU}(2)$ is a **double cover** of $\text{SO}(3)$.

Another important topological property is **simple connectedness**: A path-connected matrix Lie group G is **simply connected** if every loop in G can be continuously shrunk to a point within G .

- $\text{SU}(2) \cong S^3$ is simply connected.
- $\text{SO}(3) \cong \mathbb{RP}^3$ is not simply connected. A path from the north pole to the south pole on S^3 becomes a non-contractible loop in $\text{SO}(3)$ because the poles are identified.
- For $n \geq 1$, $\text{SU}(n)$ and $\text{Sp}(n)$ are simply connected. For $n \geq 2$, $\text{SL}(n, \mathbb{C})$ is simply connected. However, for $n \geq 2$, $\text{GL}(n, \mathbb{C})$, $\text{SL}(n, \mathbb{R})$, and $\text{SO}(n)$ are not simply connected. See [6, Chapter 13] for the argument.

6.3 General Lie Groups

So far we have focused on matrix Lie groups. The general definition requires the concept of a manifold.

- A **manifold** is a space that looks locally like a piece of $\mathbb{R}^n$.
Examples. (i) A two-dimensional torus, for example, looks locally but not globally like $\mathbb{R}^2$ and is, thus, a two-dimensional manifold. (ii) S^3 and $\mathbb{RP}^3$ are three-dimensional manifolds.
- A **smooth manifold** is a manifold M equipped with a collection of local coordinates covering M such that the change-of-coordinates map between two overlapping coordinate systems is smooth (infinitely differentiable).

Definition 6.10. A **Lie group** is a smooth manifold G that is also a group, such that the group multiplication $(g, h) \mapsto gh$ and inversion $g \mapsto g^{-1}$ are smooth maps.

Every matrix Lie group is a Lie group. However, the converse is not true; there exist Lie groups that are not isomorphic to any matrix Lie group. In this course, we will focus primarily on matrix Lie groups.

Example 6.11. (A non-matrix Lie group.) Consider the set $G = \mathbb{R} \times \mathbb{R} \times S^1 = \{(x, y, u) \mid x, y \in \mathbb{R}, u \in S^1 \subset \mathbb{C}\}$ with the group product

$$(x_1, y_1, u_1) \cdot (x_2, y_2, u_2) = (x_1 + x_2, y_1 + y_2, e^{ix_1y_2}u_1u_2).$$

The identity is $(0, 0, 1)$ and the inverse of (x, y, u) is $(-x, -y, e^{ixy}u^{-1})$. The product is associative, and both the product and inverse maps are smooth. Thus, G is a Lie group.

Writing $u = e^{ic}$, one recognises G as the quotient of the **Heisenberg group**

$$H = \left\{ \begin{pmatrix} 1 & x & c \\ 0 & 1 & y \\ 0 & 0 & 1 \end{pmatrix} \mid x, y, c \in \mathbb{R} \right\}$$

— itself a matrix Lie group — by the discrete central subgroup $\{(x, y, c) = (0, 0, 2\pi n) \mid n \in \mathbb{Z}\}$: the matrix product yields $c_1 + c_2 + x_1y_2$ in the upper-right entry, reproducing the phase $e^{ix_1y_2}u_1u_2$. For this reason, G is called the **reduced Heisenberg group**. Remarkably, taking this quotient destroys the property of being a matrix Lie group: G is not isomorphic to any matrix Lie group, as the following argument shows.

Proof. In G , every element of the central circle $\{(0, 0, e^{i\theta})\}$ is a group commutator: a direct computation with the group law gives

$$(t, 0, 1)(0, s, 1)(t, 0, 1)^{-1}(0, s, 1)^{-1} = (0, 0, e^{its}). \quad (6.1)$$

Now let $\Phi : G \rightarrow \mathrm{GL}(V)$ be any continuous homomorphism with V finite dimensional. Restricting Φ to the compact abelian centre S^1 decomposes $V = \bigoplus_k V_k$, where $(0, 0, e^{i\theta})$ acts on V_k as the scalar $e^{ik\theta}$ with $k \in \mathbb{Z}$.¹⁶ Since the centre commutes with every element of G , each V_k is invariant under all of $\Phi(G)$. We now compute the determinant of $\Phi(0, 0, e^{i\theta})|_{V_k}$ in two ways. On the one hand, it is a commutator $ABA^{-1}B^{-1}$ of matrices on V_k , so its determinant is 1. On the other hand, it is the scalar matrix $e^{ik\theta} \mathbf{1}$, whose determinant is $e^{ik\theta \dim V_k}$. Since θ is arbitrary, this forces $k = 0$. Hence, the centre acts trivially in every finite-dimensional representation of G ; in particular, no continuous homomorphism $G \rightarrow \mathrm{GL}(n, \mathbb{C})$ is injective, and G cannot be isomorphic to a matrix Lie group.¹⁷ $\square$

¹⁶This uses complete reducibility for the compact group S^1 , which we will establish in Theorem 7.13, together with the fact that the continuous homomorphisms $S^1 \rightarrow \mathbb{C}^*$ are precisely $e^{i\theta} \mapsto e^{ik\theta}$ with $k \in \mathbb{Z}$.

¹⁷This is the group-level shadow of the familiar fact that the canonical commutation relation $[\hat{x}, \hat{p}] = i\hbar$ admits no finite-dimensional realisation; there, too, the obstruction is a trace identity: $\mathrm{tr}[\hat{x}, \hat{p}] = 0 \neq \mathrm{tr}(i\hbar \mathbf{1})$.

6.4 Lie algebras

Definition 6.12. A **finite-dimensional real or complex Lie algebra** is a finite-dimensional real or complex vector space $\mathfrak{g}$, together with a map $[\cdot, \cdot] : \mathfrak{g} \times \mathfrak{g} \rightarrow \mathfrak{g}$, with the following properties:

1. $[\cdot, \cdot]$ is bilinear.
2. $[\cdot, \cdot]$ is skew-symmetric: $[X, Y] = -[Y, X]$ for any $X, Y \in \mathfrak{g}$.
3. The **Jacobi identity** holds:

$$[X, [Y, Z]] + [Y, [Z, X]] + [Z, [X, Y]] = 0 \quad (6.2)$$

for all $X, Y, Z \in \mathfrak{g}$.

Two elements X and Y of a Lie algebra $\mathfrak{g}$ **commute** if $[X, Y] = 0$. A Lie algebra $\mathfrak{g}$ is **commutative** if $[X, Y] = 0$ for all $X, Y \in \mathfrak{g}$.

The map $[\cdot, \cdot]$ is referred to as the **bracket** (or **Lie bracket**) operation on $\mathfrak{g}$. Note that the bracket operation on a Lie algebra is not, in general, associative; nevertheless, the Jacobi identity can be viewed as a substitute for associativity.

Example 6.13. Let $\mathfrak{g} = \mathbb{R}^3$ and let $[\cdot, \cdot] : \mathbb{R}^3 \times \mathbb{R}^3 \rightarrow \mathbb{R}^3$ be given by

$$[x, y] = x \times y,$$

where $x \times y$ is the cross product (or vector product). Then, $\mathfrak{g}$ is a Lie algebra. (This Lie algebra is commonly denoted as $\mathfrak{so}(3)$.)

Theorem 6.14. Let $\mathcal{A}$ be an associative algebra and let $\mathfrak{g}$ be a subspace of $\mathcal{A}$ such that $XY - YX \in \mathfrak{g}$ for all $X, Y \in \mathfrak{g}$. Then $\mathfrak{g}$ is a Lie algebra with the bracket operation given by the commutator:

$$[X, Y] = XY - YX. \quad (6.3)$$

Proof. The bilinearity and skew-symmetry of the bracket are evident. For the Jacobi identity, we can explicitly expand the left-hand side:

$$\begin{aligned} & [[X, Y], Z] + [[Y, Z], X] + [[Z, X], Y] \\ &= (XYZ - YXZ) - (ZXY - ZYX) + (YZX - ZYX) - (XYZ - XZY) \\ &+ (ZXY - XZY) - (YZX - YXZ) = 0. \end{aligned} \quad (6.4)$$

We see that the associativity of the algebra $\mathcal{A}$ is essential for the Jacobi identity to hold. (For any Lie algebra, the Jacobi identity means that the bracket operation behaves as if it were $XY - YX$ in some associative algebra, even if it is not actually defined this way.) $\square$

Example 6.15. Let $\mathfrak{sl}(n, \mathbb{C})$ be the space of matrices $X \in \text{Mat}_n(\mathbb{C})$ for which $\text{tr}(X) = 0$. Then $\mathfrak{sl}(n, \mathbb{C})$ is a Lie algebra with the bracket $[X, Y] = XY - YX$.

Proof. For any $X, Y \in \mathfrak{sl}(n, \mathbb{C})$, we have $\text{tr}(XY - YX) = 0$, so the space is closed under the commutator. Since matrix multiplication is associative, by Theorem 6.14, $\mathfrak{sl}(n, \mathbb{C})$ is a Lie algebra. $\square$

Definition 6.16. A **subalgebra** of a real or complex Lie algebra $\mathfrak{g}$ is a subspace $\mathfrak{h}$ of $\mathfrak{g}$ such that $[H_1, H_2] \in \mathfrak{h}$ for all $H_1, H_2 \in \mathfrak{h}$. If $\mathfrak{g}$ is a complex Lie algebra and $\mathfrak{h}$ is a real subspace of $\mathfrak{g}$ which is closed under brackets, then $\mathfrak{h}$ is said to be a **real subalgebra** of $\mathfrak{g}$.

Definition 6.17. If $\mathfrak{g}$ and $\mathfrak{h}$ are Lie algebras, then a linear map $\phi : \mathfrak{g} \rightarrow \mathfrak{h}$ is called a **Lie algebra homomorphism** if $\phi([X, Y]) = [\phi(X), \phi(Y)]$ for all $X, Y \in \mathfrak{g}$. If, in addition, ϕ is one-to-one and onto, then ϕ is called a **Lie algebra isomorphism**. A Lie algebra isomorphism of a Lie algebra with itself is called a **Lie algebra automorphism**.

Definition 6.18. If $\mathfrak{g}$ is a Lie algebra and X is an element of $\mathfrak{g}$, define a linear map $\text{ad}_X : \mathfrak{g} \rightarrow \mathfrak{g}$ by

$$\text{ad}_X(Y) = [X, Y].$$

The map $\text{ad} : X \mapsto \text{ad}_X$ is the **adjoint map** or **adjoint representation**.

- Although $\text{ad}_X(Y)$ is just $[X, Y]$, the alternative “ad” notation can be useful. For example, instead of writing

$$[X, [X, [X, [X, Y]]]],$$

we can now write

$$(\text{ad}_X)^4(Y).$$

- We can view ad (that is, the map $X \mapsto \text{ad}_X$) as a linear map from $\mathfrak{g}$ into $\text{End}(\mathfrak{g})$, the space of all linear operators on $\mathfrak{g}$. The Jacobi identity is then equivalent to the assertion that ad_X is a *derivation* of the bracket:

$$\text{ad}_X([Y, Z]) = [\text{ad}_X(Y), Z] + [Y, \text{ad}_X(Z)]. \quad (6.5)$$

Here are two important types of subalgebras.

Definition 6.19. A subalgebra $\mathfrak{h}$ of a Lie algebra $\mathfrak{g}$ is said to be an **ideal** in $\mathfrak{g}$ if $\text{ad}_X(H) = [X, H] \in \mathfrak{h}$ for all $X \in \mathfrak{g}$ and $H \in \mathfrak{h}$. In other words, an ideal in $\mathfrak{g}$ is a subspace that is invariant under the adjoint action of $\mathfrak{g}$.

The **centre** $\mathfrak{z}$ of a Lie algebra $\mathfrak{g}$ is defined as $\mathfrak{z} = \{Z \in \mathfrak{g} \mid \text{ad}_X(Z) = [X, Z] = 0 \text{ for all } X \in \mathfrak{g}\}$. In other words, the centre of $\mathfrak{g}$ is the kernel of the adjoint representation of $\mathfrak{g}$. By definition, the centre is an ideal.

Proposition 6.20. *If $\mathfrak{g}$ is a Lie algebra, then*

$$\text{ad}_{[X,Y]} = \text{ad}_X \text{ad}_Y - \text{ad}_Y \text{ad}_X = [\text{ad}_X, \text{ad}_Y];$$

that is, $\text{ad} : \mathfrak{g} \rightarrow \text{End}(\mathfrak{g})$ is a Lie algebra homomorphism.

Proof. Observe that for any $Z \in \mathfrak{g}$,

$$\text{ad}_{[X,Y]}(Z) = [[X, Y], Z],$$

whereas

$$[\text{ad}_X, \text{ad}_Y](Z) = \text{ad}_X(\text{ad}_Y(Z)) - \text{ad}_Y(\text{ad}_X(Z)) = [X, [Y, Z]] - [Y, [X, Z]].$$

Thus, the statement we want to show,

$$[[X, Y], Z] = [X, [Y, Z]] - [Y, [X, Z]],$$

is equivalent to the Jacobi identity. □

Definition 6.21. *If $\mathfrak{g}_1$ and $\mathfrak{g}_2$ are Lie algebras, the **direct sum** of $\mathfrak{g}_1$ and $\mathfrak{g}_2$ is the vector space direct sum $\mathfrak{g}_1 \oplus \mathfrak{g}_2$, with the bracket given by*

$$[(X_1, X_2), (Y_1, Y_2)] = ([X_1, Y_1], [X_2, Y_2]) \quad (6.6)$$

for $X_1, Y_1 \in \mathfrak{g}_1$ and $X_2, Y_2 \in \mathfrak{g}_2$.

*If $\mathfrak{g}$ is a Lie algebra and $\mathfrak{g}_1, \mathfrak{g}_2$ are subalgebras, we say that $\mathfrak{g}$ **decomposes as the Lie algebra direct sum** of $\mathfrak{g}_1$ and $\mathfrak{g}_2$ if $\mathfrak{g} = \mathfrak{g}_1 \oplus \mathfrak{g}_2$ as a vector space and $[X_1, X_2] = 0$ for all $X_1 \in \mathfrak{g}_1$ and $X_2 \in \mathfrak{g}_2$.*

- The bracket in (6.6) makes $\mathfrak{g}_1 \oplus \mathfrak{g}_2$ into a Lie algebra: bilinearity, skew-symmetry and the Jacobi identity hold componentwise.
- If $\mathfrak{g}$ decomposes as a Lie algebra direct sum of subalgebras $\mathfrak{g}_1$ and $\mathfrak{g}_2$, then $\mathfrak{g}$ is isomorphic as a Lie algebra to the “abstract” direct sum of $\mathfrak{g}_1$ and $\mathfrak{g}_2$, via the map $(X_1, X_2) \mapsto X_1 + X_2$. This would not be the case without the assumption that every element of $\mathfrak{g}_1$ commutes with every element of $\mathfrak{g}_2$.

Definition 6.22. *Let $\mathfrak{g}$ be a finite-dimensional real or complex Lie algebra, and let $\{X_1, \dots, X_N\}$ be a basis for $\mathfrak{g}$ (as a vector space). The unique constants f_{jk}^l such that*

$$[X_j, X_k] = \sum_{l=1}^N f_{jk}^l X_l \quad (6.7)$$

*are called the **structure constants** of $\mathfrak{g}$ with respect to the chosen basis.*

Structure constants appear frequently in the physics literature. (Physicists usually call the basis elements X_i the **generators of the Lie algebra $\mathfrak{g}$** .) They satisfy the following two conditions for all j, k, l, m :

$$\begin{aligned} f_{jk}^l + f_{kj}^l &= 0 \\ \sum_n (f_{jk}^n f_{nl}^m + f_{kl}^n f_{nj}^m + f_{lj}^n f_{nk}^m) &= 0. \end{aligned} \tag{6.8}$$

The first of these conditions comes from the skew-symmetry of the bracket, and the second comes from the Jacobi identity.

6.5 The matrix exponential

We will soon associate a Lie algebra with each matrix Lie group. To describe this association, we need the notion of the exponential of a matrix.

Definition 6.23. *Given a matrix $X \in \text{Mat}_n(\mathbb{C})$, we define the **matrix exponential** of X , denoted by e^X or $\exp(X)$, by the usual power series,*

$$e^X = \sum_{m=0}^{\infty} \frac{X^m}{m!}, \tag{6.9}$$

where $X^0 = \mathbf{1}$ (the identity matrix).

It can be shown that this series converges absolutely for all $X \in \text{Mat}_n(\mathbb{C})$ and that e^X is a continuous function of X ; see [6, Section 2.1].

The matrix exponential shares some, but not all, of the properties of the exponential of a number.

Theorem 6.24. *The matrix exponential has the following properties for all $X, Y \in \text{Mat}_n(\mathbb{C})$. In the following, X^T and $X^\dagger$ denote the transpose and adjoint (conjugate transpose) of X , respectively.*

1. $e^0 = \mathbf{1}$
2. $e^{X^T} = (e^X)^T$ and $e^{X^\dagger} = (e^X)^\dagger$
3. If A is an invertible $n \times n$ matrix, then $e^{AXA^{-1}} = Ae^XA^{-1}$.
4. $\det(e^X) = e^{\text{tr}(X)}$
5. If $XY = YX$ then $e^{X+Y} = e^X e^Y$
6. e^X is invertible and $(e^X)^{-1} = e^{-X}$

7. **(The Baker-Campbell-Hausdorff (BCH) formula.)** For all sufficiently small X and Y in $\text{Mat}_n(\mathbb{C})$, we have

$$e^X e^Y = \exp \left(X + Y + \frac{1}{2}[X, Y] + \frac{1}{12}[X, [X, Y]] - \frac{1}{12}[Y, [X, Y]] + \cdots \right) \quad (6.10)$$

where $\cdots$ denotes terms that are expressible in terms of repeated commutators involving X and Y , with coefficients that are “universal,” that is, independent of n (the size of the matrices) and of the choice of X and Y in $\text{Mat}_n(\mathbb{C})$.

8. **(The Lie product formula.)** For all $X, Y \in \text{Mat}_n(\mathbb{C})$, we have

$$e^{X+Y} = \lim_{m \rightarrow \infty} \left(e^{\frac{X}{m}} e^{\frac{Y}{m}} \right)^m. \quad (6.11)$$

Here are some comments:

- Property 7 is known as the Baker-Campbell-Hausdorff (BCH) formula and is a crucial result. The full proof is long and can be found in [6, Chapter 5], but the reader can verify (6.10) directly to second order. For small X and Y ,

$$e^X e^Y = \mathbf{1} + (X + Y) + \left(\frac{1}{2}X^2 + XY + \frac{1}{2}Y^2 \right) + O(3),$$

where $O(3)$ denotes terms of combined order three or higher in X, Y . Applying $\log(\mathbf{1} + A) = A - \frac{1}{2}A^2 + O(A^3)$ and keeping terms up to second order,

$$\log(e^X e^Y) = (X+Y) + \frac{1}{2}X^2 + XY + \frac{1}{2}Y^2 - \frac{1}{2}(X+Y)^2 + O(3) = X+Y + \frac{1}{2}[X, Y] + O(3),$$

in agreement with (6.10). Note how the commutator arises precisely from the failure of X and Y to commute: $XY - \frac{1}{2}(XY + YX) = \frac{1}{2}[X, Y]$. The nontrivial content of the theorem is that *all* higher-order terms can likewise be arranged into iterated commutators, with universal coefficients.

The Lie product formula (6.11) follows immediately from the BCH formula as follows:

$$(e^{X/m} e^{Y/m})^m = \exp \left(X + Y + \frac{1}{2m}[X, Y] + O(m^{-2}) \right) \rightarrow e^{X+Y} \quad \text{as } m \rightarrow \infty.$$

- Properties 1, 2, and 3 are easily verified using term-by-term computation.
- Property 6 follows from Prop. 5 by taking $Y = -X$ and applying Prop. 1.
- **Proof of Property 4.** If X is diagonalisable with eigenvalues $\lambda_1, \dots, \lambda_n$, then e^X is diagonalisable with eigenvalues $e^{\lambda_1}, \dots, e^{\lambda_n}$. Thus, $\text{tr}(X) = \sum_j \lambda_j$ and

$$\det(e^X) = e^{\lambda_1} \dots e^{\lambda_n} = e^{\lambda_1 + \dots + \lambda_n} = e^{\text{tr}(X)}.$$

If X is not diagonalisable, write $X = AJA^{-1}$ with J the Jordan canonical form of X ; then $X_k = X + \frac{1}{k}A \text{diag}(1, 2, \dots, n) A^{-1}$ has distinct eigenvalues for sufficiently large k (hence is diagonalisable), and $X_k \rightarrow X$. Since both sides of the identity are continuous in X , the result follows.

- **Proof of Property 5.** We can multiply the series expansions of e^X and e^Y as follows (this is allowed since the series converge absolutely):

$$e^X e^Y = \sum_{m=0}^{\infty} \sum_{k=0}^m \frac{X^k}{k!} \frac{Y^{m-k}}{(m-k)!} = \sum_{m=0}^{\infty} \frac{1}{m!} \sum_{k=0}^m \frac{m!}{k!(m-k)!} X^k Y^{m-k}.$$

Since X and Y commute, we have

$$(X + Y)^m = \sum_{k=0}^m \frac{m!}{k!(m-k)!} X^k Y^{m-k},$$

and so $e^X e^Y = \sum_{m=0}^{\infty} \frac{1}{m!} (X + Y)^m = e^{X+Y}$.

Computing the exponential. Suppose a matrix X is diagonalisable, meaning that

$$X = A \begin{pmatrix} \lambda_1 & & \\ & \ddots & \\ & & \lambda_n \end{pmatrix} A^{-1},$$

for some invertible matrix A and complex numbers $\lambda_1, \dots, \lambda_n$. Then,

$$e^X = A \begin{pmatrix} e^{\lambda_1} & & 0 \\ & \ddots & \\ 0 & & e^{\lambda_n} \end{pmatrix} A^{-1}.$$

If X is not diagonalisable, it can be uniquely written as $X = S + N$, where S is diagonalisable, N is nilpotent (i.e., $N^p = 0$ for some p), and $SN = NS$. (This is also known as the **SN decomposition**.) Then,

$$e^X = e^{S+N} = e^S e^N,$$

where the series expansion for e^N terminates because $N^p = 0$ for some p .

Example 6.25. *If*

$$X_1 = \begin{pmatrix} 0 & a \\ -a & 0 \end{pmatrix}, \quad X_2 = \begin{pmatrix} 0 & a & b \\ 0 & 0 & c \\ 0 & 0 & 0 \end{pmatrix}, \quad X_3 = \begin{pmatrix} a & b \\ 0 & a \end{pmatrix},$$

then

$$e^{X_1} = \begin{pmatrix} \cos a & \sin a \\ -\sin a & \cos a \end{pmatrix}, \quad e^{X_2} = \begin{pmatrix} 1 & a & b + ac/2 \\ 0 & 1 & c \\ 0 & 0 & 1 \end{pmatrix}, \quad e^{X_3} = \begin{pmatrix} e^a & e^a b \\ 0 & e^a \end{pmatrix}.$$

Proof. The eigenvalues of X_1 are $\pm ia$ with corresponding eigenvectors $(1, \pm i)$. Thus,

$$\begin{aligned} e^{X_1} &= \begin{pmatrix} 1 & 1 \\ i & -i \end{pmatrix} \begin{pmatrix} e^{ia} & 0 \\ 0 & e^{-ia} \end{pmatrix} \frac{1}{-2i} \begin{pmatrix} -i & -1 \\ -i & 1 \end{pmatrix} \\ &= -\frac{1}{2i} \begin{pmatrix} -i(e^{ia} + e^{-ia}) & -e^{ia} + e^{-ia} \\ e^{ia} - e^{-ia} & -i(e^{ia} + e^{-ia}) \end{pmatrix} = \begin{pmatrix} \cos a & \sin a \\ -\sin a & \cos a \end{pmatrix}. \end{aligned}$$

On the other hand, X_2 is nilpotent with

$$X_2^2 = \begin{pmatrix} 0 & 0 & ac \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix} \quad \text{and} \quad X_2^3 = 0.$$

Hence, $e^{X_2} = \mathbf{1} + X_2 + X_2^2/2$, which gives the required result. For X_3 , we write $X_3 = S + N$, where

$$S = \begin{pmatrix} a & 0 \\ 0 & a \end{pmatrix}, \quad N = \begin{pmatrix} 0 & b \\ 0 & 0 \end{pmatrix}.$$

Since $SN = NS$, we have

$$e^{X_3} = e^S e^N = e^S (\mathbf{1} + N) = \begin{pmatrix} e^a & 0 \\ 0 & e^a \end{pmatrix} \begin{pmatrix} 1 & b \\ 0 & 1 \end{pmatrix} = \begin{pmatrix} e^a & e^a b \\ 0 & e^a \end{pmatrix},$$

as required. $\square$

- The relation $e^{X+Y} = e^X e^Y$ does not hold for general non-commuting matrices X and Y . Nevertheless, for any $X \in \text{Mat}_n(\mathbb{C})$, we have

$$e^{(s+t)X} = e^{sX} e^{tX}$$

for all $s, t \in \mathbb{R}$, since sX commutes with tX .

- Thus, for each X , the set of matrices $\{e^{tX} \mid t \in \mathbb{R}\}$ forms a subgroup of $\text{GL}(n, \mathbb{C})$. Using term-by-term differentiation, we see that

$$\left. \frac{d}{dt} e^{tX} \right|_{t=0} = X. \quad (6.12)$$

Here, the derivative of a matrix-valued function is defined entry-wise.

Definition 6.26. A **one-parameter subgroup** of $\text{GL}(n, \mathbb{C})$ is a continuous homomorphism from $\mathbb{R}$ into $\text{GL}(n, \mathbb{C})$; that is, a continuous map $A : \mathbb{R} \rightarrow \text{GL}(n, \mathbb{C})$ such that $A(0) = \mathbf{1}$ and $A(s+t) = A(s)A(t)$ for all $s, t \in \mathbb{R}$.

Theorem 6.27. If $A(\cdot)$ is a one-parameter subgroup of $\text{GL}(n, \mathbb{C})$, there exists a unique matrix $X \in \text{Mat}_n(\mathbb{C})$ such that

$$A(t) = e^{tX}$$

for all $t \in \mathbb{R}$.

6.6 The Lie algebra of a matrix Lie group

We now associate a Lie algebra $\mathfrak{g}$ with each matrix Lie group G .

Definition 6.28. *If $G \subset \mathrm{GL}(n, \mathbb{C})$ is a matrix Lie group, then the Lie algebra $\mathfrak{g}$ of G is defined as follows:*

$$\mathfrak{g} = \{X \in \mathrm{Mat}_n(\mathbb{C}) \mid e^{tX} \in G \text{ for all } t \in \mathbb{R}\}.$$

That is to say, $X \in \mathfrak{g}$ if and only if the one-parameter subgroup generated by X lies entirely in G .

Proposition 6.29. *For any matrix Lie group G , its Lie algebra $\mathfrak{g}$ has the following properties:*

1. *The zero matrix $0 \in \mathfrak{g}$.*
2. *For any $X \in \mathfrak{g}$, $tX \in \mathfrak{g}$ for all $t \in \mathbb{R}$.*
3. *For any $X, Y \in \mathfrak{g}$, their sum $X + Y \in \mathfrak{g}$.*
4. *For any $A \in G$ and $X \in \mathfrak{g}$, the matrix $AXA^{-1} \in \mathfrak{g}$.*
5. *For any $X, Y \in \mathfrak{g}$, their commutator $[X, Y] := XY - YX \in \mathfrak{g}$.*

The first three properties establish that $\mathfrak{g}$ is a real vector space. The last property, together with the fact that $\mathrm{Mat}_n(\mathbb{C})$ is an associative algebra, shows that $\mathfrak{g}$ is a real Lie algebra.

Proof. Points 1 and 2 are elementary consequences of the definition.

To prove Point 3, we use the Lie product formula:

$$e^{t(X+Y)} = \lim_{m \rightarrow \infty} (e^{tX/m} e^{tY/m})^m.$$

For any m , the matrices $e^{tX/m}$ and $e^{tY/m}$ are in G . Since G is a group, their product and any power of it, $(e^{tX/m} e^{tY/m})^m$, must also be in G . Because G is a matrix Lie group, it is closed, so the limit of this sequence must also lie in G . This shows that $e^{t(X+Y)} \in G$ for all t , and thus $X + Y \in \mathfrak{g}$.

Point 4 follows from the identity $e^{t(AXA^{-1})} = Ae^{tX}A^{-1}$. Since $e^{tX} \in G$ and $A \in G$, the right-hand side is in G . Thus, $AXA^{-1} \in \mathfrak{g}$.

To verify Point 5, note that by the product rule,

$$\left. \frac{d}{dt} e^{tX} Y e^{-tX} \right|_{t=0} = XY - YX = [X, Y], \quad \text{i.e.,} \quad [X, Y] = \lim_{h \rightarrow 0} \frac{e^{hX} Y e^{-hX} - Y}{h}.$$

By Point 4, each difference quotient lies in $\mathfrak{g}$, and a finite-dimensional real subspace of $\mathrm{Mat}_n(\mathbb{C})$ is closed; hence the limit $[X, Y]$ also lies in $\mathfrak{g}$. $\square$

Example 6.30. Let $\mathfrak{gl}(n, \mathbb{C})$, $\mathfrak{gl}(n, \mathbb{R})$, $\mathfrak{sl}(n, \mathbb{C})$, and $\mathfrak{sl}(n, \mathbb{R})$ denote the Lie algebras of $\mathrm{GL}(n, \mathbb{C})$, $\mathrm{GL}(n, \mathbb{R})$, $\mathrm{SL}(n, \mathbb{C})$, and $\mathrm{SL}(n, \mathbb{R})$, respectively. Then we have:

$$\begin{aligned}\mathfrak{gl}(n, \mathbb{C}) &= \mathrm{Mat}_n(\mathbb{C}) \\ \mathfrak{gl}(n, \mathbb{R}) &= \mathrm{Mat}_n(\mathbb{R}) \\ \mathfrak{sl}(n, \mathbb{C}) &= \{X \in \mathrm{Mat}_n(\mathbb{C}) \mid \mathrm{tr}(X) = 0\} \\ \mathfrak{sl}(n, \mathbb{R}) &= \{X \in \mathrm{Mat}_n(\mathbb{R}) \mid \mathrm{tr}(X) = 0\}\end{aligned}$$

Proof. Let's consider the case of $\mathfrak{sl}(n, \mathbb{C})$. If $\mathrm{tr}(X) = 0$, then $\det(e^{tX}) = e^{t \cdot \mathrm{tr}(X)} = e^0 = 1$, so $e^{tX} \in \mathrm{SL}(n, \mathbb{C})$ for all t . Conversely, if X is in the Lie algebra of $\mathrm{SL}(n, \mathbb{C})$, then $e^{tX} \in \mathrm{SL}(n, \mathbb{C})$ for all $t \in \mathbb{R}$. This means $\det(e^{tX}) = 1$, so $e^{t \cdot \mathrm{tr}(X)} = 1$ for all t . This is only possible if $\mathrm{tr}(X) = 0$. The proofs for the other cases are similar. $\square$

Example 6.31. The Lie algebras of the compact groups $\mathrm{U}(n)$, $\mathrm{SU}(n)$, $\mathrm{O}(n)$, and $\mathrm{SO}(n)$ are:

$$\begin{aligned}\mathfrak{u}(n) &= \{X \in \mathrm{Mat}_n(\mathbb{C}) \mid X^\dagger = -X\} \quad (\text{skew-Hermitian matrices}) \\ \mathfrak{su}(n) &= \{X \in \mathrm{Mat}_n(\mathbb{C}) \mid X^\dagger = -X, \mathrm{tr}(X) = 0\} \\ \mathfrak{o}(n) &= \{X \in \mathrm{Mat}_n(\mathbb{R}) \mid X^T = -X\} \quad (\text{skew-symmetric matrices})\end{aligned}$$

The Lie algebra of $\mathrm{SO}(n)$ is the same as that of $\mathrm{O}(n)$, so $\mathfrak{so}(n) = \mathfrak{o}(n)$.

The Lie algebra $\mathfrak{sp}(n, \mathbb{C})$ of $\mathrm{Sp}(n, \mathbb{C})$ consists precisely of the complex matrices X satisfying $\Omega X^T \Omega = X$, where

$$\Omega = \begin{pmatrix} 0 & \mathbf{1}_n \\ -\mathbf{1}_n & 0 \end{pmatrix}.$$

The Lie algebra $\mathfrak{sp}(n)$ of $\mathrm{Sp}(n)$ consists precisely of the complex matrices X such that $\Omega X^T \Omega = X$ and $X^\dagger = -X$.

Proof. If $X^\dagger = -X$, then $(e^{tX})^\dagger = e^{tX^\dagger} = e^{-tX} = (e^{tX})^{-1}$, which shows that e^{tX} is unitary. Conversely, if e^{tX} is unitary for all $t \in \mathbb{R}$, then $(e^{tX})^\dagger = (e^{tX})^{-1}$, which means $e^{tX^\dagger} = e^{-tX}$. Differentiating at $t = 0$ gives $X^\dagger = -X$. This establishes the Lie algebra for $\mathrm{U}(n)$. For $\mathfrak{su}(n)$, we add the trace-zero condition. The calculations for $\mathfrak{o}(n)$ are similar, replacing $\dagger$ with T . Note that if a real matrix X satisfies $X^T = -X$, its diagonal entries must be zero, so $\mathrm{tr}(X) = 0$ automatically. This explains why the Lie algebras of $\mathrm{SO}(n)$ and $\mathrm{O}(n)$ are identical. $\square$

Example 6.32. The Lie algebra $\mathfrak{su}(2)$ is the space of 2×2 traceless, skew-Hermitian matrices. A basis for this real vector space is given by

$$E_1 = \frac{1}{2} \begin{pmatrix} i & 0 \\ 0 & -i \end{pmatrix}, \quad E_2 = \frac{1}{2} \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}, \quad E_3 = \frac{1}{2} \begin{pmatrix} 0 & i \\ i & 0 \end{pmatrix}. \quad (6.13)$$

These satisfy the commutation relations $[E_1, E_2] = E_3$, $[E_2, E_3] = E_1$, and $[E_3, E_1] = E_2$. The Lie algebra $\mathfrak{so}(3)$ is the space of 3×3 real skew-symmetric matrices. A standard basis is

$$F_1 = \begin{pmatrix} 0 & 0 & 0 \\ 0 & 0 & -1 \\ 0 & 1 & 0 \end{pmatrix}, \quad F_2 = \begin{pmatrix} 0 & 0 & 1 \\ 0 & 0 & 0 \\ -1 & 0 & 0 \end{pmatrix}, \quad F_3 = \begin{pmatrix} 0 & -1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}. \quad (6.14)$$

These basis elements satisfy $[F_1, F_2] = F_3$, $[F_2, F_3] = F_1$, and $[F_3, F_1] = F_2$. Since the two sets of basis vectors satisfy the same commutation relations, the two Lie algebras are isomorphic: $\mathfrak{su}(2) \cong \mathfrak{so}(3)$. This is in contrast to their corresponding Lie groups, which are not isomorphic.

Definition 6.33. A matrix Lie group G is said to be **complex** if its Lie algebra $\mathfrak{g}$ is a complex subspace of $\text{Mat}_n(\mathbb{C})$; that is, if $iX \in \mathfrak{g}$ for all $X \in \mathfrak{g}$.

Examples of complex groups include $\text{GL}(n, \mathbb{C})$, $\text{SL}(n, \mathbb{C})$, and $\text{SO}(n, \mathbb{C})$.

Proposition 6.34. If a matrix Lie group G is commutative, then its Lie algebra $\mathfrak{g}$ is also commutative.

The converse is not necessarily true in general, but it does hold if G is connected. We will see this later.

Proof. If G is commutative, then $e^{tX}e^{sY} = e^{sY}e^{tX}$ for all $s, t \in \mathbb{R}$. Differentiating with respect to s at $s = 0$ gives $e^{tX}Y = Ye^{tX}$, i.e., $e^{tX}Ye^{-tX} = Y$ for all t . Differentiating now with respect to t at $t = 0$ gives $[X, Y] = 0$. $\square$

6.7 Relationships between Lie groups and Lie algebras

We begin by showing that every Lie group homomorphism gives rise to a Lie algebra homomorphism in a natural way.

Theorem 6.35. Let G and H be matrix Lie groups, with Lie algebras $\mathfrak{g}$ and $\mathfrak{h}$, respectively. Suppose that $\Phi : G \rightarrow H$ is a Lie group homomorphism. Then there exists a unique real-linear map $\phi : \mathfrak{g} \rightarrow \mathfrak{h}$ such that

$$\Phi(e^X) = e^{\phi(X)} \quad (6.15)$$

for all $X \in \mathfrak{g}$. The map ϕ has the following additional properties:

1. $\phi(AXA^{-1}) = \Phi(A)\phi(X)\Phi(A)^{-1}$ for all $X \in \mathfrak{g}$, $A \in G$.
2. $\phi([X, Y]) = [\phi(X), \phi(Y)]$ for all $X, Y \in \mathfrak{g}$.
3. $\phi(X) = \frac{d}{dt}\Phi(e^{tX})|_{t=0}$, for all $X \in \mathfrak{g}$.

In practice, given a Lie group homomorphism Φ , the way one goes about computing ϕ is by using Property 3, which says that ϕ is the derivative (or differential) of Φ at the identity.

By Point 2, $\phi : \mathfrak{g} \rightarrow \mathfrak{h}$ is a Lie algebra homomorphism. Thus, every Lie group homomorphism gives rise to a Lie algebra homomorphism. One can ask a reverse question: If ϕ is a homomorphism between the Lie algebras of two Lie groups, is there an associated Lie group homomorphism Φ ? The answer is, in general, no, but it is yes if G is simply connected. We will discuss this in Theorem 6.42.

Proof. Since Φ is a continuous group homomorphism, $\Phi(e^{tX})$ will be a one-parameter subgroup of H , for each $X \in \mathfrak{g}$. Thus, by Theorem 6.27, there is a unique matrix Z such that

$$\Phi(e^{tX}) = e^{tZ} \quad (6.16)$$

for all $t \in \mathbb{R}$. We define $\phi(X) = Z$ and check that ϕ has the required properties. First, by putting $t = 1$ in (6.16), we see that $\Phi(e^X) = e^{\phi(X)}$ for all $X \in \mathfrak{g}$. Next, if $\Phi(e^{tX}) = e^{tZ}$ for all t , then $\Phi(e^{tsX}) = e^{tsZ}$, showing that $\phi(sX) = s\phi(X)$. Using the Lie product formula and the continuity of Φ , we then compute that

$$e^{t\phi(X+Y)} = \Phi\left(\lim_{m \rightarrow \infty} (e^{tX/m} e^{tY/m})^m\right) = \lim_{m \rightarrow \infty} (e^{t\phi(X)/m} e^{t\phi(Y)/m})^m = e^{t(\phi(X)+\phi(Y))}.$$

Differentiating this result at $t = 0$ shows that $\phi(X + Y) = \phi(X) + \phi(Y)$.

We have thus obtained a real-linear map ϕ satisfying (6.15). If there were another real-linear map ϕ' with this property, we would have

$$e^{t\phi(X)} = e^{t\phi'(X)} = \Phi(e^{tX})$$

for all $t \in \mathbb{R}$. Differentiating this result at $t = 0$ shows that $\phi(X) = \phi'(X)$. We now verify the remaining claimed properties of ϕ . For any $A \in G$, we have

$$e^{t\phi(AXA^{-1})} = \Phi(e^{tAXA^{-1}}) = \Phi(A)\Phi(e^{tX})\Phi(A)^{-1} = \Phi(A)e^{t\phi(X)}\Phi(A)^{-1};$$

differentiating at $t = 0$ gives Point 1.

Meanwhile, for any X and Y in $\mathfrak{g}$, we have, as in [6, proof of Theorem 3.20],

$$\phi([X, Y]) = \phi\left(\left.\frac{d}{dt}e^{tX}Ye^{-tX}\right|_{t=0}\right) = \left.\frac{d}{dt}\phi(e^{tX}Ye^{-tX})\right|_{t=0}$$

where we have used the fact that a derivative commutes with a linear transformation. Thus,

$$\phi([X, Y]) = \left.\frac{d}{dt}e^{t\phi(X)}\phi(Y)e^{-t\phi(X)}\right|_{t=0} = [\phi(X), \phi(Y)],$$

establishing Point 2. Finally, since $\Phi(e^{tX}) = e^{t\phi(X)}$, we can compute $\phi(X)$ as in Point 3. $\square$

If Φ is a Lie group isomorphism, then (6.16) also gives rise to a Lie algebra isomorphism. We have

Corollary 6.36. *Suppose that G and H are matrix Lie groups with Lie algebras $\mathfrak{g}$ and $\mathfrak{h}$, respectively. If G is isomorphic to H , then $\mathfrak{g}$ is isomorphic to $\mathfrak{h}$.*

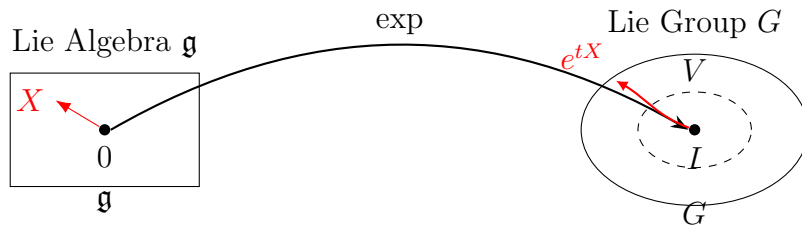
Proof. Let $\Phi : G \rightarrow H$ be a Lie group isomorphism. Applying Theorem 6.35 to Φ and to Φ^{-1} yields Lie algebra homomorphisms $\phi : \mathfrak{g} \rightarrow \mathfrak{h}$ and $\psi : \mathfrak{h} \rightarrow \mathfrak{g}$. For any $X \in \mathfrak{g}$,

$$e^{t\psi(\phi(X))} = \Phi^{-1}(e^{t\phi(X)}) = \Phi^{-1}(\Phi(e^{tX})) = e^{tX} \quad \text{for all } t ;$$

differentiating at $t = 0$ gives $\psi(\phi(X)) = X$. Similarly, $\phi(\psi(Y)) = Y$ for all $Y \in \mathfrak{h}$. Hence ϕ is a bijection, and thus a Lie algebra isomorphism. $\square$

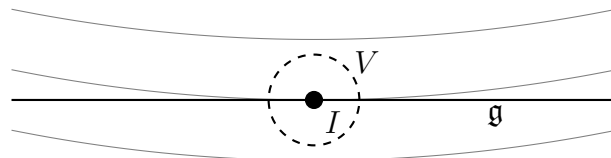
Another important fact is that, for any matrix Lie group G , the Lie algebra $\mathfrak{g}$ of G is large enough to capture what is happening in a **neighbourhood of the identity in G** .

Theorem 6.37. *Let G be a matrix Lie group with Lie algebra $\mathfrak{g}$. Then there exists a neighbourhood U of $\mathbf{0}$ in $\mathfrak{g}$ and a neighbourhood V of the identity I in G such that the exponential map takes U homeomorphically onto V .*



See [6, Corollary 3.44] for the proof. Let us discuss the consequences of this theorem.

Theorem 6.38. *Suppose $G \subset \text{GL}(n, \mathbb{C})$ is a matrix Lie group with Lie algebra $\mathfrak{g}$. Then a matrix X is in $\mathfrak{g}$ if and only if there exists a smooth curve γ in $\text{Mat}_n(\mathbb{C})$ with $\gamma(t) \in G$ for all t and such that $\gamma(0) = I$ (identity) and $d\gamma/dt|_{t=0} = X$. Thus, $\mathfrak{g}$ is the **tangent space at the identity to G** .*



Proof. If X is in $\mathfrak{g}$, then we may take $\gamma(t) = e^{tX}$ and then $\gamma(0) = I$ and $d\gamma/dt|_{t=0} = X$. In the other direction, suppose that $\gamma(t)$ is a smooth curve in G with $\gamma(0) = I$. For all sufficiently small t , we can write $\gamma(t) = e^{\delta(t)}$, where δ is a smooth curve in

$\mathfrak{g}$. Now, the derivative of $\delta(t)$ at $t = 0$ is the same as the derivative of $t \mapsto t\delta'(0)$ at $t = 0$. Thus, by the chain rule, we have

$$\gamma'(0) = \left. \frac{d}{dt} e^{\delta(t)} \right|_{t=0} = \left. \frac{d}{dt} e^{t\delta'(0)} \right|_{t=0} = \delta'(0).$$

Since $\delta(t)$ belongs to $\mathfrak{g}$ for all sufficiently small t , we conclude that $\gamma'(0) = \delta'(0) = \lim_{h \rightarrow 0} \frac{\delta(h) - \delta(0)}{h}$ belongs to $\mathfrak{g}$. $\square$

Corollary 6.39. *If G is a connected matrix Lie group, every element A of G can be written in the form*

$$A = e^{X_1} e^{X_2} \dots e^{X_m} \quad (6.17)$$

for some $X_1, X_2, \dots, X_m$ in $\mathfrak{g}$.

Proof. See [6, Corollary 3.47]. $\square$

Even if G is connected, it is in general not possible to write every $A \in G$ as a single exponential, $A = \exp X$, with $X \in \mathfrak{g}$.

Example 6.40. *There does not exist a matrix $X \in \mathfrak{sl}(2, \mathbb{C})$ with*

$$e^X = \begin{pmatrix} -1 & 1 \\ 0 & -1 \end{pmatrix} \quad (6.18)$$

even though the matrix on the right-hand side of (6.18) is in $\mathrm{SL}(2, \mathbb{C})$.

Proof. Note that the matrix on the right-hand side of (6.18) is upper-triangular with repeated eigenvalues -1 . It is not diagonalisable.

If $X \in \mathfrak{sl}(2, \mathbb{C})$ has distinct eigenvalues, then X is diagonalisable and e^X will also be diagonalisable, unlike the matrix on the right-hand side of (6.18).

If $X \in \mathfrak{sl}(2, \mathbb{C})$ has a repeated eigenvalue, this eigenvalue must be 0, otherwise the trace of X would not be zero. Thus, there is a nonzero vector v with $Xv = 0$, from which it follows that $e^X v = e^0 v = v$. We conclude that e^X has 1 as an eigenvalue, unlike the matrix on the right-hand side of (6.18). $\square$

Here is an immediate consequence of Corollary 6.39:

Corollary 6.41. *If G is a connected matrix Lie group and the Lie algebra $\mathfrak{g}$ of G is commutative, then G is commutative.*

Proof. Since $\mathfrak{g}$ is commutative, any two elements of G , when written as in Corollary 6.39, will commute. $\square$

Recall Theorem 6.35, which says that given matrix Lie groups G and H and a Lie group homomorphism $\Phi : G \rightarrow H$, we can find a Lie algebra homomorphism $\phi : \mathfrak{g} \rightarrow \mathfrak{h}$ such that $\Phi(e^X) = e^{\phi(X)}$ for all $X \in \mathfrak{g}$. The converse to this result holds in the case that G is simply connected.

Theorem 6.42. *Let G and H be matrix Lie groups with Lie algebras $\mathfrak{g}$ and $\mathfrak{h}$, respectively, and let $\phi : \mathfrak{g} \rightarrow \mathfrak{h}$ be a Lie algebra homomorphism. If G is **simply connected**, there exists a unique Lie group homomorphism $\Phi : G \rightarrow H$ such that $\Phi(e^X) = e^{\phi(X)}$ for all $X \in \mathfrak{g}$.*

We will subsequently make use of this theorem extensively. The proof of this theorem is based on the BCH formula and can be found in [6, Section 5.7]. This result has the following corollary.

Corollary 6.43. *Suppose G and H are simply connected matrix Lie groups with Lie algebras $\mathfrak{g}$ and $\mathfrak{h}$, respectively. If $\mathfrak{g}$ is isomorphic to $\mathfrak{h}$, then G is isomorphic to H .*

The proof of Corollary 6.43 can also be found in [6, Section 5.7].

6.8 The complexification of a real Lie algebra

The set of complex numbers, $\mathbb{C}$, is algebraically richer than the set of real numbers, $\mathbb{R}$. For example, any matrix in $\text{Mat}_n(\mathbb{C})$ has at least one eigenvalue, whereas a matrix in $\text{Mat}_n(\mathbb{R})$ may not.

When studying representations of a matrix Lie group G , we often work with its Lie algebra $\mathfrak{g}$. Since $\mathfrak{g}$ is generally a real Lie algebra, it is often advantageous to work with an associated complex Lie algebra, known as the complexification of $\mathfrak{g}$.

Definition 6.44. *If V is a finite-dimensional real vector space, then the **complexification** of V , denoted $V_{\mathbb{C}}$, is the space of formal linear combinations*

$$v_1 + iv_2,$$

with $v_1, v_2 \in V$. This set forms a real vector space in the obvious way. It becomes a complex vector space by defining scalar multiplication by i as

$$i(v_1 + iv_2) = -v_2 + iv_1.$$

It is straightforward to verify that this definition endows $V_{\mathbb{C}}$ with the structure of a complex vector space. We will regard V as a real subspace of $V_{\mathbb{C}}$ in the obvious way.

Proposition 6.45. *Let $\mathfrak{g}$ be a finite-dimensional real Lie algebra and let $\mathfrak{g}_{\mathbb{C}}$ be its complexification. The bracket operation on $\mathfrak{g}$ has a unique extension to $\mathfrak{g}_{\mathbb{C}}$ that makes $\mathfrak{g}_{\mathbb{C}}$ into a complex Lie algebra. The complex Lie algebra $\mathfrak{g}_{\mathbb{C}}$ is called the **complexification** of the real Lie algebra $\mathfrak{g}$.*

Proof. If the bracket on $\mathfrak{g}_{\mathbb{C}}$ is to be complex bilinear, it must be given by

$$[X_1 + iX_2, Y_1 + iY_2] = ([X_1, Y_1] - [X_2, Y_2]) + i([X_1, Y_2] + [X_2, Y_1]), \quad (6.19)$$

which proves uniqueness. For existence, the operation (6.19) is real bilinear and skew-symmetric by construction, and, using skew-symmetry, complex bilinearity reduces to the identity $[i(X_1 + iX_2), Y_1 + iY_2] = i[X_1 + iX_2, Y_1 + iY_2]$, which is verified by direct expansion. Finally, the Jacobi identity holds on $\mathfrak{g}$ by assumption and is complex-linear in each of its three arguments, so its validity extends from $\mathfrak{g}$ to $\mathfrak{g}_{\mathbb{C}}$ one argument at a time. $\square$

Proposition 6.46. *Suppose that $\mathfrak{g} \subset \text{Mat}_n(\mathbb{C})$ is a real Lie algebra and that for all nonzero X in $\mathfrak{g}$, the element iX is not in $\mathfrak{g}$. Then the abstract complexification $\mathfrak{g}_{\mathbb{C}}$ from Definition 6.44 is isomorphic to the set of matrices in $\text{Mat}_n(\mathbb{C})$ that can be expressed in the form $X + iY$ with $X, Y \in \mathfrak{g}$.*

Proof. Consider the map from the abstract complexification $\mathfrak{g}_{\mathbb{C}}$ to $\text{Mat}_n(\mathbb{C})$ that sends the formal sum $X + iY$ to the matrix sum $X + iY$. This map is easily seen to be a complex Lie algebra homomorphism. If $\mathfrak{g}$ satisfies the assumption that $\mathfrak{g} \cap i\mathfrak{g} = \{0\}$, this map is also injective and thus an isomorphism of $\mathfrak{g}_{\mathbb{C}}$ with the subspace $\mathfrak{g} + i\mathfrak{g} \subset \text{Mat}_n(\mathbb{C})$. $\square$

Corollary 6.47. *We have the following isomorphisms of complex Lie algebras:*

$$\begin{aligned} \mathfrak{gl}(n, \mathbb{C}) &\cong \mathfrak{gl}(n, \mathbb{R})_{\mathbb{C}} \cong \mathfrak{u}(n)_{\mathbb{C}} \\ \mathfrak{sl}(n, \mathbb{C}) &\cong \mathfrak{sl}(n, \mathbb{R})_{\mathbb{C}} \cong \mathfrak{su}(n)_{\mathbb{C}} \end{aligned} \tag{6.20}$$

Proof for $\mathfrak{u}(n)_{\mathbb{C}} \cong \mathfrak{gl}(n, \mathbb{C})$. Let $X \in \mathfrak{u}(n)$. Then $X^\dagger = -X$. It follows that $(iX)^\dagger = -iX^\dagger = iX$. Thus, a nonzero matrix cannot be both in $\mathfrak{u}(n)$ (skew-Hermitian) and $i\mathfrak{u}(n)$ (Hermitian). Furthermore, any matrix $Z \in \mathfrak{gl}(n, \mathbb{C})$ can be uniquely written as $Z = X + iY$ where $X = \frac{1}{2}(Z - Z^\dagger)$ and $Y = \frac{1}{2i}(Z + Z^\dagger)$ are both skew-Hermitian, i.e., in $\mathfrak{u}(n)$. This shows that $\mathfrak{gl}(n, \mathbb{C}) = \mathfrak{u}(n) \oplus i\mathfrak{u}(n)$, which is the space $\mathfrak{u}(n)_{\mathbb{C}}$. $\square$

Note that although their complexifications are isomorphic, e.g., $\mathfrak{su}(2)_{\mathbb{C}} \cong \mathfrak{sl}(2, \mathbb{R})_{\mathbb{C}} \cong \mathfrak{sl}(2, \mathbb{C})$, the real Lie algebras $\mathfrak{su}(2)$ and $\mathfrak{sl}(2, \mathbb{R})$ are not isomorphic.

Proposition 6.48. (Universal Property of Complexification) *Let $\mathfrak{g}$ be a real Lie algebra, $\mathfrak{g}_{\mathbb{C}}$ its complexification, and $\mathfrak{h}$ an arbitrary complex Lie algebra. Then every real Lie algebra homomorphism from $\mathfrak{g}$ into $\mathfrak{h}$ extends uniquely to a complex Lie algebra homomorphism from $\mathfrak{g}_{\mathbb{C}}$ into $\mathfrak{h}$.*

Proof. Let $\pi : \mathfrak{g} \rightarrow \mathfrak{h}$ be a real Lie algebra homomorphism. The unique complex-linear extension is given by defining $\pi(X + iY) = \pi(X) + i\pi(Y)$ for all $X, Y \in \mathfrak{g}$. One can then verify that this extended map is a homomorphism of complex Lie algebras. $\square$

7 Representations of Lie groups and Lie algebras

If V is a finite-dimensional real or complex vector space, we let $\mathrm{GL}(V)$ denote the group of invertible linear transformations of V . By choosing a basis for V , we can identify $\mathrm{GL}(V)$ with $\mathrm{GL}(n, \mathbb{R})$ or $\mathrm{GL}(n, \mathbb{C})$, respectively. This identification endows $\mathrm{GL}(V)$ with a topology that is independent of the choice of basis, which allows us to view $\mathrm{GL}(V)$ as a matrix Lie group. Similarly, we let $\mathfrak{gl}(V) = \mathrm{End}(V)$ denote the space of all linear operators from V to itself, which forms a Lie algebra under the commutator bracket $[X, Y] = XY - YX$.

Definition 7.1. *Let G be a matrix Lie group. A finite-dimensional **complex** (respectively, **real**) representation of G is a Lie group homomorphism $\Pi : G \rightarrow \mathrm{GL}(V)$, where V is a finite-dimensional complex (respectively, real) vector space.*

*Let $\mathfrak{g}$ be a real or complex Lie algebra. A finite-dimensional **complex** representation of $\mathfrak{g}$ is a Lie algebra homomorphism $\pi : \mathfrak{g} \rightarrow \mathfrak{gl}(V)$, where V is a finite-dimensional complex vector space; if $\mathfrak{g}$ is real, a finite-dimensional **real** representation is defined analogously, with V a real vector space.*

*A representation is called **faithful** if the corresponding homomorphism Π or π is one-to-one.*

- As with finite groups, it is useful to think of a representation as a linear action of a group or Lie algebra on a vector space. For instance, for every $g \in G$, there is an associated operator $\Pi(g)$ that acts on the vector space V . We will often adopt the notation gv in place of $\Pi(g)v$, and Xv in place of $\pi(X)v$. We may also use the shorthand “let V be a representation” to mean “let (Π, V) be a representation”.
- Even if $\mathfrak{g}$ is a real Lie algebra, we will primarily consider its complex representations.

The notions of irreducibility and isomorphism carry over from the theory of finite groups.

Definition 7.2. *Let Π be a representation of a matrix Lie group G on a space V . A subspace $W \subseteq V$ is **invariant** if $\Pi(A)w \in W$ for all $w \in W$ and all $A \in G$. An invariant subspace W is called **nontrivial** if $W \neq \{0\}$ and $W \neq V$. A representation with no nontrivial invariant subspaces is called **irreducible**. The terms are defined analogously for representations of Lie algebras.*

*A linear map $\phi : V \rightarrow W$ between two representations is called an **intertwining map** (or homomorphism of representations) if it commutes with the group action, i.e., $\phi(gv) = g\phi(v)$ for all $g \in G$ and $v \in V$. If ϕ is invertible, it is called an **isomorphism** of representations.*

Proposition 7.3. *Let G be a matrix Lie group with Lie algebra $\mathfrak{g}$, and let Π be a representation of G on a space V . Then there is a unique representation π of $\mathfrak{g}$ on the same space V such that*

$$\Pi(e^X) = e^{\pi(X)} \quad (7.1)$$

for all $X \in \mathfrak{g}$. The representation π can be computed as

$$\pi(X) = \left. \frac{d}{dt} \Pi(e^{tX}) \right|_{t=0} \quad (7.2)$$

and satisfies

$$\pi(AXA^{-1}) = \Pi(A)\pi(X)\Pi(A)^{-1} \quad (7.3)$$

for all $X \in \mathfrak{g}$ and all $A \in G$.

Proof. This is Theorem 6.35 applied to the Lie group homomorphism $\Phi = \Pi : G \rightarrow \mathrm{GL}(V)$, whose Lie algebra is $\mathfrak{gl}(V)$: the map π is the Lie algebra homomorphism ϕ of that theorem, and Points 1–3 there are precisely the stated properties. $\square$

A natural question arises: given a matrix Lie group G with Lie algebra $\mathfrak{g}$, does every representation of $\mathfrak{g}$ come from a representation of G ? This is not true in general, but the correspondence holds if G is simply connected.

Proposition 7.4. *1. Let G be a connected matrix Lie group with Lie algebra $\mathfrak{g}$. A representation Π of G is irreducible if and only if the associated representation π of $\mathfrak{g}$ is irreducible.*

2. Let G be a connected matrix Lie group. Two representations Π_1 and Π_2 of G are isomorphic if and only if their associated Lie algebra representations π_1 and π_2 are isomorphic.

Proof. 1. Suppose Π is irreducible, and let W be a subspace invariant under $\pi(X)$ for all $X \in \mathfrak{g}$. Then W is invariant under each $e^{\pi(X)} = \Pi(e^X)$ and, since G is connected, under $\Pi(A) = e^{\pi(X_1)} \cdots e^{\pi(X_m)}$ for every $A = e^{X_1} \cdots e^{X_m} \in G$ (Corollary 6.39). Irreducibility of Π then forces $W = \{0\}$ or V , so π is irreducible.

Conversely, suppose π is irreducible, and let W be a Π -invariant subspace. Then, for any $X \in \mathfrak{g}$ and $w \in W$,

$$\pi(X)w = \lim_{h \rightarrow 0} \frac{\Pi(e^{hX})w - w}{h} \in W,$$

since W is a (closed) subspace. Irreducibility of π then forces $W = \{0\}$ or V , so Π is irreducible. Point 2 is proven with a similar line of reasoning. $\square$

Proposition 7.5. *Let $\mathfrak{g}$ be a real Lie algebra and $\mathfrak{g}_{\mathbb{C}}$ its complexification. Every finite-dimensional complex representation π of $\mathfrak{g}$ has a unique extension to a complex-linear representation of $\mathfrak{g}_{\mathbb{C}}$, also denoted π . Furthermore, π is irreducible as a representation of $\mathfrak{g}_{\mathbb{C}}$ if and only if it is irreducible as a representation of $\mathfrak{g}$.*

Proof. The existence and uniqueness of the extension follow from the universal property of complexification (Proposition 6.48). The extension is given by defining its action on a general element as $\pi(X+iY) = \pi(X) + i\pi(Y)$ for all $X, Y \in \mathfrak{g}$. A complex subspace W of the representation space V is invariant under all operators $\pi(X+iY)$ if and only if it is invariant under all operators $\pi(X)$ and $\pi(Y)$ for $X, Y \in \mathfrak{g}$. Thus, the representations of $\mathfrak{g}$ and $\mathfrak{g}_{\mathbb{C}}$ have the same invariant subspaces. $\square$

Definition 7.6. *A representation $\Pi : G \rightarrow \mathrm{GL}(V)$ on a finite-dimensional inner product space V is **unitary** if $\Pi(A)$ is a unitary operator for every $A \in G$.*

Proposition 7.7. *Let G be a matrix Lie group with Lie algebra $\mathfrak{g}$. Let (Π, V) be a representation of G on a finite-dimensional inner product space, and let π be the associated representation of $\mathfrak{g}$. If Π is unitary, then $\pi(X)$ is skew-self-adjoint for all $X \in \mathfrak{g}$, i.e., $\pi(X)^{\dagger} = -\pi(X)$. Conversely, if G is connected and $\pi(X)$ is skew-self-adjoint for all $X \in \mathfrak{g}$, then Π is unitary.*

In light of this proposition, we define a representation π of a real Lie algebra $\mathfrak{g}$ on a finite-dimensional inner product space to be **unitary** if $\pi(X)^{\dagger} = -\pi(X)$ for all $X \in \mathfrak{g}$.

Proof. If Π is unitary, then $\Pi(e^{tX})$ is unitary for all $t \in \mathbb{R}$. Thus, $(\Pi(e^{tX}))^{\dagger} = (\Pi(e^{tX}))^{-1}$. This means $(e^{t\pi(X)})^{\dagger} = e^{-t\pi(X)}$, so $e^{t\pi(X)^{\dagger}} = e^{-t\pi(X)}$. Differentiating at $t = 0$ gives $\pi(X)^{\dagger} = -\pi(X)$. Conversely, if $\pi(X)^{\dagger} = -\pi(X)$, the same calculation shows that $\Pi(e^{tX}) = e^{t\pi(X)}$ is unitary. If G is connected, every element is a product of such exponentials, and a product of unitary operators is unitary. Thus, $\Pi(A)$ is unitary for all $A \in G$. $\square$

7.1 Examples

Example 7.8. *A matrix Lie group G is, by definition, a subset of some $\mathrm{GL}(n, \mathbb{C})$. A representation (Π, V) of G is called a **standard (or fundamental) representation** of G if $\Pi(A) = A$ for all $A \in G$.*

If G happens to be contained in $\mathrm{GL}(n, \mathbb{R}) \subset \mathrm{GL}(n, \mathbb{C})$, then we can also think of the standard representation as a real representation.

For example, the standard (or vector) representation V of $\mathrm{SO}(3)$ is the one in which $\mathrm{SO}(3)$ acts in the usual way on $V = \mathbb{R}^3$. The standard (or fundamental) representation W of $\mathrm{SU}(2)$ is the one in which $\mathrm{SU}(2)$ acts on $W = \mathbb{C}^2$ in the usual way.

*Similarly, if $\mathfrak{g} \subset \mathrm{Mat}_n(\mathbb{C})$ is a Lie algebra of matrices, the map $\pi(X) = X$, for all $X \in \mathfrak{g}$, is called the **standard representation** of $\mathfrak{g}$.*

Example 7.9. Consider the one-dimensional complex vector space $\mathbb{C}$. For any matrix Lie group G , we can define the **trivial representation**, $\Pi : G \rightarrow \text{GL}(1, \mathbb{C})$, by $\Pi(A) = \mathbf{1}$ for all $A \in G$. Of course, this is an irreducible representation, since $\mathbb{C}$ has no nontrivial subspaces, let alone nontrivial invariant subspaces.

Similarly, if $\mathfrak{g}$ is a Lie algebra, the **trivial representation** of $\mathfrak{g}$ is defined by $\pi : \mathfrak{g} \rightarrow \mathfrak{gl}(1, \mathbb{C})$ such that $\pi(X) = 0$ for all $X \in \mathfrak{g}$. This is an irreducible representation.

Definition 7.10. If G is a matrix Lie group with Lie algebra $\mathfrak{g}$, the **adjoint representation** of G is the map $\text{Ad} : G \rightarrow \text{GL}(\mathfrak{g})$ given by $A \mapsto \text{Ad}_A$ such that $\text{Ad}_A(X) = AXA^{-1}$ for all $X \in \mathfrak{g}$.

Similarly, the **adjoint representation** of a finite-dimensional Lie algebra $\mathfrak{g}$ is the map $\text{ad} : \mathfrak{g} \rightarrow \mathfrak{gl}(\mathfrak{g})$ given by $X \mapsto \text{ad}_X$ such that $\text{ad}_X(Y) = [X, Y]$ for all $Y \in \mathfrak{g}$.

Remark 7.11 (Gauge fields transform in the adjoint representation). In a non-abelian gauge theory (Yang–Mills theory) with gauge group G , the gauge field A_μ is a $\mathfrak{g}$ -valued field: $A_\mu(x) = \sum_a A_\mu^a(x) X_a$, where $\{X_a\}$ is a basis of $\mathfrak{g}$ and the A_μ^a are the component fields. Under a gauge transformation $g(x) \in G$, it transforms as

$$A_\mu \mapsto g A_\mu g^{-1} + g \partial_\mu g^{-1}.$$

For constant (“global”) transformations, the second term vanishes and $A_\mu \mapsto g A_\mu g^{-1} = \text{Ad}_g(A_\mu)$: the gauge field transforms precisely in the adjoint representation. The field strength $F_{\mu\nu} = \partial_\mu A_\nu - \partial_\nu A_\mu + [A_\mu, A_\nu]$ does so even under local transformations: $F_{\mu\nu} \mapsto \text{Ad}_{g(x)}(F_{\mu\nu})$. In particular, the number of gauge bosons equals $\dim \mathfrak{g}$. For quantum chromodynamics, where $G = \text{SU}(3)$ is the colour group, this gives the $8 = \dim \mathfrak{su}(3)$ gluons, transforming in the adjoint representation **8** — not to be confused with the flavour octets of the Eightfold Way (Remark 9.17), which involve a different, approximate global symmetry $\text{SU}(3)$.

7.2 Complete reducibility

Similar to the finite group case, a finite-dimensional representation of a group or Lie algebra is said to be **completely reducible** if it is isomorphic to a direct sum of a finite number of irreducible representations.

A group or Lie algebra is said to have the **complete reducibility** property if every finite-dimensional representation of it is completely reducible.

Proposition 7.12. If G is a matrix Lie group and Π is a finite-dimensional unitary representation of G , then Π is completely reducible.

Similarly, if $\mathfrak{g}$ is a real Lie algebra and π is a finite-dimensional “unitary” representation of $\mathfrak{g}$ (meaning that $\pi(X)^\dagger = -\pi(X)$ for all $X \in \mathfrak{g}$), then π is completely reducible.

Proof. Let V be an inner product space on which Π acts. If $W \subset V$ is an invariant subspace and $W^\perp$ is the orthogonal complement of W so that $V = W \oplus W^\perp$, to complete the proof it suffices to show that $W^\perp$ is an invariant subspace.

The rest of the argument is the same as that of Lemma 3.24 and the proof of Maschke's Theorem 3.14, simply replacing an element g of a finite group by $\Pi(A)$ with $A \in G$ or $\pi(X)$ with $X \in \mathfrak{g}$. $\square$

Theorem 7.13. (Complete reducibility for compact matrix Lie groups.)

If G is a compact matrix Lie group, every finite-dimensional representation of G is isomorphic to a direct sum of a finite number of irreducible representations.

Proof. The proof of this theorem can be completed using Weyl's unitary trick similarly to Theorem 3.25. However, for a compact matrix Lie group, we need to introduce a notion of integration over G . This is achieved by introducing a right-invariant measure on G , known as a **Haar measure** α . If $f : G \rightarrow \mathbb{R}$ is a smooth function, we can integrate f over all G as

$$\int_G f(A) d\alpha(A) \quad (7.4)$$

with the property that for all $B \in G$,

$$\int_G f(AB) d\alpha(A) = \int_G f(A) d\alpha(A). \quad (7.5)$$

Let us now prove the theorem. Let Π be a finite-dimensional representation of G on a space V . Choose an arbitrary inner product $(\cdot, \cdot)$ on V , and define

$$\langle v, w \rangle_G = \int_G (\Pi(A)v, \Pi(A)w) d\alpha(A), \quad (7.6)$$

where the compactness of G ensures that the integral converges. For any $B \in G$, the right-invariance of α gives

$$\langle \Pi(B)v, \Pi(B)w \rangle_G = \int_G (\Pi(AB)v, \Pi(AB)w) d\alpha(A) = \langle v, w \rangle_G, \quad (7.7)$$

so each $\Pi(B)$ is unitary with respect to $\langle \cdot, \cdot \rangle_G$. By Proposition 7.12, Π is completely reducible. $\square$

The following statements and their proofs are completely analogous to the case of finite groups.

Theorem 7.14. (Schur's Lemma.) *Let V and W be irreducible real or complex representations of a group or Lie algebra and let $\phi : V \rightarrow W$ be an intertwining map. Then the following statements are true:*

- (i) *Either ϕ is an isomorphism, or $\phi = 0$.*
- (ii) *Suppose that we work over $\mathbb{C}$. If $V = W$, then $\phi = \lambda \mathbf{1}$ for some $\lambda \in \mathbb{C}$, where $\mathbf{1}$ is the identity.*

Corollary 7.15. *An irreducible complex representation of a commutative group or Lie algebra is one-dimensional.*

Definition 7.16. *Let G be a matrix Lie group and let (Π_1, V_1) and (Π_2, V_2) be representations of G . We define the **tensor product** representation of G on $V_1 \otimes V_2$ as*

$$(\Pi_1 \otimes \Pi_2)(A) = \Pi_1(A) \otimes \Pi_2(A), \quad \text{for all } A \in G. \quad (7.8)$$

If (π_1, V_1) and (π_2, V_2) are representations of a Lie algebra $\mathfrak{g}$, we define a tensor product representation of $\mathfrak{g}$ on $V_1 \otimes V_2$ as

$$(\pi_1 \otimes \pi_2)(X) = \pi_1(X) \otimes I + I \otimes \pi_2(X), \quad \text{for all } X \in \mathfrak{g}. \quad (7.9)$$

Definition (7.9) is consistent with definition (7.8): differentiating $(\Pi_1 \otimes \Pi_2)(e^{tX})(v_1 \otimes v_2) = \Pi_1(e^{tX})v_1 \otimes \Pi_2(e^{tX})v_2$ at $t = 0$ and using the product rule gives

$$(\pi_1 \otimes \pi_2)(X)(v_1 \otimes v_2) = \pi_1(X)v_1 \otimes v_2 + v_1 \otimes \pi_2(X)v_2. \quad (7.10)$$

Definition 7.17. *Let G be a matrix Lie group and (Π, V) be a finite-dimensional representation. The **dual representation** (Π^*, V^*) of (Π, V) is the representation of G acting on V^* given by*

$$\Pi^*(g) = [\Pi(g^{-1})]^T. \quad (7.11)$$

If (π, V) is a representation of a Lie algebra $\mathfrak{g}$, then the dual representation (π^, V^*) is given by*

$$\pi^*(X) = -\pi(X)^T. \quad (7.12)$$

8 The representations of $\mathfrak{sl}(2, \mathbb{C})$

In this section, we will compute (up to isomorphism) all finite-dimensional irreducible complex representations of the Lie algebra $\mathfrak{sl}(2, \mathbb{C})$. This Lie algebra is the complexification of $\mathfrak{su}(2) \cong \mathfrak{so}(3)$, whose representations are closely related to the concepts of “angular momentum” and “spin” in quantum mechanics. Moreover, the representation theory of $\mathfrak{sl}(2, \mathbb{C})$ is a fundamental tool for determining the representations of any semisimple Lie algebra $\mathfrak{g}$, as one can apply it to various subalgebras of $\mathfrak{g}$ that are isomorphic to $\mathfrak{sl}(2, \mathbb{C})$.

We will use the following standard basis for the 3-dimensional Lie algebra $\mathfrak{sl}(2, \mathbb{C})$:

$$X = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}; \quad Y = \begin{pmatrix} 0 & 0 \\ 1 & 0 \end{pmatrix}; \quad H = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}, \quad (8.1)$$

which satisfy the commutation relations

$$[H, X] = 2X, \quad [H, Y] = -2Y, \quad [X, Y] = H. \quad (8.2)$$

If V is a finite-dimensional complex vector space, and A, B, C are operators on V satisfying these same commutation relations, then the unique linear map $\pi : \mathfrak{sl}(2, \mathbb{C}) \rightarrow \mathfrak{gl}(V)$ defined by

$$\pi(H) = A, \quad \pi(X) = B, \quad \pi(Y) = C \quad (8.3)$$

is a representation of $\mathfrak{sl}(2, \mathbb{C})$.

Theorem 8.1. (i) For each integer $m \geq 0$, there exists an irreducible complex representation of $\mathfrak{sl}(2, \mathbb{C})$ with dimension $m + 1$.

(ii) Any two irreducible complex representations of $\mathfrak{sl}(2, \mathbb{C})$ with the same dimension are isomorphic.

(iii) If (π, V) is an irreducible complex representation of $\mathfrak{sl}(2, \mathbb{C})$ with dimension $\dim V = m + 1$, then there exists a basis $\{u_0, \dots, u_m\}$ of V such that the actions of the basis elements are given by:

$$\begin{aligned} \pi(H)u_k &= (m - 2k)u_k \\ \pi(Y)u_k &= \begin{cases} u_{k+1} & \text{if } k < m \\ 0 & \text{if } k = m \end{cases} \\ \pi(X)u_k &= \begin{cases} k(m - k + 1)u_{k-1} & \text{if } k > 0 \\ 0 & \text{if } k = 0 \end{cases} \end{aligned} \quad (8.4)$$

We now introduce a simple but crucial lemma that allows us to develop this structure in an arbitrary irreducible representation of $\mathfrak{sl}(2, \mathbb{C})$.

Lemma 8.2. Let u be an eigenvector of $\pi(H)$ with eigenvalue $\alpha \in \mathbb{C}$. Then

$$\pi(H)\pi(X)u = (\alpha + 2)\pi(X)u.$$

Thus, either $\pi(X)u = 0$ or $\pi(X)u$ is an eigenvector for $\pi(H)$ with eigenvalue $\alpha + 2$. Similarly,

$$\pi(H)\pi(Y)u = (\alpha - 2)\pi(Y)u,$$

so that either $\pi(Y)u = 0$ or $\pi(Y)u$ is an eigenvector for $\pi(H)$ with eigenvalue $\alpha - 2$.

Proof. From the commutation relations, we know that $[\pi(H), \pi(X)] = \pi([H, X]) = 2\pi(X)$. Thus,

$$\begin{aligned} \pi(H)\pi(X)u &= \pi(X)\pi(H)u + 2\pi(X)u \\ &= \pi(X)(\alpha u) + 2\pi(X)u \\ &= (\alpha + 2)\pi(X)u. \end{aligned}$$

The argument with $\pi(X)$ replaced by $\pi(Y)$ is analogous. □

We call $\pi(X)$ the “raising operator” and $\pi(Y)$ the “lowering operator,” since they raise and lower the eigenvalue of $\pi(H)$ by 2, respectively.

Proof of Theorem 8.1. Let (π, V) be an irreducible representation of $\mathfrak{sl}(2, \mathbb{C})$ on a finite-dimensional complex vector space V . Our strategy is to diagonalise the operator $\pi(H)$. Since we are working over $\mathbb{C}$, $\pi(H)$ must have at least one eigenvector. Let u be an eigenvector for $\pi(H)$ with eigenvalue α . Applying Lemma 8.2 repeatedly, we find

$$\pi(H)\pi(X)^k u = (\alpha + 2k)\pi(X)^k u.$$

Since $\pi(H)$ has only finitely many eigenvalues, the vectors $\pi(X)^k u$ cannot all be nonzero: there exists $N \geq 0$ with $\pi(X)^N u \neq 0$ but $\pi(X)^{N+1} u = 0$. Let us define $u_0 = \pi(X)^N u$ and $\lambda = \alpha + 2N$. Then u_0 is a non-zero vector satisfying

$$\pi(H)u_0 = \lambda u_0 \quad \text{and} \quad \pi(X)u_0 = 0. \quad (8.5)$$

We then define a sequence of vectors $u_k = \pi(Y)^k u_0$ for $k \geq 0$. By Lemma 8.2, these are eigenvectors of $\pi(H)$:

$$\pi(H)u_k = (\lambda - 2k)u_k.$$

It can be verified by induction that the raising operator acts on these vectors as¹⁸

$$\pi(X)u_k = k(\lambda - (k - 1))u_{k-1} \quad (k \geq 1). \quad (8.6)$$

Again, since V is finite-dimensional, the sequence of non-zero vectors $\{u_k\}$ must terminate. Let m be the non-negative integer such that $u_k \neq 0$ for all $k \leq m$, but $u_{m+1} = \pi(Y)u_m = 0$. Applying $\pi(X)$ to u_{m+1} gives 0. Using (8.6), we have

$$0 = \pi(X)u_{m+1} = (m + 1)(\lambda - m)u_m.$$

Since $u_m \neq 0$ and $m + 1 > 0$, we must have $\lambda = m$: the highest eigenvalue is a non-negative integer, and the actions of the generators on $\{u_0, \dots, u_m\}$ are precisely those in (8.4).

The vectors $u_0, \dots, u_m$ are eigenvectors of $\pi(H)$ with distinct eigenvalues, hence linearly independent. Their span is invariant under $\pi(H)$, $\pi(X)$ and $\pi(Y)$, and therefore under all of $\mathfrak{sl}(2, \mathbb{C})$; by irreducibility, it is all of V . This proves that every irreducible representation is of the stated form. Conversely, one can verify that for any $m \geq 0$, the operators defined by (8.4) on an $(m + 1)$ -dimensional space satisfy the $\mathfrak{sl}(2, \mathbb{C})$ commutation relations.

Finally, this representation is irreducible: if $W \neq \{0\}$ is invariant and $w = \sum_j a_j u_j \in W$ is non-zero with $a_{j_0} \neq 0$ for the largest index j_0 , then $\pi(X)^{j_0} w$ is a non-zero multiple of u_0 , so $u_0 \in W$; applying $\pi(Y)$ repeatedly then gives $u_1, \dots, u_m \in W$, whence $W = V$. $\square$

¹⁸The inductive step can be done as follows: $\pi(X)u_{k+1} = \pi(X)\pi(Y)u_k = (\pi(H) + \pi(Y)\pi(X))u_k = (\lambda - 2k)u_k + \pi(Y)k(\lambda - (k - 1))u_{k-1} = (\lambda - 2k)u_k + k(\lambda - (k - 1))u_k = (k + 1)(\lambda - k)u_k$.

Convention 8.3. *Here is some important terminology:*

1. For the irreducible representation of $\mathfrak{sl}(2, \mathbb{C})$ of dimension $m + 1$, denoted π_m , the eigenvalues of $\pi(H)$, namely $\{m, m - 2, \dots, -m + 2, -m\}$, are called the **weights** of the representation. The largest eigenvalue, m , is called the **highest weight**. An irreducible representation is uniquely characterised by its highest weight. We sometimes denote this representation by V_m or simply (m) .
2. In the physics literature, these representations are often labeled by “spin” $l = m/2$, which can take values $l = 0, 1/2, 1, 3/2, \dots$. Physicists may also denote an $(m + 1)$ -dimensional representation by its dimension in bold, **$m + 1$** .
3. The irreducible representation with highest weight $m = 1$ (spin-1/2) is the **standard** or **fundamental** representation of $\mathfrak{sl}(2, \mathbb{C})$. In this representation, $\pi_1(Z) = Z$ for all $Z \in \mathfrak{sl}(2, \mathbb{C})$.

So far, we have classified the irreducible representations of $\mathfrak{sl}(2, \mathbb{C})$. We now relate these to the representations of the Lie group $SU(2)$.

Proposition 8.4. *The irreducible representations of the Lie group $SU(2)$ are in one-to-one correspondence with the irreducible representations of the Lie algebra $\mathfrak{sl}(2, \mathbb{C}) \cong \mathfrak{su}(2)_{\mathbb{C}} \cong \mathfrak{so}(3)_{\mathbb{C}}$.*

Proof. The correspondence follows from our previous results:

- Since $SU(2)$ is a connected Lie group, every representation of $SU(2)$ gives rise to a representation of its Lie algebra $\mathfrak{su}(2)$, and Proposition 7.4 shows that this correspondence preserves irreducibility and isomorphisms.
- By Proposition 7.5, there is a one-to-one correspondence between the complex representations of the real Lie algebra $\mathfrak{su}(2)$ and its complexification $\mathfrak{su}(2)_{\mathbb{C}} \cong \mathfrak{sl}(2, \mathbb{C})$, which also preserves irreducibility.
- Since $SU(2)$ is simply connected (it is homeomorphic to S^3), we can go in the opposite direction by restricting any representation of $\mathfrak{sl}(2, \mathbb{C}) \cong \mathfrak{su}(2)_{\mathbb{C}}$ to $\mathfrak{su}(2)$ and then applying Theorem 6.42 to obtain a representation of $SU(2)$.

This establishes the one-to-one correspondence. □

This correspondence does not hold for the group $SO(3)$, which is not simply connected. Specifically, we will show that a representation of the Lie algebra $\mathfrak{so}(3) \cong \mathfrak{su}(2)$ lifts to a representation of the group $SO(3)$ only if its highest weight m is an even integer. In the language of physics, this means that representations with integer spin ($l = m/2 \in \mathbb{Z}$) are true representations of $SO(3)$, while those with half-integer spin are not.

As mentioned earlier, the representation theory of $\mathfrak{sl}(2, \mathbb{C})$ is crucial for studying other Lie algebras, such as $\mathfrak{sl}(3, \mathbb{C})$, because these algebras contain subalgebras isomorphic to $\mathfrak{sl}(2, \mathbb{C})$. For such applications, we need a few results about finite-dimensional representations of $\mathfrak{sl}(2, \mathbb{C})$ that are not necessarily irreducible.

Theorem 8.5. *If (π, V) is any finite-dimensional representation of $\mathfrak{sl}(2, \mathbb{C})$, the following results hold:*

1. *Every eigenvalue of $\pi(H)$ is an integer. Furthermore, if v is an eigenvector of $\pi(H)$ with eigenvalue λ and $\pi(X)v = 0$, then λ is a non-negative integer.*
2. *The operators $\pi(X)$ and $\pi(Y)$ are nilpotent.*
3. *If we define the operator $S = e^{\pi(X)}e^{-\pi(Y)}e^{\pi(X)}$, then $S\pi(H)S^{-1} = -\pi(H)$.*
4. *The set of eigenvalues of $\pi(H)$ is symmetric about the origin. That is, if k is an eigenvalue, then $-k$ is also an eigenvalue, and so is each of the numbers*

$$-|k|, -|k| + 2, \dots, |k| - 2, |k|.$$

Proof. Since $SU(2)$ is compact and simply connected, every finite-dimensional representation of its Lie algebra $\mathfrak{su}(2)$, and thus of $\mathfrak{sl}(2, \mathbb{C})$, is completely reducible. We can therefore decompose V into a direct sum of irreducible representations V_m , and each of the stated properties can be deduced from its irreducible components, using the explicit formulas (8.4).¹⁹ $\square$

8.1 Characters of representations

Suppose (π, V) is a finite-dimensional representation of $\mathfrak{sl}(2, \mathbb{C})$. The **character** of π is the function $\chi_\pi : \mathfrak{g} \rightarrow \mathbb{C}$ given by

$$\chi_\pi(Z) = \text{tr}(e^{\pi(Z)}). \quad (8.7)$$

It is often sufficient to evaluate the character on a single element, such as aH , for a parameter a .

¹⁹Here is the argument for Points 3 and 4. Using $e^A B e^{-A} = e^{\text{ad}_A}(B)$ and the fact that π is a Lie algebra homomorphism, so that $\text{ad}_{\pi(Z)}(\pi(W)) = \pi(\text{ad}_Z(W))$, we have $S\pi(H)S^{-1} = e^{\text{ad}_{\pi(X)}}e^{-\text{ad}_{\pi(Y)}}e^{\text{ad}_{\pi(X)}}(\pi(H)) = \pi(e^{\text{ad}_X}e^{-\text{ad}_Y}e^{\text{ad}_X}(H))$. The argument of π can now be computed with 2×2 matrices: it equals sHs^{-1} , where $s = e^X e^{-Y} e^X = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}$, and a direct computation gives $sHs^{-1} = -H$. Hence $S\pi(H)S^{-1} = \pi(-H) = -\pi(H)$, which is Point 3. For Point 4, if $\pi(H)v = kv$, then $\pi(H)(Sv) = -S\pi(H)v = -k(Sv)$, so the invertible operator S maps the eigenspace of k onto that of $-k$. The intermediate eigenvalues $-|k| + 2, \dots, |k| - 2$ occur because any irreducible component on which $|k|$ appears is some V_m with $m \geq |k|$ and $m - |k|$ even, whose spectrum $\{m, m - 2, \dots, -m\}$ contains all of them.

Example 8.6. Let (π_m, V_m) be the irreducible representation of $\mathfrak{sl}(2, \mathbb{C})$ of dimension $m + 1$. The eigenvalues of $\pi_m(H)$ are $m, m - 2, \dots, -m$. The character of this representation, evaluated at aH , is

$$\begin{aligned}\chi_{V_m}(z) &:= \chi_{\pi_m}(aH) = \text{tr}(e^{a\pi_m(H)}) = e^{ma} + e^{(m-2)a} + \dots + e^{-ma} \\ &= z^m + z^{m-2} + \dots + z^{-m+2} + z^{-m},\end{aligned}\tag{8.8}$$

where $z = e^a$. This can be written in closed form as a geometric series:

$$\chi_{V_m}(z) = \frac{z^{m+1} - z^{-(m+1)}}{z - z^{-1}} = \frac{\sinh((m+1)a)}{\sinh(a)},\tag{8.9}$$

provided $z \neq \pm 1$.

The character of a tensor product is the product of the characters. This allows us to decompose tensor product representations.

Example 8.7. (i) For $V_1 \otimes V_1$, the character is

$$\begin{aligned}\chi_{V_1 \otimes V_1}(z) &= (z + z^{-1})^2 = z^2 + 2 + z^{-2} \\ &= (z^2 + 1 + z^{-2}) + 1 = \chi_{V_2}(z) + \chi_{V_0}(z).\end{aligned}\tag{8.10}$$

This implies the decomposition $V_1 \otimes V_1 \cong V_2 \oplus V_0$. In physics, this corresponds to combining two spin-1/2 particles to get a total spin of 1 (a triplet state, V_2) or 0 (a singlet state, V_0).

(ii) Similarly, for $V_2 \otimes V_1$:

$$\begin{aligned}\chi_{V_2 \otimes V_1}(z) &= (z^2 + 1 + z^{-2})(z + z^{-1}) \\ &= (z^3 + z^1 + z^{-1} + z^{-3}) + (z + z^{-1}) = \chi_{V_3}(z) + \chi_{V_1}(z).\end{aligned}\tag{8.11}$$

This gives the decomposition $V_2 \otimes V_1 \cong V_3 \oplus V_1$.

Theorem 8.8. (The Clebsch-Gordan rule.) Let $m, n \in \mathbb{Z}_{\geq 0}$ with $m \geq n$. The tensor product of the irreducible representations V_m and V_n of $\mathfrak{sl}(2, \mathbb{C})$ decomposes as

$$V_m \otimes V_n \cong V_{m+n} \oplus V_{m+n-2} \oplus \dots \oplus V_{m-n}.\tag{8.12}$$

Proof. We verify this by computing

$$\begin{aligned}\chi_{V_m}(z)\chi_{V_n}(z) &= \frac{z^{m+1} - z^{-m-1}}{z - z^{-1}}(z^n + z^{n-2} + \dots + z^{-n}) \\ &= \sum_{j=0}^n \frac{z^{m+n+1-2j} - z^{2j-m-n-1}}{z - z^{-1}} = \sum_{j=0}^n \chi_{V_{m+n-2j}}(z).\end{aligned}\tag{8.13}$$

Note that the condition $m \geq n$ ensures that there are no cancellations in the sum. $\square$

Example 8.9. Using characters, we can check the following statements:

1. $\text{Sym}^2 V_1 = V_2$, since

$$\begin{aligned}\chi_{\text{Sym}^2 V_1}(z) &= \frac{1}{2} [\chi_{V_1}(z)^2 + \chi_{V_1}(z^2)] = \frac{1}{2} [(z + z^{-1})^2 + (z^2 + z^{-2})] \\ &= z^2 + 1 + z^{-2} = \chi_{V_2}(z).\end{aligned}\tag{8.14}$$

2. In general, $V_n = \text{Sym}^n V_1$. Any irreducible representation of $\mathfrak{sl}(2, \mathbb{C})$ is a symmetric power of the standard (fundamental) representation $V_1 \cong \mathbb{C}^2$.

3. $\text{Sym}^2(\text{Sym}^2 V_1) = \text{Sym}^2(V_2) = V_4 \oplus V_0$.

8.2 The irreducible representations of $\text{SO}(3)$

As established, every irreducible representation π_m of $\mathfrak{sl}(2, \mathbb{C})$ corresponds to a representation of $\text{SU}(2)$. We now determine which of these descend to representations of $\text{SO}(3)$.

Proposition 8.10. An irreducible representation π_m of the Lie algebra $\mathfrak{so}(3) \cong \mathfrak{su}(2)$ gives rise to a well-defined representation of the Lie group $\text{SO}(3)$ if and only if its highest weight m is an even integer.

Proof. Let $\sigma_m = \pi_m \circ \phi^{-1}$ be the representation of $\mathfrak{so}(3)$, where $\phi : \mathfrak{su}(2) \rightarrow \mathfrak{so}(3)$ is the Lie algebra isomorphism. A representation Σ_m of $\text{SO}(3)$ exists if $\Sigma_m(e^X) = e^{\sigma_m(X)}$ is well-defined for any $X \in \mathfrak{so}(3)$.

Consider the basis element $F_1 = \begin{pmatrix} 0 & 0 & 0 \\ 0 & 0 & -1 \\ 0 & 1 & 0 \end{pmatrix}$ of $\mathfrak{so}(3)$, which generates rotations

about the x_1 -axis. The group element $e^{2\pi F_1}$ is the rotation by 2π , i.e., the identity matrix $\mathbf{1} \in \text{SO}(3)$. If a representation Σ_m of $\text{SO}(3)$ exists, we must therefore have $\Sigma_m(e^{2\pi F_1}) = e^{2\pi \sigma_m(F_1)} = \Sigma_m(\mathbf{1}) = \mathbf{1}$.

Note that $\sigma_m(F_1) = \pi_m(\phi^{-1}(F_1)) = \pi_m(E_1) = \frac{i}{2}\pi_m(H)$, with E_1 the following basis element of $\mathfrak{su}(2)$:

$$E_1 = \frac{1}{2} \begin{pmatrix} i & 0 \\ 0 & -i \end{pmatrix} = \frac{1}{2} iH. \tag{8.15}$$

The eigenvalues of $\sigma_m(F_1) = \frac{i}{2}\pi_m(H)$ are $\frac{i}{2}(m - 2j)$ for $j = 0, \dots, m$, so the eigenvalues of $e^{2\pi \sigma_m(F_1)}$ are $e^{i\pi(m-2j)} = (-1)^m$. Hence $e^{2\pi \sigma_m(F_1)} = (-1)^m \mathbf{1}$, and the requirement above forces $(-1)^m = 1$: for odd m , no representation of $\text{SO}(3)$ exists.

If m is even and Π_m is the representation of $\text{SU}(2)$, then $\Pi_m(-\mathbf{1}) \stackrel{(8.15)}{=} \Pi_m(e^{2\pi E_1}) = e^{\pi_m(2\pi E_1)} = \mathbf{1}$. Since $\text{SU}(2)$ is a double cover of $\text{SO}(3)$, there is a unique pair of elements $\{U, -U\}$ in $\text{SU}(2)$ that is mapped to a given $R \in \text{SO}(3)$ under a homomorphism $\text{SU}(2) \rightarrow \text{SO}(3)$. It follows that $\Pi_m(U) = \Pi_m(-U)$ for any $U \in \text{SU}(2)$. Therefore, we obtain a well-defined representation Σ_m of $\text{SO}(3)$ by $\Sigma_m(R) = \Pi_m(U)$. $\square$

An electron is a “spin-1/2” particle, meaning its quantum state transforms according to the two-dimensional representation π_1 of $\mathfrak{so}(3)$ (where $m = 1$). Since $m = 1$ is odd, this representation does not come from a representation of $\mathrm{SO}(3)$. This is the origin of the statement that a 360° rotation of an electron’s wavefunction results in a sign change, reflecting the fact that the path of rotation corresponds to a path from I to $-I$ in the covering group $\mathrm{SU}(2)$.

9 The representations of $\mathfrak{sl}(3, \mathbb{C})$

We will now compute (up to isomorphism) all finite-dimensional irreducible complex representations of the Lie algebra $\mathfrak{sl}(3, \mathbb{C})$. This Lie algebra is the complexification of $\mathfrak{su}(3)$, and its representation theory is of great physical importance, particularly in the quark model of particle physics. Moreover, the methods used here serve as a prototype for classifying the representations of all complex simple Lie algebras.

We will use the following basis for the 8-dimensional Lie algebra $\mathfrak{sl}(3, \mathbb{C})$:

$$\begin{aligned} H_1 &= \begin{pmatrix} 1 & 0 & 0 \\ 0 & -1 & 0 \\ 0 & 0 & 0 \end{pmatrix}, & H_2 &= \begin{pmatrix} 0 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & -1 \end{pmatrix} \\ X_1 &= \begin{pmatrix} 0 & 1 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}, & X_2 &= \begin{pmatrix} 0 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \end{pmatrix}, & X_3 &= \begin{pmatrix} 0 & 0 & 1 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix} \\ Y_1 &= \begin{pmatrix} 0 & 0 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}, & Y_2 &= \begin{pmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 1 & 0 \end{pmatrix}, & Y_3 &= \begin{pmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 1 & 0 & 0 \end{pmatrix}. \end{aligned} \tag{9.1}$$

Note that the span of $\{H_1, X_1, Y_1\}$ forms a subalgebra isomorphic to $\mathfrak{sl}(2, \mathbb{C})$, which can be seen by ignoring the third row and column of each matrix. Similarly, the subalgebra spanned by $\{H_2, X_2, Y_2\}$ is isomorphic to $\mathfrak{sl}(2, \mathbb{C})$, and so is the one spanned by $\{H_1 + H_2, X_3, Y_3\}$. The commutation relations between the basis elements are computed directly from the matrices; those we shall need are

$$\begin{aligned} [H_1, H_2] &= 0, & [X_1, Y_1] &= H_1, & [X_2, Y_2] &= H_2, \\ [H_1, X_1] &= 2X_1, & [H_1, X_2] &= -X_2, & [H_1, X_3] &= X_3, \\ [H_2, X_1] &= -X_1, & [H_2, X_2] &= 2X_2, & [H_2, X_3] &= X_3, \\ [H_1, Y_1] &= -2Y_1, & [H_1, Y_2] &= Y_2, & [H_1, Y_3] &= -Y_3, \\ [H_2, Y_1] &= Y_1, & [H_2, Y_2] &= -2Y_2, & [H_2, Y_3] &= -Y_3, \\ [X_1, X_2] &= X_3, & [Y_1, Y_2] &= -Y_3, & [X_3, Y_3] &= H_1 + H_2. \end{aligned} \tag{9.2}$$

Our strategy for classifying the representations of $\mathfrak{sl}(3, \mathbb{C})$ is to simultaneously diagonalise the operators $\pi(H_1)$ and $\pi(H_2)$. Since $[H_1, H_2] = 0$, their images under

any representation π will also commute, so it is possible to find a basis of simultaneous eigenvectors.²⁰

Definition 9.1. If (π, V) is a representation of $\mathfrak{sl}(3, \mathbb{C})$, an ordered pair $\mu = (m_1, m_2) \in \mathbb{C}^2$ is called a **weight** for π if there exists a nonzero vector $v \in V$ such that

$$\pi(H_1)v = m_1v \quad \text{and} \quad \pi(H_2)v = m_2v. \quad (9.3)$$

A nonzero vector v satisfying (9.3) is called a **weight vector** corresponding to the weight μ . The **weight space** for a weight μ is the space of all vectors satisfying (9.3), and its dimension is the **multiplicity** of the weight.

Thus, a weight is simply a pair of simultaneous eigenvalues for $\pi(H_1)$ and $\pi(H_2)$. Isomorphic representations have the same weights and multiplicities, since an intertwiner maps a weight vector to a weight vector of the same weight.

Proposition 9.2. Every finite-dimensional complex representation of $\mathfrak{sl}(3, \mathbb{C})$ has at least one weight.

Proof. Since we work over $\mathbb{C}$, the operator $\pi(H_1)$ has an eigenvalue m_1 , with eigenspace $W \subset V$. Since $\pi(H_2)$ commutes with $\pi(H_1)$, it preserves W , and its restriction to W has an eigenvector w , which is then a simultaneous eigenvector for $\pi(H_1)$ and $\pi(H_2)$. $\square$

Every representation of $\mathfrak{sl}(3, \mathbb{C})$ can be viewed, by restriction, as a representation of the subalgebras $\langle H_1, X_1, Y_1 \rangle$ and $\langle H_2, X_2, Y_2 \rangle$, both of which are isomorphic to $\mathfrak{sl}(2, \mathbb{C})$.

Proposition 9.3. If (π, V) is a finite-dimensional representation of $\mathfrak{sl}(3, \mathbb{C})$ and $\mu = (m_1, m_2)$ is a weight of V , then both m_1 and m_2 must be integers.

Proof. This follows from applying the results for $\mathfrak{sl}(2, \mathbb{C})$ representations (Theorem 8.5) to the restrictions of π to the two $\mathfrak{sl}(2, \mathbb{C})$ subalgebras mentioned above. The eigenvalues of the operators corresponding to H_1 and H_2 must be integers. $\square$

Our strategy is to identify a simultaneous eigenvector for $\pi(H_1)$ and $\pi(H_2)$ and then to analyze the effect of applying the other operators $\pi(X_j)$ and $\pi(Y_j)$.

Definition 9.4. An ordered pair $\alpha = (a_1, a_2) \in \mathbb{C}^2$ is called a **root** if

1. a_1 and a_2 are not both zero, and

²⁰If two diagonalisable matrices A and B commute ($AB = BA$), then they are simultaneously diagonalisable; that is, there exists an invertible matrix P such that both $P^{-1}AP$ and $P^{-1}BP$ are diagonal. (The converse also holds, since simultaneously diagonalisable matrices commute.) However, there exist commuting matrices that are not diagonalisable.

2. there exists a nonzero $Z \in \mathfrak{sl}(3, \mathbb{C})$ such that

$$[H_1, Z] = a_1 Z \quad \text{and} \quad [H_2, Z] = a_2 Z. \quad (9.4)$$

The element Z is called a **root vector** corresponding to the root α .

The second condition means that Z is a simultaneous eigenvector for the adjoint actions of H_1 and H_2 . Thus, we can say:

The roots are precisely the nonzero weights of the adjoint representation.

Using the commutation relations (9.2), we find the six roots of $\mathfrak{sl}(3, \mathbb{C})$:

Root α	Root Vector Z	Root α	Root Vector Z
$(2, -1)$	X_1	$(-2, 1)$	Y_1
$(-1, 2)$	X_2	$(1, -2)$	Y_2
$(1, 1)$	X_3	$(-1, -1)$	Y_3

(9.5)

The elements H_1 and H_2 are simultaneous eigenvectors with weight $(0, 0)$ and are thus not root vectors. The six roots form a root system known as A_2 . We single out two roots:

$$\alpha_1 = (2, -1) \quad \text{and} \quad \alpha_2 = (-1, 2), \quad (9.6)$$

which we call the **positive simple roots**. All other roots can be written as integer linear combinations of these two, where the coefficients are either all non-negative or all non-positive. The significance of roots is revealed in the following lemma.

Lemma 9.5. *Let $\alpha = (a_1, a_2)$ be a root with corresponding root vector Z_α . Let π be a representation of $\mathfrak{sl}(3, \mathbb{C})$, let $\mu = (m_1, m_2)$ be a weight, and let $v \neq 0$ be a corresponding weight vector. Then*

$$\begin{aligned} \pi(H_1)\pi(Z_\alpha)v &= (m_1 + a_1)\pi(Z_\alpha)v, \\ \pi(H_2)\pi(Z_\alpha)v &= (m_2 + a_2)\pi(Z_\alpha)v. \end{aligned}$$

Thus, either $\pi(Z_\alpha)v = 0$ or $\pi(Z_\alpha)v$ is a new weight vector with weight $\mu + \alpha$.

Proof. By definition of a root, $[\pi(H_i), \pi(Z_\alpha)] = a_i \pi(Z_\alpha)$ for $i = 1, 2$. The computation is then identical to the proof of Lemma 8.2: $\pi(H_i)\pi(Z_\alpha)v = \pi(Z_\alpha)\pi(H_i)v + a_i \pi(Z_\alpha)v = (m_i + a_i)\pi(Z_\alpha)v$. $\square$

9.1 The theorem of the highest weight

Applying root vectors to a weight vector generates new weight vectors. Since a representation is finite-dimensional, this process must terminate. This suggests a structure analogous to the $\mathfrak{sl}(2, \mathbb{C})$ case, organised around a “highest” weight.

Definition 9.6. Let $\alpha_1 = (2, -1)$ and $\alpha_2 = (-1, 2)$ be the positive simple roots. A weight μ_1 is **higher** than a weight μ_2 (denoted $\mu_1 \geq \mu_2$) if their difference can be written as a non-negative real linear combination of the positive simple roots:

$$\mu_1 - \mu_2 = a\alpha_1 + b\alpha_2, \quad \text{with } a \geq 0, b \geq 0. \quad (9.7)$$

A weight μ_0 for a representation π is a **highest weight** if $\mu \leq \mu_0$ for all weights μ of π .

Note that the relation of “higher” is only a partial ordering; for example, $\alpha_1 - \alpha_2$ is neither higher nor lower than 0. In particular, a finite set of weights need not have a highest element. Note also that the coefficients a and b do not have to be integers, even if both μ_1 and μ_2 have integer entries. For example, $(1, 0)$ is higher than $(0, 0)$ since $(1, 0) = \frac{2}{3}\alpha_1 + \frac{1}{3}\alpha_2$.

We are now ready to state the main theorem regarding the irreducible representations of $\mathfrak{sl}(3, \mathbb{C})$.

Theorem 9.7. (The Theorem of the Highest Weight for $\mathfrak{sl}(3, \mathbb{C})$)

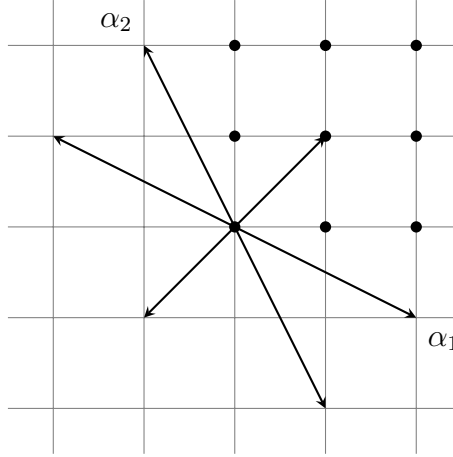
1. Every irreducible representation of $\mathfrak{sl}(3, \mathbb{C})$ is the direct sum of its weight spaces.
2. Every irreducible representation of $\mathfrak{sl}(3, \mathbb{C})$ has a unique highest weight μ .
3. Two irreducible representations of $\mathfrak{sl}(3, \mathbb{C})$ with the same highest weight are isomorphic.
4. The highest weight μ of an irreducible representation must be of the form $\mu = (m_1, m_2)$, where m_1 and m_2 are non-negative integers.
5. For every pair (m_1, m_2) of non-negative integers, there exists an irreducible representation of $\mathfrak{sl}(3, \mathbb{C})$ with highest weight (m_1, m_2) .

A similar classification holds for the group $SU(3)$.

Theorem 9.8. For every pair of non-negative integers (m_1, m_2) , there exists an irreducible representation Π of $SU(3)$ whose associated Lie algebra representation π of $\mathfrak{sl}(3, \mathbb{C})$ has highest weight (m_1, m_2) .

The possible highest weights, (m_1, m_2) with $m_1, m_2 \in \mathbb{Z}_{\geq 0}$, are known as the **dominant integral elements**.

In the diagram below, the roots (arrows) and dominant integral elements (black dots) are drawn in the “obvious basis” for the space of weights; that is, the x -coordinate is the eigenvalue of H_1 and the y -coordinate is the eigenvalue of H_2 .²¹



Once we have introduced the Weyl group, we will see the same picture rendered using a Weyl-invariant inner product, which will give a more symmetric view of the situation.

A detailed proof of Theorem 9.7, based on the notion of a *highest-weight cyclic representation*, is given in Appendix C. We will not need this machinery in the remainder of the course; the key input is the explicit construction of the irreducible representations, to which we now turn.

9.2 Examples of the irreducible representations of $\mathfrak{sl}(3, \mathbb{C})$

We now construct the most important representations of $\mathfrak{sl}(3, \mathbb{C})$.

- The irreducible representation with highest weight $(1, 0)$ is the **standard** or **fundamental representation**. This is the representation where $\mathfrak{sl}(3, \mathbb{C})$ acts on $\mathbb{C}^3$ in the usual way: $\pi(Z) = Z$, for all $Z \in \mathfrak{sl}(3, \mathbb{C})$. The standard basis vectors e_1, e_2, e_3 are weight vectors with weights $(1, 0)$, $(-1, 1)$, and $(0, -1)$, respectively.²² The highest weight is $(1, 0)$.
- The **dual of the standard representation**, given by $\pi(Z) = -Z^T$ for all $Z \in \mathfrak{sl}(3, \mathbb{C})$, is also irreducible and is known as the **antifundamental representation**. It also has weight vectors e_1, e_2 , and e_3 , with corresponding weights $(-1, 0)$, $(1, -1)$, and $(0, 1)$ and with highest weight $(0, 1)$.

²¹In the literature, this basis is sometimes referred to as the **fundamental weight basis**, or the **Dynkin basis**, or the ω -**basis**, where ω stands for the fundamental weights.

²²Recall that $H_1 = \text{diag}(1, -1, 0)$ and $H_2 = \text{diag}(0, 1, -1)$. For example, the weight $(-1, 1)$ arises from $H_1 e_2 = -e_2$ and $H_2 e_2 = e_2$.

- Let (V_1, π_1) be the fundamental representation and (V_2, π_2) be the antifundamental representation. Let $v_1 = e_1$ and $v_2 = e_3$ be their respective highest-weight vectors. Consider the tensor product representation on the space

$$(V_1^{\otimes m_1}) \otimes (V_2^{\otimes m_2}). \quad (9.8)$$

The vector

$$v = v_1^{\otimes m_1} \otimes v_2^{\otimes m_2} = e_1^{\otimes m_1} \otimes e_3^{\otimes m_2} \quad (9.9)$$

is a weight vector with weight (m_1, m_2) and is annihilated by the raising operators X_1, X_2, X_3 , since for example, $\pi_1(X_j)e_1 = X_j e_1 = 0$ and $\pi_2(X_j)e_3 = -X_j^T e_3 = 0$.

Note that both the fundamental and antifundamental representations of $\mathfrak{su}(3)$ are unitary, since $\pi_1(X)^\dagger = X^\dagger = -X = -\pi_1(X)$ and $\pi_2(X)^\dagger = -(X^T)^\dagger = -X^* = X^T = -\pi_2(X)$, for all $X \in \mathfrak{su}(3)$. Since the tensor product of unitary representations is unitary, and thus completely reducible, the space $(V_1^{\otimes m_1}) \otimes (V_2^{\otimes m_2})$ decomposes as a direct sum of irreducible representations of $\mathfrak{su}(3)$, and hence of $\mathfrak{su}(3)_\mathbb{C} \cong \mathfrak{sl}(3, \mathbb{C})$.

Let W be an irreducible component of this direct sum such that the projection P onto W satisfies $Pv \neq 0$; such a component exists since $v \neq 0$, as in (9.9). Since P commutes with the action of $\mathfrak{sl}(3, \mathbb{C})$, the vector $Pv \in W$ is again a weight vector with weight (m_1, m_2) that is annihilated by all raising operators. Moreover, since

1. every weight of the fundamental (respectively, antifundamental) representation is obtained from $(1, 0)$ (respectively, $(0, 1)$) by *subtracting* $c_1\alpha_1 + c_2\alpha_2$, for some $c_1, c_2 \geq 0$, where α_1, α_2 are the positive simple roots, and
2. the weights of the tensor product $R_1 \otimes R_2$ are the sums of the weights of R_1 and the weights of R_2 ,

it follows that every weight of $(V_1^{\otimes m_1}) \otimes (V_2^{\otimes m_2})$, and in particular every weight of W , is lower than or equal to (m_1, m_2) . It follows that W is an irreducible representation of $\mathfrak{sl}(3, \mathbb{C})$ with highest weight (m_1, m_2) . (For a more systematic argument, based on highest-weight cyclic representations, see Appendix C.)

Proof of Theorem 9.8. For every pair (m_1, m_2) of non-negative integers, there exists an irreducible representation Π of $\mathrm{SU}(3)$ such that the associated representation π of $\mathfrak{sl}(3, \mathbb{C})$ has highest weight (m_1, m_2) .

The standard representation $\pi_{(1,0)}$ of $\mathfrak{sl}(3, \mathbb{C})$ corresponds to the standard representation $\Pi_{(1,0)}$ of $\mathrm{SU}(3)$, and likewise for the dual representation. By taking tensor products, we obtain a representation Π_{m_1, m_2} of $\mathrm{SU}(3)$ for any non-negative integers

m_1, m_2 . The irreducible subspace W constructed above is also invariant under the action of $\text{SU}(3)$, and this restriction of Π_{m_1, m_2} to W is the desired irreducible representation of $\text{SU}(3)$. $\square$

Highest weight $(1, 1)$: the adjoint representation

To obtain the irreducible representation with highest weight $(1, 1)$, we consider the tensor product $V_{(1,0)} \otimes V_{(0,1)}$.

- The highest weight vector is $e_1 \otimes f_1$ (using a basis $\{f_i\}$ for the dual space). By applying the lowering operators $\pi_{1,1}(Y_1)$ and $\pi_{1,1}(Y_2)$ repeatedly, we generate the entire representation space.

(Since $Y_3 = -[Y_1, Y_2]$, it suffices to apply only $\pi_{1,1}(Y_1)$ and $\pi_{1,1}(Y_2)$.)

- For the fundamental representation $V_{(1,0)}$, the highest weight vector is e_1 and the actions of the operators $\pi(Y_1) = Y_1$ and $\pi(Y_2) = Y_2$ are

$$\begin{aligned} Y_1 e_1 &= e_2, & Y_1 e_2 &= 0, & Y_1 e_3 &= 0, \\ Y_2 e_1 &= 0, & Y_2 e_2 &= e_3, & Y_2 e_3 &= 0. \end{aligned} \quad (9.10)$$

- For the antifundamental representation $V_{(0,1)}$ (dual of $V_{(1,0)}$), we have $\pi(Z) = \bar{Z} \equiv -Z^T$ for any $Z \in \mathfrak{sl}(3, \mathbb{C})$. We introduce the new basis

$$f_1 = e_3, \quad f_2 = -e_2, \quad f_3 = e_1. \quad (9.11)$$

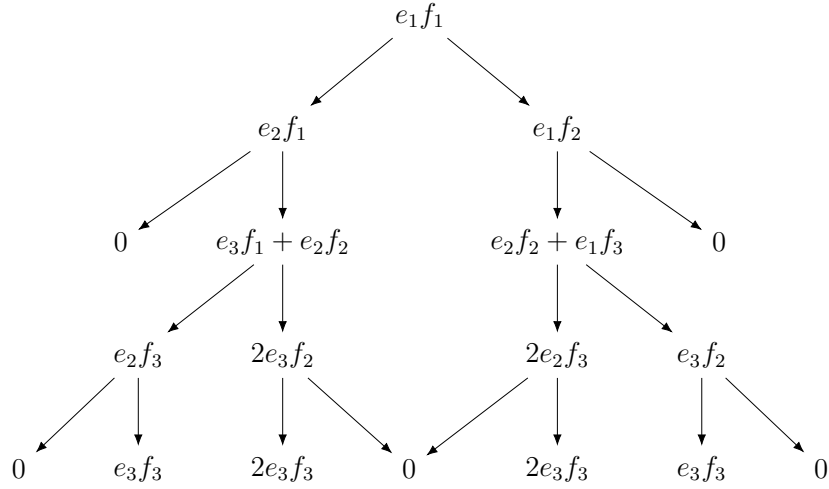
Then, the highest weight vector is f_1 and

$$\begin{aligned} \bar{Y}_1 f_1 &= 0, & \bar{Y}_1 f_2 &= f_3, & \bar{Y}_1 f_3 &= 0, \\ \bar{Y}_2 f_1 &= f_2, & \bar{Y}_2 f_2 &= 0, & \bar{Y}_2 f_3 &= 0. \end{aligned} \quad (9.12)$$

- We now repeatedly apply the following operators, starting from $e_1 \otimes f_1$ until we get zero,

$$\begin{aligned} \pi_{1,1}(Y_1) &= Y_1 \otimes I + I \otimes \bar{Y}_1, \\ \pi_{1,1}(Y_2) &= Y_2 \otimes I + I \otimes \bar{Y}_2. \end{aligned} \quad (9.13)$$

Below we abbreviate $e_i \otimes f_j = e_i f_j$. Moreover, for each element, the left arrow indicates the action of $\pi_{1,1}(Y_1)$ and the right arrow indicates the action of $\pi_{1,1}(Y_2)$.



In summary, a basis for the space spanned by these vectors is

$$\begin{array}{cccc} e_1f_1, & e_2f_1, & e_1f_2, & e_3f_1 + e_2f_2, \\ e_2f_2 + e_1f_3, & e_2f_3, & e_3f_2, & e_3f_3. \end{array} \quad (9.14)$$

- Now e_1, e_2, e_3 have weights $(1, 0), (-1, 1), (0, -1)$ respectively, and f_1, f_2, f_3 have weights $(0, 1), (1, -1), (-1, 0)$. The weight of $e_i f_j$ is the sum of that of e_i and that of f_j . Therefore, the weights of the basis elements listed above are

$$\begin{array}{cccc} (1, 1), & (-1, 2), & (2, -1), & (0, 0), \\ (0, 0), & (-2, 1), & (1, -2), & (-1, -1). \end{array} \quad (9.15)$$

These are precisely the six roots of $\mathfrak{sl}(3, \mathbb{C})$. Together with the weight $(0, 0)$ with multiplicity 2, this is actually the 8-dimensional **adjoint representation** of $\mathfrak{sl}(3, \mathbb{C})$.

9.3 Characters of representations: the Weyl formula

For a finite-dimensional representation (π, V) of $\mathfrak{sl}(3, \mathbb{C})$, the character is the function $\chi_\pi(X) = \text{tr}(e^{\pi(X)})$. We typically evaluate it on the Cartan subalgebra by taking an element $X = a_1 H_1 + a_2 H_2$, where

$$H_1 = \begin{pmatrix} 1 & 0 & 0 \\ 0 & -1 & 0 \\ 0 & 0 & 0 \end{pmatrix}, \quad H_2 = \begin{pmatrix} 0 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & -1 \end{pmatrix}. \quad (9.16)$$

Writing $z_1 = e^{a_1}$ and $z_2 = e^{a_2}$, the character of the representation (π_μ, V_μ) with highest weight $\mu = (\mu_1, \mu_2)$ is a function of z_1 and z_2 :

$$\chi_{(\mu_1, \mu_2)}(z_1, z_2) = \text{tr}(e^{a_1 \pi(H_1) + a_2 \pi(H_2)}) = \sum_{(m_1, m_2)} z_1^{m_1} z_2^{m_2}, \quad (9.17)$$

where the sum is over all weights (m_1, m_2) of the representation, counted with multiplicity.

The weights of the fundamental (standard) representation $V_{(1,0)}$ are $(1, 0)$, $(-1, 1)$, and $(0, -1)$. Its character is therefore

$$\chi_{(1,0)}(z_1, z_2) = z_1 + z_1^{-1}z_2 + z_2^{-1}. \quad (9.18)$$

Similarly, the character of the antifundamental representation $V_{(0,1)}$ (with weights $(-1, 0)$, $(1, -1)$, $(0, 1)$) is

$$\chi_{(0,1)}(z_1, z_2) = z_1^{-1} + z_1z_2^{-1} + z_2. \quad (9.19)$$

For the adjoint representation $V_{(1,1)}$, the weights are the six roots of $\mathfrak{sl}(3, \mathbb{C})$ and the weight $(0, 0)$ with multiplicity 2:

$$\chi_{(1,1)}(z_1, z_2) = z_1z_2 + z_1^2z_2^{-1} + z_1^{-1}z_2^{-2} + 2 + z_1^{-1}z_2^2 + z_1^{-2}z_2 + z_1^{-1}z_2^{-1}. \quad (9.20)$$

From these expressions, one can verify that $\chi_{(1,0)}\chi_{(0,1)} = \chi_{(1,1)} + 1$, which corresponds to the tensor product decomposition

$$V_{(1,0)} \otimes V_{(0,1)} \cong V_{(1,1)} \oplus V_{(0,0)}, \quad (9.21)$$

where $V_{(0,0)}$ is the one-dimensional trivial representation with character $\chi_{(0,0)} = 1$.

The characters of the symmetric and exterior squares of a representation V can be computed using the formulas:

$$\begin{aligned} \chi_{\text{Sym}^2 V}(z_1, z_2) &= \frac{1}{2} [\chi_V(z_1, z_2)^2 + \chi_V(z_1^2, z_2^2)], \\ \chi_{\wedge^2 V}(z_1, z_2) &= \frac{1}{2} [\chi_V(z_1, z_2)^2 - \chi_V(z_1^2, z_2^2)]. \end{aligned} \quad (9.22)$$

Using these character formulas, we can identify the following isomorphisms for $\mathfrak{sl}(3, \mathbb{C})$:

$$\text{Sym}^2 V_{(1,0)} \cong V_{(2,0)}, \quad \text{Sym}^2 V_{(0,1)} \cong V_{(0,2)}, \quad \wedge^2 V_{(1,0)} \cong V_{(0,1)}, \quad \wedge^2 V_{(0,1)} \cong V_{(1,0)}. \quad (9.23)$$

From this, it follows that $V_{(1,0)}^{\otimes 2} = \text{Sym}^2 V_{(1,0)} \oplus \wedge^2 V_{(1,0)} \cong V_{(2,0)} \oplus V_{(0,1)}$. In general, we have

$$\text{Sym}^m V_{(1,0)} \cong V_{(m,0)}, \quad \text{Sym}^m V_{(0,1)} \cong V_{(0,m)}. \quad (9.24)$$

Two important and powerful theorems allow for the computation of the character and dimension of any irreducible representation of $\mathfrak{sl}(n, \mathbb{C})$. For more details, we refer the reader to [4, Chapter 24] and [6, Chapter 10].

Theorem 9.9. (The Weyl character formula for $\mathfrak{sl}(n, \mathbb{C})$.) Let $\mu = (\mu_1, \dots, \mu_{n-1})$ be the highest weight of an irreducible representation V_μ of $\mathfrak{sl}(n, \mathbb{C})$. We define

$$\lambda = \left(\sum_{j=1}^{n-1} \mu_j, \sum_{j=2}^{n-1} \mu_j, \sum_{j=3}^{n-1} \mu_j, \dots, \mu_{n-1}, 0 \right). \quad (9.25)$$

Then, the character of V_μ is

$$\chi_\mu(x_1, \dots, x_n) = \frac{\det_{i,j}(x_j^{\lambda_i + n - i})}{\det_{i,j}(x_j^{n - i})} \quad (9.26)$$

where

$$x_1 = z_1, \quad x_j = z_j z_{j-1}^{-1} \text{ (with } 2 \leq j \leq n-1 \text{)}, \quad x_n = z_{n-1}^{-1}. \quad (9.27)$$

For example, let us consider $\mathfrak{sl}(3, \mathbb{C})$ and $\mu = (1, 1)$. In this case $\lambda = (2, 1, 0)$. Formula (9.26) is

$$\begin{aligned} \chi_{(1,1)}(x_1, x_2, x_3) &= \frac{x_2^2 x_1^4 - x_2^4 x_1^2 + x_3^4 x_1^2 - x_3^2 x_1^4 + x_2^4 x_3^2 - x_2^2 x_3^4}{x_2 x_1^2 - x_2^2 x_1 + x_3^2 x_1 - x_3 x_1^2 + x_2^2 x_3 - x_2 x_3^2} \\ &= \sum_{1 \leq i < j \leq 3} (x_i x_j^2 + x_j x_i^2) + 2x_1 x_2 x_3 = (9.20), \end{aligned} \quad (9.28)$$

upon taking $x_1 = z_1$, $x_2 = z_2 z_1^{-1}$ and $x_3 = z_2^{-1}$.

Theorem 9.10. (The Weyl dimension formula for $\mathfrak{sl}(n, \mathbb{C})$) Let $\mu = (\mu_1, \dots, \mu_{n-1})$ be the highest weight of an irreducible representation V_μ of $\mathfrak{sl}(n, \mathbb{C})$. The dimension of V_μ is given by

$$\dim V_\mu = \prod_{1 \leq i < j \leq n} \frac{(\mu_i + \dots + \mu_{j-1}) + (j - i)}{j - i}. \quad (9.29)$$

For example, for an irreducible representation of $\mathfrak{sl}(3, \mathbb{C})$ with highest weight $\mu = (\mu_1, \mu_2)$, the dimension formula simplifies to

$$\dim V_{(\mu_1, \mu_2)} = \frac{1}{2}(\mu_1 + 1)(\mu_2 + 1)(\mu_1 + \mu_2 + 2). \quad (9.30)$$

9.4 Notation in physics and generalisation to $\mathfrak{sl}(n, \mathbb{C})$

In the physics literature, an irreducible representation is often denoted by its dimension in boldface. For $\mathfrak{sl}(3, \mathbb{C})$, this includes:

$$\begin{aligned} V_{(1,0)} &= \mathbf{3}, & V_{(0,1)} &= \bar{\mathbf{3}}, & V_{(2,0)} &= \mathbf{6}, & V_{(0,2)} &= \bar{\mathbf{6}}, & V_{(1,1)} &= \mathbf{8}, \\ V_{(2,1)} &= \mathbf{15}, & V_{(4,0)} &= \mathbf{15}', & \mathbf{3} \otimes \bar{\mathbf{3}} &= \mathbf{8} \oplus \mathbf{1}, & \mathbf{3} \otimes \mathbf{3} &= \mathbf{6} \oplus \bar{\mathbf{3}}. \end{aligned} \quad (9.31)$$

This notation generalises to $\mathfrak{sl}(n, \mathbb{C})$ as follows:

$$\begin{aligned} V_{(1,0,\dots,0)} &= \mathbf{n}, & V_{(0,\dots,0,1)} &= \bar{\mathbf{n}}, & V_{(1,0,\dots,0,1)} &= \mathbf{n}^2 - \mathbf{1} \text{ (adjoint)}, \\ V_{(2,0,\dots,0)} &= \text{Sym}^2 V_{(1,0,\dots,0)} = \frac{1}{2} \mathbf{n}(\mathbf{n} + \mathbf{1}), \\ V_{(0,1,0,\dots,0)} &= \wedge^2 V_{(1,0,\dots,0)} = \frac{1}{2} \mathbf{n}(\mathbf{n} - \mathbf{1}). \end{aligned} \tag{9.32}$$

The following points explain this notation in the context of tensor calculus.

- An element of the fundamental representation $V_{(1,0,\dots,0)} \cong \mathbb{C}^n$ is denoted by a tensor with one upper index, φ^i (for $i = 1, \dots, n$). An element of the anti-fundamental representation $V_{(0,\dots,0,1)} \cong (\mathbb{C}^n)^*$ is denoted by a tensor with one lower index, ψ_i .
- Indices can be contracted using the invariant tensors δ_j^i , $\epsilon^{i_1 \dots i_n}$, and $\epsilon_{i_1 \dots i_n}$.
- A rank-two tensor φ^{ij} transforms in the representation $V_{(1,0,\dots,0)} \otimes V_{(1,0,\dots,0)}$. This space decomposes into symmetric and antisymmetric parts:

$$V_{(1,0,\dots,0)}^{\otimes 2} \cong \text{Sym}^2 V_{(1,0,\dots,0)} \oplus \wedge^2 V_{(1,0,\dots,0)}. \tag{9.33}$$

This corresponds to decomposing any rank-two tensor into its symmetric and antisymmetric parts: $\varphi^{ij} = \frac{1}{2}(\varphi^{ij} + \varphi^{ji}) + \frac{1}{2}(\varphi^{ij} - \varphi^{ji})$.

In the special case of $\mathfrak{sl}(3, \mathbb{C})$, the antisymmetric tensor representation $\wedge^2 V_{(1,0)}$ is isomorphic to the antifundamental representation $V_{(0,1)} = \bar{\mathbf{3}}$, an equivalence established via the Levi-Civita symbol: $\psi_k = \epsilon_{ijk} \varphi^{[ij]}$.

- A tensor with one upper and one lower index, φ_j^i , transforms in the representation $\mathbf{n} \otimes \bar{\mathbf{n}}$ and has n^2 components. This representation is reducible. The trace, $\text{tr}(\varphi) = \varphi_i^i$, transforms as a scalar in the trivial representation ($\mathbf{1}$). The traceless part,

$$\tilde{\varphi}_j^i = \varphi_j^i - \frac{1}{n}(\text{tr } \varphi) \delta_j^i, \tag{9.34}$$

has $n^2 - 1$ components and transforms in the adjoint representation, $\mathbf{n}^2 - \mathbf{1}$. This corresponds to the decomposition $\mathbf{n} \otimes \bar{\mathbf{n}} \cong (\mathbf{n}^2 - \mathbf{1}) \oplus \mathbf{1}$.

- For $\mathfrak{sl}(3, \mathbb{C})$, the representation $V_{(4,0)}$ corresponds to the space of rank-four totally symmetric tensors $\varphi^{(ijkl)}$. Its dimension is $\binom{3+4-1}{4} = 15$, so it is denoted $\mathbf{15}'$.

On the other hand, the representation $V_{(2,1)} = \mathbf{15}$ can be constructed from the tensor product $V_{(2,0)} \otimes V_{(0,1)} = V_{(2,1)} \oplus V_{(1,0)}$, which corresponds to tensors of the form $\varphi_k^{(ij)}$. Imposing a condition $\varphi_j^{(ij)} = 0$ projects out the $V_{(1,0)}$ component, leaving the irreducible $\mathbf{15}$, as required.

9.4.1 Summary of characters

Using the Weyl character formula, we have the following characters for some important representations of $\mathfrak{sl}(n, \mathbb{C})$:

$$\begin{aligned}\chi_{\mathbf{n}} &= \sum_{j=1}^n x_j \text{ (with } x_1 \cdots x_n = 1) = z_1 + \sum_{j=2}^{n-1} z_j z_{j-1}^{-1} + z_{n-1}^{-1} \\ \chi_{\bar{\mathbf{n}}} &= \sum_{j=1}^n x_j^{-1} \text{ (with } x_1 \cdots x_n = 1) = z_1^{-1} + \sum_{j=2}^{n-1} z_j^{-1} z_{j-1} + z_{n-1} \\ \chi_{\mathbf{n}^2 - \mathbf{1}} &= \chi_{\mathbf{n}} \chi_{\bar{\mathbf{n}}} - 1 = (n-1) + \sum_{1 \leq i < j \leq n} (x_i x_j^{-1} + x_i^{-1} x_j) .\end{aligned}\tag{9.35}$$

9.4.2 Restriction to subalgebras: branching rules

Given a Lie algebra $\mathfrak{g}$, we can consider its restriction to a subalgebra $\mathfrak{g}'$ and ask how representations of $\mathfrak{g}$ decompose into representations of $\mathfrak{g}'$. This process is known as **decomposition of representations** or determining the **branching rules**.

Example: $\mathfrak{su}(n) \rightarrow \mathfrak{su}(n-1) \oplus \mathfrak{u}(1)$

The character of the fundamental representation of $\mathfrak{su}(n)$ is $\chi_{\mathbf{n}}^{\mathfrak{su}(n)} = \sum_{j=1}^n x_j$ (with $\prod x_j = 1$), and that of $\mathfrak{su}(n-1)$ is $\chi_{\mathbf{n}-\mathbf{1}}^{\mathfrak{su}(n-1)} = \sum_{j=1}^{n-1} w_j$ (with $\prod w_j = 1$). We use q^m to denote the character of the representation with charge m of $\mathfrak{u}(1)$. We can define an embedding by relating the character variables as follows:

$$x_j = q w_j \text{ for } j = 1, \dots, n-1; \quad x_n = q^{-(n-1)}.\tag{9.36}$$

This map preserves the unimodular condition: $\prod x_j = (q^{n-1} \prod w_j) q^{-(n-1)} = 1$. Substituting this into the character formulas gives:

$$\begin{aligned}\chi_{\mathbf{n}}^{\mathfrak{su}(n)} &= q \chi_{\mathbf{n}-\mathbf{1}}^{\mathfrak{su}(n-1)} + q^{-(n-1)}, \\ \chi_{\bar{\mathbf{n}}}^{\mathfrak{su}(n)} &= q^{-1} \chi_{\bar{\mathbf{n}-\mathbf{1}}}^{\mathfrak{su}(n-1)} + q^{n-1}.\end{aligned}\tag{9.37}$$

From these character relations, we can read off the branching rules for how representations of $\mathfrak{su}(n)$ decompose into representations of $\mathfrak{su}(n-1) \oplus \mathfrak{u}(1)$. The notation $(\mathbf{r}; m)$ indicates an irreducible representation $\mathbf{r}$ of $\mathfrak{su}(n-1)$ with $\mathfrak{u}(1)$ charge m .

$$\begin{aligned}\mathbf{n} &\rightarrow (\mathbf{n}-\mathbf{1}; 1) \oplus (\mathbf{1}; -(n-1)) \\ \bar{\mathbf{n}} &\rightarrow (\overline{\mathbf{n}-\mathbf{1}}; -1) \oplus (\mathbf{1}; n-1)\end{aligned}\tag{9.38}$$

From the character relation $\chi_{\mathbf{n}^2 - \mathbf{1}} = \chi_{\mathbf{n}} \chi_{\bar{\mathbf{n}}} - 1$, we find the decomposition of the adjoint representation:

$$\mathbf{n}^2 - \mathbf{1} \rightarrow ((\mathbf{n}-\mathbf{1})^2 - \mathbf{1}; 0) \oplus (\mathbf{n}-\mathbf{1}; n) \oplus (\overline{\mathbf{n}-\mathbf{1}}; -n) \oplus (\mathbf{1}; 0).\tag{9.39}$$

Similarly, for the symmetric square representation, the character is

$$\chi_{\text{Sym}^2 \mathbf{n}}^{\mathfrak{su}(n)}(x_1, \dots, x_n) = \frac{1}{2} [(\chi_{\mathbf{n}}^{\mathfrak{su}(n)}(x_1, \dots, x_n))^2 + \chi_{\mathbf{n}}^{\mathfrak{su}(n)}(x_1^2, \dots, x_n^2)]. \quad (9.40)$$

Using the map (9.36), we obtain the branching rule:

$$\text{Sym}^2 \mathbf{n} \rightarrow (\text{Sym}^2(\mathbf{n} - \mathbf{1}); 2) \oplus (\mathbf{n} - \mathbf{1}; -(n - 2)) \oplus (\mathbf{1}; -2(n - 1)). \quad (9.41)$$

Example: $\mathfrak{su}(2n) \rightarrow \mathfrak{su}(n)_A \oplus \mathfrak{su}(n)_B \oplus \mathfrak{u}(1)$

Let the characters for the fundamental representations of $\mathfrak{su}(2n)$, the first $\mathfrak{su}(n)_A$, and the second $\mathfrak{su}(n)_B$ be $\chi_{\mathbf{2n}}^{\mathfrak{su}(2n)} = \sum_{j=1}^{2n} x_j$, $\chi_{\mathbf{n}}^{\mathfrak{su}(n)_A} = \sum_{j=1}^n v_j$, and $\chi_{\mathbf{n}}^{\mathfrak{su}(n)_B} = \sum_{j=1}^n w_j$, respectively, each with the appropriate unimodular condition. Let q^m denote the character for the $\mathfrak{u}(1)$ factor. We define the map:

$$x_j = qv_j \text{ for } j = 1, \dots, n; \quad x_{n+j} = q^{-1}w_j \text{ for } j = 1, \dots, n. \quad (9.42)$$

The characters of the fundamental and anti-fundamental representations of $\mathfrak{su}(2n)$ then decompose as:

$$\begin{aligned} \chi_{\mathbf{2n}}^{\mathfrak{su}(2n)} &= q\chi_{\mathbf{n}}^{\mathfrak{su}(n)_A} + q^{-1}\chi_{\mathbf{n}}^{\mathfrak{su}(n)_B}, \\ \chi_{\overline{\mathbf{2n}}}^{\mathfrak{su}(2n)} &= q^{-1}\chi_{\overline{\mathbf{n}}}^{\mathfrak{su}(n)_A} + q\chi_{\overline{\mathbf{n}}}^{\mathfrak{su}(n)_B}. \end{aligned} \quad (9.43)$$

From these relations, we can determine the branching rules for several representations of $\mathfrak{su}(2n)$ under the subalgebra $\mathfrak{su}(n)_A \oplus \mathfrak{su}(n)_B \oplus \mathfrak{u}(1)$:

$$\begin{aligned} \mathbf{2n} &\rightarrow (\mathbf{n}, \mathbf{1}; 1) \oplus (\mathbf{1}, \mathbf{n}; -1) \\ \overline{\mathbf{2n}} &\rightarrow (\overline{\mathbf{n}}, \mathbf{1}; -1) \oplus (\mathbf{1}, \overline{\mathbf{n}}; 1) \\ (\mathbf{2n})^2 - \mathbf{1} &\rightarrow (\mathbf{n}^2 - \mathbf{1}, \mathbf{1}; 0) \oplus (\mathbf{1}, \mathbf{n}^2 - \mathbf{1}; 0) \oplus (\mathbf{1}, \mathbf{1}; 0) \\ &\quad \oplus (\mathbf{n}, \overline{\mathbf{n}}; 2) \oplus (\overline{\mathbf{n}}, \mathbf{n}; -2) \\ \text{Sym}^2(\mathbf{2n}) &\rightarrow (\text{Sym}^2 \mathbf{n}, \mathbf{1}; 2) \oplus (\mathbf{1}, \text{Sym}^2 \mathbf{n}; -2) \oplus (\mathbf{n}, \mathbf{n}; 0). \end{aligned} \quad (9.44)$$

Example: $\mathfrak{su}(5) \rightarrow \mathfrak{su}(3) \oplus \mathfrak{su}(2) \oplus \mathfrak{u}(1)$

Another important example in physics, relevant to Grand Unified Theories, is the decomposition of $\mathfrak{su}(5)$ representations under its subalgebra $\mathfrak{su}(3) \oplus \mathfrak{su}(2) \oplus \mathfrak{u}(1)$. Let the characters of the fundamental representations be

$$\begin{aligned} \chi_{\mathbf{5}}^{\mathfrak{su}(5)} &= x_1 + x_2 + \dots + x_5, \quad x_1 \cdots x_5 = 1, \\ \chi_{\mathbf{3}}^{\mathfrak{su}(3)} &= v_1 + v_2 + v_3, \quad v_1 v_2 v_3 = 1, \\ \chi_{\mathbf{2}}^{\mathfrak{su}(2)} &= w_1 + w_2, \quad w_1 w_2 = 1. \end{aligned} \quad (9.45)$$

We use q^m to denote the character of the representation with charge m of $\mathfrak{u}(1)$. We define the embedding by relating the character variables:

$$x_1 = q^2 v_1, \quad x_2 = q^2 v_2, \quad x_3 = q^2 v_3, \quad x_4 = q^{-3} w_1, \quad x_5 = q^{-3} w_2. \quad (9.46)$$

This map preserves the unimodular condition: $\prod x_j = 1$. The characters for the $\mathbf{5}$ and $\bar{\mathbf{5}}$ representations of $\mathfrak{su}(5)$ become:

$$\begin{aligned}\chi_{\mathbf{5}}^{\mathfrak{su}(5)} &= q^2 \chi_{\mathbf{3}}^{\mathfrak{su}(3)} + q^{-3} \chi_{\mathbf{2}}^{\mathfrak{su}(2)}, \\ \chi_{\bar{\mathbf{5}}}^{\mathfrak{su}(5)} &= q^{-2} \chi_{\bar{\mathbf{3}}}^{\mathfrak{su}(3)} + q^3 \chi_{\mathbf{2}}^{\mathfrak{su}(2)}.\end{aligned}\tag{9.47}$$

This leads to the following branching rules:

$$\begin{aligned}\mathbf{5} &\rightarrow (\mathbf{3}, \mathbf{1}; 2) \oplus (\mathbf{1}, \mathbf{2}; -3) \\ \bar{\mathbf{5}} &\rightarrow (\bar{\mathbf{3}}, \mathbf{1}; -2) \oplus (\mathbf{1}, \mathbf{2}; 3) \\ \mathbf{24} &\rightarrow (\mathbf{8}, \mathbf{1}; 0) \oplus (\mathbf{1}, \mathbf{3}; 0) \oplus (\mathbf{1}, \mathbf{1}; 0) \oplus (\mathbf{3}, \mathbf{2}; 5) \oplus (\bar{\mathbf{3}}, \mathbf{2}; -5) \\ \wedge^2 \mathbf{5} = \mathbf{10} &\rightarrow (\bar{\mathbf{3}}, \mathbf{1}; 4) \oplus (\mathbf{3}, \mathbf{2}; -1) \oplus (\mathbf{1}, \mathbf{1}; -6),\end{aligned}\tag{9.48}$$

where we recall that the character of the antisymmetric square representation is

$$\chi_{\wedge^2 \mathbf{n}}^{\mathfrak{su}(n)}(x_1, \dots, x_n) = \frac{1}{2} \left[(\chi_{\mathbf{n}}^{\mathfrak{su}(n)}(x_1, \dots, x_n))^2 - \chi_{\mathbf{n}}^{\mathfrak{su}(n)}(x_1^2, \dots, x_n^2) \right]. \tag{9.49}$$

9.5 The Weyl group

We now discuss a symmetry among the weights of a given representation, which gives rise to the notion of the **Weyl group**.

Definition 9.11. Let $\mathfrak{h}$ be the two-dimensional subspace of $\mathfrak{sl}(3, \mathbb{C})$ spanned by H_1 and H_2 . Let N be the subgroup of $\mathrm{SU}(3)$ consisting of those $A \in \mathrm{SU}(3)$ such that $\mathrm{Ad}_A(H) = AHA^{-1} \in \mathfrak{h}$ for all $H \in \mathfrak{h}$. Let Z be the subgroup of $\mathrm{SU}(3)$ consisting of those $A \in \mathrm{SU}(3)$ such that $\mathrm{Ad}_A(H) = AHA^{-1} = H$ for all $H \in \mathfrak{h}$.

The space $\mathfrak{h}$ is called a **Cartan subalgebra** of $\mathfrak{sl}(3, \mathbb{C})$. By definition, N and Z are subgroups of $\mathrm{SU}(3)$, and Z is a normal subgroup of N .²³ Here, N is called the **normaliser** of $\mathfrak{h}$ in $\mathrm{SU}(3)$, and Z is the **centraliser** of $\mathfrak{h}$ in $\mathrm{SU}(3)$.

This leads to the definition of the Weyl group.

Definition 9.12. The **Weyl group** of $\mathrm{SU}(3)$, denoted W , is the quotient group $W = N/Z$.

The primary significance of the Weyl group for representation theory is that it acts as a symmetry group on the set of weights of any representation.

²³**Proof.** We want to show that for any $n \in N$ and $z \in Z$, the element $nz n^{-1}$ is in Z . We consider its action on an arbitrary $H \in \mathfrak{h}$: $\mathrm{Ad}_{nz n^{-1}}(H) = (nz n^{-1})H(nz n^{-1})^{-1} = n(z(n^{-1}Hn)z^{-1})n^{-1}$. Since $n \in N$, $n^{-1}Hn$ is in $\mathfrak{h}$. Since $z \in Z$, it commutes with all elements of $\mathfrak{h}$, so $z(n^{-1}Hn)z^{-1} = n^{-1}Hn$. The expression simplifies to $n(n^{-1}Hn)n^{-1} = H$. Thus, $nz n^{-1}$ commutes with all elements of $\mathfrak{h}$, which means $nz n^{-1} \in Z$.

The Weyl group action. We can define an action of W on $\mathfrak{h}$ as follows. For each element $w \in W$, choose a coset element $A \in nZ$ for some $n \in N$. Then, for any $H \in \mathfrak{h}$, we define the action of w on H by

$$w \cdot H = \text{Ad}_A(H).$$

This is well-defined: any other representative of the coset has the form $B = AC$ with $C \in Z$, and $\text{Ad}_B(H) = \text{Ad}_A(\text{Ad}_C(H)) = \text{Ad}_A(H)$ by the definition of Z . Note that if $w \cdot H = H$ for all $H \in \mathfrak{h}$, then the representative A must be in Z , so w is the identity element of W . This allows us to identify W with the group of linear transformations of $\mathfrak{h}$ of the form $H \mapsto w \cdot H$.

Proposition 9.13. *The group Z consists precisely of the diagonal matrices in $\text{SU}(3)$. The group N consists of matrices $A \in \text{SU}(3)$ such that for each standard basis vector e_j , Ae_j is a scalar multiple of some other standard basis vector e_{k_j} . The Weyl group $W = N/Z$ is isomorphic to the symmetric group S_3 .*

Proof. If $A \in Z$, then A commutes with H_1 , whose eigenspaces are the one-dimensional spans of e_1, e_2, e_3 (the eigenvalues $1, -1, 0$ are distinct); preserving them forces A to be diagonal. Conversely, any diagonal matrix in $\text{SU}(3)$ commutes with $\mathfrak{h}$.

If $A \in N$, then for a generic $H \in \mathfrak{h}$ (with distinct diagonal entries), AHA^{-1} is again diagonal with the same eigenvalues, and its eigenvectors Ae_1, Ae_2, Ae_3 must be scalar multiples of the standard basis vectors: $Ae_j = \lambda_j e_{k_j}$ for some permutation $\sigma : j \mapsto k_j$ and unimodular scalars λ_j .

Finally, the action of Ad_A on a diagonal matrix $H = \text{diag}(h_1, h_2, h_3)$ results in another diagonal matrix whose entries are a permutation of the original ones: $\text{Ad}_A(H) = AHA^{-1} = \text{diag}(h_{\sigma^{-1}(1)}, h_{\sigma^{-1}(2)}, h_{\sigma^{-1}(3)})$. It is clear that every permutation in S_3 can be realised by some $A \in N$. The elements of Z (the diagonal matrices in $\text{SU}(3)$) correspond to the identity permutation. Thus, the quotient group $W = N/Z$ is isomorphic to S_3 . $\square$

We now show that the Weyl group acts as a symmetry on the set of weights of any finite-dimensional representation of $\mathfrak{sl}(3, \mathbb{C})$. To understand this, we need a more abstract view of weights. A simultaneous eigenvector for $\pi(H_1)$ and $\pi(H_2)$ is an eigenvector for every $\pi(H)$ with $H \in \mathfrak{h}$, with an eigenvalue depending linearly on H : **a weight is naturally a linear functional on $\mathfrak{h}$, i.e., an element of the dual space $\mathfrak{h}^*$** . We can further identify $\mathfrak{h}^*$ with $\mathfrak{h}$ itself via the inner product

$$\langle H, H' \rangle = \text{tr}(H^\dagger H') , \tag{9.50}$$

under which every $\phi \in \mathfrak{h}^*$ takes the form $\phi(H) = \langle \lambda, H \rangle$ for a unique $\lambda \in \mathfrak{h}$.

This leads to a basis-independent notion of a weight.

Definition 9.14. Let $\mathfrak{h}$ be the Cartan subalgebra of $\mathfrak{sl}(3, \mathbb{C})$ and let (π, V) be a representation. An element $\lambda \in \mathfrak{h}$ is called a **weight** for π if there exists a nonzero vector $v \in V$ such that

$$\pi(H)v = \langle \lambda, H \rangle v$$

for all $H \in \mathfrak{h}$. Such a vector v is called a **weight vector** with weight λ .

If λ is a weight in this new sense, the original pair (m_1, m_2) is recovered by $m_1 = \langle \lambda, H_1 \rangle$ and $m_2 = \langle \lambda, H_2 \rangle$. The action of the Weyl group on $\mathfrak{h}$ is unitary with respect to this inner product: $\langle w \cdot H, w \cdot H' \rangle = \langle H, H' \rangle$. Since roots are the nonzero weights of the adjoint representation, we now also think of **roots as elements of $\mathfrak{h}$** .

Theorem 9.15. Let (Π, V) be a finite-dimensional representation of $\mathrm{SU}(3)$ with associated Lie algebra representation (π, V) of $\mathfrak{sl}(3, \mathbb{C})$. If $\lambda \in \mathfrak{h}$ is a weight for V , then for any $w \in W$, $w \cdot \lambda$ is also a weight for V with the same multiplicity. In particular, the set of roots is invariant under the action of the Weyl group.

Proof. Suppose λ is a weight for V with weight vector v . For any $U \in N$ and $H \in \mathfrak{h}$, we have

$$\begin{aligned} \pi(H)\Pi(U)v &= \Pi(U)(\Pi(U)^{-1}\pi(H)\Pi(U))v \\ &= \Pi(U)\pi(U^{-1}HU)v \\ &= \langle \lambda, U^{-1}HU \rangle \Pi(U)v. \end{aligned}$$

Here, we have used the fact that $U \in N$, which guarantees that $U^{-1}HU \in \mathfrak{h}$. If w is the Weyl group element represented by U , then $U^{-1}HU = w^{-1} \cdot H$. Thus,

$$\pi(H)\Pi(U)v = \langle \lambda, w^{-1} \cdot H \rangle \Pi(U)v = \langle w \cdot \lambda, H \rangle \Pi(U)v.$$

This shows that $\Pi(U)v$ is a weight vector with weight $w \cdot \lambda$. Since $\Pi(U)$ is an invertible map from the weight space of λ to the weight space of $w \cdot \lambda$, the two spaces must have the same dimension, and thus the weights have the same multiplicity. $\square$

Example 9.16. Let us consider the two fundamental weights, corresponding to $(m_1, m_2) = (1, 0)$ and $(0, 1)$. We seek diagonal, traceless matrices $\mu_1, \mu_2 \in \mathfrak{h}$ such that

$$\langle \mu_1, H_1 \rangle = 1, \quad \langle \mu_1, H_2 \rangle = 0 \quad \text{and} \quad \langle \mu_2, H_1 \rangle = 0, \quad \langle \mu_2, H_2 \rangle = 1.$$

These are found to be

$$\mu_1 = \mathrm{diag}(2/3, -1/3, -1/3); \quad \mu_2 = \mathrm{diag}(1/3, 1/3, -2/3).$$

The angle θ between them satisfies $\cos \theta = 1/2$, so $\theta = \pi/3$. The positive simple roots $\alpha_1 = (2, -1)$ and $\alpha_2 = (-1, 2)$ are represented in this basis as

$$\alpha_1 = 2\mu_1 - \mu_2 = \mathrm{diag}(1, -1, 0); \quad \alpha_2 = -\mu_1 + 2\mu_2 = \mathrm{diag}(0, 1, -1).$$

The angle ψ between α_1 and α_2 satisfies $\cos \psi = -1/2$, so $\psi = 2\pi/3$.

We show the roots (arrows) and dominant integral elements (black dots) for $\mathfrak{sl}(3, \mathbb{C})$, computed with respect to a Weyl-invariant inner product.²⁴

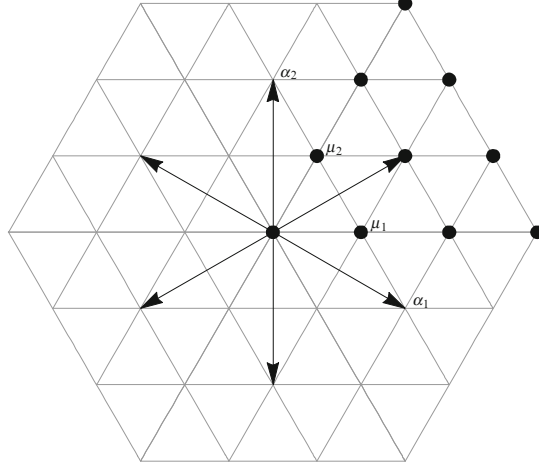


Figure 3: The roots (arrows) and dominant integral elements (black dots) of $\mathfrak{sl}(3, \mathbb{C})$, drawn with respect to a Weyl-invariant inner product.

Let w_σ be a Weyl group element associated to the permutation $\sigma \in S_3$ acting on the **diagonal entries** of each of $H_1 = \text{diag}(1, -1, 0)$ and $H_2 = \text{diag}(0, 1, -1)$ in $\mathfrak{h}$. We consider the following examples:

- If $\sigma = (123)$, we see that $(H_1, H_2, -H_1 - H_2)$ are cyclically permuted; this also means that $(\alpha_1, \alpha_2, -\alpha_1 - \alpha_2)$ are cyclically permuted, corresponding to a counterclockwise rotation by $2\pi/3$ in the root diagram in Figure 3.
- Another example: $\sigma = (12)$. We see that $H_1 \leftrightarrow -H_1$ and $H_2 \leftrightarrow H_1 + H_2$, meaning $\alpha_1 \leftrightarrow -\alpha_1$ and $\alpha_2 \leftrightarrow \alpha_1 + \alpha_2$; this corresponds to the reflection across the line perpendicular to α_1 (i.e. passing through 0 and μ_2).
- It can be checked, also in the other cases, that the Weyl group is the symmetry group of the equilateral triangle depicted below.

²⁴We draw only the two-dimensional *real* subspace of $\mathfrak{h}$ consisting of those elements μ such that $\langle \mu, H_1 \rangle$ and $\langle \mu, H_2 \rangle$ are real, since all roots and weights have this property.

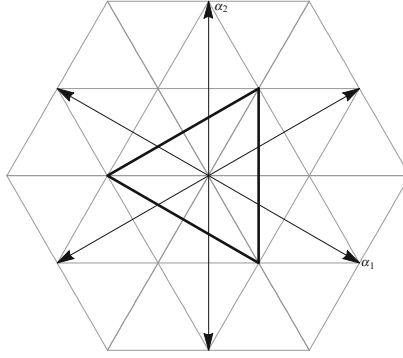


Figure 4: The Weyl group $W \cong S_3$ of $\mathfrak{sl}(3, \mathbb{C})$ is the symmetry group of the equilateral triangle shown.

Previously, we defined a pair (m_1, m_2) to be *integral* if $m_1, m_2 \in \mathbb{Z}$ and *dominant* if $m_1 \geq 0$ and $m_2 \geq 0$. These concepts translate into the new terminology as follows. If $\lambda \in \mathfrak{h}$, then λ is **integral** if $\langle \lambda, H_1 \rangle \in \mathbb{Z}$ and $\langle \lambda, H_2 \rangle \in \mathbb{Z}$ and λ is **dominant** if $\langle \lambda, H_1 \rangle \geq 0$ and $\langle \lambda, H_2 \rangle \geq 0$. Geometrically, the set of dominant elements is a sector spanning an angle of $\pi/3$.

Remark 9.17 (The Eightfold Way). *The hexagonal diagram in Figure 3 — equivalently, the weight diagram (9.15) of the adjoint representation $\mathbf{8}$, whose six nonzero weights are the roots and whose central weight $(0, 0)$ is doubly degenerate — is celebrated in particle physics. Gell-Mann and Ne’eman (1961) observed that the light hadrons organise themselves into representations of an approximate **flavour** $SU(3)$ symmetry; in modern language, the up, down and strange quarks span the fundamental $\mathbf{3}$. The lightest pseudoscalar mesons $(\pi^\pm, \pi^0, K^\pm, K^0, \bar{K}^0, \eta)$ and the lightest spin- $\frac{1}{2}$ baryons $(p, n, \Sigma^\pm, \Sigma^0, \Lambda, \Xi^-, \Xi^0)$ each fill out an octet $\mathbf{8} = V_{(1,1)}$: plotted in the plane of the two commuting conserved charges — the third component of isospin I_3 and the hypercharge Y , which form a basis of the Cartan subalgebra — each family reproduces exactly this hexagonal weight diagram, with two particles (π^0 and η for the mesons; Σ^0 and Λ for the baryons) occupying the doubly-degenerate central weight. This organisation is Gell-Mann’s **Eightfold Way**. The spin- $\frac{3}{2}$ baryons similarly fill the decuplet $\mathbf{10} = V_{(3,0)}$; the particle at its then-unobserved corner, the Ω^- , was predicted by Gell-Mann in 1962, with approximately the correct mass, and discovered in 1964 — a triumph of representation theory that earned Gell-Mann the 1969 Nobel Prize.*

10 Semisimple Lie Algebras

Definition 10.1. *A complex Lie algebra $\mathfrak{g}$ is **reductive** if there is a compact matrix Lie group K with Lie algebra $\mathfrak{k}$ such that*

$$\mathfrak{g} \cong \mathfrak{k}_{\mathbb{C}}. \quad (10.1)$$

A complex Lie algebra $\mathfrak{g}$ is **semisimple** if it is reductive and its centre is trivial.

Definition 10.2. If $\mathfrak{g}$ is a semisimple Lie algebra, a real subalgebra $\mathfrak{k}$ of $\mathfrak{g}$ is a **compact real form** of $\mathfrak{g}$ if

1. $\mathfrak{k}$ is the Lie algebra of some compact matrix Lie group, and
2. every element Z of $\mathfrak{g}$ can be expressed uniquely as $Z = X + iY$ with $X, Y \in \mathfrak{k}$.

For example, $\mathfrak{g} = \mathfrak{sl}(n, \mathbb{C})$ has a compact real form $\mathfrak{k} = \mathfrak{su}(n)$, and $\mathfrak{g} = \mathfrak{so}(n, \mathbb{C})$ has a compact real form $\mathfrak{k} = \mathfrak{so}(n)$. For any complex semisimple Lie algebra $\mathfrak{g}$, we will always assume we have chosen a compact real form $\mathfrak{k}$ such that $\mathfrak{g} = \mathfrak{k}_{\mathbb{C}}$.

Example 10.3. The following Lie algebras are semisimple:

$$\mathfrak{sl}(n, \mathbb{C}), n \geq 2; \quad \mathfrak{so}(n, \mathbb{C}), n \geq 3; \quad \mathfrak{sp}(n, \mathbb{C}), n \geq 1. \quad (10.2)$$

The Lie algebras $\mathfrak{gl}(n, \mathbb{C})$ and $\mathfrak{so}(2, \mathbb{C})$ are reductive but not semisimple.

Proof. It is clear that the listed Lie algebras are reductive, with the corresponding compact groups K being $SU(n)$, $SO(n)$, $Sp(n)$, $U(n)$, and $SO(2)$, respectively.

The Lie algebra $\mathfrak{gl}(n, \mathbb{C})$ has a nontrivial centre consisting of scalar multiples of the identity, while the Lie algebra $\mathfrak{so}(2, \mathbb{C})$ is commutative and thus its centre is $\mathfrak{so}(2, \mathbb{C})$ itself.

We now show that the centre of $\mathfrak{sl}(n, \mathbb{C})$ with $n \geq 2$ is trivial. Let X be central, and for $j \neq k$ let E_{jk} be the matrix with a 1 in the (j, k) position and zeros elsewhere; note that $E_{jk} \in \mathfrak{sl}(n, \mathbb{C})$. The (a, b) entry of the condition $0 = [X, E_{jk}]$ reads $X_{aj}\delta_{kb} = \delta_{aj}X_{kb}$ for all a, b . Taking $a \neq j$, $b = k$ gives $X_{aj} = 0$, and taking $a = j$, $b = k$ gives $X_{jj} = X_{kk}$. Ranging over all pairs $j \neq k$, we conclude that X is diagonal with all diagonal entries equal; since $\text{tr}(X) = 0$, we have $X = 0$.

The calculations for the remaining semisimple Lie algebras can be carried out using a similar argument. $\square$

Definition 10.4. A Lie algebra $\mathfrak{g}$ is said to be **simple** if it has no nontrivial ideals²⁵ and $\dim \mathfrak{g} \geq 2$.

The following Lie algebras are simple:

$$\mathfrak{sl}(n, \mathbb{C}), n \geq 2; \quad \mathfrak{so}(3, \mathbb{C}), \mathfrak{so}(n, \mathbb{C}), n \geq 5; \quad \mathfrak{sp}(n, \mathbb{C}), n \geq 1. \quad (10.3)$$

- A one-dimensional Lie algebra, such as $\mathfrak{so}(2, \mathbb{C})$, is, by definition, not simple.

²⁵Recall that an ideal in $\mathfrak{g}$ is an invariant subspace for the adjoint action of $\mathfrak{g}$.

- $\mathfrak{so}(4, \mathbb{C})$ is not simple, since $\mathfrak{so}(4, \mathbb{C}) \cong \mathfrak{sl}(2, \mathbb{C}) \oplus \mathfrak{sl}(2, \mathbb{C})$. This follows by complexifying the isomorphism $\mathfrak{so}(4) \cong \mathfrak{so}(3) \oplus \mathfrak{so}(3)$, which can be seen as follows. Let $J_{ab} = E_{ab} - E_{ba}$ (with $1 \leq a < b \leq 4$) be the standard basis of the real 4×4 skew-symmetric matrices, and set $F_1 = J_{32}$, $F_2 = J_{13}$, $F_3 = J_{21}$ (the rotation generators of Example 6.32, embedded in the upper-left 3×3 block) and $G_i = J_{i4}$. A direct computation gives

$$[F_i, F_j] = \sum_k \epsilon_{ijk} F_k, \quad [F_i, G_j] = \sum_k \epsilon_{ijk} G_k, \quad [G_i, G_j] = \sum_k \epsilon_{ijk} F_k. \quad (10.4)$$

The **self-dual and anti-self-dual combinations** $L_i = \frac{1}{2}(F_i + G_i)$ and $R_i = \frac{1}{2}(F_i - G_i)$ then satisfy

$$[L_i, L_j] = \sum_k \epsilon_{ijk} L_k, \quad [R_i, R_j] = \sum_k \epsilon_{ijk} R_k, \quad [L_i, R_j] = 0, \quad (10.5)$$

so $\mathfrak{so}(4) = \text{span}\{L_i\} \oplus \text{span}\{R_i\} \cong \mathfrak{so}(3) \oplus \mathfrak{so}(3)$.

Proposition 10.5. *Let $\mathfrak{g} := \mathfrak{k}_{\mathbb{C}}$ be a reductive Lie algebra. Then there exists an inner product on $\mathfrak{g}$ that is real-valued on $\mathfrak{k}$ and such that the adjoint action of $\mathfrak{k}$ on $\mathfrak{g}$ is “unitary,” meaning that*

$$\langle \text{ad}_X(Y), Z \rangle = -\langle Y, \text{ad}_X(Z) \rangle \quad (10.6)$$

for all $X \in \mathfrak{k}$ and all $Y, Z \in \mathfrak{g}$. If we define an operation $X \mapsto X^\dagger$ on $\mathfrak{g}$ by the formula

$$(X_1 + iX_2)^\dagger = -X_1 + iX_2 \quad (10.7)$$

for $X_1, X_2 \in \mathfrak{k}$, then any inner product satisfying (10.6) also satisfies

$$\langle \text{ad}_X(Y), Z \rangle = \langle Y, \text{ad}_{X^\dagger}(Z) \rangle \quad (10.8)$$

for all X, Y , and Z in $\mathfrak{g}$.

In particular, it follows from (10.6) that

$$\text{ad}_X^\dagger = -\text{ad}_X, \quad \text{for all } X \in \mathfrak{k} \quad (10.9)$$

and from (10.8) that

$$\text{ad}_X^\dagger = \text{ad}_{X^\dagger}, \quad \text{for all } X \in \mathfrak{g} = \mathfrak{k}_{\mathbb{C}}. \quad (10.10)$$

The motivation for the definition (10.7) of $X^\dagger$ is that if $\mathfrak{g} = \mathfrak{gl}(n, \mathbb{C})$ and $\mathfrak{k} = \mathfrak{u}(n)$, then $X^\dagger$ is the usual matrix adjoint (conjugate transpose) of X .

Proof. Using Weyl’s unitary trick, we can construct a real-valued inner product on $\mathfrak{k}$ that is invariant under the adjoint action of the compact group K . This inner product then extends to a complex inner product on $\mathfrak{g} = \mathfrak{k}_{\mathbb{C}}$ for which the adjoint action of K is unitary. Thus, by Proposition 7.7, the adjoint action of $\mathfrak{k}$ on $\mathfrak{g}$ is “unitary” (skew-self-adjoint) in the sense of (10.6). The relation (10.8) then follows from the properties of the inner product and the definition of the adjoint action: $\langle [X, Y], Z \rangle = \langle XY - YX, Z \rangle = \langle Y, X^\dagger Z \rangle - \langle Y, ZX^\dagger \rangle = \langle Y, [X^\dagger, Z] \rangle$. $\square$

The following theorem, which we state without proof, gives an equivalent definition of a semisimple Lie algebra (see [4, Appendix C]).

Theorem 10.6. *A Lie algebra $\mathfrak{g}$ is semisimple in the sense of Definition 10.1 if and only if $\mathfrak{g}$ decomposes as a Lie algebra direct sum*

$$\mathfrak{g} = \bigoplus_{j=1}^m \mathfrak{g}_j, \quad (10.11)$$

where each $\mathfrak{g}_j \subset \mathfrak{g}$ is a simple Lie algebra.

Moreover, if $\mathfrak{g}$ is a complex semisimple Lie algebra, then the subalgebras $\mathfrak{g}_j$ appearing in the decomposition (10.11) are unique up to reordering.

Our plan for this section parallels the analysis of $\mathfrak{sl}(3, \mathbb{C})$ in Section 9: we introduce a **Cartan subalgebra** (a maximal set of commuting elements, to be simultaneously diagonalised in every representation), whose nonzero adjoint eigenvalues and eigenvectors are the **roots** and **root vectors**; the root vectors raise and lower weights in any representation, and each root yields an $\mathfrak{sl}(2, \mathbb{C})$ subalgebra, through which Theorem 8.5 delivers integrality; finally, the **Weyl group** acts as a symmetry of the roots and of the weights.

10.1 Cartan subalgebras

Definition 10.7. *If $\mathfrak{g}$ is a complex semisimple Lie algebra, then a **Cartan subalgebra** of $\mathfrak{g}$ is a complex subspace $\mathfrak{h}$ of $\mathfrak{g}$ with the following properties:*

1. *For all H_1 and H_2 in $\mathfrak{h}$, $[H_1, H_2] = 0$.*
2. *If, for some $X \in \mathfrak{g}$, we have $[H, X] = 0$ for all H in $\mathfrak{h}$, then X is in $\mathfrak{h}$.*
3. *For all H in $\mathfrak{h}$, ad_H is diagonalisable.*

Condition 1 says that $\mathfrak{h}$ is a **commutative subalgebra** of $\mathfrak{g}$. Condition 2 says that $\mathfrak{h}$ is a **maximal commutative subalgebra** (i.e., not contained in any larger commutative subalgebra). Condition 3 says that each ad_H ($H \in \mathfrak{h}$) is diagonalisable. Since the elements H in $\mathfrak{h}$ commute, the operators ad_H also commute, and thus they are simultaneously diagonalisable.

Proposition 10.8. *Let $\mathfrak{g} = \mathfrak{k}_{\mathbb{C}}$ be a complex semisimple Lie algebra and let $\mathfrak{t}$ be any maximal commutative subalgebra of $\mathfrak{k}$. Define $\mathfrak{h} \subset \mathfrak{g}$ by*

$$\mathfrak{h} = \mathfrak{t}_{\mathbb{C}} = \mathfrak{t} + i\mathfrak{t}.$$

Then $\mathfrak{h}$ is a Cartan subalgebra of $\mathfrak{g}$.

Proof. It is clear that $\mathfrak{h}$ is a commutative subalgebra of $\mathfrak{g}$. We must first show that $\mathfrak{h}$ is maximal commutative. Suppose that $X \in \mathfrak{g}$ commutes with every element of $\mathfrak{h}$. Then it commutes with every element of $\mathfrak{t}$. If we write $X = X_1 + iX_2$ with X_1 and X_2 in $\mathfrak{k}$, then for any H in $\mathfrak{t}$, we have

$$[H, X_1 + iX_2] = [H, X_1] + i[H, X_2] = 0,$$

where $[H, X_1]$ and $[H, X_2]$ are in $\mathfrak{k}$. Since every element of $\mathfrak{g}$ has a unique decomposition into a real and an imaginary part with components in $\mathfrak{k}$, we see that $[H, X_1]$ and $[H, X_2]$ must be separately zero. Since this holds for all H in $\mathfrak{t}$ and since $\mathfrak{t}$ is maximal commutative in $\mathfrak{k}$, we must have $X_1, X_2 \in \mathfrak{t}$. This means that $X = X_1 + iX_2$ is in $\mathfrak{h}$, which shows that $\mathfrak{h}$ is maximal commutative.

If $\langle \cdot, \cdot \rangle$ is an inner product on $\mathfrak{g}$ as in Proposition 10.5, then for each $X \in \mathfrak{k}$, the operator $\text{ad}_X : \mathfrak{g} \rightarrow \mathfrak{g}$ is skew-self-adjoint. In particular, each ad_H (with $H \in \mathfrak{t}$) is skew-self-adjoint and thus diagonalisable.²⁶

Finally, if $H \in \mathfrak{h} = \mathfrak{t} + i\mathfrak{t}$, then $H = H_1 + iH_2$, with H_1 and H_2 in $\mathfrak{t}$. Since H_1 and H_2 commute, ad_{H_1} and ad_{H_2} also commute. As both are diagonalisable, they are simultaneously diagonalisable. It follows that $\text{ad}_H = \text{ad}_{H_1} + i\text{ad}_{H_2}$ is also diagonalisable. $\square$

Throughout this course, we consider only Cartan subalgebras of the form $\mathfrak{h} = \mathfrak{t}_{\mathbb{C}}$, where $\mathfrak{t}$ is a maximal commutative subalgebra of some compact real form $\mathfrak{k}$ of $\mathfrak{g}$.

Definition 10.9. *If $\mathfrak{g}$ is a complex semisimple Lie algebra, the **rank** of $\mathfrak{g}$ is the dimension of any of its Cartan subalgebras.*

We have seen that $\mathfrak{sl}(2, \mathbb{C})$ and $\mathfrak{sl}(3, \mathbb{C})$ have rank 1 and 2, respectively. In general, the classical Lie algebras $\mathfrak{sl}(n+1, \mathbb{C})$, $\mathfrak{so}(2n+1, \mathbb{C})$, $\mathfrak{sp}(n, \mathbb{C})$, and $\mathfrak{so}(2n, \mathbb{C})$ have rank n .

10.2 Roots and root spaces

From now on, we assume the following:

²⁶Recall that a matrix $A \in \text{Mat}_n(\mathbb{C})$ is said to be normal if $A^\dagger A = AA^\dagger$. A fundamental result of linear algebra is that any normal matrix is diagonalisable by a unitary matrix. Since skew-self-adjoint matrices ($A^\dagger = -A$) are normal, they are diagonalisable and have purely imaginary eigenvalues.

- We have chosen a compact real form $\mathfrak{k}$ of $\mathfrak{g}$ and a maximal commutative subalgebra $\mathfrak{t}$ of $\mathfrak{k}$, and we consider the Cartan subalgebra $\mathfrak{h} = \mathfrak{t} + i\mathfrak{t}$.
- We have chosen an inner product on $\mathfrak{g}$ that is real on $\mathfrak{k}$ and invariant under the adjoint action of K (as in Proposition 10.5).

Since the operators ad_H for $H \in \mathfrak{h}$ commute and are each diagonalisable, they are simultaneously diagonalisable. If $X \in \mathfrak{g}$ is a simultaneous eigenvector for each ad_H , then the corresponding eigenvalues depend linearly on H . If this linear functional is nonzero, we call it a **root**. As before, it is convenient to represent this linear functional as $H \mapsto \langle \alpha, H \rangle$ for some unique $\alpha \in \mathfrak{h}$.

This leads to the following definition.

Definition 10.10. *A nonzero element α of $\mathfrak{h}$ is a **root** (for $\mathfrak{g}$ relative to $\mathfrak{h}$) if there exists a nonzero $X \in \mathfrak{g}$ such that*

$$\text{ad}_H(X) = [H, X] = \langle \alpha, H \rangle X \quad \text{for all } H \in \mathfrak{h}.$$

*The set of all roots is denoted by R . We call X a **root vector** for the root α .*

Recall that our inner products are linear in the second factor, so that $\langle \alpha, H \rangle$ depends linearly on H for a fixed α .

Proposition 10.11. *Each root α is an element of $i\mathfrak{t} \subset \mathfrak{h}$.*

Proof. As we have noted, each ad_H (with $H \in \mathfrak{t}$) is a skew-self-adjoint operator on $\mathfrak{h}$, i.e. $\text{ad}_H^\dagger = -\text{ad}_H$, which means that ad_H has purely imaginary eigenvalues. It follows that if α is a root, the eigenvalue $\langle \alpha, H \rangle$ must be purely imaginary for all $H \in \mathfrak{t}$. Since our inner product is real-valued on $\mathfrak{t} \subset \mathfrak{k}$, this can only happen if $\alpha \in i\mathfrak{t}$. $\square$

Definition 10.12. *If α is a root, then the **root space** $\mathfrak{g}_\alpha$ is the space of all $X \in \mathfrak{g}$ for which $[H, X] = \langle \alpha, H \rangle X$ for all $H \in \mathfrak{h}$. A nonzero element of $\mathfrak{g}_\alpha$ is called a **root vector** for α .*

More generally, if α is any element of $\mathfrak{h}$, we define $\mathfrak{g}_\alpha$ to be the space of all $X \in \mathfrak{g}$ for which $[H, X] = \langle \alpha, H \rangle X$ for all $H \in \mathfrak{h}$ (but we do not call $\mathfrak{g}_\alpha$ a root space unless α is actually a root).

For $\alpha = 0$, the space $\mathfrak{g}_0$ is the set of all elements of $\mathfrak{g}$ that commute with every element of $\mathfrak{h}$. Since $\mathfrak{h}$ is a maximal commutative subalgebra, we conclude that $\mathfrak{g}_0 = \mathfrak{h}$. If α is not zero and not a root, then $\mathfrak{g}_\alpha = \{0\}$.

As noted, the operators ad_H (with $H \in \mathfrak{h}$) are simultaneously diagonalisable. As a result, $\mathfrak{g}$ can be decomposed as the direct sum of $\mathfrak{h}$ and the root spaces $\mathfrak{g}_\alpha$.

Proposition 10.13. *The Lie algebra $\mathfrak{g}$ can be decomposed as a direct sum of vector spaces as follows:*

$$\mathfrak{g} = \mathfrak{h} \oplus \bigoplus_{\alpha \in R} \mathfrak{g}_\alpha.$$

That is, every element of $\mathfrak{g}$ can be written uniquely as a sum of an element of $\mathfrak{h}$ and an element from each root space. This is not a Lie algebra direct sum, since elements of $\mathfrak{h}$ do not, in general, commute with elements of $\mathfrak{g}_\alpha$.

Proposition 10.14. *For any $\alpha, \beta \in \mathfrak{h}$, we have*

$$[\mathfrak{g}_\alpha, \mathfrak{g}_\beta] \subset \mathfrak{g}_{\alpha+\beta}.$$

In particular, if $X \in \mathfrak{g}_\alpha$ and $Y \in \mathfrak{g}_{-\alpha}$, then $[X, Y] \in \mathfrak{g}_0 = \mathfrak{h}$. Furthermore, if $X \in \mathfrak{g}_\alpha$, $Y \in \mathfrak{g}_\beta$, and $\alpha + \beta$ is neither zero nor a root, then $[X, Y] = 0$.

Proof. We start from the Jacobi identity: $[H, [X, Y]] = [[H, X], Y] + [X, [H, Y]]$. If $X \in \mathfrak{g}_\alpha$ and $Y \in \mathfrak{g}_\beta$, we have

$$\begin{aligned} [H, [X, Y]] &= [\langle \alpha, H \rangle X, Y] + [X, \langle \beta, H \rangle Y] \\ &= \langle \alpha, H \rangle [X, Y] + \langle \beta, H \rangle [X, Y] = \langle \alpha + \beta, H \rangle [X, Y] \end{aligned}$$

for all $H \in \mathfrak{h}$, which shows that $[X, Y] \in \mathfrak{g}_{\alpha+\beta}$. $\square$

Proposition 10.15. *The roots have the following properties:*

1. *If $\alpha \in \mathfrak{h}$ is a root, so is $-\alpha$. Specifically, if $X \in \mathfrak{g}_\alpha$, then $X^\dagger \in \mathfrak{g}_{-\alpha}$, where $X^\dagger$ is defined by (10.7).*
2. *The roots span $\mathfrak{h}$.*

Proof. For Point 1, if $X = X_1 + iX_2$ with $X_1, X_2 \in \mathfrak{k}$, let $\overline{X} = X_1 - iX_2$. Since $\mathfrak{k}$ is closed under brackets, if $H \in \mathfrak{t} \subset \mathfrak{k}$ and $X \in \mathfrak{g}$, we have

$$[\overline{H}, \overline{X}] = [H, X_1] - i[H, X_2] = [H, \overline{X}].$$

In particular, if X is a root vector with root $\alpha \in i\mathfrak{t}$, then for all $H \in \mathfrak{t}$ we have

$$[H, \overline{X}] = \overline{[H, X]} = \overline{\langle \alpha, H \rangle X} = -\langle \alpha, H \rangle \overline{X}, \quad (10.12)$$

since $\langle \alpha, H \rangle$ is purely imaginary for $H \in \mathfrak{t}$. Extending by linearity, we see that $[H, \overline{X}] = -\langle \alpha, H \rangle \overline{X}$ for all $H \in \mathfrak{h}$. Thus, $\overline{X}$ is a root vector for the root $-\alpha$. It follows that $X^\dagger = -\overline{X}$ belongs to $\mathfrak{g}_{-\alpha}$.

For Point 2, suppose that the roots did not span $\mathfrak{h}$. Then there would exist a nonzero $H \in \mathfrak{h}$ such that $\langle \alpha, H \rangle = 0$ for all $\alpha \in R$. Then we would have $[H, H'] = 0$ for all $H' \in \mathfrak{h}$, and also

$$[H, X] = \langle \alpha, H \rangle X = 0$$

for any $X \in \mathfrak{g}_\alpha$. Thus, by Proposition 10.13, H would be in the centre of $\mathfrak{g}$, contradicting the assumption that $\mathfrak{g}$ is semisimple. $\square$

10.3 An $\mathfrak{sl}(2, \mathbb{C})$ subalgebra of a semisimple Lie algebra

A key tool in the study of a semisimple Lie algebra $\mathfrak{g}$ is the existence of certain subalgebras isomorphic to $\mathfrak{sl}(2, \mathbb{C})$.

Theorem 10.16. *For each root α , we can find linearly independent elements $X_\alpha \in \mathfrak{g}_\alpha$, $Y_\alpha \in \mathfrak{g}_{-\alpha}$, and $H_\alpha \in \mathfrak{h}$ such that H_α is a multiple of α and*

$$[H_\alpha, X_\alpha] = 2X_\alpha, \quad [H_\alpha, Y_\alpha] = -2Y_\alpha, \quad [X_\alpha, Y_\alpha] = H_\alpha. \quad (10.13)$$

Furthermore, Y_α can be chosen to equal $X_\alpha^\dagger$.

If $X_\alpha, Y_\alpha, H_\alpha$ are as in the theorem, then on the one hand, $[H_\alpha, X_\alpha] = 2X_\alpha$, while on the other hand, $[H_\alpha, X_\alpha] = \langle \alpha, H_\alpha \rangle X_\alpha$. Thus,

$$\langle \alpha, H_\alpha \rangle = 2. \quad (10.14)$$

Since H_α is a multiple of α , the unique such multiple compatible with (10.14) is

$$H_\alpha = 2 \frac{\alpha}{\langle \alpha, \alpha \rangle}. \quad (10.15)$$

This is called the **coroot** associated to the root α .

Here is one of the consequences of Theorem 10.16.

Corollary 10.17. *For each root α , let X_α, Y_α , and H_α be as in Theorem 10.16, with $Y_\alpha = X_\alpha^\dagger$. Then the elements*

$$E_1^\alpha := \frac{i}{2}H_\alpha; \quad E_2^\alpha := \frac{i}{2}(X_\alpha + Y_\alpha); \quad E_3^\alpha := \frac{1}{2}(Y_\alpha - X_\alpha)$$

are linearly independent elements of $\mathfrak{k}$ and satisfy the commutation relations

$$[E_1^\alpha, E_2^\alpha] = E_3^\alpha; \quad [E_2^\alpha, E_3^\alpha] = E_1^\alpha; \quad [E_3^\alpha, E_1^\alpha] = E_2^\alpha.$$

Thus, the span of E_1^α, E_2^α , and E_3^α is a subalgebra of $\mathfrak{k}$ isomorphic to $\mathfrak{su}(2)$.

Proof. Since $\alpha \in i\mathfrak{t}$ and H_α is a real multiple of α , the element $E_1^\alpha = (i/2)H_\alpha$ belongs to $\mathfrak{t} \subset \mathfrak{k}$. Meanwhile, one may check that $(E_2^\alpha)^\dagger = -E_2^\alpha$ and $(E_3^\alpha)^\dagger = -E_3^\alpha$. From the definition of the $\dagger$ operation in (10.7), it follows that E_2^α and E_3^α also belong to $\mathfrak{k}$. Furthermore, since X_α, Y_α , and H_α are linearly independent, so are E_1^α, E_2^α , and E_3^α . Finally, direct computation with the commutation relations in (10.13) confirms the claimed commutation relations for the E_j^α 's. $\square$

We begin the proof of Theorem 10.16 with a lemma.

Lemma 10.18. *Suppose that $X \in \mathfrak{g}_\alpha$, $Y \in \mathfrak{g}_{-\alpha}$, and $H \in \mathfrak{h}$. Then $[X, Y] \in \mathfrak{h}$ and*

$$\langle [X, Y], H \rangle = \langle \alpha, H \rangle \langle Y, X^\dagger \rangle, \quad (10.16)$$

where $X^\dagger$ is defined as in (10.7).

Proof. That $[X, Y] \in \mathfrak{h}$ follows from Proposition 10.14. Then, using Proposition 10.5, we compute that

$$\langle [X, Y], H \rangle = \langle \text{ad}_X(Y), H \rangle = \langle Y, \text{ad}_{X^\dagger}(H) \rangle = -\langle Y, [H, X^\dagger] \rangle. \quad (10.17)$$

Since $X \in \mathfrak{g}_\alpha$, Proposition 10.15 tells us that $X^\dagger \in \mathfrak{g}_{-\alpha}$, so that $[H, X^\dagger] = -\langle \alpha, H \rangle X^\dagger$. Substituting this into (10.17) gives the desired result. $\square$

Proof of Theorem 10.16. Choose a nonzero $X \in \mathfrak{g}_\alpha$, so that $X^\dagger = -\overline{X}$ is a nonzero element of $\mathfrak{g}_{-\alpha}$. Applying Lemma 10.18 with $Y = X^\dagger$ gives

$$\langle [X, X^\dagger], H \rangle = \langle \alpha, H \rangle \langle X^\dagger, X^\dagger \rangle. \quad (10.18)$$

This shows that $[X, X^\dagger]$ is a multiple of α . Furthermore, if we choose H such that $\langle \alpha, H \rangle \neq 0$, we see that $\langle [X, X^\dagger], H \rangle \neq 0$, so $[X, X^\dagger] \neq 0$. Let $H_0 = [X, X^\dagger]$. Evaluating (10.18) with $H = H_0$ gives

$$\langle H_0, H_0 \rangle = \langle \alpha, H_0 \rangle \langle X^\dagger, X^\dagger \rangle.$$

Since $\langle H_0, H_0 \rangle$ and $\langle X^\dagger, X^\dagger \rangle$ are positive, $\langle \alpha, H_0 \rangle$ is real and strictly positive. We can now define the elements of our $\mathfrak{sl}(2, \mathbb{C})$ subalgebra by rescaling:

$$H_\alpha = \frac{2}{\langle \alpha, H_0 \rangle} H_0, \quad X_\alpha = \sqrt{\frac{2}{\langle \alpha, H_0 \rangle}} X, \quad Y_\alpha = \sqrt{\frac{2}{\langle \alpha, H_0 \rangle}} X^\dagger, \quad (10.19)$$

where we recall that $H_0 = [X, X^\dagger]$. These elements satisfy the required commutation relations: since $X \in \mathfrak{g}_\alpha$ and $X^\dagger \in \mathfrak{g}_{-\alpha}$, we have $[H_0, X] = \langle \alpha, H_0 \rangle X$, giving $[H_\alpha, X_\alpha] = 2X_\alpha$ and, similarly, $[H_\alpha, Y_\alpha] = -2Y_\alpha$; moreover, $[X_\alpha, Y_\alpha] = \frac{2[X, X^\dagger]}{\langle \alpha, H_0 \rangle} = \frac{2H_0}{\langle \alpha, H_0 \rangle} = H_\alpha$. Since X_α, Y_α and H_α are eigenvectors of ad_{H_α} with distinct eigenvalues 2, -2, 0, they must be linearly independent. $\square$

We now use these subalgebras to obtain results about roots and root spaces. Note that

$$\mathfrak{u}^\alpha := \langle X_\alpha, Y_\alpha, H_\alpha \rangle \quad (10.20)$$

acts on $\mathfrak{g}$ via the adjoint representation.

Theorem 10.19. *The roots have the following properties:*

1. *For each root α , the only multiples of α that are roots are α and $-\alpha$.*
2. *For each root α , the root space $\mathfrak{g}_\alpha$ is one-dimensional.*

Lemma 10.20. *If α and $c\alpha$ are both roots with $|c| \geq 1$, then $c = \pm 1$ or ± 2 .*

Proof. If $\beta = c\alpha$ is also a root and $X \in \mathfrak{g}_\beta$ is nonzero, then

$$[H_\alpha, X] = \langle \beta, H_\alpha \rangle X = c \langle \alpha, H_\alpha \rangle X = 2cX.$$

Thus, $2c$ is an eigenvalue for the adjoint action of H_α on $\mathfrak{g}$. By Theorem 8.5, any such eigenvalue must be an integer. By reversing the roles of α and β , we find that $2/c$ must also be an integer. The only possibilities with $|c| \geq 1$ are $c = \pm 1$ or ± 2 . $\square$

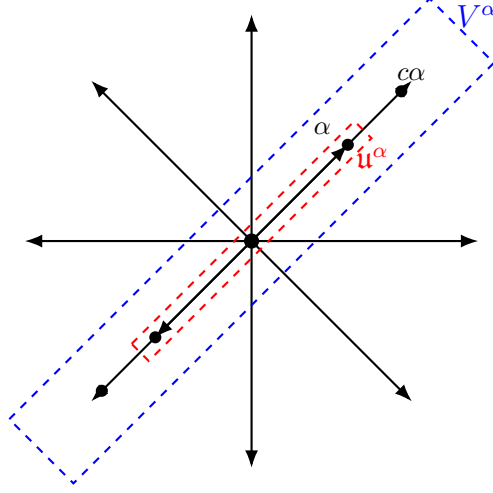


Figure 5: The subalgebras $\mathfrak{u}^\alpha \subseteq V^\alpha$ (dashed strips) along the line through $\pm\alpha$. The proof of Theorem 10.19 shows that $V^\alpha = \mathfrak{u}^\alpha$, so that no root of the form $c\alpha$ with $c \neq \pm 1$ exists.

Proof of Theorem 10.19. Let α be a root, and consider the subspace $V^\alpha = \text{span}(H_\alpha) \oplus \bigoplus_{c \in \mathbb{Z} \setminus \{0\}} \mathfrak{g}_{c\alpha}$. This space is a subalgebra of $\mathfrak{g}$. The situation to be ruled out — a would-be root $c\alpha$ on the line through $\pm\alpha$, together with the subalgebras $\mathfrak{u}^\alpha \subseteq V^\alpha$ drawn as dashed strips along that line — is depicted in Figure 5. By Lemma 10.20, $c\alpha$ (with $c \neq 0$) can be a root only if $c = \pm 1$ or ± 2 , so the eigenvalues of ad_{H_α} on V^α lie in $\{0, \pm 2, \pm 4\}$.

V^α is certainly invariant under the adjoint action of $\mathfrak{u}^\alpha$, and $\mathfrak{u}^\alpha$ itself is an invariant subspace of the adjoint action of $\mathfrak{u}^\alpha$. By Proposition 10.5, there is an inner product on $\mathfrak{g}$ such that the adjoint action of $\mathfrak{u}^\alpha$ on V^α is “unitary”. It follows that V^α is completely reducible. Thus, the orthogonal complement of $\mathfrak{u}^\alpha$ in V^α , call it W^α , is also an invariant subspace.

If $W^\alpha \neq \{0\}$, it would have to contain an eigenvector of ad_{H_α} with eigenvalue 0, ± 2 or ± 4 . Since these are even numbers, by the properties of $\mathfrak{sl}(2, \mathbb{C})$ representations (Point 4 of Theorem 8.5), 0 must also be an eigenvalue for ad_{H_α} in W^α . However, the space of vectors with eigenvalue 0 is just $\mathfrak{h} \cap V^\alpha = \text{span}(H_\alpha) \subset \mathfrak{u}^\alpha$, which is orthogonal to W^α . Thus, W^α cannot have an eigenvector with eigenvalue 0, and we reach a contradiction. Therefore, $W^\alpha = \{0\}$ and $V^\alpha = \mathfrak{u}^\alpha$. This implies that **the**

only multiples of α that are roots are $\pm\alpha$, and the root spaces with the roots $\pm\alpha$ are one-dimensional. $\square$

10.4 The Weyl group

We now discuss an important symmetry of the set R of roots, which gives rise to the notion of the **Weyl group**. Here we follow the Lie algebra approach to the Weyl group, as opposed to the Lie group approach presented earlier in the $SU(3)$ case. It can be shown that the two approaches are equivalent (see [6, Section 11.7]).

Definition 10.21. For each root $\alpha \in R$, define a linear map $s_\alpha : \mathfrak{h} \rightarrow \mathfrak{h}$ by the formula

$$s_\alpha \cdot H = H - 2 \frac{\langle \alpha, H \rangle}{\langle \alpha, \alpha \rangle} \alpha. \quad (10.21)$$

The **Weyl group** of the root system R , denoted W , is the subgroup of $GL(\mathfrak{h})$ generated by all the reflections s_α for $\alpha \in R$.

Since each root α is in $i\mathfrak{t}$ and our inner product is real on $\mathfrak{t}$, if H is in $i\mathfrak{t}$, then $s_\alpha \cdot H$ is also in $i\mathfrak{t}$. As a map of $i\mathfrak{t}$ to itself,

s_α is the reflection about the hyperplane orthogonal to α .

This is because $s_\alpha \cdot H = H$ whenever H is orthogonal to α , and $s_\alpha \cdot \alpha = -\alpha$. Since each reflection is an orthogonal linear transformation, we see that W is a subgroup of the orthogonal group $O(i\mathfrak{t})$.

Theorem 10.22. The action of the Weyl group W on $i\mathfrak{t}$ preserves the set of roots R . That is, if α is a root, then $w \cdot \alpha$ is also a root for all $w \in W$.

Proof. For each $\alpha \in R$, consider the invertible linear operator S_α on $\mathfrak{g}$ given by

$$S_\alpha = e^{\text{ad}_{X_\alpha}} e^{-\text{ad}_{Y_\alpha}} e^{\text{ad}_{X_\alpha}}.$$

Now, if $H \in \mathfrak{h}$ satisfies $\langle \alpha, H \rangle = 0$, then $[H, X_\alpha] = \langle \alpha, H \rangle X_\alpha = 0$. Thus, H and X_α commute, which means that ad_H and ad_{X_α} also commute, and similarly for ad_H and ad_{Y_α} . Thus, if $\langle \alpha, H \rangle = 0$, the operator S_α will commute with ad_H , so that

$$S_\alpha \text{ad}_H S_\alpha^{-1} = \text{ad}_H, \quad \text{if } \langle \alpha, H \rangle = 0. \quad (10.22)$$

On the other hand, if we apply Point 3 of Theorem 8.5 to the adjoint action of $\mathfrak{u}^\alpha = \langle X_\alpha, Y_\alpha, H_\alpha \rangle \cong \mathfrak{sl}(2, \mathbb{C})$ on $\mathfrak{g}$, we see that

$$S_\alpha \text{ad}_{H_\alpha} S_\alpha^{-1} = -\text{ad}_{H_\alpha}. \quad (10.23)$$

By combining (10.22) and (10.23), we see that for all $H \in \mathfrak{h}$, we have

$$S_\alpha \text{ad}_H S_\alpha^{-1} = \text{ad}_{s_\alpha \cdot H}. \quad (10.24)$$

Now if β is any root and X is an associated root vector, consider the vector $S_\alpha(X) \in \mathfrak{g}$. We compute that

$$\begin{aligned}\mathrm{ad}_H(S_\alpha(X)) &= (S_\alpha \mathrm{ad}_H S_\alpha^{-1})(S_\alpha(X)) \\ &= \mathrm{ad}_{s_\alpha \cdot H}(S_\alpha(X)) \\ &= \langle \beta, s_\alpha \cdot H \rangle S_\alpha(X) \\ &= \langle s_\alpha \cdot \beta, H \rangle S_\alpha(X).\end{aligned}$$

Thus, $S_\alpha(X)$ is a root vector with root $s_\alpha \cdot \beta$. This shows that each reflection s_α maps R into itself and, since $s_\alpha \circ s_\alpha$ is the identity, onto itself; hence every $w \in W$ maps R onto R . $\square$

Corollary 10.23. *The Weyl group is finite.*

Proof. Since the roots span $\mathfrak{h}$, each element $w \in W$ is uniquely determined by its action on the set of roots R . Since w maps R to itself bijectively, we can view W as a subgroup of the permutation group on the finite set of roots. Thus, W must be finite. $\square$

10.5 Root systems

We now discuss several important properties of roots that follow from the preceding results.

Recall from Theorem 10.16 that for each root α , we have an element $H_\alpha \in \mathfrak{h}$ contained in the subspace $[\mathfrak{g}_\alpha, \mathfrak{g}_{-\alpha}]$. As we saw in (10.14) and (10.15), H_α satisfies $\langle \alpha, H_\alpha \rangle = 2$ and is related to α by the formula $H_\alpha = 2\alpha / \langle \alpha, \alpha \rangle$. In particular, the coroot H_α is independent of the choice of X_α and Y_α in Theorem 10.16.

Proposition 10.24. *For all roots α and β , the number*

$$\langle \beta, H_\alpha \rangle = 2 \frac{\langle \alpha, \beta \rangle}{\langle \alpha, \alpha \rangle} \tag{10.25}$$

is an integer.

Proof. Let $\mathfrak{u}^\alpha = \langle X_\alpha, Y_\alpha, H_\alpha \rangle$ be the subalgebra from Theorem 10.16. If X is a root vector associated with the root β , then $[H_\alpha, X] = \langle \beta, H_\alpha \rangle X$. Thus, $\langle \beta, H_\alpha \rangle$ is an eigenvalue for the adjoint action of $\mathfrak{u}^\alpha \cong \mathfrak{sl}(2, \mathbb{C})$ on $\mathfrak{g}$. Point 1 of Theorem 8.5 then shows that $\langle \beta, H_\alpha \rangle$ must be an integer. $\square$

Note that if α and β are elements of an inner product space, the orthogonal projection of β onto the line spanned by α is given by

$$\frac{\langle \alpha, \beta \rangle}{\langle \alpha, \alpha \rangle} \alpha.$$

Proposition 10.24 implies that twice this projection must be an integer multiple of α . Alternatively, using (10.21), we may interpret Proposition 10.24 as stating that **the difference between β and its reflection $s_\alpha \cdot \beta$ must be an integer multiple of α .**

If we consider the set of roots R as a subset of the real inner product space $E := \mathfrak{t}$, we can summarise the properties of R as follows.

Theorem 10.25. *The set R of roots is a finite set of nonzero elements of a real inner product space E , and R has the following additional properties:*

1. *The roots span E .*
2. *If $\alpha \in R$ then $-\alpha \in R$, and the only multiples of α in R are α and $-\alpha$.*
3. *If α and β are in R , so is $s_\alpha \cdot \beta$, where s_α is the reflection defined by (10.21).*
4. *For all α and β in R , the quantity*

$$2 \frac{\langle \alpha, \beta \rangle}{\langle \alpha, \alpha \rangle} \quad (10.26)$$

is an integer.

Any such collection of vectors is called a **root system**.

10.6 Examples: The root system of the classical Lie algebras

We now consider, via explicit examples, how the preceding discussion applies to the classical Lie algebras $\mathfrak{sl}(n+1, \mathbb{C})$, $\mathfrak{so}(2n, \mathbb{C})$, $\mathfrak{so}(2n+1, \mathbb{C})$, and $\mathfrak{sp}(n, \mathbb{C})$, each of which has rank n .

10.6.1 The special linear algebras $\mathfrak{sl}(n+1, \mathbb{C})$

We work with the compact real form $\mathfrak{k} = \mathfrak{su}(n+1)$ and its commutative subalgebra $\mathfrak{t}$, which is the intersection of the set of diagonal matrices with $\mathfrak{su}(n+1)$; that is,

$$\mathfrak{t} = \left\{ \begin{pmatrix} ia_1 & & \\ & \ddots & \\ & & ia_{n+1} \end{pmatrix} \middle| a_j \in \mathbb{R}, a_1 + \cdots + a_{n+1} = 0 \right\}. \quad (10.27)$$

We then consider its complexification $\mathfrak{h} := \mathfrak{t}_{\mathbb{C}}$, which is the space of all traceless diagonal matrices in $\text{Mat}_{n+1}(\mathbb{C})$:

$$\mathfrak{h} = \left\{ \begin{pmatrix} \lambda_1 & & \\ & \ddots & \\ & & \lambda_{n+1} \end{pmatrix} \middle| \lambda_j \in \mathbb{C}, \lambda_1 + \cdots + \lambda_{n+1} = 0 \right\}. \quad (10.28)$$

We will show below that $\mathfrak{t}$ is a maximal commutative subalgebra of $\mathfrak{su}(n+1)$, which implies that $\mathfrak{h}$ is a Cartan subalgebra.

- Let E_{jk} denote the matrix with a 1 in the (j, k) position and zeros elsewhere. For any $H = \text{diag}(\lambda_1, \dots, \lambda_{n+1}) \in \mathfrak{h}$, we have $HE_{jk} = \lambda_j E_{jk}$ and $E_{jk}H = \lambda_k E_{jk}$. Thus,

$$\text{ad}_H(E_{jk}) = [H, E_{jk}] = (\lambda_j - \lambda_k)E_{jk}. \quad (10.29)$$

If $j \neq k$, then $E_{jk} \in \mathfrak{sl}(n+1, \mathbb{C})$. Equation (10.29) shows that E_{jk} is a simultaneous eigenvector for all ad_H with $H \in \mathfrak{h}$. Any matrix $X \in \mathfrak{sl}(n+1, \mathbb{C})$ can be uniquely decomposed into its diagonal part (an element of $\mathfrak{h}$) and its off-diagonal part (a linear combination of the E_{jk} for $j \neq k$).

- If we initially think of roots as linear functionals in $\mathfrak{h}^*$, then (10.29) shows that the roots are the functionals α_{jk} that map a matrix $H \in \mathfrak{h}$ to the scalar $\lambda_j - \lambda_k$. Using the inner product $\langle X, Y \rangle = \text{tr}(X^\dagger Y)$ on $\mathfrak{h}$, we can identify $\mathfrak{h}$ with a subspace of $\mathbb{C}^{n+1}$ via the map $H \mapsto (\lambda_1, \dots, \lambda_{n+1})$. This inner product corresponds to the standard inner product on $\mathbb{C}^{n+1}$.

The roots of $\mathfrak{sl}(n+1, \mathbb{C})$, viewed as vectors in this space, are then

$$\alpha_{jk} := e_j - e_k \quad (j \neq k), \quad (10.30)$$

where $\{e_i\}$ is the standard orthonormal basis for $\mathbb{R}^{n+1}$.

Each root has length $\sqrt{2}$.

- We now show that $\mathfrak{t}$ is a maximal commutative subalgebra of $\mathfrak{su}(n+1)$. Suppose a matrix $X \in \mathfrak{su}(n+1)$ commutes with every element of $\mathfrak{t}$. Then it also commutes with every element of $\mathfrak{h}$. Any $X \in \mathfrak{sl}(n+1, \mathbb{C})$ can be written as $X = H_X + \sum_{j \neq k} c_{jk} E_{jk}$, where $H_X \in \mathfrak{h}$. For any $H \in \mathfrak{h}$, we have $0 = [X, H] = -\sum_{j \neq k} c_{jk} (\lambda_j - \lambda_k) E_{jk}$. Since the linear functionals $\lambda_j - \lambda_k$ are not identically zero, we must have $c_{jk} = 0$ for all $j \neq k$. Thus, X must be in $\mathfrak{h}$. If X is also in $\mathfrak{su}(n+1)$, it must be a diagonal skew-Hermitian matrix, which means $X \in \mathfrak{t}$. Hence, $\mathfrak{t}$ is maximal commutative.
- The root system of $\mathfrak{sl}(n+1, \mathbb{C})$ is conventionally called A_n , where the subscript n denotes the rank of the Lie algebra.
- For any two roots α_{jk} and $\alpha_{j'k'}$, the inner product $\langle \alpha_{jk}, \alpha_{j'k'} \rangle = 0, \pm 1$, or ± 2 . Thus, the quantity $2\langle \alpha_{jk}, \alpha_{j'k'} \rangle / \langle \alpha_{jk}, \alpha_{jk} \rangle = 0, \pm 1$, or ± 2 , satisfying the root system axioms.

In particular, if α and β are roots and $\alpha \neq \beta$ and $\alpha \neq -\beta$, then the angle between α and β is either $\pi/3$, $\pi/2$, or $2\pi/3$, depending on whether $\langle \alpha, \beta \rangle$ has the value 1, 0, or -1 .

- For any j and k , the reflection $s_{\alpha_{jk}}$ acts on $\mathbb{C}^{n+1}$ by interchanging the j th and k th entries of each vector in $\mathbb{C}^{n+1}$. It follows that **the Weyl group of the A_n root system is the permutation group on $n+1$ elements.**

10.6.2 The orthogonal algebras $\mathfrak{so}(2n, \mathbb{C})$

The root system for $\mathfrak{so}(2n, \mathbb{C})$ is denoted D_n . We consider its compact real form $\mathfrak{k} = \mathfrak{so}(2n)$.

- We choose the commutative subalgebra $\mathfrak{t} \subset \mathfrak{so}(2n)$ consisting of block-diagonal matrices where the j -th block is of the form

$$B_j = \begin{pmatrix} 0 & a_j \\ -a_j & 0 \end{pmatrix}, \quad a_j \in \mathbb{R}. \quad (10.31)$$

- Consider $\mathfrak{h} = \mathfrak{t} + i\mathfrak{t}$, which consists of block-diagonal matrices of the same form, but with $a_j \in \mathbb{C}$.

As we will show below, $\mathfrak{t}$ is actually a maximal commutative subalgebra of $\mathfrak{so}(2n)$ so that $\mathfrak{h}$ is a Cartan subalgebra of $\mathfrak{so}(2n, \mathbb{C})$.

- The root vectors $E_{jk}^{(s)}$ (for $j < k$) are $2n \times 2n$ matrices with a 2×2 matrix C_s in the (j, k) -th block, $-C_s^T$ in the (k, j) -th block, and zeros elsewhere. The matrices C_s are:

$$C_1 = \begin{pmatrix} 1 & i \\ i & -1 \end{pmatrix}, \quad C_2 = \begin{pmatrix} 1 & -i \\ -i & -1 \end{pmatrix}, \quad C_3 = \begin{pmatrix} 1 & -i \\ i & 1 \end{pmatrix}, \quad C_4 = \begin{pmatrix} 1 & i \\ -i & 1 \end{pmatrix}. \quad (10.32)$$

- A direct calculation shows that $[H, E_{jk}^{(s)}] = A_{jk}^{(s)} E_{jk}^{(s)}$, where the roots (as linear functionals on $\mathfrak{h}$) are given by $A_{jk}^{(s)} = i(\pm a_j \pm a_k)$, with $s = 1, 2, 3, 4$ labelling the four possible signs.
- Using the inner product $\langle X, Y \rangle = \frac{1}{2} \text{tr}(X^\dagger Y)$ and identifying $\mathfrak{h}$ with $\mathbb{C}^n$ via the map $H \mapsto i(a_1, \dots, a_n)$, the roots become vectors in $\mathbb{C}^n$. (The factor $\frac{1}{2}$, which differs from the normalisation used for $\mathfrak{sl}(n+1, \mathbb{C})$ above, compensates for the fact that each a_j appears twice in H ; it is chosen so that the roots take the simple form below. The same normalisation is used for $\mathfrak{so}(2n+1, \mathbb{C})$ and $\mathfrak{sp}(n, \mathbb{C})$.)

The roots of $\mathfrak{so}(2n, \mathbb{C})$ are the vectors

$$\pm e_j \pm e_k, \quad 1 \leq j < k \leq n, \quad (10.33)$$

where $\{e_j\}$ is the standard basis for $\mathbb{R}^n$.

- We now demonstrate that $\mathfrak{t}$ is a maximal commutative subalgebra of $\mathfrak{so}(2n)$.

Recall from Proposition 10.13 that any $X \in \mathfrak{g}$ can be expressed as an element H of $\mathfrak{h}$ plus a linear combination of the root vectors described above:

$$X = H + \sum_{s=1}^4 \sum_{j < k} c_{jk}^{(s)} E_{jk}^{(s)}.$$

If X commutes with every element of $\mathfrak{t}$, then X also commutes with every element of $\mathfrak{h}$, and so $[X, H] = -\sum_{s=1}^4 \sum_{j < k} c_{jk}^{(s)} A_{jk}^{(s)} E_{jk}^{(s)} = 0$. Since each of $A_{jk}^{(s)} = i(\pm a_j \pm a_k)$, with $j < k$, is nonzero on $\mathfrak{h}$, the coefficients $c_{jk}^{(s)}$ of the root vectors in the expansion of X must be zero; that is, X must be in $\mathfrak{h}$. If, also, X is in $\mathfrak{so}(2n)$, then X must be in $\mathfrak{h} \cap \mathfrak{so}(2n) = \mathfrak{t}$. This shows that $\mathfrak{t}$ is maximal commutative in $\mathfrak{so}(2n)$.

- We now show that the centre of $\mathfrak{so}(2n, \mathbb{C})$ is trivial for $n \geq 2$, as claimed in Example 10.3. Indeed, if X is in the centre of $\mathfrak{so}(2n, \mathbb{C})$, then as shown in the previous paragraph, X must belong to $\mathfrak{h}$. But then for each root vector X_α with root α , we must have

$$0 = [X, X_\alpha] = \langle \alpha, X \rangle X_\alpha,$$

so that $\langle \alpha, X \rangle = 0$. If $n \geq 2$, the roots in (10.33) span $\mathfrak{h} \cong \mathbb{C}^n$ — for instance, $\frac{1}{2}[(e_j + e_k) + (e_j - e_k)] = e_j$ — and we conclude that X must be zero. (If, on the other hand, $n = 1$, then there are no roots and $\mathfrak{so}(2, \mathbb{C}) = \mathfrak{h}$ is commutative.)

10.6.3 The orthogonal algebras $\mathfrak{so}(2n+1, \mathbb{C})$

The root system for $\mathfrak{so}(2n+1, \mathbb{C})$ is denoted B_n .

- We choose a Cartan subalgebra $\mathfrak{h}$ that is identifiable with the Cartan subalgebra of $\mathfrak{so}(2n, \mathbb{C})$. Thus, $\mathfrak{so}(2n+1, \mathbb{C})$ also has rank n .
- Every root for $\mathfrak{so}(2n, \mathbb{C})$ is also a root for $\mathfrak{so}(2n+1, \mathbb{C})$.
- There are $2n$ additional roots for $\mathfrak{so}(2n+1, \mathbb{C})$. The corresponding root vectors are the following $2n$ $(2n+1) \times (2n+1)$ matrices, denoted by $E_k^{(p)}$ (with $k = 1, \dots, n$ and $p = 1, 2$), defined in block form as follows:

$$E_k^{(p)} = \begin{cases} B_p & \text{block in rows } 2k-1, 2k \text{ of column } 2n+1 \\ -B_p^T & \text{block in row } 2n+1, \text{ columns } 2k-1, 2k \\ 0 & \text{elsewhere} \end{cases} \quad (10.34)$$

where B_1 and B_2 are the 2×1 blocks given by

$$B_1 = \begin{pmatrix} 1 \\ i \end{pmatrix}, \quad B_2 = \begin{pmatrix} 1 \\ -i \end{pmatrix}. \quad (10.35)$$

Direct calculation shows that

$$[H, E_k^{(p)}] = (-1)^{p+1} i a_k E_k^{(p)}, \quad (10.36)$$

where H is an element of $\mathfrak{h}$ described in the preceding paragraph. The corresponding roots, viewed as elements of $\mathfrak{h}^*$, are therefore given by

$$ia_k \quad \text{and} \quad -ia_k, \quad k = 1, \dots, n. \quad (10.37)$$

- If we use the inner product $\langle X, Y \rangle = \text{tr}(X^\dagger Y)/2$ to identify the roots with elements of $\mathfrak{h}$ and then identify $\mathfrak{h}$ with $\mathbb{C}^n$, we have

the roots for $\mathfrak{so}(2n+1, \mathbb{C})$ consist of the roots $\pm e_j \pm e_k$ (with $1 \leq j < k \leq n$), for $\mathfrak{so}(2n, \mathbb{C})$, together with additional roots of the form

$$\pm e_j, \quad j = 1, \dots, n. \quad (10.38)$$

These additional roots are shorter by a factor of $\sqrt{2}$ than the roots $\pm e_j \pm e_k$ for $\mathfrak{so}(2n, \mathbb{C})$.

10.6.4 The symplectic algebras $\mathfrak{sp}(n, \mathbb{C})$

The root system for $\mathfrak{sp}(n, \mathbb{C})$ is denoted C_n .

- The Lie algebra $\mathfrak{sp}(n, \mathbb{C})$ consists of $2n \times 2n$ complex matrices of the form $X = \begin{pmatrix} A & B \\ C & -A^T \end{pmatrix}$, where B and C are symmetric $n \times n$ matrices.

We consider the compact real form $\mathfrak{sp}(n) = \mathfrak{sp}(n, \mathbb{C}) \cap \mathfrak{u}(2n)$.

- We consider the commutative subalgebra $\mathfrak{t}$ of $\mathfrak{sp}(n)$ consisting of matrices of the form $\text{diag}(a_1, \dots, a_n, -a_1, \dots, -a_n)$, with a_j purely imaginary.

The subalgebra $\mathfrak{h} = \mathfrak{t} + i\mathfrak{t}$ of $\mathfrak{sp}(n, \mathbb{C})$ consists of matrices of the same form, but where each a_j is now an arbitrary complex number.

As in the previous cases, the calculations below will show that $\mathfrak{t}$ is maximal commutative, so that $\mathfrak{h}$ is a Cartan subalgebra.

- Let E_{jk} denote the matrix that has a one in the j th row and k th column and has zeros elsewhere.

The $2n \times 2n$ matrices of the block form

$$\begin{pmatrix} 0 & E_{jk} + E_{kj} \\ 0 & 0 \end{pmatrix}, \quad \begin{pmatrix} 0 & 0 \\ E_{jk} + E_{kj} & 0 \end{pmatrix}, \quad j \neq k \quad (10.39)$$

are root vectors for which the corresponding roots are $(a_j + a_k)$ and $-(a_j + a_k)$, respectively.

Next, matrices of the block form

$$\begin{pmatrix} E_{jk} & 0 \\ 0 & -E_{kj} \end{pmatrix}, \quad j \neq k \quad (10.40)$$

are root vectors for which the corresponding roots are $(a_k - a_j)$.

Finally, matrices of the block form

$$\begin{pmatrix} 0 & E_{jj} \\ 0 & 0 \end{pmatrix}, \quad \begin{pmatrix} 0 & 0 \\ E_{jj} & 0 \end{pmatrix} \quad (10.41)$$

are root vectors for which the corresponding roots are $2a_j$ and $-2a_j$.

- If we use the inner product $\langle X, Y \rangle = \text{tr}(X^\dagger Y)/2$ to identify the roots with elements of $\mathfrak{h}$ and then identify $\mathfrak{h}$ with $\mathbb{C}^n$ via the map $H \mapsto (a_1, \dots, a_n)$, we find:

The roots of $\mathfrak{sp}(n, \mathbb{C})$ are the vectors

$$\pm e_j \pm e_k, \quad j < k, \quad \text{and} \quad \pm 2e_j, \quad j = 1, \dots, n.$$

This root system is similar to that of $\mathfrak{so}(2n+1, \mathbb{C})$, but the roles of long and short roots are reversed.

10.7 Further properties of the root system

We now discuss several rich properties of root systems.

Proposition 10.26. *Suppose α and β are roots, α is not a multiple of β , and $\langle \alpha, \alpha \rangle \geq \langle \beta, \beta \rangle$. Then one of the following holds:*

1. $\langle \alpha, \beta \rangle = 0$ and the angle between α and β is $\pi/2$.
2. $\langle \alpha, \alpha \rangle = \langle \beta, \beta \rangle$ and the angle between α and β is $\pi/3$ or $2\pi/3$.
3. $\langle \alpha, \alpha \rangle = 2\langle \beta, \beta \rangle$ and the angle between α and β is $\pi/4$ or $3\pi/4$.
4. $\langle \alpha, \alpha \rangle = 3\langle \beta, \beta \rangle$ and the angle between α and β is $\pi/6$ or $5\pi/6$.

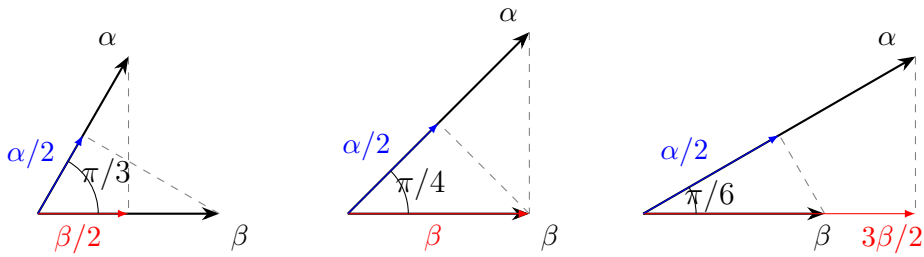


Figure 6: The three acute configurations allowed by Proposition 10.26, with angles $\pi/3$, $\pi/4$ and $\pi/6$. In each case the projection of β onto α is $\alpha/2$, while the projection of α onto β is $\beta/2$, β and $3\beta/2$, respectively.

Figure 6 shows the allowed angles and length ratios for the case of an acute angle. In each case, $2\langle \alpha, \beta \rangle / \langle \alpha, \alpha \rangle = 1$, whereas $2\langle \beta, \alpha \rangle / \langle \beta, \beta \rangle$ takes the values 1, 2, and 3 in the three successive cases.

Proof. Let $m_1 = 2\langle\alpha, \beta\rangle/\langle\alpha, \alpha\rangle$ and $m_2 = 2\langle\beta, \alpha\rangle/\langle\beta, \beta\rangle$. Both m_1 and m_2 are integers. Note that $m_1 m_2 = 4\cos^2 \theta$, where θ is the angle between α and β . Since $0 \leq \cos^2 \theta \leq 1$, we have $0 \leq m_1 m_2 \leq 4$. Also, $|m_2|/|m_1| = \langle\alpha, \alpha\rangle/\langle\beta, \beta\rangle \geq 1$. The only integer pairs (m_1, m_2) satisfying these conditions correspond to the cases listed in the proposition.

If $m_1 m_2 = 0$, then $\cos \theta = 0$: Case 1. If $m_1 m_2 = 4$, then $\cos^2 \theta = 1$ and α, β are multiples of one another, which is excluded. If $m_1 m_2 = 1$, then $\cos^2 \theta = 1/4$ and $m_1 = m_2 = \pm 1$, so $\theta = \pi/3$ (for the sign $+$) or $2\pi/3$ (for $-$), with $\langle\alpha, \alpha\rangle = \langle\beta, \beta\rangle$: Case 2. If $m_1 m_2 = 2$, then $\cos^2 \theta = 1/2$ and, since $|m_2| \geq |m_1|$, we have $(m_1, m_2) = \pm(1, 2)$, so $\theta = \pi/4$ or $3\pi/4$, with $\langle\alpha, \alpha\rangle = 2\langle\beta, \beta\rangle$: Case 3. The case $m_1 m_2 = 3$ is analogous and gives Case 4. $\square$

Corollary 10.27. *Suppose α and β are roots. If the angle between α and β is strictly obtuse, then $\alpha + \beta$ and $-\alpha - \beta$ are roots. If the angle is strictly acute, then $\alpha - \beta$ and $\beta - \alpha$ are roots.*

Proof. This follows from examining the allowed geometries from Proposition 10.26. The reflection of β across the hyperplane perpendicular to α is $s_\alpha \cdot \beta = \beta - (2\langle\alpha, \beta\rangle/\langle\alpha, \alpha\rangle)\alpha = \beta - m_1\alpha$. In all acute cases, $m_1 = 1$, so $\beta - \alpha$ (and hence $\alpha - \beta$) is a root. For all obtuse cases, $m_1 = -1$, so $\beta + \alpha$ (and hence $-\alpha - \beta$) is a root. $\square$

We illustrate the acute case below, where the projection of β onto α is equal to $\alpha/2$ and $s_\alpha \cdot \beta = \beta - \alpha$.

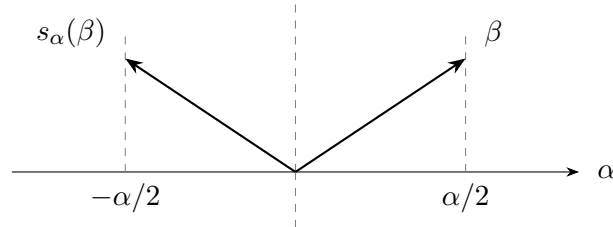


Figure 7: The acute case: the projection of β onto α is $\alpha/2$, and the reflection is $s_\alpha \cdot \beta = \beta - \alpha$.

10.7.1 Rank-one and rank-two examples

If the rank of the root system is one, then there is only one possibility: R must consist of a pair $\{\alpha, -\alpha\}$, where α is a nonzero element of E . This root system is known as A_1 . We also know that this arises as the root system of $\mathfrak{sl}(2, \mathbb{C})$.

In rank two, there are 4 possibilities listed below with their conventional names and the corresponding Lie algebras.

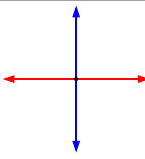
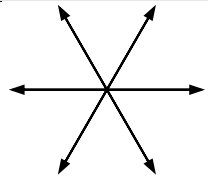
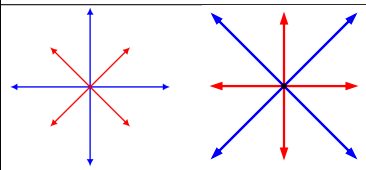
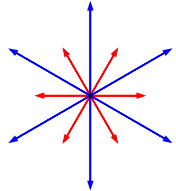
Root system	Diagram	Lie algebra
$A_1 \times A_1$		$\mathfrak{so}(4, \mathbb{C}) \cong \mathfrak{sl}(2, \mathbb{C}) \oplus \mathfrak{sl}(2, \mathbb{C})$
A_2		$\mathfrak{sl}(3, \mathbb{C})$
B_2 or C_2		$\mathfrak{so}(5, \mathbb{C}) \cong \mathfrak{sp}(2, \mathbb{C})$
G_2		G_2

Table 1: The four rank-two root systems, with the corresponding Lie algebras.

- In the case of $A_1 \times A_1$, the lengths of the horizontal roots are unrelated to the lengths of the vertical roots.
- In A_2 , all roots have the same length.
- In B_2 or C_2 , the length of the longer roots is $\sqrt{2}$ times the length of the shorter roots.
- In G_2 , the length of the longer roots is $\sqrt{3}$ times the length of the shorter roots.

The angle between successive roots is

$$\begin{aligned}
& \pi/2 \text{ for } A_1 \times A_1; & \pi/3 \text{ for } A_2; \\
& \pi/4 \text{ for } B_2 \text{ or } C_2; & \pi/6 \text{ for } G_2.
\end{aligned} \tag{10.42}$$

It is easy to check that Proposition 10.26 is satisfied for each pair of roots.

- The case of $A_1 \times A_1$ is an example of a **reducible** root system.

In general, if a root system (E, R) can be decomposed into an orthogonal decomposition $E = E_1 \oplus E_2$ with $\dim E_1, \dim E_2 > 0$ such that every element

of R is either in E_1 or in E_2 , then the root system (E, R) is said to be **reducible**. If such a decomposition does not exist, then (E, R) is **irreducible**.

Our task will be to classify all irreducible root systems.

- Here is an important statement:

In the Lie algebra setting, the root system associated to a complex semisimple Lie algebra $\mathfrak{g}$ is irreducible precisely if $\mathfrak{g}$ is simple.

For $A_1 \times A_1$, $\mathfrak{so}(4, \mathbb{C}) \cong \mathfrak{sl}(2, \mathbb{C}) \oplus \mathfrak{sl}(2, \mathbb{C})$ is semisimple but not simple.

- There are two ways of drawing the root system for B_2 or C_2 . We will soon see explicitly that they are related by a rotation and rescaling. We say that the root systems of B_2 and C_2 are **isomorphic**.

In general, two root systems (E, R) and (F, S) are said to be **isomorphic** if there exists an invertible linear map $A : E \rightarrow F$ such that A maps R onto S and such that for all $\alpha \in R$, $\beta \in E$, we have

$$A(s_\alpha \cdot \beta) = s_{A\alpha} \cdot (A\beta) ,$$

where s_α is the reflection about the hyperplane orthogonal to α . The map A is called an **isomorphism**.

In other words, the linear map A preserves the reflections about the roots.

- Using the results of Section 10.6, we can see that the root systems $A_1 \times A_1$, A_2 , and B_2 arise as root systems of $\mathfrak{so}(4, \mathbb{C}) \cong \mathfrak{sl}(2, \mathbb{C}) \oplus \mathfrak{sl}(2, \mathbb{C})$, $\mathfrak{sl}(3, \mathbb{C})$ and $\mathfrak{so}(5, \mathbb{C}) \cong \mathfrak{sp}(2, \mathbb{C})$, as indicated in Table 1. Note that the root system G_2 , meanwhile, is the root system of an “exceptional” Lie algebra, which is also referred to as G_2 .

Proposition 10.28. *Every rank-two root system is isomorphic to one of the systems listed in Table 1.*

Proof. Let R be a rank-two root system in the plane $E = \mathbb{R}^2$. Let θ be the minimum positive angle between any two roots in R . Since the roots must span the plane, we can choose two linearly independent roots α and β . If the angle between them is greater than $\pi/2$, the angle between α and $-\beta$ will be less than $\pi/2$. Thus, the minimum angle θ must be in the range $(0, \pi/2]$. According to Proposition 10.26, the only possible values for θ are $\pi/2, \pi/3, \pi/4$, or $\pi/6$.

Let α and β be two elements of R such that the angle between them is this minimum angle θ . The vector $-s_\beta \cdot \alpha$ is another root. Geometrically, the angle between β and $-s_\beta \cdot \alpha$ is also θ , as shown below:

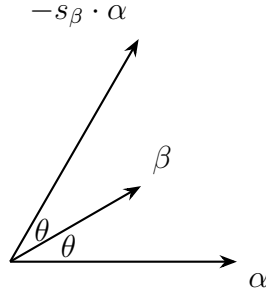


Figure 8: Two roots α and β at the minimal angle θ , and the root $-s_\beta \cdot \alpha$, which makes the angle θ with β and 2θ with α .

The angle between α and $-s_\beta \cdot \alpha$ is 2θ . By repeatedly applying reflections, we can generate a chain of roots, each separated by the angle θ . Since there are a finite number of roots, this chain must close on itself. Furthermore, because θ is the minimum angle, there can be no roots lying between these generated “spokes.”

Thus, R must consist of a set of vectors spaced by the angle θ . If, for example, $\theta = \pi/4$, then to satisfy the length requirements of Proposition 10.26, the roots must alternate between a shorter length and a length that is greater by a factor of $\sqrt{2}$. This configuration corresponds to the root system B_2 . Similar reasoning shows that the other possible values of θ yield the other root systems in Table 1. $\square$

Proposition 10.29. *If R is a rank-two root system with minimum angle $\theta = 2\pi/k$ for $k \in \{4, 6, 8, 12\}$, then the Weyl group of R is the symmetry group of a regular $k/2$ -gon (the dihedral group $D_{k/2}$).*

Proof. The group W contains a reflection s_α for each root α . If α and β are roots with an angle ϕ between them, the composition $s_\beta s_\alpha$ is a rotation by an angle $\pm 2\phi$. To see this, note that reflections have determinant -1 , so their product has determinant 1 and must be a rotation. The angle can be found by applying the rotation to a vector, for example, α as follows: $s_\beta s_\alpha \cdot \alpha = s_\beta \cdot (-\alpha)$. From Figure 8, it is clear that the resulting vector is at an angle of 2ϕ from the original vector α .

Since the angle ϕ between any two roots in the generated set is an integer multiple of the minimum angle θ , the composition of reflections generates rotations by all integer multiples of 2θ . Moreover, the composition of a reflection s_α and a rotation by angle 2ϕ is another reflection s_β , where β is a root at angle ϕ to α .

The set of all reflections and the rotations they generate forms a group, which is precisely the Weyl group. This group is also the symmetry group of a regular $2\pi/2\theta = k/2$ -gon, where $\theta = 2\pi/k$. $\square$

Note that in the case of A_2 , the Weyl group is $S_3 \cong D_3$. This group is generated by three reflections and contains rotations by multiples of $2\pi/3$. The full symmetry group of the A_2 root diagram is D_6 , which includes rotations by $\pi/3$. These additional symmetries are not in the Weyl group.

10.8 Dual root systems

We now introduce an important duality operation on root systems.

Definition 10.30. *If (E, R) is a root system, then for each root $\alpha \in R$, the **coroot** H_α is the vector given by*

$$H_\alpha = \frac{2\alpha}{\langle \alpha, \alpha \rangle}.$$

*The set of all coroots, denoted $R^\vee$, is called the **dual root system** to R .*

Proposition 10.31. *If R is a root system, then $R^\vee$ is also a root system. The Weyl group for $R^\vee$ is the same as the Weyl group for R , and the dual of the dual is the original root system: $(R^\vee)^\vee = R$.*

Proof. We compute that

$$\langle H_\alpha, H_\alpha \rangle = \frac{4\langle \alpha, \alpha \rangle}{\langle \alpha, \alpha \rangle^2} = \frac{4}{\langle \alpha, \alpha \rangle}, \quad (10.43)$$

and therefore,

$$\frac{2H_\alpha}{\langle H_\alpha, H_\alpha \rangle} = \frac{2 \cdot \frac{2\alpha}{\langle \alpha, \alpha \rangle}}{\frac{4}{\langle \alpha, \alpha \rangle}} = \alpha. \quad (10.44)$$

Taking the inner product of (10.44) with H_β yields:

$$\frac{2\langle H_\alpha, H_\beta \rangle}{\langle H_\alpha, H_\alpha \rangle} = \langle \alpha, H_\beta \rangle = \frac{2\langle \alpha, \beta \rangle}{\langle \beta, \beta \rangle}, \quad (10.45)$$

which implies the left-hand side of (10.45) is an integer, since the right-hand side is.

Furthermore, since H_α is a scalar multiple of α , the reflection s_{H_α} is the same as s_α . Also, since s_α is an orthogonal transformation,

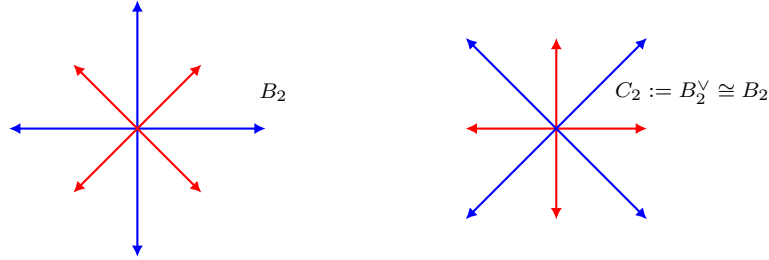
$$s_\alpha(H_\beta) = \frac{2s_\alpha(\beta)}{\langle \beta, \beta \rangle} = \frac{2s_\alpha(\beta)}{\langle s_\alpha(\beta), s_\alpha(\beta) \rangle} = H_{s_\alpha(\beta)}.$$

Thus, the set of coroots is invariant under each reflection s_α . This fact, together with the integer property shown in (10.45), confirms that $R^\vee$ satisfies the axioms of a root system. The remaining properties for $R^\vee$ follow immediately from those for R .

Since $s_\alpha = s_{H_\alpha}$, R and $R^\vee$ have the same Weyl group. Finally, (10.44) shows that the formula for α in terms of H_α is the same as for H_α in terms of α . Hence, $H_{H_\alpha} = \alpha$ and $(R^\vee)^\vee = R$. $\square$

- Note from (10.45) that the integer associated with the pair (H_α, H_β) in the dual root system $R^\vee$ is the same as the integer associated with the pair (β, α) in the original root system R .

- If all roots in R have the same length, then $R^\vee$ is proportional to R , and we say R is self-dual.
- Even if the roots have different lengths, R and $R^\vee$ can still be isomorphic. For instance, the dual of the root system B_2 is C_2 . However, C_2 can be rotated by $\pi/4$ and rescaled to become identical to B_2 , so the two root systems are isomorphic.



From (10.43), we see that if α is a long root, then the coroot H_α is a short root, and vice versa.

- On the other hand, for $n \geq 3$, the root system B_n and its dual C_n are not isomorphic. This distinction becomes clear when we examine their Dynkin diagrams.

10.9 Base and positive simple roots

Definition 10.32. If (E, R) is a root system, a subset Δ of R is called a **base** if the following conditions hold:

1. Δ is a basis for E as a vector space.
2. Each root $\alpha \in R$ can be expressed as a linear combination of elements of Δ with integer coefficients, such that the coefficients are either all non-negative or all non-positive.

The roots whose coefficients in this expansion are all non-negative are called **positive roots**, and the others are called **negative roots** (relative to the chosen base Δ). The set of positive roots is denoted R^+ and the set of negative roots is denoted R^- . The elements of Δ are called the **positive simple roots**.

Proposition 10.33. If α and β are distinct positive simple roots, then $\langle \alpha, \beta \rangle \leq 0$.

Geometrically, this means that either α and β are orthogonal or the angle between them is **obtuse**.

Proof. Since $\alpha \neq \beta$, if we had $\langle \alpha, \beta \rangle > 0$, the angle between them would be strictly acute. By Corollary 10.27, $\alpha - \beta$ would have to be a root. However, the expansion of $\alpha - \beta$ in the base Δ has one positive coefficient (for α) and one negative coefficient (for β), which violates the definition of a base. Thus, we must have $\langle \alpha, \beta \rangle \leq 0$. $\square$

As an example, for the A_2 root system, the positive simple roots are denoted by α_1 and α_2 below.

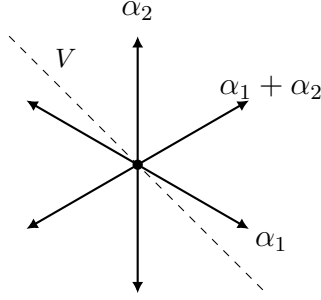


Figure 9: The A_2 root system with positive simple roots α_1, α_2 , and a hyperplane V (dashed) containing no root.

Proposition 10.34. *There exists a hyperplane V through the origin in E that does not contain any element of R .*

Proof. The finitely many hyperplanes $V_\alpha = \{H \in E \mid \langle \alpha, H \rangle = 0\}$, with $\alpha \in R$, cannot cover E , so there exists H' lying in none of them, i.e., with $\langle \alpha, H' \rangle \neq 0$ for all $\alpha \in R$. The hyperplane V through the origin orthogonal to H' then contains no root. $\square$

Definition 10.35. *Let (E, R) be a root system. Let V be a hyperplane through the origin in E such that V does not contain any root. Choose one “side” of V and let R^+ denote the set of roots on this side of V . An element α of R^+ is **decomposable** if there exist β and γ in R^+ such that $\alpha = \beta + \gamma$; if no such elements exist, α is **indecomposable**.*

In other words, if H is a nonzero vector orthogonal to V , then V is the set of $\mu \in E$ for which $\langle \mu, H \rangle = 0$. The two “sides” of V are then the sets $\{\mu \in E \mid \langle \mu, H \rangle > 0\}$ and $\{\mu \in E \mid \langle \mu, H \rangle < 0\}$.

Theorem 10.36. *Suppose (E, R) is a root system, V is a hyperplane through the origin in E not containing any element of R , and R^+ is the set of roots lying on a fixed side of V . Then the set of indecomposable elements of R^+ is a base for R , i.e. they are positive simple roots.*

This construction motivates the terminology: the positive roots are those on the chosen side of V , and the positive *simple* roots are the indecomposable elements among them.

Proof. Let Δ denote the set of indecomposable elements in R^+ . Choose a nonzero vector H orthogonal to V so that the chosen side of V is the set of $\mu \in E$ for which $\langle H, \mu \rangle > 0$.

Step 1: Every $\alpha \in R^+$ can be expressed as a linear combination of elements of Δ with non-negative integer coefficients. If not, choose a counterexample α with $\langle H, \alpha \rangle$ as small as possible. Then $\alpha \notin \Delta$, so α is decomposable: $\alpha = \beta_1 + \beta_2$ with $\beta_1, \beta_2 \in R^+$. Since $\langle H, \beta_1 \rangle$ and $\langle H, \beta_2 \rangle$ are positive and sum to $\langle H, \alpha \rangle$, both are smaller than $\langle H, \alpha \rangle$; by minimality, each β_i is a non-negative integer combination of elements of Δ , and hence so is α — a contradiction.

Step 2: If α and β are distinct elements of Δ , then $\langle \alpha, \beta \rangle \leq 0$. (Since Δ is not yet known to be a base, Proposition 10.33 does not apply.) If $\langle \alpha, \beta \rangle > 0$, then by Corollary 10.27 one of $\alpha - \beta$ and $\beta - \alpha$ lies in R^+ ; then $\alpha = (\alpha - \beta) + \beta$ or $\beta = (\beta - \alpha) + \alpha$ exhibits α or β as decomposable, contradicting indecomposability.

Step 3: The elements of Δ are linearly independent. Suppose we have

$$\sum_{\alpha \in \Delta} c_\alpha \alpha = 0 \quad (10.46)$$

for some collection of constants c_α . Separate the sum into those terms where $c_\alpha \geq 0$ and those where $c_\alpha = -d_\alpha < 0$, so that

$$\sum c_\alpha \alpha = \sum d_\beta \beta \quad (10.47)$$

where the sums range over disjoint subsets of Δ . If u denotes the vector in (10.47), we have

$$\langle u, u \rangle = \left\langle \sum c_\alpha \alpha, \sum d_\beta \beta \right\rangle = \sum \sum c_\alpha d_\beta \langle \alpha, \beta \rangle.$$

However, c_α and d_β are non-negative and (by Step 2) $\langle \alpha, \beta \rangle \leq 0$. Thus, $\langle u, u \rangle \leq 0$ and u must be zero. Now, if $u = 0$, then $\langle H, u \rangle = \sum c_\alpha \langle H, \alpha \rangle = 0$, which implies that all the c_α are zero since $c_\alpha \geq 0$ and $\langle H, \alpha \rangle > 0$. The same argument then shows that all the d_β are zero as well.

Step 4: Δ is a base. We have shown that Δ is linearly independent and that all of the elements of R^+ can be expressed as linear combinations of elements of Δ with non-negative integer coefficients. The remaining elements of R , namely the elements of R^- , are simply the negatives of the elements of R^+ , and so they can be expressed as linear combinations of elements of Δ with nonpositive integer coefficients. Since the elements of R span E , Δ must also span E , and it is a base. $\square$

Figure 10 illustrates the above theorem in the case of the G_2 root system, with α_1 and α_2 being the indecomposable roots (and, hence, positive simple roots) on one side of the dashed line.

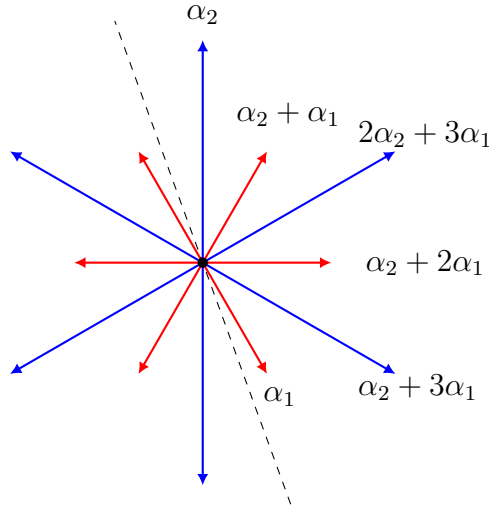


Figure 10: The G_2 root system. The roots α_1 and α_2 are the indecomposable roots on one side of the dashed line.

We also illustrate the root systems of B_2 and its dual ($C_2 = B_2^\vee \cong B_2$) with the positive roots labelled below. Note that $H_\alpha = 2\alpha/\langle\alpha, \alpha\rangle$.

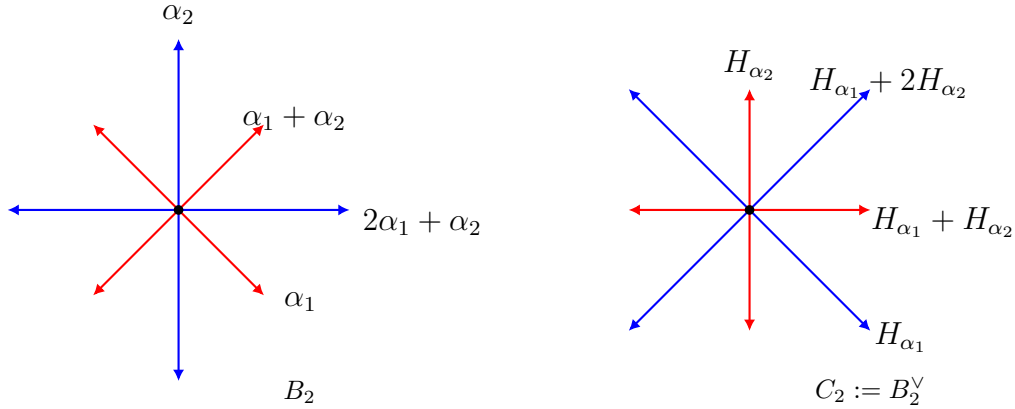


Figure 11: The root systems B_2 and $C_2 = B_2^\vee$, with the positive roots labelled. Long roots are drawn in blue and short roots in red.

10.10 Dynkin diagrams

A Dynkin diagram is a convenient graphical way of encoding the structure of a base for a root system, and thus of the root system itself.

Definition 10.37. If $\Delta = \{\alpha_1, \dots, \alpha_r\}$ is a base for a root system R , the **Dynkin diagram** for R is a graph having vertices $v_1, \dots, v_r$ corresponding to each positive simple root $\alpha_1, \dots, \alpha_r$. Between two distinct vertices v_j and v_k , we place a number of edges according to the angle between the corresponding roots:

- 0 edges for an angle of $\pi/2$.
- 1 edge for an angle of $2\pi/3$.
- 2 edges for an angle of $3\pi/4$.
- 3 edges for an angle of $5\pi/6$.

In addition, if α_j and α_k have different lengths, we add an arrow (the “greater than” sign) to the edges pointing from the vertex associated with the longer root to the vertex associated with the shorter root.

By Proposition 10.26, we see that

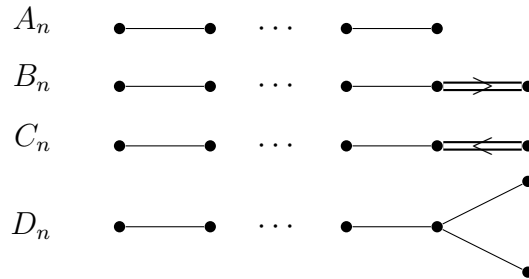
No. of edges between two vertices	Angle between roots	Length ratio of the roots (longer to shorter)	(10.48)
1	$2\pi/3$	1	
2	$3\pi/4$	$\sqrt{2}$	
3	$5\pi/6$	$\sqrt{3}$	

Here are the Dynkin diagrams for the rank-two root systems:



In the case of the root system G_2 , for example, a base consists of two roots at an angle of $5\pi/6$, with a length ratio of $\sqrt{3}$. Thus, the Dynkin diagram consists of two vertices connected by three edges, with an arrow pointing from the longer root (α_2) to the shorter one (α_1). One can similarly read off the Dynkin diagrams for A_2 from Figure 9 and B_2 from Figure 11. For $A_1 \times A_1$, the two elements of the base are orthogonal, yielding two disconnected vertices.

From Section 10.6, we can read off the Dynkin diagrams for the **classical Lie algebras**, $\mathfrak{sl}(n+1, \mathbb{C})$, $\mathfrak{so}(2n+1, \mathbb{C})$, $\mathfrak{sp}(n, \mathbb{C})$, and $\mathfrak{so}(2n, \mathbb{C})$. The results are shown below, where each diagram has n nodes:



For small values of n , some of these diagrams coincide, reflecting exceptional isomorphisms between the low-dimensional Lie algebras.

- The Dynkin diagram for D_n is ill-defined for $n = 1$, reflecting the fact that the Lie algebra $\mathfrak{so}(2, \mathbb{C})$ is not semisimple.
- The coincidences $A_1 = B_1 = C_1$ reflect the isomorphisms

$$\mathfrak{sl}(2, \mathbb{C}) \cong \mathfrak{so}(3, \mathbb{C}) \cong \mathfrak{sp}(1, \mathbb{C}), \quad \bullet \quad (10.49)$$

- The diagram $D_2 = A_1 \times A_1$ consists of two disconnected nodes, corresponding to the isomorphism

$$\mathfrak{so}(4, \mathbb{C}) \cong \mathfrak{sl}(2, \mathbb{C}) \oplus \mathfrak{sl}(2, \mathbb{C}), \quad \bullet \quad \bullet \quad (10.50)$$

This shows that the algebra $\mathfrak{so}(4, \mathbb{C})$ is semisimple but not simple.

- The coincidence $B_2 = C_2$ corresponds to the isomorphism

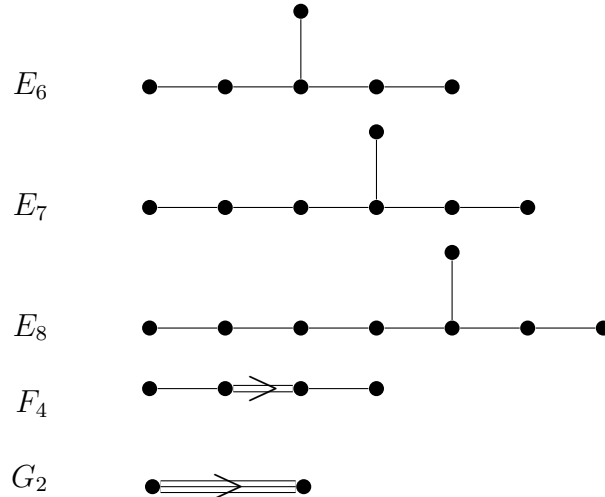
$$\mathfrak{so}(5, \mathbb{C}) \cong \mathfrak{sp}(2, \mathbb{C}), \quad \bullet \rightleftarrows \bullet = \bullet \rightleftarrows \bullet \quad (10.51)$$

since the dual of the B_2 root system is isomorphic to itself.

- The coincidence $D_3 = A_3$ corresponds to the isomorphism

$$\mathfrak{so}(6, \mathbb{C}) \cong \mathfrak{sl}(4, \mathbb{C}), \quad \begin{array}{c} \bullet \\ \diagup \quad \diagdown \\ \bullet \end{array} = \bullet - \bullet - \bullet \quad (10.52)$$

In addition to the four infinite families of classical algebras, there are exactly five other possibilities, known as the **exceptional Lie algebras**. Their Dynkin diagrams are listed below.



The classification of simple Lie algebras is elegantly summarised by the classification of their Dynkin diagrams. We state the main results here without proof.

- A semisimple Lie algebra is simple if and only if its associated Dynkin diagram is connected.
- Every irreducible root system is isomorphic to exactly one of the types listed above: $A_n(n \geq 1)$, $B_n(n \geq 2)$, $C_n(n \geq 3)$, $D_n(n \geq 4)$, or one of the five exceptional types G_2, F_4, E_6, E_7, E_8 .
- Since all bases of a fixed root system are equivalent under the action of the Weyl group, there is only one Dynkin diagram associated with each semisimple Lie algebra $\mathfrak{g}$.
- The classification of irreducible root systems therefore gives rise to a complete classification of simple complex Lie algebras.

We thus have the following important theorem.

Theorem 10.38. (The Cartan Classification.) *Every simple Lie algebra over $\mathbb{C}$ is isomorphic to precisely one of the following algebras:*

$$\mathfrak{sl}(n+1, \mathbb{C}), n \geq 1; \quad \mathfrak{so}(2n+1, \mathbb{C}), n \geq 2; \quad \mathfrak{sp}(n, \mathbb{C}), n \geq 3; \quad \mathfrak{so}(2n, \mathbb{C}), n \geq 4,$$

or one of the five exceptional Lie algebras G_2, F_4, E_6, E_7 , and E_8 .

Summary: a dictionary for $\mathfrak{g} = \mathfrak{sl}(3, \mathbb{C})$

The general structures introduced in this section were all constructed by hand for $\mathfrak{g} = \mathfrak{sl}(3, \mathbb{C})$ in Section 9. The following dictionary summarises the correspondence; the reader is encouraged to revisit Section 9 with the general theory in hand.

Structure	Realisation for $\mathfrak{g} = \mathfrak{sl}(3, \mathbb{C})$
compact real form $\mathfrak{k}$	$\mathfrak{su}(3)$
maximal commutative $\mathfrak{t} \subset \mathfrak{k}$	traceless diagonal matrices in $\mathfrak{su}(3)$
Cartan subalgebra $\mathfrak{h} = \mathfrak{t} + i\mathfrak{t}$	traceless diagonal matrices = $\text{span}_{\mathbb{C}}\{H_1, H_2\}$
rank = $\dim_{\mathbb{C}} \mathfrak{h}$	2
roots $\alpha \in R$	$\pm\alpha_1, \pm\alpha_2, \pm(\alpha_1+\alpha_2)$; cf. (9.5)
root spaces $\mathfrak{g}_{\alpha}$	$\mathbb{C}X_1, \mathbb{C}X_2, \mathbb{C}X_3, \mathbb{C}Y_1, \mathbb{C}Y_2, \mathbb{C}Y_3$ ²⁷
$\mathfrak{sl}(2, \mathbb{C})$ subalgebras $\mathfrak{u}^{\alpha} = \langle X_{\alpha}, Y_{\alpha}, H_{\alpha} \rangle$	$\langle X_i, Y_i, H_i \rangle$ ($i = 1, 2$); $\langle X_3, Y_3, H_1+H_2 \rangle$
coroots $H_{\alpha} = 2\alpha/\langle \alpha, \alpha \rangle$	H_1, H_2, H_1+H_2 for $\alpha_1, \alpha_2, \alpha_1+\alpha_2$
Weyl group W	S_3 , the symmetry group of the root hexagon
root system (E, R)	A_2
base Δ (positive simple roots)	$\{\alpha_1, \alpha_2\}$
Dynkin diagram	two nodes joined by a single edge (A_2)

(10.53)

²⁷Here $\mathbb{C}X$ denotes the one-dimensional complex span $\{cX \mid c \in \mathbb{C}\}$ of X : each root space consists of all scalar multiples of a single root vector, in accordance with Theorem 10.19.

A More character tables

This appendix collects the character tables of finite groups that are not discussed in the main text.

A.1 The character table of the quaternion group $\mathbb{H}$

“Here as he walked by on the 16th of October 1843 Sir William Rowan Hamilton in a flash of genius discovered the fundamental formula for quaternion multiplication $i^2 = j^2 = k^2 = ijk = -\mathbf{1}$ and cut it on a stone of this bridge.”

(Quaternion plaque on Brougham (Broom) Bridge, Dublin.)

The quaternion group can be written as

$$\mathbb{H} = \{\pm \mathbf{1}, \pm i, \pm j, \pm k\},$$

with the relations $i^2 = j^2 = k^2 = ijk = -\mathbf{1}$.

A.1.1 The conjugacy classes

There are 5 conjugacy classes in the quaternion group $\mathbb{H}$, namely

$$\{\mathbf{1}\}, \{i, -i\}, \{j, -j\}, \{k, -k\}, \{-\mathbf{1}\}.$$

A.1.2 The irreducible representations

The one-dimensional representations. In a one-dimensional representation, each $g \in \mathbb{H}$ acts by a scalar λ_g , and scalars commute. Note that $ij = k$ (multiply $ijk = -\mathbf{1} = k^2$ on the right by k^{-1}) and $ji = -k$ (multiply $ijij = (ij)^2 = k^2 = -\mathbf{1}$ by $i^{-1} = -i$ on the left and by $j^{-1} = -j$ on the right). Hence

$$\lambda_k = \lambda_i \lambda_j = \lambda_j \lambda_i = \lambda_{-\mathbf{1}} \lambda_k ,$$

which forces $\lambda_{-\mathbf{1}} = 1$. Then $i^2 = j^2 = -\mathbf{1}$ gives $\lambda_i = \pm 1$ and $\lambda_j = \pm 1$, with $\lambda_k = \lambda_i \lambda_j$. The four sign choices $(\lambda_i, \lambda_j) = (\pm 1, \pm 1)$ define four one-dimensional representations — the **trivial representation** U_1 (both signs $+$) and U_2, U_3, U_4 — with characters

$$\chi_U = (1, \lambda_i, \lambda_j, \lambda_i \lambda_j, 1)$$

on the conjugacy classes $[\mathbf{1}], [i], [j], [k], [-\mathbf{1}]$; these are the first four rows of the character table below.

The ‘spin’ representation. Since $\mathbb{H}$ has 5 conjugacy classes, there must be 5 irreducible representations. We have found 4 one-dimensional representations so far. The sum-of-squares formula, (3.8), requires that the dimension n of the remaining irreducible representation satisfies $8 = 1^2 + 1^2 + 1^2 + 1^2 + n^2$, which implies $n = 2$. This two-dimensional representation V is known as the **spin representation**.

The character χ_V follows immediately from the regular representation. On the classes $[\mathbf{1}]$, $[i]$, $[j]$, $[k]$, $[-\mathbf{1}]$, the characters of the four one-dimensional representations sum to $(4, 0, 0, 0, 4)$, while $\sum_m (\dim V_m) \chi_{V_m}$ equals the character $(8, 0, 0, 0, 0)$ of the regular representation (Lemma 3.41). Hence

$$2\chi_V = (8, 0, 0, 0, 0) - (4, 0, 0, 0, 4), \quad \text{i.e.,} \quad \chi_V = (2, 0, 0, 0, -2).$$

The complete character table of $\mathbb{H}$ is therefore

$\mathbb{H}$	$\mathbf{1}$	i	j	k	$-\mathbf{1}$
U_1	1	1	1	1	1
U_2	1	-1	-1	1	1
U_3	1	1	-1	-1	1
U_4	1	-1	1	-1	1
V	2	0	0	0	-2

(A.1)

Observe that the character table of D_4 is the same as that of the quaternion group $\mathbb{H}$. However, these two groups are not isomorphic, as D_4 has four elements of order 2, while $\mathbb{H}$ has only one $(-\mathbf{1})$.

A.2 The character table of D_5

The symmetry group of a regular pentagon is the dihedral group of order 10, denoted by D_5 . Let a be a rotation by $2\pi/5$ and b be a reflection. Then, D_5 is generated by a and b with the relations

$$a^5 = e, \quad b^2 = e, \quad ab = ba^{-1}. \quad (\text{A.2})$$

A.2.1 The conjugacy classes

There are 4 conjugacy classes:

$$\{e\}, \quad \{b, ab, a^2b, a^3b, a^4b\}, \quad \{a, a^4\}, \quad \{a^2, a^3\}. \quad (\text{A.3})$$

The sizes of these conjugacy classes are 1, 5, 2, and 2, respectively.

A.2.2 The irreducible representations

Since there are 4 conjugacy classes, there must be 4 irreducible representations.

- The group D_5 acts on the plane, giving a natural **two-dimensional representation**, which we denote by V_1 . In a suitable basis, the generators a and b can be represented by matrices:

$$A = \begin{pmatrix} \omega & 0 \\ 0 & \omega^{-1} \end{pmatrix}, \quad B = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \quad (\text{A.4})$$

where $\omega = e^{2\pi i/5}$. One can check that this representation is irreducible by computing $(\chi_{V_1}, \chi_{V_1}) = 1$. The character χ_{V_1} is found by taking the traces of the matrices representing the conjugacy class elements:

$$\chi_{V_1} = (2, 0, \omega + \omega^{-1}, \omega^2 + \omega^{-2}). \quad (\text{A.5})$$

- The **trivial representation**, U_1 , is a one-dimensional representation where every element acts as the identity. Its character is $\chi_{U_1} = (1, 1, 1, 1)$.
- The **sign representation**, U_2 , is a one-dimensional representation defined by the action of the generators: $a \mapsto 1$ and $b \mapsto -1$. The character is $\chi_{U_2} = (1, -1, 1, 1)$.
- We have found three irreducible representations so far. The sum-of-squares formula, $|G| = \sum (\dim V_i)^2$, requires the dimension d of the final irreducible representation to satisfy $10 = 2^2 + 1^2 + 1^2 + d^2$, so $d = 2$; call this representation V_2 . It is obtained from V_1 by replacing ω with ω^2 , i.e., $a \mapsto \text{diag}(\omega^2, \omega^{-2})$ and $b \mapsto B$; this again satisfies the relations (A.2), is irreducible since $(\chi_{V_2}, \chi_{V_2}) = 1$, and is not isomorphic to V_1 , since the characters differ:

$$\chi_{V_2} = (2, 0, \omega^2 + \omega^{-2}, \omega^4 + \omega^{-4}) = (2, 0, \omega^2 + \omega^{-2}, \omega + \omega^{-1}). \quad (\text{A.6})$$

The complete character table of D_5 is given as follows:

D_5	1 e	5 b	2 a	2 a^2
U_1	1	1	1	1
U_2	1	-1	1	1
V_1	2	0	$\omega + \omega^{-1}$	$\omega^2 + \omega^{-2}$
V_2	2	0	$\omega^2 + \omega^{-2}$	$\omega + \omega^{-1}$

(A.7)

where $\omega = e^{2\pi i/5}$ and

$$\begin{aligned} \omega + \omega^{-1} &= 2 \cos\left(\frac{2\pi}{5}\right) = \frac{1}{2}(\sqrt{5} - 1), \\ \omega^2 + \omega^{-2} &= 2 \cos\left(\frac{4\pi}{5}\right) = -\frac{1}{2}(\sqrt{5} + 1). \end{aligned} \quad (\text{A.8})$$

A.3 The character table of A_4

The alternating group A_4 has 4 conjugacy classes:

$$[e], [(123)], [(132)], [(12)(34)]. \quad (\text{A.9})$$

There are 1, 4, 4, 3 elements in each conjugacy class, respectively. Since there are 4 conjugacy classes, there must be 4 irreducible representations.

As always, we start with the **trivial representation** U_1 , whose character on these conjugacy classes is

$$\chi_{U_1} = (1, 1, 1, 1). \quad (\text{A.10})$$

Similar to the case of S_4 , the character of the permutation representation on $\mathbb{C}^4$ is $\chi_{\mathbb{C}^4} = (4, 1, 1, 0)$. The character of the **standard representation** V is therefore given by

$$\chi_V = \chi_{\mathbb{C}^4} - \chi_{U_1} = (3, 0, 0, -1). \quad (\text{A.11})$$

The sum of the squares of the dimensions of the representations found so far is $(\dim U_1)^2 + (\dim V)^2 = 1^2 + 3^2 = 10$. The sum-of-squares formula, (3.8), tells us that the sum of the squares of the dimensions of the other two irreducible representations must be $|A_4| - 10 = 12 - 10 = 2$. Thus, the remaining two representations must both be one-dimensional. Let us denote them by U_2 and U_3 .

To determine their characters, note that in a one-dimensional representation U , each g acts by the scalar $\chi_U(g)$. Since $(123)^3 = e$, the value $\chi_U((123))$ is a cube root of unity, $\chi_U((123)) \in \{1, \omega, \omega^2\}$ with $\omega = e^{2\pi i/3}$, and $(132) = (123)^2$ gives $\chi_U((132)) = \chi_U((123))^2$. Moreover, $(123)(234) = (12)(34)$, and (234) lies in the conjugacy class of (132) ; hence

$$\chi_U((12)(34)) = \chi_U((123)) \chi_U((132)) = \chi_U((123))^3 = 1. \quad (\text{A.12})$$

The three choices of cube root give the trivial representation U_1 and the two remaining one-dimensional representations,

$$\chi_{U_2} = (1, \omega, \omega^2, 1), \quad \chi_{U_3} = (1, \omega^2, \omega, 1). \quad (\text{A.13})$$

The character table of A_4 is therefore (with $\omega = e^{2\pi i/3}$)

A_4	1 e	4 (123)	4 (132)	3 $(12)(34)$
U_1	1	1	1	1
U_2	1	ω	ω^2	1
U_3	1	ω^2	ω	1
V	3	0	0	-1

(A.14)

A.4 The character tables of D_n

The dihedral group D_n has order $2n$ (for $n \geq 3$) and can be presented explicitly with generators a (a rotation) and b (a reflection) as

$$D_n = \{e, a, \dots, a^{n-1}, b, ab, a^2b, \dots, a^{n-1}b\}, \quad (\text{A.15})$$

subject to the relations

$$a^n = e, \quad b^2 = e, \quad ab = ba^{-1}. \quad (\text{A.16})$$

Since each element of D_n corresponds to a symmetry of a regular n -gon in a plane, we can construct a family of two-dimensional representations V_j for $1 \leq j < n/2$. In a suitable basis, the generators a and b can be represented by the matrices:

$$A_j = \begin{pmatrix} \omega^j & 0 \\ 0 & \omega^{-j} \end{pmatrix}, \quad B_j = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \quad (\text{A.17})$$

where $\omega = e^{2\pi i/n}$. One can verify that these matrices satisfy the relations in (A.16). To check for irreducibility, we compute the inner product of the character with itself:

$$(\chi_{V_j}, \chi_{V_j}) = \frac{1}{2n} \sum_{g \in D_n} |\chi_{V_j}(g)|^2 = \frac{1}{2n} \left(\sum_{r=0}^{n-1} |\text{tr}(A_j^r)|^2 + \sum_{r=0}^{n-1} |\text{tr}(B_j A_j^r)|^2 \right). \quad (\text{A.18})$$

Since $\text{tr}(B_j A_j^r) = 0$ and $\text{tr}(A_j^r) = \omega^{jr} + \omega^{-jr} = 2 \cos(2\pi jr/n)$, this becomes

$$(\chi_{V_j}, \chi_{V_j}) = \frac{1}{2n} \sum_{r=0}^{n-1} |2 \cos(2\pi jr/n)|^2 = \frac{2}{n} \sum_{r=0}^{n-1} \cos^2\left(\frac{2\pi jr}{n}\right) = 1. \quad (\text{A.19})$$

Thus, each V_j for $1 \leq j < n/2$ is an irreducible representation.

We now consider the cases of odd and even n separately.

Case I: n odd

For odd n , there are $(n+3)/2$ conjugacy classes:

$$\{e\}, \quad \{b, ab, \dots, a^{n-1}b\}, \quad \{a^r, a^{-r}\} \text{ for } 1 \leq r \leq (n-1)/2. \quad (\text{A.20})$$

The discussion above provides $(n-1)/2$ distinct two-dimensional irreducible representations V_j (for $j = 1, \dots, (n-1)/2$). The sum of the squares of their dimensions is $(n-1)/2 \cdot 2^2 = 2(n-1)$. By the sum-of-squares formula (3.8), the two remaining irreducible representations must satisfy $d_1^2 + d_2^2 = 2n - 2(n-1) = 2$, which implies they are both one-dimensional. One of these is the **trivial representation** U . The other is the **sign representation** U' , where rotations act trivially and reflections

act as multiplication by -1 . Their characters on the conjugacy classes $[e], [b], [a^r]$ are:

$$\chi_U = (1, 1, 1, \dots, 1), \quad \chi_{U'} = (1, -1, 1, \dots, 1). \quad (\text{A.21})$$

The complete character table for odd n is given below

D_n (odd n)	1 e	n b	2 a	$\dots$ $\dots$	2 $a^{(n-1)/2}$
Trivial U	1	1	1	$\dots$	1
Sign U'	1	-1	1	$\dots$	1
V_j ($1 \leq j \leq (n-1)/2$)	2	0	$2 \cos(\frac{2\pi j}{n})$	$\dots$	$2 \cos(\frac{2\pi j(n-1)}{2n})$

(A.22)

Case II: n even

Let $n = 2m$. There are $m + 3$ conjugacy classes:

$$\{e\}, \quad \{a^m\}, \quad \{a^r, a^{-r}\}_{1 \leq r \leq m-1}, \quad \{a^{2k}b\}_{0 \leq k \leq m-1}, \quad \{a^{2k+1}b\}_{0 \leq k \leq m-1}. \quad (\text{A.23})$$

The discussion above provides $m - 1$ two-dimensional irreducible representations V_j (for $j = 1, \dots, m - 1$). The sum of the squares of their dimensions is $(m - 1) \cdot 2^2 = 4m - 4$. The sum-of-squares formula requires that the four remaining irreducible representations satisfy $d_1^2 + d_2^2 + d_3^2 + d_4^2 = 2n - (4m - 4) = 4m - (4m - 4) = 4$. Thus, all four must be one-dimensional.

Let U be any one-dimensional representation. The action of a and b must be by scalars. The relations $b^2 = e$ and $a^n = e$ imply that the action of b is ± 1 and the action of a is an n -th root of unity. The relation $ab = ba^{-1}$ implies $ab = a^{-1}b$, so $a^2 = 1$. Since $n = 2m$ is even, this means the action of a must be ± 1 . This gives four possible one-dimensional representations, based on the choices for the actions of a and b :

- U_1 (trivial): $a \rightarrow 1, b \rightarrow 1$.
- U_2 : $a \rightarrow 1, b \rightarrow -1$.
- U_3 : $a \rightarrow -1, b \rightarrow 1$.
- U_4 : $a \rightarrow -1, b \rightarrow -1$.

The characters for these representations can be calculated for each conjugacy class. The complete character table for even n is given below

D_n ($n = 2m$ even)	1 e	1 a^m	m b	m ab	2 for each r a^r ($1 \leq r \leq m-1$)
U_1	1	1	1	1	1
U_2	1	1	-1	-1	1
U_3	1	$(-1)^m$	1	-1	$(-1)^r$
U_4	1	$(-1)^m$	-1	1	$(-1)^r$
V_j ($1 \leq j \leq m-1$)	2	$2(-1)^j$	0	0	$2 \cos(\frac{2\pi jr}{n})$

(A.24)

B Reality and pseudoreality of irreducible representations

Let $R : G \rightarrow \text{GL}(V)$ be a finite-dimensional unitary irreducible representation of a compact group G , acting on a complex vector space V . Define the conjugate representation R^* on $\bar{V}$ by $R^*(g) := R(g)^*$. We say R is *self-conjugate* if $R \simeq R^*$, i.e. there exists an invertible intertwiner S such that

$$S R(g) S^{-1} = R(g)^* \quad \forall g \in G. \quad (\text{B.1})$$

Theorem B.1. *Let R be a unitary irreducible self-conjugate representation, and let S satisfy (B.1). Then $S^T = \lambda S$ with $\lambda \in \{+1, -1\}$, and the sign λ depends only on R , not on the choice of S . We call R*

- real (equivalently, orthogonal) if $\lambda = +1$;
- pseudoreal (equivalently, symplectic / quaternionic) if $\lambda = -1$.

Proof. The argument proceeds in three steps. The key input is Schur's lemma together with the observation that transposition is an involution on the one-dimensional space of intertwiners.

Step 1: S^T is also an intertwiner of R with R^* . Taking the transpose of (B.1),

$$(S^{-1})^T R(g)^T S^T = R(g)^{*T} = R(g)^\dagger.$$

Since R is unitary, $R(g)^\dagger = R(g)^{-1} = R(g^{-1})$; similarly $R(g)^T = R(g^{-1})^*$. Substituting and replacing $g \mapsto g^{-1}$ yields

$$(S^{-1})^T R(g)^* S^T = R(g),$$

which rearranges to

$$S^T R(g) (S^T)^{-1} = R(g)^*. \quad (\text{B.2})$$

Thus S^T satisfies the same intertwining relation as S .

Step 2: Schur's lemma forces $S^T = \lambda S$ with $\lambda = \pm 1$. Both S and S^T intertwine R with R^* , so $S^{-1}S^T$ intertwines R with itself. Since R is irreducible, Schur's lemma gives $S^{-1}S^T = \lambda \mathbf{1}$ for some $\lambda \in \mathbb{C}$, i.e.,

$$S^T = \lambda S. \quad (\text{B.3})$$

Transposing (B.3) gives $S = \lambda S^T = \lambda^2 S$, so $\lambda \in \{+1, -1\}$.

Step 3: λ is an invariant of R . Suppose S' is another invertible intertwiner satisfying (B.1). Then $S^{-1}S'$ intertwines R with itself, so by Schur's lemma $S' = \mu S$ for some $\mu \in \mathbb{C}^*$. Hence

$$(S')^T = \mu S^T = \mu \lambda S = \lambda S',$$

which exhibits the same sign λ : “real vs. pseudoreal” is a property of the representation itself, independent of the chosen intertwiner. $\square$

Remark B.2. $\text{Hom}_G(R, R^*)$ is one-dimensional by Schur's lemma, and transposition defines an involution on it; an involution on a one-dimensional space acts as ± 1 . The two eigenvalues correspond to the two possible G -invariant nondegenerate bilinear forms on V :

λ	invariant form	structure
+1	symmetric $B(u, v) = B(v, u)$	orthogonal (real)
-1	antisymmetric $B(u, v) = -B(v, u)$	symplectic (pseudoreal)

The correspondence is $B(u, v) := u^T S v$, which is G -invariant by (B.1) and has the symmetry dictated by $S^T = \lambda S$.

Remark B.3. When $\lambda = -1$, the antilinear map $J_R : V \rightarrow V$ defined by $J_R(\psi) := S \psi^*$ commutes with $R(g)$ and satisfies

$$J_R^2 = S S^* = -\mathbf{1},$$

using $S^T = -S$ together with unitarity of S (which may be assumed without loss of generality by rescaling). Squaring to $-\mathbf{1}$ is precisely the data of a quaternionic structure on V , which is why pseudoreal representations are often called quaternionic.²⁸

Remark B.4 (Frobenius–Schur indicator). The sign λ is computed by the Frobenius–Schur indicator

$$\nu(R) = \int_G \chi_R(g^2) dg,$$

where dg is the normalised Haar measure and $\chi_R = \text{tr } R$ is the character. One has

$$\nu(R) = \begin{cases} +1 & R \text{ real,} \\ -1 & R \text{ pseudoreal,} \\ 0 & R \not\cong R^* \text{ (complex).} \end{cases}$$

²⁸This is also what obstructs a Majorana-type reality condition: $\psi = J_R \psi$ would force $\psi = J_R^2 \psi = -\psi$.

Sketch. We decompose $\chi_R(g^2) = \chi_{\text{Sym}^2 R}(g) - \chi_{\wedge^2 R}(g)$ and use the fact that the trivial representation appears in $\text{Sym}^2 R$ iff there is a G -invariant symmetric bilinear form on V , and in $\wedge^2 R$ iff there is a G -invariant antisymmetric one. Schur's lemma guarantees that, for an irreducible self-conjugate R , exactly one of these occurs. $\square$

Application: the fundamental representation of $\text{USp}(2N)$ is pseudoreal

Realise $\text{USp}(2N)$ as the subgroup of $U(2N)$ preserving the symplectic form

$$J = \begin{pmatrix} 0 & \mathbf{1}_N \\ -\mathbf{1}_N & 0 \end{pmatrix}, \quad J^T = -J, \quad J^2 = -\mathbf{1}.$$

That is, $U \in \text{USp}(2N)$ iff $U^T J U = J$ and $U^\dagger U = \mathbf{1}$. Combining these two conditions,

$$U^T J = J U^{-1} = J U^\dagger = J (U^*)^T,$$

and transposing while using $J^T = -J$ gives

$$J U J^{-1} = U^* \quad \forall U \in \text{USp}(2N). \quad (\text{B.4})$$

Thus $S = J$ intertwines the fundamental representation with its conjugate, and $S^T = -S$. By Theorem B.1, $\lambda = -1$: **the fundamental representation of $\text{USp}(2N)$ is pseudoreal**. The case $N = 1$ recovers the well-known pseudoreality of the fundamental of $\text{SU}(2) = \text{USp}(2)$, with $S = i\sigma^2$.

C Highest-weight cyclic representations of $\mathfrak{sl}(3, \mathbb{C})$

In this appendix, we provide detailed proofs of the statements of the Theorem of the Highest Weight (Theorem 9.7), based on the notion of a highest-weight cyclic representation.

Proposition C.1. *Every irreducible representation (π, V) of $\mathfrak{sl}(3, \mathbb{C})$ is the direct sum of its weight spaces.*

Proof. Let W be the sum of all weight spaces in V . By Proposition 9.2, W is non-zero. Lemma 9.5 shows that applying any root vector operator $\pi(Z_\alpha)$ to a weight vector in W produces another vector in W . Since the root vectors and the elements H_1, H_2 span $\mathfrak{sl}(3, \mathbb{C})$, the subspace W is invariant under the entire Lie algebra. By irreducibility, we must have $W = V$. Since eigenvectors with distinct weights are linearly independent, this sum is a direct sum. $\square$

Definition C.2. *A representation (π, V) of $\mathfrak{sl}(3, \mathbb{C})$ is a **highest-weight cyclic representation with weight** $\mu = (m_1, m_2)$ if there exists a non-zero vector $v \in V$ such that:*

1. v is a weight vector with weight μ .
2. v is annihilated by all raising operators: $\pi(X_j)v = 0$ for $j = 1, 2, 3$.
3. The smallest invariant subspace of V containing v is all of V .

The vector v is also known as a **cyclic vector**.

Proposition C.3. *Let (π, V) be a highest-weight cyclic representation of $\mathfrak{sl}(3, \mathbb{C})$ with weight μ . Then:*

1. μ is the unique highest weight of the representation π .
2. The weight space corresponding to μ is one-dimensional.

Proof. Let v be the highest weight vector. Consider the subspace W of V spanned by elements of the form

$$w = \pi(Y_{j_1})\pi(Y_{j_2}) \cdots \pi(Y_{j_N})v \quad (\text{C.1})$$

with each j_l equal to 1, 2, or 3 and $N \geq 0$. (If $N = 0$, then $w = v$.) We claim that W is invariant. Applying any basis element of $\mathfrak{sl}(3, \mathbb{C})$ to w and using the commutation relations, we can rewrite the result as a linear combination of terms in which the $\pi(X_j)$'s act first (directly on v), the $\pi(H_j)$'s act second, and the $\pi(Y_j)$'s act last. Any term containing a $\pi(X_j)$ vanishes, since $\pi(X_j)v = 0$; the factors of $\pi(H_j)$ acting on v become constants; what remains is a linear combination of vectors of the form (C.1). Thus, W is invariant and contains v , so $W = V$.

Each application of a lowering operator $\pi(Y_j)$ subtracts a positive root from the weight. Thus, each element of the form (C.1) with $N > 0$ is a weight vector with weight lower than μ . This establishes that μ is the unique highest weight. Furthermore, any vector with weight μ must be a multiple of v , making the weight space one-dimensional. $\square$

Proposition C.4. *Every irreducible representation of $\mathfrak{sl}(3, \mathbb{C})$ is a highest-weight cyclic representation with a unique highest weight.*

Proof. Since the representation is finite-dimensional, the set of weights is finite and non-empty (Proposition 9.2), so there is a maximal weight μ_0 , i.e., one for which no other weight is strictly higher. Any weight vector v_0 for μ_0 satisfies $\pi(X_j)v_0 = 0$, since otherwise μ_0 plus a positive root would be a weight, by Lemma 9.5. By irreducibility, the smallest invariant subspace containing v_0 is all of V : the representation is highest-weight cyclic. Uniqueness of the highest weight follows from Proposition C.3. $\square$

Proposition C.5. *A completely reducible, highest-weight cyclic representation of $\mathfrak{sl}(3, \mathbb{C})$ is irreducible.*

Proof. Let (π, V) be a highest-weight cyclic representation with highest weight vector v . Since it is completely reducible, V decomposes into a direct sum of irreducible representations, $V \cong \bigoplus_j V_j$. The projections of v onto the components V_j are weight- μ vectors; since the μ -weight space of V is one-dimensional (Proposition C.3) and spanned by v , exactly one projection is non-zero, i.e., v lies in a single component V_k . But since V_k is an invariant subspace containing the cyclic vector v , it must be that $V_k = V$. Thus, there is only one term in the sum, and V is irreducible. $\square$

Proposition C.6. *Two irreducible representations of $\mathfrak{sl}(3, \mathbb{C})$ with the same highest weight are isomorphic.*

Proof. Suppose (π_1, V) and (π_2, W) are irreducible representations with the same highest weight μ . Let $v \in V$ and $w \in W$ be their respective highest-weight vectors. Consider the representation $V \oplus W$. The vector (v, w) is a highest-weight vector in this space. Let U be the smallest invariant subspace of $V \oplus W$ containing (v, w) . Then U is a highest-weight cyclic representation. Since $V \oplus W$ is completely reducible (being a direct sum of irreducibles), so is U . By Proposition C.5, U is irreducible.

Let $P_1 : V \oplus W \rightarrow V$ and $P_2 : V \oplus W \rightarrow W$ be the projections. Their restrictions to U are intertwining maps between irreducible representations, and they are non-zero, since $P_1(v, w) = v \neq 0$ and $P_2(v, w) = w \neq 0$. By Schur's Lemma, both are isomorphisms; thus, $V \cong U \cong W$. $\square$

Proposition C.7. *If π is an irreducible representation of $\mathfrak{sl}(3, \mathbb{C})$ with highest weight $\mu = (m_1, m_2)$, then m_1 and m_2 must be non-negative integers.*

Proof. By Proposition 9.3, m_1 and m_2 are integers. Let v be the highest-weight vector. Then $\pi(X_1)v = 0$ and $\pi(X_2)v = 0$. Applying Theorem 8.5 to the $\mathfrak{sl}(2, \mathbb{C})$ subalgebras generated by $\{H_1, X_1, Y_1\}$ and $\{H_2, X_2, Y_2\}$, we conclude that the eigenvalues m_1 and m_2 must be non-negative. $\square$

Proposition C.8. *For any pair of non-negative integers (m_1, m_2) , there exists an irreducible representation of $\mathfrak{sl}(3, \mathbb{C})$ with highest weight (m_1, m_2) .*

Proof. The trivial representation has highest weight $(0, 0)$. For the other highest weights, the representations are constructed explicitly in Section 9.2, using tensor products of the fundamental and antifundamental representations. $\square$

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